



UNIVERSITY OF SAN CARLOS
SCHOOL OF ENGINEERING

PHYSICS ENFORCEMENT ON MACHINE
LEARNING SURROGATES OF THE ACTIVATED
SLUDGE PROCESS
DISSERTATION

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Cebu City, Philippines, 2026
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Declaration

I hereby declare that the work presented in the dissertation entitled “Physics Enforcement on Machine Learning Surrogates of the Activated Sludge Process” is original and the result of my own research. Formulations and ideas taken from other sources are cited as such.

Cebu City, Philippines, 2026

Egberto F. Selerio Jr.

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Abstract

Activated sludge treatment links biological conversion, nutrient transformations, solids separation, and recirculation. Mechanistic models describe these relationships, but repeated simulation can be costly when many operating conditions are examined. Statistical surrogates offer faster response evaluation. Their usefulness depends on more than agreement with reference data because a prediction can have a small average error while violating mass conservation or producing negative component concentrations. This research develops a common framework for prediction, physical enforcement, interpretation, and operating assessment. A steady-state reactor benchmark compares several learning families through training-size studies, nested validation, and separately simulated beyond-domain conditions. A common post prediction procedure imposes stoichiometric mass conservation and non-negativity while allowing paired assessment of prediction error and projection size. A structured second-order regression separates operating effects, influent effects, curvature, and component coupling. Its checked output procedure distinguishes invariant-aware fitting from final-state feasibility. A connected plant surrogate extends enforcement to the mixer, reactor train, clarifier outlets, and solids inventory. Surrogate-selected operating controls are assessed through independent mechanistic solution and comparison with direct optimization under common engineering priorities. The theoretical and methodological framework treats predictive accuracy, physical admissibility, interpretability, and decision quality as distinct requirements. Its guarantees concern the declared process representation and imposed constraints. Empirical treatment outcomes and model performance are reserved for subsequent evaluation.

Keywords. Activated sludge process, machine learning surrogate, wastewater treatment, mass conservation, non-negativity, interpretable regression, operating optimization.

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Principal Symbols

Symbol	Meaning
F, P, K	Numbers of component states, reaction processes, and independent stoichiometric invariants
ν	Stoichiometric matrix with processes in rows and components in columns
ρ	Vector of mechanistic process rates
A	Full-row-rank stoichiometric invariant operator
V_0, Z	Orthonormal bases of the stoichiometric right null space and the null space of A
c_{in}, c_{out}	Influent state and mechanistic reference effluent state
$c_{raw}, \tilde{c}$	Raw component prediction and projected component prediction
I_{comp}	Fixed component-to-reported-composite map
D_i	Hydraulic dilution rate for the stated reactor and flow basis
ω, e_{SO}	Net dissolved-oxygen transfer rate and dissolved-oxygen coordinate selector
u	Standalone-reactor operating input vector
ϕ, B, Γ, R	Second-order feature vector, driver coefficients, learned component coupling, and coupled-system matrix
$\hat{C}, \Phi$	Auxiliary fitted component states and training feature matrix
N	Number of observations in the stated training-size design
n_r, L	Numbers of biological reactor stages and clarifier layers
ϑ	Connected-plant control vector
q_P, q_C, q_E, q_U	Flow ratios for the reactor train, clarifier feed, overflow, and underflow relative to fresh flow
m, c_i, c_E, c_U	Mixer, reactor-stage, overflow, and underflow concentrations
g_E, g_U	Flow-weighted clarifier outlet component responses
χ, M_{cl}	Joint plant response and clarifier solids inventory
D_χ	Positive scaling matrix for the plant response
J	Dimensionless engineering objective

Symbols specific to a formulation are defined with their equations. A numerical

coordinate is interpreted using the unit of its corresponding component. The plant projection uses u locally for its standardized response projection.

Chapter 1

Introduction: Research Problem and Purpose

1.1 Background and rationale

1.1.1 Wastewater treatment as an engineering need

Wastewater treatment is an engineering response to the movement of pollutants from communities and industries into receiving waters. Organic matter, nutrients, and suspended particles create different treatment requirements. Their effects depend on the receiving environment and the conditions of exposure. Madhav et al. (2020) describe the sources and environmental effects of water pollutants. Lin et al. (2022) examine their different health consequences. These concerns establish why treatment decisions require more than a single measure of effluent quality.

The need is relevant to the Philippines, where water management connects treatment infrastructure to institutions, financing, and local conditions. Tabios III (2020) discusses the physical setting and governance of Philippine water resources. Damarlerio et al. (2022b) identify technical, financial, social, and institutional barriers to wastewater management in developing countries and relate their discussion to Philippine technology selection. Their work supports a practical motivation for methods that help engineers compare treatment alternatives transparently. A faster calculation cannot resolve infrastructure or governance barriers by itself. It can support the technical assessment that informs planning and operation.

Treatment must also fit the resources available to the community or operator. The guide edited by Ulrich et al. (2009) links decentralized treatment to technical requirements and local implementation conditions. Jafarinejad (2020) places energy efficiency, cost, reliability, and sustainability within the same treatment-design problem. These sources explain why engineering relevance concerns the balance among treatment requirements and resource use. It does not follow from prediction accuracy alone.

Biological wastewater treatment uses microorganisms to transform organic matter

and nutrients. In activated sludge treatment, suspended biomass interacts with the liquid phase and is retained through solids separation and recirculation. Organic matter can be oxidized or incorporated into biomass. Nitrogen can move among ammonium, nitrite, nitrate, biomass, and nitrogen gas. Phosphorus can move among dissolved, stored, and particulate forms. Duan et al. (2022) describe biological nitrogen pathways, while Wentzel et al. (1992) explain the connections among nitrogen and phosphorus removal. The interactions make treatment a problem of several component responses.

The activated sludge model (ASM) family represents these responses through component balances and process rates (Henze et al., 1987, 1999, 2015). Chemical oxygen demand (COD) describes oxygen-demand equivalents associated with oxidizable material. Total nitrogen (TN), total phosphorus (TP), and total suspended solids (TSS) provide other engineering summaries. These reported quantities are important, but they do not identify a unique component state. Two effluents can have the same TN while containing different proportions of ammonium and nitrate. The same numerical total can therefore describe different treatment conditions.

Operating choices influence the entire response. Hydraulic retention time (HRT) relates reactor volume to the specified flow basis. Aeration supplies oxygen for biological reactions. Recycle and wasting influence the material returned to the treatment train and the solids retained in the plant (Henze et al., 2008). An operating decision consequently affects several components and locations at once. Understanding these relationships is necessary when a predicted response is used to compare candidate settings.

1.1.2 From process representation to repeated decision calculations

Mechanistic models connect operating inputs to treatment outcomes through the represented biological, chemical, hydraulic, and settling processes. Their value is the explicit relation between a response and the assumptions used to calculate it. Marais and Ekama (1976) establish steady-state activated sludge analysis, and Hauduc et al. (2013) review the process knowledge and limitations of activated sludge modeling. A model remains conditional on its component definitions, parameters, and system boundary.

Simulation supports design and operating analysis because it permits engineers to examine conditions that have not been directly observed. Calise et al. (2020) address model validation for an activated sludge process. RIVAS et al. (2008) connect model-based calculations to treatment-plant design. These applications show how a process representation becomes a decision tool. A candidate operating setting must be evaluated, its treatment response interpreted, and its feasibility compared with the intended engineering requirements.

Several routes can support these comparisons. An engineer can examine settings through repeated simulation and judgment. A numerical optimizer can search the mechanistic model directly. A statistical surrogate can approximate its input–output response and reduce the work of repeated evaluation (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019). Each route has a different relation to the reference equations. An acceptable simulated setting is not necessarily an optimal setting. An optimizer’s termination does not establish global optimality in a nonconvex treatment problem.

Aeration illustrates the need for repeated calculations. More oxygen supply can change substrate removal and nitrification, but it also changes a resource demand. de Araujo et al. (2013) examine operating sensitivities in an activated sludge optimization problem. Gu et al. (2023) review aeration optimization and control strategies. Choosing a setting requires assessment of the joint treatment response and the declared priorities. The preferred setting cannot be determined from an aeration variable or one effluent quantity alone.

Statistical surrogates make repeated evaluation less expensive by learning from paired inputs and reference responses. Durkin et al. (2024) use surrogates in wastewater resource-recovery optimization. Sundui et al. (2021) review biological wastewater applications of machine learning. These developments motivate the use of learning methods within engineering analysis. Their use also creates a need to assess the physical meaning and decision relevance of the predicted component state.

1.1.3 Why accurate prediction is insufficient

A prediction can be close to a reference response while violating a physical requirement. Consider a hypothetical component with a reference concentration of 0.2 g m^{-3} and a predicted concentration of -0.1 g m^{-3} . Its absolute error is only 0.3 g m^{-3} , but the prediction violates non-negativity. Large components can dominate an aggregate error score and conceal this problem. A small average error can also conceal a violation of mass conservation distributed across several components (D. Hansen et al., 2024; Kircher & Votsmeier, 2025).

Mass conservation and non-negativity therefore need separate, explicit checks. Mass conservation concerns the material relations implied by the represented reactions and system boundary. Non-negativity requires each concentration to be zero or greater. They must hold for the same returned state. Setting a negative component to zero can violate mass conservation, while a projection that imposes only mass conservation can leave a negative component.

Projection provides a defined procedure for imposing these requirements on a completed prediction. It uses the component state and stated constraints to calculate a returned state. Kircher and Votsmeier (2025) develop projection for simultaneous mass conservation and non-negativity in mechanistic kinetics surrogates. A projection can

be used with different underlying learners, but its guarantee concerns the imposed constraints. It does not establish that the returned state reproduces all reaction rates or that projection improves every prediction error.

Training and validation also require a clear distinction between available inputs and unknown responses. An effluent quantity that is unavailable when an operating decision is made cannot be supplied as an ordinary predictive input. Training-dependent transformations and model selection must also remain separate from final assessment information (Kaufman et al., 2012; Newhart et al., 2019). Preventing information leakage makes the assessment more credible. It does not by itself establish accurate extrapolation or agreement with an operating plant.

Interpretability addresses another requirement. A structured regression can expose operating terms, influent terms, curvature, and component coupling. Such a representation allows engineers to inspect the basis of a prediction (Brunton et al., 2016; Rudin, 2019). The terms describe the fitted response and need not be uniquely identified biological mechanisms. Projection can change that response and its local sensitivity. Interpretation therefore needs to distinguish the raw regression from the final projected state.

The decision problem adds a further concern. An optimizer searches deliberately for favorable predicted behavior. It can select a condition at which the surrogate is inaccurate even when average held-out error is small. Alexandrov et al. (1998) develop approximation management for optimization, and Pedrozo et al. (2025) compare surrogate optimization with reference process calculations. Their work supports independent assessment at selected controls. Mass conservation and non-negativity remain necessary properties of a component prediction, while decision quality requires additional evidence.

1.1.4 Rationale for the research

The research is motivated by the need for a fast treatment prediction that remains understandable and usable as a component state. Existing literature supplies methods for statistical approximation, projection, and process optimization. The engineering problem is to connect these methods while preserving the different questions each one answers. Prediction error describes agreement with a reference. Mass conservation and non-negativity describe specified physical requirements. Interpretability concerns the basis of the returned response. Decision quality concerns the operating choice evaluated with the reference model.

This distinction matters at both reactor and plant scale. For one reactor, projection can impose mass conservation and non-negativity on the full component prediction. For a connected plant, mixing, reactor, clarifier, recycle, and inventory relationships must also agree across the returned response. A physically acceptable state at one unit does not establish consistency of its connections to other units.

The research therefore uses a common mechanistic foundation to examine three linked capabilities. The first is statistical approximation and projection of the complete reactor component response. The second is an interpretable second-order regression with a defined projection procedure. The third is a connected plant surrogate that supports operating analysis and independent evaluation of selected controls. These capabilities address a progression from prediction to physical constraints and then to engineering decisions.

The research is intended to support more transparent treatment analysis. Researchers can distinguish prediction error from the effect of projection. Engineers can inspect the returned component state and evaluate a selected operating setting with the original process model. A valid projection establishes mass conservation and non-negativity under its stated assumptions. Accuracy, computational cost, and decision quality remain separate assessment questions.

1.2 Statement of the problem

The central problem is how to use machine learning surrogates for activated sludge analysis when accurate approximation alone is insufficient for a usable engineering response. The research addresses four connected limitations.

1.2.1 Uncertain agreement with the complete component response

An error score for one reported effluent quantity does not establish agreement with the full component state. Small components and compensating component errors can remain hidden. Performance can also depend on training size and on whether the input lies within the sampled domain. A common component-level assessment is needed to distinguish these effects without changing the prediction target among models.

1.2.2 Mass conservation and non-negativity are not established by training alone

Ordinary regression objectives do not impose every sample’s mass conservation equality or concentration bound. Physical residual penalties can guide a fit, but a finite penalty does not establish exact compliance at every new input (D. Hansen et al., 2024; Raissi et al., 2019). A defined projection and independent numerical checks are needed to assess both requirements on each returned state.

1.2.3 A visible regression structure can still be difficult to interpret

Coupled coefficients, nonlinear terms, and projection all influence a returned prediction. A named coefficient cannot be interpreted reliably without identifying its input basis and its position in that prediction procedure. The model also requires a fitting and validation design that uses only information available for the intended prediction. Interpretability and information separation must therefore be addressed together.

1.2.4 Unit predictions do not establish the quality of plant decisions

Separately predicted unit responses can disagree at their physical connections. Joint projection can impose specified plant relationships, but it does not replace all nonlinear reaction and settling equations. An operating search can exploit these remaining approximation errors. Selected controls consequently require independent mechanistic evaluation under common engineering priorities.

1.3 Research questions and objectives

The general objective is to develop a unified framework for activated sludge surrogate prediction, projection, interpretation, and operating assessment. The framework treats mass conservation and non-negativity as explicit requirements and keeps predictive error and decision quality as separate endpoints.

The first question asks how accurately different statistical learners approximate the complete reactor component response as training data change and inputs move beyond the sampled domain. Its objective is to establish a common prediction benchmark with shared targets, data partitions, and evaluation scales.

The second question asks how projection affects mass conservation, non-negativity, error, displacement, and computational cost for the same raw predictions. Its objective is to compare raw and projected states in paired assessments. The comparison identifies the effect of projection without attributing it to a different fitted learner.

The third question asks how an explicit second-order driver and learned component coupling can support interpretable prediction with mass conservation and non-negativity. Its objective is to formulate and examine the regression, its fitting problem, and its projection procedure. Interpretation distinguishes the driver, the raw component map, and the final projected response.

The fourth question asks how joint prediction and projection can support operating choices for a connected activated sludge plant. Its objective is to impose the declared unit connections, formulate common engineering priorities, and independently evalu-

ate selected controls. Surrogate-assisted search is compared with direct mechanistic optimization on the original reference basis.

1.4 Significance of the study

1.4.1 Scientific significance

The scientific value is a clear connection between learned treatment responses and the component accounting of the mechanistic model. Mass conservation follows from stated stoichiometric and boundary relations. Non-negativity applies to the concentrations represented by that model. Their explicit treatment makes the physical meaning of a predicted state available for examination.

This contribution also clarifies the limit of a physical guarantee. A state that satisfies mass conservation and non-negativity need not satisfy the full nonlinear kinetic equations. Likewise, a reported nutrient total need not be identical to an inventory used for mass conservation. Identifying these distinctions supports a more precise interpretation of scientific machine learning in biological wastewater treatment (Henze et al., 2015; Takács & Vanrolleghem, 2006).

1.4.2 Methodological significance

For researchers comparing activated sludge surrogates, the study provides separate evidence for prediction and projection. Each comparison uses common inputs, targets, and assessment scales within its stated design. Training-dependent preprocessing and tuning preserve information separation. Paired raw and projected predictions identify which changes arise from projection. The reactor and plant assessments retain their own validation designs rather than implying one common model ranking.

The design also separates component errors, reported-composite errors, mass conservation residuals, non-negativity violations, projection displacement, and computational effort. These quantities describe different aspects of model performance. Their joint reporting can prevent an accurate-looking aggregate score from being interpreted as sufficient evidence of physical or engineering usefulness.

The structured regression adds an inspectable model representation. Its named terms and component coupling permit direct coefficient and sensitivity calculations. Projection is then analyzed as a separate part of the returned response. The benefit is transparency of the calculation and its limits, rather than an unsupported claim that fitted coefficients are causal biological parameters.

1.4.3 Engineering and practical significance

Engineers need to compare operating conditions while maintaining a consistent account of material across the treatment plant. The connected surrogate represents mixer, reactor, clarifier-outlet, and inventory responses together. Joint projection addresses contradictions that separate unit predictions can leave unresolved. The resulting numerical checks provide a basis for detecting specified mass conservation and non-negativity violations before a prediction enters an operating search.

Independent mechanistic evaluation gives the operating assessment a practical reference. It determines treatment responses at the controls actually selected, checks the original engineering requirements, and distinguishes available decisions from failed or ineligible cases. This is useful for assessing whether a faster search supports the intended decision. It does not establish lower operating costs or better field treatment before those outcomes are evaluated.

The framework is also relevant to technical capacity in wastewater planning. Philippine and other resource-limited settings require decisions that connect treatment needs with the resources and conditions of implementation (Damalerio et al., 2022b; Tabios III, 2020; Ulrich et al., 2009). A transparent, computationally manageable assessment can support such decisions. Plant calibration, reliable measurements, operating expertise, and institutional capacity remain necessary for practical application.

1.4.4 Significance for further research

The formulations provide a defined starting point for examining uncertainty, dynamic operation, and other biological treatment configurations. Their value is the explicit separation of physical constraints, statistical agreement, interpretation, and decision evidence. Extensions must define their own state representation and boundary conditions. They cannot inherit the reactor or plant guarantees solely because they use the same learning method.

1.5 Contributions of the research

The first contribution is a common reactor assessment that retains the complete component target. It examines learning behavior under specified data availability and input-domain conditions. Component and composite scores have distinct roles, so a favorable reported total cannot conceal every defect in the component state.

The second contribution is a paired projection assessment for mass conservation and non-negativity. It adopts the projection principles of Kircher and Votsmeier (2025) and applies a common physical requirement across different learners. Mathematical feasibility, projection displacement, prediction error, and cost are evaluated separately.

The contribution concerns their organized use in activated sludge analysis and does not claim invention of the underlying projection method.

The third contribution is an interpretable second-order regression with learned component coupling and checked projection. The explicit feature groups support structural inspection. The fitting formulation distinguishes soft physical penalties from the projection used to impose mass conservation and non-negativity on the returned state. Interpretation follows the complete response calculation rather than an isolated coefficient block.

The fourth contribution is a connected plant formulation with joint projection and independent operating verification. It links unit responses through the declared mixing, reaction-inventory, clarification, and stored-solids relations. Selected controls are then assessed with the original mechanistic equations and a common objective. This connects the physical meaning of a surrogate state to the quality of the decision made with it.

Table 1.1 makes the problem–contribution relationship explicit. The evidence column identifies how each contribution is assessed. It does not report an empirical outcome.

Table 1.1: Research problems, methodological contributions, and assessment evidence.

Research problem	Contribution	Assessment evidence
Complete response is hidden by a few reported totals	Common component-level reactor benchmark	Held-out component and composite errors, learning curves, and beyond-domain assessment
Training does not establish mass conservation and non-negativity	Common projection with paired raw and projected assessment	Equality residuals, concentration checks, error differences, displacement, and cost
Model terms do not directly explain the final response	Explicit second-order regression and component coupling with checked projection	Named coefficient groups, full response sensitivities, and stage-specific diagnostics
Consistent predictions do not establish useful operating decisions	Joint plant projection and original-model verification of selected controls	Unit and inventory checks, common reference objectives, engineering eligibility, and failure accounting

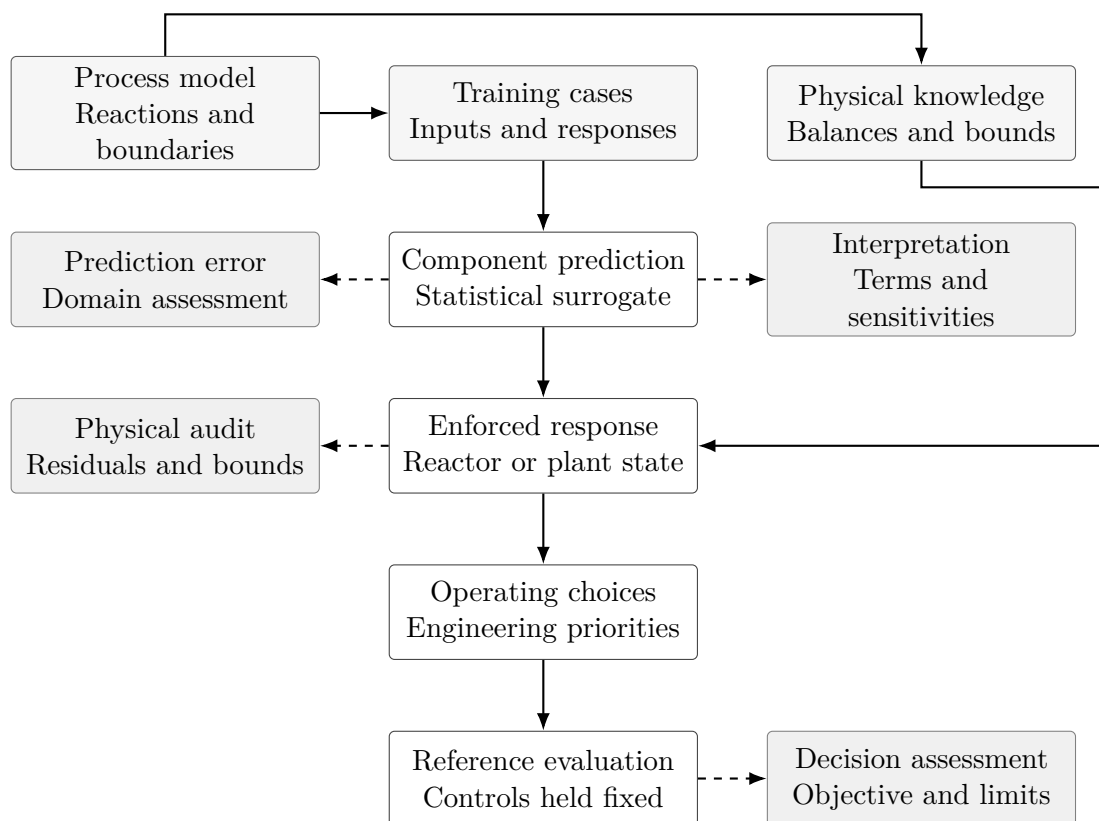
These contributions are mathematical and methodological. Their significance is the capacity to address the stated limitations through a coherent assessment. Empirical superiority over another learner or optimization route is not presumed.

1.6 Research framework

The research logic begins with a mechanistic representation and a declared operating domain. Paired inputs and component responses support statistical fitting. Reaction structure and boundary accounting supply mass conservation relations. Non-negativity supplies the concentration requirement. Projection connects a raw prediction to a returned state that is checked against those requirements.

The logic extends to interpretation and operating analysis. A structured predictor exposes terms and sensitivities. A connected plant formulation adds relationships among unit responses and stored solids. An operating search selects controls, and independent mechanistic evaluation establishes their reference treatment response. Each step addresses a specific claim. The evidence required for one claim does not substitute for the evidence required for another.

Figure 1.1 links the information used to construct the prediction procedure to the evidence used to assess its engineering value. The central sequence concerns prediction, projection, operating choice, and reference evaluation. The side boxes retain the different assessment questions.



Solid arrows connect modeling steps. Dashed arrows identify assessment.

Figure 1.1: Conceptual research logic linking process knowledge, component prediction, projection, interpretation, and independent assessment of operating choices.

Mechanistic cases supply supervised targets, while physical knowledge supplies the

requirements used by projection. Error assessment tests agreement with the reference. Physical audits test mass conservation, non-negativity, and the additional declared plant constraints. Structural inspection examines a transparent fitted representation. Decision assessment evaluates selected controls with the original mechanistic model. The reactor and connected plant use this logic with separately specified predictors and physical boundaries.

1.7 Scope and limitations

The study concerns simulated steady-state activated sludge responses. It includes a standalone reactor and a connected plant with biological reactors, secondary clarification, and recycle streams. The mechanistic equations supply reference responses under fixed process definitions and parameter values. The reactor and connected plant have separately specified operating domains.

The state includes organic, nitrogen, phosphorus, oxygen, alkalinity, biomass, and solids coordinates within the adopted model basis. Mass conservation is imposed through that basis and the stated system boundary. Non-negativity is imposed on the represented concentrations. The guarantees do not establish complete elemental accounting for unrepresented species, full kinetic feasibility of every projected state, or dynamic stability beyond the specified checks.

Agreement with simulated responses establishes agreement with the selected mechanistic reference. It does not establish performance at an operating treatment plant. Calibration uncertainty, measurement error, microbial adaptation, and unrepresented reactions require additional field evidence (Machado et al., 2014; Newhart et al., 2019). Reliable calibration and process understanding remain essential (Meijer, 2004; Petersen et al., 2003).

Exploratory simulation informed the connected plant assessment design. Its separate holdout therefore supports descriptive post-selection assessment rather than independent external confirmation. Fitting and tuning still exclude the holdout responses. Independent mechanistic evaluation checks the selected controls, but it does not remove this statistical limitation.

Attached-growth treatment, spatial transport, resource-recovery networks, and dynamic control provide relevant background and possible extensions. They are not additional evaluated case studies. The connected operating problem uses the declared topology and local numerical assessment. It does not certify a global optimum or a new network configuration.

Numerical examples and conceptual figures explain the mathematical operations and are explicitly hypothetical. They do not report treatment, prediction, optimization, or timing findings. The empirical assessment remains separate from the present theoretical and methodological presentation.

Chapter 2

Review of Related Literature

2.1 Treatment mechanisms and the meaning of model outputs

Water quality is shaped by pollutant sources, transport, exposure, and the receiving environment. Madhav et al. (2020) organize pollutants by their sources and effects, while Lin et al. (2022) examine the different health consequences associated with polluted water. These works support a treatment assessment that considers several pollutant groups. They also explain why reducing a single effluent quantity cannot describe every environmental consequence of a treatment decision.

Biological treatment makes this assessment a problem of interacting transformations. Duan et al. (2022) review heterotrophic nitrification and its role within biological nitrogen removal. The review shows that a process label can contain several microbial pathways. Wentzel et al. (1992) connect nitrification, denitrification, and excess biological phosphorus removal through process conditions and model structure. For a surrogate, the implication is that nitrogen and phosphorus predictions belong to a coupled treatment response. Learning each reported total independently can omit relationships that matter when operating conditions change.

The ASM family provides a systematic framework for these relationships. Henze et al. (1999) describe the component and process structure of the model that includes biological phosphorus removal and denitrification by phosphorus-accumulating organisms. The collected models of Henze et al. (2015) place this development alongside other activated sludge formulations. Henze et al. (2008) provide the broader treatment principles needed to relate biological models to engineering design. These sources anchor the separation among the component state, the reaction model, and reported treatment indicators.

Carbon availability provides a concrete example of coupling. Guerrero et al. (2011) examine competition between phosphorus-accumulating organisms and denitrifiers under different carbon sources. Their work makes the identity and availability of carbon

relevant to nutrient removal, rather than treating organic loading as a single interchangeable number. A component surrogate should therefore retain distinctions among fermentable material, readily available substrates, storage products, and biomass. The inclusion of interaction terms has a process rationale because the effect of one input can depend on another.

Oxygen supply has a similar role. Stenstrom and Song (1991) study the effect of oxygen transport limitations on nitrification. This distinguishes oxygen transfer from the biological capacity to consume oxygen. A model can include both and still give different responses when their relative limitations change. An educational explanation of aeration should therefore identify what is controlled, how oxygen enters the balance, and which reaction rates depend on dissolved oxygen.

The state of the biomass also depends on its operating history. Yilmaz et al. (2007) examine how alternating aerobic and anoxic or anaerobic conditions affect biomass activity during starvation. Their setting differs from a steady-state surrogate, but it identifies a limit of that surrogate. A mapping from current influent and operating conditions to one steady-state response does not represent all effects of prior exposure or prolonged substrate deprivation. Such evidence supports a precise boundary on claims about dynamic operation.

Model detail creates a calibration problem as well as a prediction problem. Machado et al. (2014) compare parameter identifiability with uncertainty in influent and operating conditions during full-scale model calibration. A close match to measurements does not uniquely identify every kinetic or stoichiometric parameter. This matters when interpreting a fitted surrogate. The surrogate approximates a response conditioned on the parameter values used to generate its training data. Its coefficients cannot be taken as newly identified biological constants.

2.2 Development and calibration of activated sludge models

The history of activated sludge modeling explains why a component response has an explicit scientific meaning. Marais and Ekama (1976) develop steady-state treatment relationships that connect biomass, substrate, and operating conditions. Dold et al. (1981) extend the representation toward a general process model. These works establish a distinction between an observed effluent total and the internal quantities used to calculate that total. A surrogate intended for component prediction inherits that distinction.

Henze et al. (1987) describe a general single-sludge formulation associated with Activated Sludge Model No. 1 (ASM1). Its process structure separates substrate use, biomass growth, decay, and nitrogen transformations. “Activated Sludge Model No. 3” (1999) develop Activated Sludge Model No. 3 (ASM3) with an explicit role for intra-

cellular storage. The choice between these representations changes which intermediate states are available for prediction. A statistical model cannot infer a uniquely defined hidden storage state unless the supervised representation defines that state.

Nutrient-removal models extend the same principle. Barker and Dold (1997) present a general biological nutrient-removal model, while Hu et al. (2007) develop a kinetic treatment of the linked processes. The phosphorus-removal formulation of Henze et al. (1999) adds another established component and process description. These sources support joint treatment of carbon, nitrogen, and phosphorus. They also show why a model name does not make every parameter or microbial pathway interchangeable.

Rieger et al. (2001) connect biological phosphorus removal to the storage-based formulation. Fall et al. (2015) modify an activated sludge model for low solids production. These contributions identify different purposes for model extension. Adding a process or changing a yield can alter the component relationships and the corresponding mass conservation conditions. An invariant operator must therefore be derived from the adopted stoichiometric matrix rather than borrowed from a model with similar names.

Model development has a practical history as well as a mathematical one. Van Loosdrecht et al. (2015) review the development and use of ASM1 over twenty-five years. Hauduc et al. (2013) examine the state of process knowledge, modeling concepts, and limitations. Their discussions support deliberate selection of a representation for a defined question. More states can describe additional mechanisms, but they also require additional information for calibration and assessment.

Petersen et al. (2003) review experimental designs for activated sludge calibration. Vanrolleghem et al. (1999) use respirometry to estimate model parameters and components. These studies connect a model quantity to the measurements that can support it. An oxygen-use observation can inform biological activity, but it is not a direct observation of every modeled component. The same issue arises when a surrogate is assessed against a few reported water quality measurements.

Meijer (2004) examines the practical validation and calibration of a metabolic phosphorus-removal model. The work distinguishes steady-state and dynamic information in testing a model. Nelson and Sidhu (2009) analyze ASM1 mathematically. Together, these sources make clear that calibration, solution of the equations, and analysis of the resulting state are separate responsibilities. A good fit, a converged numerical solve, and an accepted physical state need not provide the same evidence.

The empirical biomass formula discussed by Hoover and Porges (1952) supplies a useful historical example of constituent accounting. Biomass growth and decay redistribute organic matter and nitrogen, so a solids concentration cannot be treated as chemically empty material. Takács and Vanrolleghem (2006) examine elemental balances in activated sludge models and the role of omitted reaction products. Their contribution is especially relevant to projection. Mass conservation in an adopted lumped-component basis does not automatically establish complete elemental account-

ing for unrepresented water, carbon dioxide, or other boundary terms.

2.3 Biological pathways and the boundaries of a model

The functional role of a microorganism depends on both its energy source and its carbon source. Heterotrophic growth uses organic material, while nitrifying chemolithoautotrophs obtain energy from inorganic nitrogen transformations and use inorganic carbon for synthesis. Narayanan and Narayan (2019) review biological treatment and bioreactor design. Preena et al. (2021) review nitrification and denitrification in recirculating aquaculture. The latter setting differs from municipal activated sludge, but the distinction among electron donors, electron acceptors, and nitrogen forms remains useful.

Two-step nitrification separates ammonium oxidation from nitrite oxidation. It is an engineering representation of functional conversions rather than a universal rule about how many organisms perform them. van Kessel et al. (2015) establish complete nitrification within one microorganism. Their work explains why an explicit two-population model is a chosen process representation. It should not be interpreted as excluding every biological route that combines those functions.

Lu et al. (2014) review the microbial ecology of denitrification in biological wastewater treatment. Carbon availability, electron-acceptor conditions, and microbial populations jointly influence the nitrogen response. This supports retaining carbon and nitrogen components together when learning a surrogate. It also explains why the amount of reported nitrogen can change through conversion and separation even when the complete modeled nitrogen inventory satisfies mass conservation.

Anaerobic ammonium oxidation provides a different nitrogen pathway. Strous et al. (1998) study slowly growing anaerobic ammonium-oxidizing microorganisms. Dietl et al. (2015) investigate the hydrazine synthase complex, while Maalcke et al. (2016) examine the enzyme involved in nitrogen-gas production from hydrazine. These works clarify that a shared final nitrogen product does not make the underlying process identical to heterotrophic denitrification. The adopted reactor model does not contain an explicit anaerobic ammonium-oxidation population or its full kinetic pathway.

Temperature and treatment purpose create further boundaries. Vandekerckhove (2018) investigate thermophilic nitrogen removal, and Xie (2021) develops strategies for nitrogen conversion, recovery, and removal. Winkler and Straka (2019) review newer directions in biological nitrogen management. Their relevance is the need to identify which nitrogen transformation and engineering goal a model represents. A parameterized steady-state activated sludge surrogate does not acquire these additional capabilities through mass conservation and non-negativity alone.

2.4 Suspended-growth and attached-growth treatment

Suspended-growth and attached-growth systems organize biological activity differently. A completely mixed activated sludge reactor uses one bulk concentration vector for its control volume. A biofilm or packed treatment medium can contain spatial gradients and retained biomass at different locations. Wik (2003) reviews trickling filters and biofilm modeling. T. C. Zhang and Bishop (1994) examine biofilm density, porosity, and pore structure. These contributions explain why transport and geometry can affect reaction rates even when the same nitrogen or organic-matter conversion is considered.

“A novel and cost-effective sewage treatment system consisting of UASB pre-treatment and aerobic post-treatment units for developing countries” (1997) develop a combined anaerobic and aerobic treatment setting for developing countries. Tawfik et al. (2006) examine combined anaerobic treatment and a down-flow sponge system. Tawfik et al. (2010) investigate the role of sponge volume in that downstream process. These studies connect treatment configuration and retained medium to the response being predicted. A medium volume or transport property is a different input from the hydraulic and aeration controls of a completely mixed reactor.

Mahmoud et al. (2010) study a naturally ventilated bio-tower for organic and nutrient treatment, and Mahmoud et al. (2011) examine a sponge reactor for municipal wastewater post-treatment. Hatamoto et al. (2018) relate sponge-reactor structure to process function and microbial communities. These contributions establish attached-growth treatment as a relevant engineering context. They do not supply additional evaluated cases for the present activated sludge benchmark.

Hellal and Abou-Elela (2021) simulate a passively aerated biological filter, while Liang et al. (2021) model a three-stage biological trickling filter. “Dynamic simulation of a low loaded trickling filter for nitrification” (1999) examine dynamic nitrification in a low-loaded trickling filter. Their different configurations show why stage structure, spatial transport, and time dependence need to be identified before model responses are compared. A surrogate can be fast while approximating a different physical problem from the one intended.

Nutrient form also matters in these systems. Kasi et al. (2011) model dissolved organic nitrogen in a two-stage trickling filter, and Simsek et al. (2012) examine its fate. Bressani-Ribeiro et al. (2021) investigate inorganic carbon limitation during nitrogen conversions in sponge-bed treatment. These works connect prediction targets to constituent forms and growth requirements. They reinforce the need to retain the component basis rather than treating one measured nitrogen total as a complete process state.

Díez-Montero et al. (2019) assess a trickling-filter upgrade for nutrient removal.

Luan et al. (2023) examine nitrification and denitrification in a rotating self-aerated biofilm reactor. Both illustrate the engineering importance of configuration-specific process relationships. A statistical predictor trained for one hydraulic and biological arrangement requires new evidence before transfer to another.

The resource setting is also relevant. Tyagi et al. (2021) review the energy-saving motivation for sponge treatment, and Nasr et al. (2022) discuss its decentralized applications. Rapi et al. (2021) apply biological sponge treatment to palm oil mill effluent. These sources broaden the context of treatment selection while keeping the wastewater type explicit. Industrial and municipal influents can differ in their constituent distributions and treatment demands.

Mahmoud et al. (2018) consider post-treatment of anaerobic effluent containing a specific organic contaminant and heavy metals. Their setting illustrates a further limitation of a declared component model. The presence of metal-hydroxide and metal-phosphate coordinates does not make an activated sludge surrogate a general predictor of toxic-metal removal or every organic contaminant. The represented species and reactions determine that capability.

Transport modeling can join reaction kinetics with flow and diffusion. Sadino-Riquelme et al. (2023) review how biological kinetics are integrated with computational fluid dynamics. The concern is the spatial location and transport of reacting material. The current standalone reactor and connected plant use their declared completely mixed stages and clarifier representation. Spatial treatment models provide a useful boundary comparison and a possible extension, rather than a replacement for those reference equations.

2.5 Why treatment analysis calls for surrogate models

Treatment design often contains competing objectives and coupled decisions. Padrón-Páez et al. (2020) use multiobjective optimization to address sustainable wastewater treatment plant design. Aboagye et al. (2021) link wastewater network design to optimization and sustainability assessment. Their contributions show why treatment evaluation involves repeated comparisons among alternatives. The computational burden belongs to the decision problem as well as to an individual simulation.

Sensitivity analysis gives another reason for repeated evaluation. de Araujo et al. (2013) examine optimal operation and the selection of economic controlled variables in an activated sludge model. The work connects operating sensitivities to the quantities that are useful for control. A fast approximation can assist this analysis, but it must retain the correct input definitions and reporting basis. A sensitivity with respect to reactor volume is different from a sensitivity with respect to a residence time that uses

a recycle-adjusted flow.

The review of Bhosekar and Ierapetritou (2018) treats surrogate construction, feasibility analysis, and optimization as connected tasks. It emphasizes the importance of sampling, model selection, and validation for the intended application. McBride and Sundmacher (2019) place surrogate modeling within chemical process engineering and distinguish different ways of approximating an expensive response. These reviews support a broad comparison of learning methods. They do not imply that one statistical family is best for every response or data regime.

Fang et al. (2011) develop a simulation-based integrated approach to biological nutrient-removal optimization in a full-scale wastewater treatment plant. Their contribution links operating choices to the coupled nitrogen and phosphorus response through a process model. It also identifies the reference calculation that a statistical surrogate seeks to reduce. The surrogate can approximate repeated evaluations, but the original component and treatment relationships remain the basis for assessing an operating choice.

Wastewater resource recovery provides a direct application. Durkin et al. (2024) combine process surrogates with optimization of resource recovery from wastewater. The statistical approximation is useful because it participates in a larger constrained decision problem. He et al. (2023) use a Kriging surrogate in a papermaking wastewater treatment application with environmental objectives. Both studies connect faster response evaluation to engineering choice. Their relevance is methodological even when the treatment processes and outputs differ from an activated sludge component surrogate.

Recent network work extends this concern across treatment stages. Tang et al. (2026) develop surrogate-based design for a multistage coal chemical wastewater treatment network. A network surrogate must preserve the meaning of the connections among units as well as the response of each unit. Tsochatzidi et al. (2025) develop a surrogate optimization framework for pharmaceutical process systems. That work is outside wastewater treatment, but it illustrates the same distinction between approximation of process responses and optimization over the resulting model. Transfer to activated sludge requires the treatment-specific component and hydraulic constraints.

Reduced models can also be chosen around quantities that matter to operation. Straus and Skogestad (2018) generate surrogates using self-optimizing variables. Their contribution concerns the connection between reduced representation and decision purpose. A surrogate that is adequate for one operating objective can be inadequate for another. This supports retaining complete component predictions when the later objective may combine nutrients, organic matter, solids, and resource use.

2.6 From local approximations to nonlinear data-driven prediction

Local approximation has a clear engineering role. M. Li et al. (2020) use fuzzy clustering in dissolved oxygen model predictive control, while T. Yang et al. (2014) develop fuzzy model-based predictive control for the activated sludge process. Such approaches organize nonlinear behavior through local models or operating regimes. Their usefulness depends on how well the local representation covers the conditions encountered in operation. They provide an alternative to fitting one highly flexible global predictor.

Polynomial and response surface methods make nonlinear terms explicit. Kim et al. (2009) develop a dual optimization strategy for nitrogen and phosphorus removal. Lim et al. (2011) examine operating conditions for organic matter and nitrogen removal in a membrane bioreactor. Ahn et al. (2014) compare modeled and experimental operating factors for livestock wastewater treatment. These works demonstrate why transparent approximations remain useful in treatment analysis. The terms have an inspectable structure, but the selected polynomial degree and sampled domain limit the behaviors they can represent.

Nazlabadi et al. (2021) review the application of response surface methodology to biological wastewater treatment. The review places polynomial approximations within a wider design and analysis practice. A low-order surface can reveal curvature and input interaction without reproducing every mechanistic detail. It should therefore be assessed as an approximation over a stated domain. This perspective supports second-order regression while preventing a claim that a quadratic structure is a complete reaction model.

Partial least squares provides another transparent starting point. Wold et al. (2001) explain how latent variables relate predictors to responses in chemometric regression. Khatri et al. (2021) review its use in water quality analysis. The method is relevant when influent variables are correlated and when several outputs are modeled together. Its basic linear relationship still depends on whether the predictor representation captures the nonlinear behavior of interest.

Extensions address this limitation in different ways. Woo et al. (2009) use kernel partial least squares to estimate process variables in industrial wastewater treatment. Liu et al. (2021) use a dynamic concurrent kernel approach for wastewater process monitoring. C. Yang et al. (2021) embed a relevance vector model within a partial least squares framework for adaptive effluent prediction. These contributions show that latent structure, nonlinearity, and changing process behavior can be combined. They also separate dynamic monitoring from the steady-state input–output task considered here.

The wider evidence on wastewater data use is reviewed by Newhart et al. (2019).

Their discussion makes data quality, process knowledge, and validation important parts of performance analysis. Ba-Alawi et al. (2021) address influent sensor validation through stacked denoising autoencoders. Their contribution concerns the credibility of the inputs before a prediction is made. An accurate model cannot compensate for every error in a sensor signal or a mislabeled input quantity.

Bagherzadeh et al. (2021) examine total nitrogen prediction and feature selection in a wastewater treatment plant. Sadri Moghaddam and Mesghali (2023) develop a hybrid ensemble for total nitrogen prediction in a combined trickling filter and activated sludge system. These studies demonstrate the value of statistical prediction for an operational indicator. Their output target also clarifies a limitation. Evidence for one total nitrogen model does not establish mass conservation or non-negativity of the complete component vector.

Recent studies connect prediction to other practical indicators. Y. Wang et al. (2025) compare interpretable learning models with a mechanistic model for effluent nitrogen prediction. S. Yang et al. (2025) examine waste activated sludge generation for operational efficiency. Xiao et al. (2026) combine mechanistic dynamics and an interpretable surrogate in activated sludge management. Together, these applications broaden the role of learning from effluent estimation to sludge and operating analysis. Their different targets require careful comparison because an error measure over one indicator is not an error measure over the full state.

Asadi et al. (2017) use data mining in aeration optimization, and Mao et al. (2024) use artificial intelligence to balance effluent quality and aeration energy. These works give a concrete decision context for prediction. Oxygen delivery can support treatment while consuming resources. A useful operating surrogate must represent this relationship well enough at the selected decision, rather than only on average over randomly held-out inputs.

2.7 Prediction targets and information available to a decision

Wastewater prediction studies often address a reported quantity rather than a complete mechanistic state. Ching et al. (2022) develop a soft sensor for five-day biochemical oxygen demand. Manav-Demir et al. (2024) compare learning methods for nutrient-removal effluent parameters. These applications demonstrate the relevance of prediction to treatment monitoring. Their supervised target determines what an error score can establish. Agreement with one reported quantity does not establish mass conservation or non-negativity across the full component vector.

Ekinici et al. (2023) examine feature selection for sludge-production prediction. P. Wang et al. (2024) predict COD component parameters for activated sludge modeling. These contributions concern different output roles. A sludge production quantity, a

wastewater fractionation parameter, and an effluent concentration are not interchangeable labels. The predictor inputs must also match the information available when each quantity is needed.

Poch (1993) provides an observational treatment-plant dataset for classifying plant state and detecting faults. It contains measurements from different stages and performance quantities. A measurement can be a legitimate input for an operational diagnosis while remaining unavailable for prospective design or steady-state prediction at an untested setting. The dataset consequently helps explain information availability. It is not the training collection used for the current reactor or plant surrogates.

Caro et al. (2024) describe automated acquisition of simulator responses. Their contribution is relevant to the organization of paired model-generated observations. Such acquisition does not establish agreement with field measurements or eliminate the need for independent physical and numerical checks. The review of Sundui et al. (2021) places learning applications across biological wastewater tasks and similarly supports identifying the data source and intended output before comparing models.

Full-scale mechanistic applications provide complementary evidence. Wu et al. (2016) simulate and optimize a coking wastewater process with activated sludge models. Sadri Moghaddam et al. (2021) address calibration of a full-scale plant model, and Jasim (2020) develops a treatment-plant design model. These studies connect a model to a particular process setting. Their calibration does not remove the need to assess a surrogate separately on its sampled operating domain.

Szeląg et al. (2022) examine nutrient removal and energy consumption under uncertainty. Nazif et al. (2023) formulate practical operating analysis with attention to influent characteristics. These contributions strengthen the decision context of model assessment. The operating response depends on the influent and parameter assumptions, so a selected decision should be evaluated at the stated conditions. Physical constraints do not remove every uncertainty in those conditions.

2.8 Learning families and what each comparison can reveal

Different learning methods offer different ways to approximate the same response. Hastie et al. (2009) provide the statistical foundations for regularization, ensembles, kernels, and model assessment. Shwartz-Ziv and Armon (2022) examine tabular data and caution against assuming that deep learning is necessary for every structured dataset. Borisov et al. (2024) review deep neural networks for tabular learning. These contributions motivate a comparison that includes simple, nonlinear, and neural methods under the same prediction target.

Tree ensembles form one important group. Breiman (2001) develops random forests by aggregating randomized trees. Averaging can reduce the instability of a single tree

while retaining nonlinear input partitioning. Geurts et al. (2006) introduce extremely randomized trees with stronger randomization of split choices. Their different construction makes both methods useful in a benchmark. Neither construction imposes a coupled material balance across outputs unless that relation is added explicitly.

Boosting builds an ensemble sequentially. Drucker (1997) adapts boosting to regression by increasing attention to difficult observations. Chen and Guestrin (2016) develop extreme gradient boosting (XGBoost) with regularized tree boosting and efficient treatment of large datasets. Ke et al. (2017) develop the light gradient boosting machine (LightGBM) with methods for efficient gradient boosting. Prokhorenkova et al. (2018) develop CatBoost with an ordered approach that addresses prediction shift and categorical variables. The component data considered here are numeric, but these algorithms still provide different nonlinear fitting procedures. Their names do not establish physical admissibility or superiority in advance.

Neighbor and kernel methods give complementary comparisons. Mack (1981) studies the local properties of nearest-neighbor regression estimates. The prediction is strongly connected to the distribution of nearby training inputs. This makes domain distance relevant when the requested input moves outside the sampled range. Smola and Schölkopf (2004) explain support vector regression and its control of model complexity through an error-insensitive fitting objective and regularization. A kernel can represent nonlinear relationships, but the learned relationship still requires separate assessment in an extrapolation regime.

Multiresponse regularization addresses structure across outputs. Zou and Hastie (2005) combine shrinkage and variable selection through the elastic net. Yuan and Lin (2006) develop regression selection for grouped variables. Argyriou et al. (2008) treat feature learning across several tasks. Liu et al. (2009) provide blockwise procedures for the multi-task lasso, and Simon et al. (2013) develop blockwise descent for group-penalized multiresponse regression. These works justify comparisons that can share information across component outputs. A shared sparsity pattern is a statistical relation and should not be mistaken for stoichiometric mass conservation.

Neural regression offers a flexible function representation. Rumelhart et al. (1986) develop learning through back propagation, and Hornik et al. (1989) establish approximation properties of multilayer feedforward networks. Such approximation results concern representational capacity under stated conditions. They do not provide an automatic guarantee about finite-data accuracy, optimization success, extrapolation, or material balance. Curteanu and Cartwright (2011) discuss network topology in chemical applications, making architecture selection part of the modeling problem.

Arik and Pfister (2021) introduce TabNet with sequential attention over tabular inputs. Its feature masks can support inspection of which inputs are used. Attention-based selection still differs from a named algebraic coefficient or a component balance for mass conservation. It is useful to compare such a model with an explicit second-

order regression because the two forms of interpretability answer different questions.

2.9 Learning behavior, sampling, and fair assessment

Training error alone gives incomplete evidence of approximation quality. Geman et al. (1992) analyze the bias and variance dilemma in neural learning. A flexible model can reduce systematic approximation error while becoming more sensitive to the training sample. Belkin et al. (2019) show that modern learning behavior can be more complex than a single classical tradeoff curve. These contributions support direct learning-curve assessment rather than assigning a fixed sample-efficiency ranking to a model family.

Viering and Loog (2023) review the shapes of learning curves and the conditions that influence them. The curve relates available training data to predictive performance under an explicit evaluation design. Its interpretation depends on the error scale, spacing of sample sizes, and variability of the data partitions. A normalized area under such a curve is a useful summary only when those choices are common across the compared models.

Training inputs must also cover the intended process domain. McKay et al. (1979) compare sampling procedures and motivate stratified coverage of input ranges. Latin hypercube sampling can spread observations across each selected input range without enumerating every combination. It does not guarantee equal coverage of every joint response regime, nor does it make independently sampled influent components representative of every real wastewater composition. These limitations need to remain visible when simulated data define the benchmark.

Active learning raises a related question about the cost of obtaining informative targets. Guyon et al. (2011) organize an active learning challenge in which labels are limited. Cawley (2011) provides simple regularized baseline strategies and emphasizes the value of comparing an elaborate selection method with simpler alternatives. Their contribution is relevant to simulator data because each reference response also has a cost. The present learning-curve design measures dependence on available labels. It does not claim to implement or validate an active selection strategy.

Hyperparameter selection is part of model fitting. Bergstra et al. (2011) develop model-based methods for searching hyperparameter spaces. Akiba et al. (2019) provide a framework for organized hyperparameter optimization. These methods can improve the use of a search budget, but they require separation between the observations used to select settings and the observations used to assess the selected model. The same requirement applies when choosing a projection threshold or a trust criterion.

Kaufman et al. (2012) formalize leakage in data mining and explain how information that should be unavailable can enter the fitted model. In this setting, scaling all

observations before cross-validation would allow validation inputs to influence training transformations. Selecting hyperparameters against an outer assessment set would allow the same set to guide and judge the model. Nested partitioning and training-only transformations are therefore part of the evidence design rather than minor computational details.

Pinheiro et al. (2025) examine how feature scaling affects learning tasks. Their contribution supports an explicit account of which methods use scaling and how those scales are fitted. Scaling also changes the numerical meaning of a coefficient. A standardized second-order term must be transformed back to physical input units before an engineering interpretation is attempted.

Comparison must account for variation across assessment problems. Demšar (2006) develops statistical comparisons for classifiers over multiple datasets. The present application concerns regression and repeated data partitions, so its recommendations require adaptation to the unit of independent evidence. Shevchenko et al. (2024) examine benchmarking variability in recommender systems. Their setting is also different, but the concern transfers. Reported rankings can depend on splits and evaluation choices. Paired model comparisons and transparent variability are more defensible than a universal ranking inferred from one split.

Error normalization deserves its own explanation. Tofallis (2015) examines relative prediction accuracy and the problems created by some relative measures. Near-zero component concentrations make ordinary percentage error unstable. A small absolute error can become an enormous percentage or remain undefined at zero. Componentwise normalization using a declared training reference provides a different comparison basis. It does not eliminate the need to report physically meaningful absolute errors and constraint violations.

Shahbazi et al. (2026) address imbalanced regression through data-level and algorithm-level balancing. This concern is relevant when rare operating conditions or low residual concentrations matter more to a decision than their frequency suggests. An average error over the sampled domain can conceal such behavior. Component-resolved errors and selected-decision evaluation are therefore needed alongside aggregate scores.

2.10 Physical constraints in scientific machine learning

Scientific machine learning combines data-based approximation with knowledge of physical structure. Karniadakis et al. (2021) review this combination across several forms of physics-informed learning. Kashinath et al. (2021) discuss applications in weather and climate modeling. These reviews show that physics can enter through inputs, architecture, training, or outputs. The practical question is which relation is enforced, at what stage, and under what conditions.

Raissi et al. (2019) develop physics-informed neural networks that include differential equation residuals in a learning objective. This approach can use governing equations even when direct response data are limited. A finite residual penalty still permits a tradeoff with the data-fitting term and other penalties. Exact mass conservation of every returned component state requires an additional structural argument or an explicit feasibility procedure.

Wastewater-specific work makes the same distinction useful. H. Li et al. (2024) apply physics-informed learning to unit operations and processes, including an activated sludge reactor. The governing-equation information supports the approximation task. Its inclusion in training does not automatically establish every plant-wide hydraulic or clarifier balance. Those relations must be defined at the level where the coupled outputs are returned.

Architectural enforcement supplies one stronger route. Beucler et al. (2021) impose analytic constraints in neural models by arranging the output relationship so that the required equalities hold. Some outputs can be reconstructed from others to complete a constrained state. This depends on which quantities are independent and whether the reconstructed quantities remain within their allowed bounds. Equality completion alone does not settle non-negativity.

Sturm and Wexler (2020) develop a method that enforces mass conservation in photochemical calculations. Sturm and Wexler (2022) embed atom balance in a neural architecture through the stoichiometric relationship. These contributions connect learned transformations to reaction structure. Sturm et al. (2023) use dimensionality reduction with mass conservation for transport of organic aerosol. Their scaling can also maintain non-negative concentrations in the reduced representation. This provides a precedent for addressing mass conservation and non-negativity together while reducing the number of transported quantities. Its transport setting differs from steady-state effluent projection. In activated sludge, an analogous invariant construction must respect the native component basis and the external forcing terms.

Reconciliation provides another route from a statistical estimate to a consistent state. D. Hansen et al. (2024) develop learning models that respect mass conservation through a probabilistic reconciliation perspective. Mass conservation supplies information that can be combined with the learned prediction. This differs from assuming that a low training loss has already made the prediction consistent with mass conservation. The connection to activated sludge is the availability of known linear inventory relations. The probabilistic interpretation remains distinct from a deterministic projection that returns one state without a calibrated uncertainty distribution.

Ruckstuhl et al. (2021) learn a projection for a data assimilation procedure from the difference between an unconstrained analysis and a more expensive constrained analysis. The training target contains information about the projection that the physical constraints require. This is a different task from learning a reactor effluent directly

from influent and operating conditions. It also illustrates a separate computational choice. One can solve a constrained problem at prediction time or train another model to approximate its projection. If the projection is approximated, the returned state still requires a feasibility check. The information carried by constrained training examples should not be confused with exact enforcement of the constraints on every new prediction.

Geiss et al. (2022) address mass conservation when a neural model converts coarse atmospheric chemistry fields to a finer spatial grid. Their formulation accounts for unequal grid-cell areas and consistency between coarse inputs and fine outputs. It also treats the normalization of quantities whose distributions span large concentration ranges. This application clarifies why a physical operator must match the representation on which it acts. A spatial average needs area weights, while a treatment inventory needs native-unit constituent weights and a specified boundary. Directly borrowing a layer for mass conservation without transforming those weights would enforce a different relation. The wastewater application therefore adopts the principle of explicit physical accounting while deriving its own component operator.

Harder et al. (2022) emulate aerosol microphysics by learning changes in state variables from mechanistic input–output pairs. They compare physical information in soft loss terms with completion and projection layers, and examine non-negativity enforcement. Their tendency target gives an additional lesson. A negative change can be physically meaningful when it represents consumption, even though the resulting concentration must remain non-negative. The sign requirement belongs to the appropriate quantity. In activated sludge, driver coefficients and reaction-induced changes can have either sign, while the returned component state is subject to non-negativity. These distinctions help specify what a constrained surrogate actually guarantees across different scientific tasks.

Output non-negativity is another physical requirement. Haddad et al. (2010) provide a mathematical treatment of non-negative and compartmental dynamical systems. In a concentration model, non-negativity belongs to the state description itself. Clipping a negative value can satisfy this bound while changing the total represented material. A projection that enforces mass conservation alone can redistribute material into a negative component. Simultaneous enforcement must therefore handle both requirements together.

Hard constraint layers also raise questions about expressive capacity and uncertainty. Min et al. (2024) study an input-dependent enforcement layer for affine and convex output constraints with neural approximation guarantees. These guarantees depend on the stated architecture and assumptions. They do not establish the approximation capacity of a finite second-order regression. Utkarsh et al. (2025) develop a probabilistic projection that uses prediction-variance information to impose hard constraints and assess uncertainty. These methods incorporate feasibility into the learning

architecture or prediction distribution. The deterministic projections used here have different weighting choices and do not inherit a probabilistic uncertainty guarantee.

Lakshminarayanan et al. (2017) develop deep ensembles for predictive uncertainty estimation. Spread among models can help describe uncertainty in a prediction. It does not certify a component balance or establish the validity of an extrapolated state. Likewise, a deterministic projection does not by itself produce a calibrated uncertainty distribution. Physical admissibility and uncertainty assessment are separate parts of model credibility.

2.11 Projection and the geometry of admissible states

Projection restricts a completed prediction to a set defined by physical relations. Sturm and Silva (2025) use a species-weighted projection for mass conservation through atom balances in atmospheric composition. The weighting matters because species can differ greatly in concentration. An equal absolute adjustment can be small for a major component and large for a trace component. This motivates an explicit choice of projection geometry rather than treating every projection for mass conservation as equivalent.

Valente et al. (2025) project predictions onto physical manifolds defined by mass conservation laws. Their work gives output projection a general geometric interpretation. A predicted point is moved to an admissible set according to a selected notion of distance. The interpretation of the projection depends on the distance, the set, and the existence of a feasible state. A closer point in one weighted distance need not have a smaller error in another evaluation metric.

Kircher and Votsmeier (2025) combine mass conservation and non-negativity for mechanistic kinetics surrogates. The procedure uses a positive provisional prediction to define relative weighting, obtains a candidate satisfying mass conservation, and enforces non-negativity through interpolation with an admissible reference when needed. This construction can be applied after different statistical predictors have been fitted. It is valuable for isolating the effect of physical enforcement from the effect of changing the learner.

The reference state has a practical meaning. If the influent is non-negative and has the required inventories used for mass conservation, it lies in the feasible set for the standalone reactor invariant constraints. Interpolating toward this state can enforce non-negativity without changing those inventories. The resulting state is physically admissible under the imposed constraints. It need not reproduce the reference reactor kinetics. This distinction prevents a guarantee about mass conservation from becoming an unsupported claim about treatment accuracy.

The geometry also limits claims about accuracy improvement. Projection onto a

closed convex set in a fixed Euclidean norm has useful distance properties when the true state lies in that set (Boyd & Vandenberghe, 2004). A relative-weighted candidate satisfying mass conservation followed by boundary interpolation is a different map. Its weights can depend on the prediction, and its final interpolation is not generally the nearest feasible point under the reported error metric. Paired empirical assessment is therefore necessary even when feasibility follows from the construction.

2.12 Interpretable constrained regression

Rudin (2019) argues for interpretable models in high-consequence settings when such models are available. The argument concerns the ability to inspect the model itself, rather than relying only on an explanation added afterward. For treatment surrogates, an explicit regression can identify operating terms, loading terms, nonlinear terms, and component coupling. This creates a structure that an engineer can inspect in relation to the process representation.

Brunton et al. (2016) identify governing relationships through sparse candidate functions in dynamical systems. Their contribution shows how named terms can connect learned structure to scientific interpretation. A second-order treatment surrogate uses named terms for a different purpose. It approximates a steady-state response and does not claim to discover the biological reaction laws. Similar predictions can arise from different coefficient combinations when inputs are correlated or when several blocks compensate for one another.

Shen et al. (2023) discuss differentiable modeling that joins physical and learned components. This supports inspection of the entire prediction chain. For a coupled regression, the coefficients of the driver are transformed by the coupling solution. For a projected model, the final output also depends on the active constraints. An explanation that ignores these transformations describes only part of the returned prediction.

The published formulation of E. F. Selerio Jr. (2026) supplies a specific activated sludge example. It combines an invariant-aware second-order regression with a checked output procedure that imposes mass conservation and component non-negativity. The important theoretical distinction is between the unconstrained regression response, the fitted state variables used during training, and the final deployed state. Soft invariant-aware fitting guides the regression. The checked projection establishes the stated final-state constraints. These are different mechanisms and require different diagnostics.

Numerical optimization underlies both mechanisms. Boyd and Vandenberghe (2004) provide the convex optimization foundations for affine constraints and quadratic objectives. Razaviyayn et al. (2013) analyze block successive minimization under specified regularity conditions. A regression objective that jointly fits coupling and state variables can be nonconvex even when individual updates are convex. Decreasing the objective at each accepted update does not establish a globally optimal fit. Condi-

tioning restrictions also need their own treatment because they are not ordinary box constraints.

Stellato et al. (2018) present an operator-splitting approach for quadratic programs, and Stellato et al. (2020) give its fuller mathematical and computational development. Huangfu and Hall (2018) develop a parallel dual revised simplex method for linear programming. These works support the numerical treatment of regression subproblems and output projections. Their role is to solve specified optimization problems. Numerical termination still needs explicit residual and bound checks before a predicted state is accepted.

2.13 Connected plants and the move from prediction to decisions

A connected activated sludge plant contains more physical structure than one reactor. Härtel and Pöpel (1992) develop a dynamic secondary clarifier model with sludge thickening. Takács et al. (1991) model clarification and thickening through a settling relationship. Jeppsson and Diehl (1996) examine the propagation of biological components through the secondary clarifier. These contributions explain why clarification must preserve component relationships and why overflow, underflow, and stored solids cannot be predicted as unrelated quantities.

Alex et al. (2008) define a benchmark simulation setting that connects biological treatment, clarification, and recirculation. Its value is a common process context for comparing operating strategies. A plant with an extended component basis remains distinct from that benchmark, but can use the same discipline of declared hydraulics, unit connections, and assessment conditions. In particular, a recycle stream affects both concentration mixing and the flow passing through each reactor.

Solids retention adds another connection. Leu et al. (2012) examine the implications of long solids retention time for energy use, effluent quality, and stability. The retained solids depend on both reactor and clarifier inventories and on external solids losses. A surrogate that predicts effluent quality accurately but neglects inventory can therefore misrepresent a solids-age constraint. Miederer et al. (2025) develop an energy management model for wastewater treatment plants, further connecting operating choices to resource use.

The U.S. Environmental Protection Agency (2009) review provides engineering guidance for nutrient control technologies. Its role is to identify treatment considerations and design relationships. Site-specific effluent limits and safety thresholds still belong to the declared optimization problem. They should be supplied as explicit inputs rather than presented as universal regulatory limits.

Control studies show how a prediction model enters operation. Sadeghassadi et al. (2018) use neural models for setpoint design and predictive control. Bernardelli et

al. (2020) combine a learning model with real-time predictive control of a wastewater treatment plant. Ren et al. (2026) study ammonium-based aeration scheduling under influent disturbances. Croll et al. (2023) review reinforcement learning for wastewater process control. These contributions demonstrate a range of operating applications. Their dynamic decisions remain distinct from a steady-state operating search, which evaluates equilibrium responses under fixed controls.

Joint output projection offers a way to preserve the specified connections in a statistical plant model. Mixer equations, reactor invariants, clarifier outlet balances, and inventory bounds can be imposed together. At fixed controls, the relevant relations can form a convex quadratic projection problem (Boyd & Vandenberghe, 2004; Stellato et al., 2020). The projection can remove contradictions among separately predicted unit responses. It does not impose every kinetic rate equation or every layerwise settling equation of the mechanistic plant.

This creates an optimization problem with a nested prediction procedure. Colson et al. (2007) review bilevel optimization, which helps explain the distinction between an outer control choice and an inner constrained response calculation. The plant projection has a strongly convex inner objective when the weights are positive and the feasible set is nonempty. Tøndel et al. (2003) analyze parametric quadratic programming and its relation to piecewise solution structure. Their results motivate attention to active constraints, but an operating problem with control-dependent balance coefficients need not have the same globally piecewise-affine structure.

2.14 Water networks and the distinction between topology and operation

Water-network design separates the origins, combinations, and destinations of streams. A source supplies water with specified quantities and quality. A mixer combines compatible streams. A sink accepts water for a use or discharge requirement. Galan and Grossmann (1998) optimize distributed wastewater treatment networks, and Galán and Grossmann (1999) examine design and synthesis strategies for those networks. Their contribution concerns the choice of connected treatment and routing alternatives. The present connected plant uses a fixed topology and varies declared operating controls.

Regeneration and constituent-specific treatment complicate those choices. Guelli Ulson de Souza et al. (2011) consider differentiated regeneration of contaminants in chemical-industry water reuse. L. Yang et al. (2014) include wastewater regeneration models in network optimization. The treatment response influences which stream can reach which destination. These works show why a constant removal fraction can be an insufficient description when concentrations and operating conditions change.

Industry-specific applications make the boundary explicit. Gutterres et al. (2010)

examine water reuse in tannery operations, É. Hansen et al. (2018) minimize water and wastewater in a petrochemical setting, and Xu et al. (2015) integrate a yeast-production water system. The application determines the relevant contaminants, unit operations, and water-use requirements. Their network methods are useful conceptual precedents, while their process states are not the activated sludge component targets.

Rubio-Castro et al. (2013) develop global optimization for property-based interplant water integration. Their result concerns the specified mathematical formulation and the conditions required by its optimization method. McCormick (1976) provides convex underestimating relations for nonconvex factorable expressions. Such relations can support bounds on an optimization problem. They do not establish that a projected surrogate response satisfies every original nonlinear process equation.

Tosarkani and Amin (2020) design a wastewater treatment network under uncertainty through a multiobjective robust formulation. Ye et al. (2019) use a hybrid particle-swarm approach for network planning. These contributions address uncertainty and search over a planning problem. The numerical route and stopping evidence must be distinguished from a global optimality certificate. A faster operating search in a fixed plant is a different contribution from selecting a globally optimal treatment network.

Damalerio et al. (2022a) connect phosphorus-removal priorities with treatment-configuration and cost objectives. E. F. Selerio (2022) develops regression-based optimization for urban water eutrophication abatement. Both connect response approximation to a declared decision purpose. Their relevance is the organized treatment of several engineering quantities. The current study retains a component-based activated sludge response, projection for mass conservation and non-negativity, and independent verification of selected controls.

2.15 Approximation and the structure of an optimization claim

Kusiak and Wei (2013) examine data-based optimization of the activated sludge process. Their contribution connects a learned response to operating selection. Niu et al. (2022) combine deep learning and multiobjective dynamic wastewater optimization. These studies illustrate different response representations and search settings. A steady-state operating comparison should identify its time basis and objective rather than treating all learning-based optimization as one task.

Jana et al. (2022) combine neural and support-vector regression methods in detergent-industry effluent optimization. Waqas et al. (2022) model membrane permeability for a rotating biological contactor, while B. Li et al. (2022) join response-surface and neural methods for membrane fabrication. These applications show how statistical approximation can enter an engineering search. They also show why evidence for one

target cannot be transferred directly to a complete reactor component state.

Shojaei and Shojaei (2021) discuss the optimization of conditions in wastewater degradation. Poh et al. (2016) review mechanistic and metaheuristic optimization in anaerobic digestion. The process models and decision variables differ from those of activated sludge, but the common issue is the relation between an inexpensive response calculation and a constrained operating choice. Model accuracy, search completeness, and the chosen priorities remain distinct.

Jain and Kar (2017) examine nonconvex optimization in machine learning. A model can be linear in its fitted coefficients while remaining nonlinear in its operating inputs. A convex projection at fixed inputs can also be embedded in a nonconvex outer search. This distinction is essential to the present contribution. Neither transparent regression terms nor an exactly solved projection alone establish a globally optimal operating decision.

2.16 Managing approximation during operating searches

An optimizer can exploit approximation error in a favorable region. Alexandrov et al. (1998) develop a trust-region framework for managing approximation models. The central idea is to judge the approximation where it is being used and control the search accordingly. Sahigara et al. (2012) compare applicability-domain methods for quantitative structure–activity models. Their application differs from treatment modeling, but it supports the general need to distinguish nearby training support from distant prediction requests.

Distance restrictions and projection size answer different questions. An input can be near the training observations while its raw output requires a large projection. Conversely, a distant input can produce an apparently admissible state. Both indicators can be useful screening criteria, but neither is a calibrated probability of prediction accuracy. Hameed et al. (2026) develop trust-region filter methods for gray-box optimization with derivative information and feasibility considerations. Their contribution reinforces the need to separate approximation agreement, feasibility, and progress in the original objective.

Direct optimization provides a meaningful comparator only when its numerical formulation is explicit. Buzzi-Ferraris and Manenti (2013) explain nonlinear systems and optimization in chemical engineering. Kraft (1988) develops sequential quadratic programming, while Wächter and Biegler (2006) develop an interior-point filter line-search method for nonlinear programming. These approaches support different numerical routes for the operating problem. A local termination condition does not establish the global optimum of a nonconvex plant model.

The underlying steady-state solution also needs attention. Curtiss and Hirschfelder

(1952) establish the problem of stiff differential equations. Biological treatment contains processes with different response scales, which can make steady-state initialization difficult. Kelley and Keyes (1998) analyze pseudo-transient continuation as a route toward a steady solution. Time relaxation followed by an algebraic solve can provide a better initial state, but acceptance still depends on checking the original equations and component bounds.

Audet and Dennis (2006) develop mesh adaptive direct search for constrained optimization. Ragonneau (2022) studies model-based derivative-free methods. These works explain why an operating search can remain possible when the projected prediction map is nonsmooth or reliable derivatives are unavailable. The comparison with a smooth direct formulation should account for the differences in numerical structure and stopping rules.

Blik et al. (2023) benchmark surrogate optimization algorithms on expensive black-box functions. Their contribution makes assessment design and evaluation cost central to a performance comparison. Pedrozo et al. (2025) compare surrogate optimization with a reference process optimization setting. Both support evaluating the objective and constraints at the selected decisions using the original model. Search time, development time, and verification time should be reported separately because they answer different cost questions.

X. Zhang et al. (2026) develop an agile benchmarking framework for wastewater resource recovery technologies. Their emphasis on comparable assessment strengthens the need for common engineering priorities and clear process boundaries. Route-specific numerical scales can help solve the equations, but both routes must ultimately be judged using the same physical outputs and objective definition.

2.17 The integrated research gap

The literature provides strong foundations for component modeling, nonlinear statistical prediction, physical constraint enforcement, interpretable regression, and surrogate-assisted optimization. It also shows why these foundations cannot be reduced to one error score. A component surrogate must represent the response that is being assessed. A projection must enforce the relations that are being claimed. An interpretation must describe the prediction map that is actually returned. An operating decision must be evaluated against the original engineering problem.

The research gap concerns their integration in activated sludge analysis. A common reactor benchmark can compare raw and projected predictions across learning families. A structured regression can make influent, operating, and coupling terms inspectable while preserving a separate feasibility procedure. A connected plant formulation can extend enforcement to mixing, treatment stages, clarification, and solids inventory. Independent mechanistic evaluation can then assess the selected controls. This sequence

supplies a coherent basis for studying predictive accuracy, physical admissibility, interpretability, and decision quality without treating any one of them as evidence for all the others.

Chapter 3

Mechanistic Foundations of Physically Admissible Prediction

3.1 The component state as the object of prediction

An activated sludge reactor changes the form and distribution of material. Organic substrate can become biomass, ammonium can become nitrite and nitrate, and soluble phosphate can enter biological storage or a precipitate. A description of treatment therefore needs more than one aggregate effluent concentration. The ASM framework separates material into components and connects their changes through biological and chemical processes (Henze et al., 2015). This separation provides the physical meaning of a surrogate prediction. The surrogate estimates a complete component state from the influent state and the operating conditions.

The mechanistic representation is Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN). It combines the phosphorus-removal framework of Henze et al. (2015) with the nitrogen-transformation formulation used by Guerrero et al. (2011). The state contains 20 components and the reaction description contains 28 processes. Ten components are dissolved and ten are particulate.

3.1.1 Reading component vectors and matrix operations

A concentration is first a scalar, which is one numerical value with a specified unit. For example, 5 g N m^{-3} means 5 grams of nitrogen per cubic metre. Its numerical value is also 5 mg N L^{-1} because both the mass and volume units change by a factor of one thousand. Oxygen, COD, nitrogen, phosphorus, solids, and alkalinity have different reporting bases. A unit written as g COD m^{-3} refers to oxygen-demand equivalents rather than grams of one named organic molecule. These bases determine which coefficients can be used when quantities are combined.

A vector collects related scalar values in a fixed order. A three-component column

vector and its transpose can be written as

$$c = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}, \quad c^\top = \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix}.$$

The superscript $\top$ means transpose. It changes a column into a row, or interchanges the rows and columns of a matrix. It does not change the component values. The notation $c \in \mathbb{R}^3$ says that the vector has three real-valued coordinates. The notation $c \geq 0$ means that c_1 , c_2 , and c_3 are each non-negative. It does not mean only that their sum is non-negative.

A strictly positive coordinate satisfies $c_j > 0$, whereas non-negativity permits $c_j = 0$. The distinction matters when a logarithm or a reciprocal is evaluated. Such an operation can require a strictly positive argument even though a physical concentration may legitimately be zero.

Subscripts identify the coordinate being discussed. The symbol c_j means component j . When several observations are involved, c_{ij} means component j in observation i . The labels *in* and *out* identify the side of the reactor boundary. Thus $c_{out,ij}$ is the effluent concentration of component j in observation i . Parenthetical superscripts used for a surrogate or prediction type are identifiers rather than powers.

A matrix arranges coefficients in rows and columns. If a matrix has two rows and three columns, its dimension is 2×3 . A matrix product is defined when the number of columns of the first factor equals the number of rows of the second factor. Each output entry is obtained by multiplying corresponding entries in a row and a column and adding the products. For hypothetical numerical values,

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 1(3) + 2(4) + 0(5) \\ 0(3) + 1(4) + 1(5) \end{bmatrix} = \begin{bmatrix} 11 \\ 9 \end{bmatrix}.$$

The product has two coordinates because the matrix has two rows. In a physical calculation, each coefficient also converts the unit of its associated input into the required output unit. The numerical example explains the operation without assigning these coefficients to an activated sludge process.

A row-column product gives a weighted sum. For a coefficient column a , the scalar product $a^\top c$ means $a_1c_1 + a_2c_2 + a_3c_3$. The summation notation $\sum_{j=1}^3 a_jc_j$ writes the same operation by identifying its index, starting value, and ending value. An identity matrix, denoted I , leaves a vector unchanged. A zero matrix has every entry equal to zero. These operations allow many component equations to be recorded in one expression while preserving the scalar accounting behind them.

The magnitude of a vector is described by a norm. The Euclidean norm is the square root of the sum of squared coordinates. A hypothetical vector $(3, -4)^\top$ has Euclidean norm $\sqrt{3^2 + (-4)^2} = 5$. The sum of absolute coordinates gives its absolute-sum norm,

$|3| + |-4| = 7$. Squaring a norm and squaring each coordinate are therefore different steps that must be applied in the stated order. When coordinates have different physical bases, a numerical norm is meaningful only with the adopted unit convention or an explicit standardization.

The 20-component state uses the same column-vector convention. Its fixed ordering is

$$c = (S_O, S_F, S_A, S_{NH_4}, S_{NO_2}, S_{NO_3}, S_{N_2}, S_{PO_4}, S_I, S_{ALK}, X_I, X_S, X_H, X_{PAO}, X_{PP}, X_{PHA}, X_{AOB}, X_{NOB}, X_{MeP}, X_{MeOH})^\top. \quad (3.1)$$

Every concentration is represented by its numerical value in the native unit assigned to that component. A vector of these values combines several constituent bases. An organic component expressed as g COD m⁻³ is not expressed on the same material basis as ammonium expressed as g N m⁻³. The coefficients of every physical operator must account for this distinction.

Table 3.1: Component meanings, native units, and fixed coefficients for the four reported water quality quantities. Each coefficient converts the native component basis to the corresponding reported basis.

Component	Meaning	Native unit	COD	TN	TP	TSS
S_O	Dissolved oxygen	g O ₂ m ⁻³	0	0	0	0
S_F	Fermentable substrate	g COD m ⁻³	1	0	0	0
S_A	Acetate and fermentation products	g COD m ⁻³	1	0	0	0
S_{NH_4}	Ammonium nitrogen	g N m ⁻³	0	1	0	0
S_{NO_2}	Nitrite nitrogen	g N m ⁻³	0	1	0	0
S_{NO_3}	Nitrate nitrogen	g N m ⁻³	0	1	0	0
S_{N_2}	Dissolved dinitrogen	g N m ⁻³	0	0	0	0
S_{PO_4}	Soluble phosphate	g P m ⁻³	0	0	1	0
S_I	Soluble inert organic matter	g COD m ⁻³	1	0.01	0	0
S_{ALK}	Alkalinity	mol m ⁻³	0	0	0	0
X_I	Particulate inert organic matter	g COD m ⁻³	1	0.02	0.01	0.75
X_S	Slowly biodegradable substrate	g COD m ⁻³	1	0	0	0.75
X_H	Heterotrophic biomass	g COD m ⁻³	1	0.07	0.02	0.90
X_{PAO}	Phosphate-accumulating biomass	g COD m ⁻³	1	0.07	0.02	0.90
X_{PP}	Stored polyphosphate	g P m ⁻³	0	0	1	3.23
X_{PHA}	Stored polyhydroxyalkanoates	g COD m ⁻³	1	0	0	0.60
X_{AOB}	Ammonia-oxidizing biomass	g COD m ⁻³	1	0.07	0.02	0.90
X_{NOB}	Nitrite-oxidizing biomass	g COD m ⁻³	1	0.07	0.02	0.90
X_{MeP}	Precipitated metal phosphate	g TSS m ⁻³	0	0	1/4.87	1
X_{MeOH}	Metal hydroxide	g TSS m ⁻³	0	0	0	1

The symbols in Table 3.1 identify state variables. They do not introduce additional process abbreviations. For example, X_{AOB} is the concentration coordinate assigned to ammonia-oxidizing biomass. The numerical constituent contents and solids conversion factors are fixed assumptions of the adopted reporting map. They are not estimated from the surrogate fit. Their meaning follows the component accounting used in activated sludge modeling (Henze et al., 2015).

The full state preserves distinctions that aggregate quantities cannot recover. Ammonium, nitrite, and nitrate contribute to the same reported nitrogen total, but their conversion involves different microbial populations and oxygen requirements. Dissolved dinitrogen remains a component of the modeled nitrogen inventory even though it has zero weight in reported TN. Similarly, soluble phosphate, stored polyphosphate, and metal phosphate have different process roles even when all contribute to TP. These differences matter when a predicted state is used to interpret an operating decision.

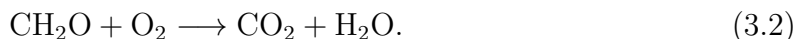
3.2 Biological meaning of the component transformations

3.2.1 Electron donors, electron acceptors, and biomass

Microorganisms use part of a substrate to obtain energy and another part to form cellular material. An electron donor supplies electrons in an oxidation reaction. An electron acceptor receives them in a reduction reaction. Under aerobic conditions, organic material can serve as the donor and oxygen as the acceptor. Under anoxic conditions, some organisms use nitrate or nitrite instead of oxygen. Naming the donor and acceptor helps explain why the same organic substrate can support different treatment pathways (Narayanan & Narayan, 2019; Preena et al., 2021).

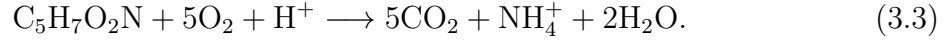
The carbon source and the energy source describe different properties. Heterotrophic biomass uses organic carbon to form cells. Chemolithoautotrophic nitrifying biomass obtains energy from inorganic nitrogen oxidation and uses inorganic carbon for cell synthesis. Its growth is therefore not simply another form of organic-substrate removal. The separate biomass coordinates in the adopted model retain this distinction (Guerrero et al., 2011; Henze et al., 2015). Complete ammonia oxidation can also occur within one microbial organism (van Kessel et al., 2015). That biological observation does not replace the separate ammonium- and nitrite-oxidation processes in the adopted model.

An idealized organic unit, CH_2O , illustrates oxidation without assigning one chemical formula to all wastewater organic matter. Its complete aerobic oxidation is



Both sides contain one carbon atom, two hydrogen atoms, and three oxygen atoms. Using approximate molar masses of 30 g mol^{-1} for the organic unit and 32 g mol^{-1} for oxygen gives $32/30 = 1.067 \text{ g oxygen per g}$ of this idealized material. A hypothetical 30 g of this material would require 32 g oxygen for complete oxidation. The example explains the oxygen-demand basis of organic components. It does not prescribe the oxygen demand of an arbitrary wastewater mixture.

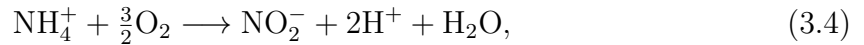
Cell synthesis changes that accounting because part of the carbon and nitrogen enters biomass. The empirical formula $C_5H_7O_2N$ is a useful teaching representation of biomass composition, as discussed in the assimilation study of Hoover and Porges (1952). It describes an average composition rather than a single molecular species. For example, an idealized complete oxidation of this biomass with nitrogen released as ammonium can be written as



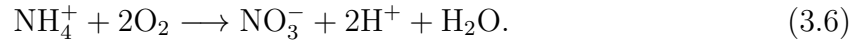
The carbon, hydrogen, oxygen, and nitrogen counts are respectively 5, 8, 12, and 1 on each side. The net charge is +1 on each side. Five moles of oxygen correspond to 160 g, while the empirical biomass unit has an approximate mass of 113 g. The ratio is therefore about 1.416 g oxygen per g of this idealized biomass. These calculations distinguish dry biomass mass from biomass COD. The adopted model uses its specified yields, constituent contents, and lysis processes rather than substituting these teaching reactions for its stoichiometric entries.

3.2.2 Nitrogen transformations and their oxygen requirements

Separating nitrification into two steps makes the oxygen demand and the nitrite intermediate visible. Ignoring cell synthesis for this teaching calculation, the ammonium-oxidation and nitrite-oxidation reactions are

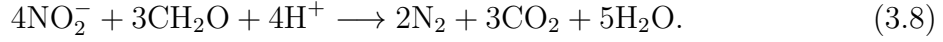
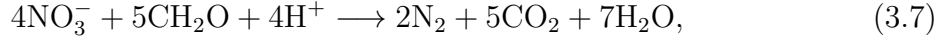


The first reaction preserves one nitrogen atom, four hydrogen atoms, and three oxygen atoms. Its right-hand charge is $-1 + 2 = +1$, which equals its left-hand charge. The second reaction preserves one nitrogen atom, three oxygen atoms, and a net charge of -1 . Adding the reactions and cancelling the intermediate nitrite gives



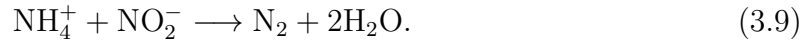
Each mole of nitrogen has an approximate mass of 14 g. The first step requires $1.5(32)/14 = 3.43$ g oxygen per g nitrogen. The second requires $0.5(32)/14 = 1.14$ g oxygen per g nitrogen. Their sum is 4.57 g oxygen per g nitrogen. A hypothetical conversion of 10 g ammonium nitrogen to nitrate would therefore require about 45.7 g oxygen under these idealized assumptions. Actual process demand also depends on biomass synthesis and the adopted yields. The hydrogen ions in the first step explain why nitrification also draws on the buffering capacity represented by alkalinity (Guerrero et al., 2011; Henze et al., 2015).

Denitrification uses oxidized nitrogen as an electron acceptor and can convert it to dinitrogen. With CH_2O as the illustrative donor and with cell synthesis omitted, two balanced net reactions are



For the nitrate reaction, both sides contain four nitrogen atoms, five carbon atoms, fourteen hydrogen atoms, and seventeen oxygen atoms. Its left-hand charge is $-4 + 4 = 0$, matching the uncharged products. The nitrite reaction can be checked in the same way. The donor requirements are $5/4$ and $3/4$ moles of the illustrative organic unit per mole of nitrogen. This difference explains why the pathway through nitrite can require less organic donor than the pathway through nitrate under the same simplified assumptions. Carbon supply and donor identity also affect denitrification kinetics, as reviewed by Lu et al. (2014). A stoichiometric requirement alone does not establish a reaction rate.

Anaerobic ammonium oxidation provides a different nitrogen pathway. An idealized net reaction that omits biomass synthesis and minor products is



It preserves two nitrogen atoms, four hydrogen atoms, two oxygen atoms, and zero net charge. It couples ammonium and nitrite without an organic electron donor in this idealized balance. The cultivation work of Strous et al. (1998) and the pathway studies of Dietl et al. (2015); Maalcke et al. (2016) explain why it differs biologically from ordinary heterotrophic denitrification. Its application requires its own organisms and operating conditions (Vandekerckhove, 2018; Winkler & Straka, 2019; Xie, 2021). It is included here to clarify the boundary of the nitrogen discussion. It is not an additional process in the adopted 20-component, 28-process reactor model.

3.2.3 From elemental accounting to the adopted state basis

The teaching reactions permit direct checks of atom counts and charge because every displayed species has a chemical formula. An activated sludge state instead combines oxygen-demand equivalents, nitrogen mass, phosphorus mass, solids mass, and alkalinity. A biomass coordinate in g COD m^{-3} cannot be added directly to a nitrogen coordinate in g N m^{-3} to obtain a material total. Its nitrogen-content coefficient must first convert its contribution to the nitrogen basis.

Takács and Vanrolleghem (2006) explain the distinction between balances on lumped wastewater components and complete elemental accounting. A full elemental balance may require water, carbon dioxide, gas exchange, and other species outside a particular activated sludge state. The mass conservation conditions in this dissertation

are established from the adopted stoichiometric matrix and the declared flow and transfer boundary. They certify the quantities represented by that model. They do not imply that every unrepresented chemical species or external flux has been measured. The derivation below identifies the precise quantities that can be imposed without adding an unsupported physical condition.

3.3 Process transformations and their rates

Hydrolysis transfers slowly biodegradable material toward forms that microorganisms can use more readily. Heterotrophic metabolism links substrate consumption to biomass growth and electron-acceptor use. Under aerobic conditions oxygen acts as an electron acceptor. Under anoxic conditions nitrate or nitrite can support the corresponding modeled transformations. Fermentation supplies acetate from suitable organic material. Biomass lysis redistributes material from active populations to substrate and inert pools. Biological phosphorus removal adds storage, growth, phosphate uptake, and release mechanisms. Chemical precipitation and redissolution redistribute phosphorus between soluble and particulate forms (Guerrero et al., 2011; Henze et al., 2015).

Two-step nitrification represents ammonium oxidation to nitrite separately from nitrite oxidation to nitrate. It therefore retains both ammonia-oxidizing and nitrite-oxidizing biomass. A one-step description can reproduce an overall ammonium-to-nitrate conversion while hiding a nitrite intermediate. Retaining the intermediate allows the state to distinguish a low-ammonium effluent that contains nitrite from one in which nitrite has been converted further. The distinction concerns the reaction representation rather than an assurance that every nitrogen pathway occurring in a real plant is included. The broader diversity of microbial nitrogen-removal pathways reviewed by Duan et al. (2022) sets that limitation.

The stoichiometric matrix, also called the Petersen matrix, records which components are consumed and produced by each process. A small hypothetical reaction network illustrates the entries before the full model is written. Let three material pools be denoted by A , B , and C . One unit of B contains twice the tracked inventory of one unit of A or C . Consider the transformations



Figure 3.1 connects these two transformations to their component changes. The coefficients below each arrow count changes in the three pools, while the inventory weights explain why both processes preserve the same material quantity.

Two units of A contain the same inventory as one unit of B . That unit of B contains the same inventory as two units of C . The middle pool receives material from the first process and supplies it to the second. Its net concentration change therefore depends

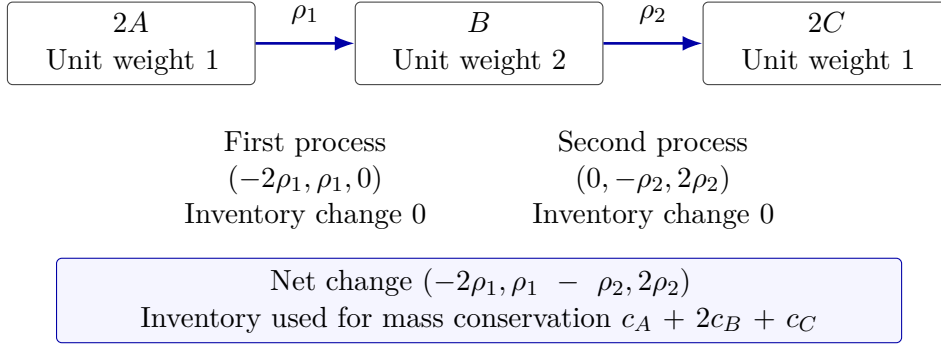


Figure 3.1: Hypothetical reaction accounting for three pools with inventory weights 1, 2, and 1. Each process redistributes the tracked inventory without creating or removing it.

on the difference between the two rates, even though each process separately preserves the weighted inventory.

The first process consumes two units of A and produces one unit of B . It has no direct effect on C . The second consumes one unit of B and produces two units of C . If the two process rates are ρ_1 and ρ_2 , their scalar contributions are

$$\begin{aligned}
 \dot{c}_A|_{\text{reaction}} &= -2\rho_1, \\
 \dot{c}_B|_{\text{reaction}} &= \rho_1 - \rho_2, \\
 \dot{c}_C|_{\text{reaction}} &= 2\rho_2.
 \end{aligned}$$

The dot indicates a time derivative. These equations isolate reaction contributions and contain no hydraulic transport. They can be organized into a process-by-component matrix and a rate column,

$$\nu_{\text{toy}} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \end{bmatrix}, \quad \rho_{\text{toy}} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}.$$

Transposing the matrix puts each component's coefficients in one row. Multiplication then recovers all three scalar rates,

$$\nu_{\text{toy}}^\top \rho_{\text{toy}} = \begin{bmatrix} -2 & 0 \\ 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix} = \begin{bmatrix} -2\rho_1 \\ \rho_1 - \rho_2 \\ 2\rho_2 \end{bmatrix}.$$

For hypothetical rates of 3 and 1 in compatible amount-per-time units, the contributions are -6 , 2 , and 2 . The middle pool increases because its production rate exceeds its consumption rate. A positive value in one component does not imply that either process creates the complete tracked inventory.

The full model uses the same convention. Its orientation is fixed as

$$\nu \in \mathbb{R}^{P \times F}, \quad P = 28, \quad F = 20. \quad (3.10)$$

Row p describes process p and column j describes component j . A negative entry ν_{pj} represents consumption and a positive entry represents production. A zero entry means that the process has no direct stoichiometric contribution to that component. Coefficients contain yields and constituent conversions appropriate to the native units.

For component j of the full model, the individual process contributions are $\nu_{1j}\rho_1$, $\nu_{2j}\rho_2$, and so on through $\nu_{28j}\rho_{28}$. Adding them gives the net reaction rate of that component. The expression $\rho(c, u)$ indicates that process rates depend on the current component state and operating inputs. For a rate vector $\rho(c, u) \in \mathbb{R}^{28}$, the reaction contribution to the component balance is

$$r_{\text{reaction}}(c, u) = \nu^\top \rho(c, u), \quad r_{\text{reaction},j}(c, u) = \sum_{p=1}^{28} \nu_{pj} \rho_p(c, u). \quad (3.11)$$

Stoichiometry specifies the proportions of a transformation. Kinetics specifies its speed at a particular state. These are different responsibilities. Increasing the concentration of biomass can change a process rate while leaving its stoichiometric row unchanged. Likewise, changing aeration can alter the oxygen limitation of a process without changing the material accounting of that process.

A simple saturation expression makes this distinction concrete. Suppose, only for illustration, that the substrate concentration and its saturation constant are both 4 g COD m⁻³. Dividing the concentration by their sum gives $4/(4 + 4) = 0.5$. The unit cancels because numerator and denominator use the same constituent basis. If a limiting dissolved substrate has concentration S and saturation constant $K_S > 0$, the factor

$$\frac{S}{K_S + S} \quad (3.12)$$

equals one half at $S = K_S$, approaches zero as S approaches zero, and approaches one as S becomes large. A representative growth expression combines a maximum specific growth rate, biomass abundance, and two limitation factors. For hypothetical values $\mu = 2 \text{ d}^{-1}$, $X = 10 \text{ g biomass COD m}^{-3}$, a substrate factor of 0.5, and an oxygen factor of 0.25, the growth contribution is $2(10)(0.5)(0.25) = 2.5 \text{ g biomass COD m}^{-3} \text{ d}^{-1}$. Multiplication by dimensionless limitation factors reduces the unrestricted contribution $\mu X = 20$ to 2.5. A representative growth expression can combine substrate and oxygen limitation with biomass abundance,

$$\rho_{\text{growth}} = \mu X \frac{S}{K_S + S} \frac{S_O}{K_O + S_O}. \quad (3.13)$$

Here μ is a maximum specific growth rate, X is the associated biomass coordinate, and K_O is an oxygen saturation constant. Equation (3.13) illustrates the role of multiplicative limitation in activated sludge kinetics. It does not replace the complete 28-process rate formulation, whose process-specific yields, inhibition terms, and storage dependencies follow Henze et al. (2015) and Guerrero et al. (2011).

A hypothetical substrate concentration of $K_S/9$ gives a substrate factor of 0.1. Raising the concentration to $9K_S$ gives a factor of 0.9. The growth rate changes by a factor of nine if all other factors stay fixed, although the substrate concentration changes by a factor of 81. This calculation explains why a linear relationship between influent load and effluent state is not generally sufficient. Several limiting factors and competing populations can act together. It also explains why an accurate surrogate needs to represent interactions without confusing them with a new stoichiometric law.

3.4 The steady-state reactor boundary

The single-reactor prediction problem uses a completely mixed reactor at steady state. Perfect mixing makes the reactor concentration equal to the effluent concentration. The standalone boundary contains hydraulic inflow and outflow, the modeled reaction processes, and dissolved-oxygen transfer. It excludes a clarifier, solids return, and other external additions. These exclusions define the balance being enforced. They do not imply that those processes are absent from wastewater treatment plants.

Let c_{in} and c_{out} be the influent and effluent component vectors. Let $D > 0$ be the hydraulic dilution rate. For component j , hydraulic entry contributes $Dc_{in,j}$ and hydraulic exit contributes $-Dc_{out,j}$. Reaction contributes the sum of the 28 process terms. Oxygen transfer contributes only to the oxygen coordinate. The scalar balance is therefore

$$0 = Dc_{in,j} - Dc_{out,j} + \sum_{p=1}^{28} \nu_{pj} \rho_p + \omega(e_{S_O})_j.$$

The selector has first coordinate one because S_O is first in the chosen component order. Its other nineteen coordinates are zero. Thus the oxygen equation contains $+\omega$, while the ammonium equation contains no direct oxygen-transfer term. Writing the oxygen equation separately makes this distinction explicit,

$$0 = D(S_{O,in} - S_{O,out}) + \sum_{p=1}^{28} \nu_{p,S_O} \rho_p + \omega.$$

All terms are numerical rates on the oxygen basis for this row. For another component, the units change to that component's native basis. With $e_{S_O} \in \mathbb{R}^{20}$ selecting dissolved oxygen, the same twenty scalar equations are collected into the steady balance

$$0 = D(c_{in} - c_{out}) + \nu^\top \rho(c_{out}, u) + \omega e_{S_O}, \quad (3.14)$$

Moving the hydraulic difference to the opposite side changes its sign. For one component, this operation gives $D(c_{out,j} - c_{in,j}) = \sum_p \nu_{pj} \rho_p + \omega(e_{S_O})_j$. The vector equation is equivalently

$$D(c_{out} - c_{in}) = \nu^\top \rho(c_{out}, u) + \omega e_{S_O}. \quad (3.15)$$

The hydraulic term has the same native component-unit-per-day basis as the corresponding reaction and transfer terms. Treating the balance as a numerical vector equation requires a fixed unit convention throughout.

The operating inputs are HRT and a dimensionless aeration coordinate u_{air} . HRT specifies the time needed to replace one reactor volume on the stated flow basis. A 12-hour residence time corresponds to two reactor-volume replacements per day because $24/12 = 2$. The dilution rate is consequently an inverse time. Oxygen transfer uses a different inverse-time coefficient. Multiplying that coefficient by the oxygen concentration deficit produces a concentration-per-time rate. The adopted parameterization is

$$D = \frac{24 \text{ h d}^{-1}}{\text{HRT}}, \quad (3.16)$$

$$k_{La} = (2 + 18u_{\text{air}}) \text{ d}^{-1}, \quad (3.17)$$

$$\omega = k_{La}(S_{O,\text{sat}} - S_{O,\text{out}}), \quad S_{O,\text{sat}} = 8.5 \text{ g O}_2 \text{ m}^{-3}. \quad (3.18)$$

The dilution rate connects the time basis of the hydraulic specification to that of the process rates. The oxygen-transfer term depends on both the aeration setting and the difference between saturation and the reactor oxygen concentration.

For a hypothetical HRT of 12 h, Equation (3.16) gives $D = 2 \text{ d}^{-1}$. A component difference of 5 g m^{-3} then corresponds to a net transformation or transfer of $10 \text{ g m}^{-3} \text{ d}^{-1}$ in that coordinate. For a hypothetical aeration setting of 1, the transfer coefficient is 20 d^{-1} . If dissolved oxygen is $2 \text{ g O}_2 \text{ m}^{-3}$, the oxygen-transfer rate is $130 \text{ g O}_2 \text{ m}^{-3} \text{ d}^{-1}$. These calculations illustrate the accounting. They are not simulated treatment outcomes.

Steady state means that each modeled inventory has zero time derivative. It does not establish a unique steady state, dynamic stability, or a field-calibrated description. Sensitivity of operating choices to the process representation remains relevant even when a balance closes (de Araujo et al., 2013). In particular, a small numerical residual certifies agreement with the stated equations at the returned state. It cannot certify the completeness of the equations for a real wastewater system.

3.5 From stoichiometry to mass conservation invariants

A stoichiometric invariant is a linear combination of component concentrations that the modeled reactions cannot change. In the hypothetical three-pool network, adding the concentration values without weights would count one unit of B as if it contained the same inventory as one unit of A . Its material basis instead requires

$$I_{\text{toy}} = c_A + 2c_B + c_C.$$

Substitution of the three scalar reaction rates shows directly that its derivative is zero,

$$\frac{dI_{\text{toy}}}{dt} = (-2\rho_1) + 2(\rho_1 - \rho_2) + (2\rho_2) = 0.$$

The ρ_1 and ρ_2 terms each cancel separately. The identity therefore holds at every combination of process rates, including the hypothetical pair that gives component changes $(-6, 2, 2)^\top$. Its weighted change is $-6 + 2(2) + 2 = 0$.

The weights can be found instead of guessed. Let an unknown coefficient column be $a = (a_A, a_B, a_C)^\top$. Requiring zero weighted contribution from each process gives

$$\begin{aligned} -2a_A + a_B &= 0, \\ -a_B + 2a_C &= 0. \end{aligned}$$

The first equation gives $a_B = 2a_A$. The second then gives $a_C = a_A$. Choosing $a_A = 1$ produces $(1, 2, 1)^\top$. Choosing $a_A = 3$ produces $(3, 6, 3)^\top$ and the same mass conservation law multiplied by three. These columns lie in the null space of ν_{toy} , which is the set of columns sent to zero by that matrix.

For the full component basis, $\nu a = 0$ imposes the same zero-contribution condition on every modeled process. If $a \in \mathbb{R}^{20}$ satisfies this condition, then

$$a^\top \nu^\top \rho = (\nu a)^\top \rho = 0 \tag{3.19}$$

for every process-rate vector. This identity does not depend on the accuracy of an estimated rate or on the value of an operating input. It follows from the reaction accounting alone. Mass conservation-aware surrogate constructions use this separation to preserve balances without solving all of the original rate equations during inference (D. Hansen et al., 2024; Sturm & Wexler, 2020, 2022).

The number of independent mass conservation laws follows from the number of independent reaction directions. In the toy matrix, neither reaction row is a multiple of the other. More explicitly, suppose $s(-2, 1, 0) + t(0, -1, 2) = (0, 0, 0)$. The first coordinate gives $s = 0$ and the third gives $t = 0$. There are therefore two independent rows. Their rank is two. Three component coordinates subject to two independent zero-contribution equations leave one free invariant coefficient. This relation is called rank-nullity,

$$\text{number of component coordinates} = \text{rank} + \text{null-space dimension}.$$

The one free coefficient in the toy calculation is the choice of a_A . A basis is a collection of independent vectors from which all vectors in the space can be formed by weighted addition. One basis column suffices for that one-dimensional invariant space.

Singular-value decomposition determines such directions without solving every null-space equation manually. It factors a matrix into orthonormal directions and non-negative singular values. Orthonormal columns have unit Euclidean norm and zero

scalar product with each other. A zero singular value marks a direction that the matrix sends to zero. For the toy matrix, multiplying it by its transpose gives

$$\nu_{\text{toy}}\nu_{\text{toy}}^\top = \begin{bmatrix} 5 & -1 \\ -1 & 5 \end{bmatrix}.$$

This square matrix has eigenvalues 4 and 6, so the two nonzero singular values of ν_{toy} are 2 and $\sqrt{6}$. The remaining component direction is the invariant direction. The term eigenvalue means the factor by which a square matrix stretches a corresponding direction. Singular values describe the corresponding non-negative stretching magnitudes for a matrix that can be rectangular.

For the full decomposition, U has dimension 28×28 , Σ has dimension 28×20 , and V has dimension 20×20 . The rectangular Σ contains singular values on its diagonal and zeros elsewhere. Multiplication in the order $U\Sigma V^\top$ recovers a 28×20 matrix. An independent invariant basis is obtained by singular-value decomposition of the 28×20 matrix ν . The factorization is

$$\nu = U\Sigma V^\top, \quad (3.20)$$

In finite numerical arithmetic, an exactly zero singular value may appear as a very small number. The rank calculation therefore uses a threshold rather than testing exact numerical equality with zero. The largest singular value sets the matrix scale. Machine precision sets the scale of rounding error, and the larger matrix dimension accounts for the calculation size. The numerical rank counts singular values exceeding

$$\tau_{\text{rank}} = \max(P, F)\epsilon_{\text{mach}}\|\nu\|_2. \quad (3.21)$$

Here ϵ_{mach} is the machine precision used for the calculation and $\|\nu\|_2$ is the largest singular value. The adopted stoichiometric representation has rank 15. The rank-nullity identity therefore gives five independent invariants. This is a structural property of the specified reaction matrix.

The toy invariant column $(1, 2, 1)^\top$ has length $\sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}$. Dividing each entry by that length produces the normalized column

$$V_{0,\text{toy}} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

Its two process dot products are $(-2 + 2 + 0)/\sqrt{6} = 0$ and $(0 - 2 + 2)/\sqrt{6} = 0$. Its transpose gives an invariant row with length one. For the full matrix, five orthonormal right singular vectors play this role. Let $V_0 \in \mathbb{R}^{20 \times 5}$ contain the right singular vectors spanning $\text{null}(\nu)$. Define

$$A = V_0^\top \in \mathbb{R}^{5 \times 20}. \quad (3.22)$$

The first product below tests every combination of an invariant row and a reaction row. Its (h, p) entry is the scalar sum $\sum_{j=1}^{20} A_{hj}\nu_{pj}$. The second product tests the dot

products of invariant rows. A diagonal entry is their squared length and an off-diagonal entry is the dot product of two different rows. Unit lengths and mutual orthogonality therefore give the identity matrix. The resulting operator obeys

$$A\nu^\top = 0, \quad AA^\top = I_5. \quad (3.23)$$

Each row represents an independent component combination defining mass conservation. The orthonormal rows need not correspond individually to named elemental inventories. Their purpose is to provide a numerically specified basis for the same invariant space.

The matrix orientation matters. Because processes are rows of ν , the component combinations defining mass conservation lie in its right null space. Taking the right null space of $\nu^\top$ instead produces vectors in process coordinates. Those vectors describe dependencies among process contributions and have a different interpretation. Dimension checks prevent the two constructions from being confused.

Mass conservation of reactions does not by itself establish mass conservation across an open reactor boundary. For invariant row h , multiply the scalar component balance by A_{hj} and add it over all twenty components. The reaction sum can be grouped by process,

$$\begin{aligned} D \sum_{j=1}^{20} A_{hj}(c_{out,j} - c_{in,j}) &= \sum_{p=1}^{28} \rho_p \sum_{j=1}^{20} A_{hj} \nu_{pj} + \omega \sum_{j=1}^{20} A_{hj}(e_{s_O})_j \\ &= 0 + \omega A_{h1}. \end{aligned}$$

The first sum vanishes by the invariant construction. The selector leaves only the oxygen-column coefficient A_{h1} . Reaction cancellation and forcing cancellation are consequently separate checks. Premultiplying Equation (3.15) by A gives

$$DA(c_{out} - c_{in}) = \omega Ae_{s_O}. \quad (3.24)$$

The aeration forcing must therefore be compatible with the selected invariants. The distinction can be checked in the toy network. Adding one unit to pool A without another change gives a forcing column $(1, 0, 0)^\top$. Its invariant product is $1/\sqrt{6}$, so it changes the tracked inventory. A forcing column $(-2, 1, 0)^\top$ instead gives $(-2 + 2)/\sqrt{6} = 0$ and lies in the reaction-change space. The example concerns an abstract inventory and does not identify either forcing with oxygen transfer.

The range of a matrix is the set of vectors produced by multiplying it by all possible coordinate columns. A forcing in $\text{range}(\nu^\top)$ can be represented as a combination of reaction-change columns and is annihilated by every invariant row. For the adopted matrix, the oxygen selector lies in $\text{range}(\nu^\top)$. There is a process-coordinate vector $\lambda_O \in \mathbb{R}^{28}$ such that

$$e_{s_O} = \nu^\top \lambda_O, \quad Ae_{s_O} = A\nu^\top \lambda_O = 0. \quad (3.25)$$

The coordinates in λ_O establish a subspace relation. They do not state that aeration is itself a biological reaction or that these coordinates are non-negative process rates. The

scalar consequence is $A_{h1} = 0$ for every invariant row. Dividing $D \sum_j A_{hj}(c_{out,j} - c_{in,j}) = 0$ by the positive dilution rate gives equality of the influent and effluent invariant sums. Since $D > 0$, the invariant boundary relation reduces to

$$Ac_{out} = Ac_{in}. \quad (3.26)$$

The relation is justified jointly by stoichiometry, the forcing direction, and the reactor boundary. It should not be imposed on a system with an unaccounted external phosphorus dose, sludge withdrawal, or another source that changes the selected inventories.

More generally, an additional component source g contributes $\sum_j A_{hj}g_j$ to row h . Dividing this contribution by D converts its concentration-per-time basis into a concentration-equivalent invariant change. An additional component source g gives

$$A(c_{out} - c_{in}) = D^{-1}Ag \quad (3.27)$$

when aeration remains compatible. A hypothetical phosphate addition with a nonzero phosphorus-inventory contribution changes the right-hand side. Forcing equality with the original influent would then remove a real external contribution. This example shows why a physical constraint must be derived from the actual system boundary rather than chosen solely because it improves a statistical score.

The full-precision operator is used for all calculations. Its construction is checked through $A\nu^\top$, $AA^\top - I_5$, and Ae_{SO} . Rounding coefficients for display cannot replace the original operator in a numerical calculation. Any nonsingular row transformation T defines $A' = TA$ and the same equality set. An arbitrary transformation can change the magnitude of a residual. The orthonormal construction fixes the basis and scaling used for the mass conservation diagnostic.

3.6 Reaction progress and the feasible state geometry

The invariant equation can also be understood through reaction progress. In the hypothetical three-pool network, let the first process advance by one unit and the second by one half-unit on a compatible concentration-equivalent basis. The accumulated component changes are

$$\Delta c_A = -2(1) + 0(0.5) = -2,$$

$$\Delta c_B = 1(1) - 1(0.5) = 0.5,$$

$$\Delta c_C = 0(1) + 2(0.5) = 1.$$

Their weighted sum is $-2 + 2(0.5) + 1 = 0$. Starting from the hypothetical state $(8, 1, 1)^\top$ gives $(6, 1.5, 2)^\top$, and both states have weighted inventory 11. The amount of

progress differs from the instantaneous rate. Multiplication by a time scale converts a rate into a concentration-equivalent change.

In a steady reactor balance, the relevant time scale is D^{-1} . Substituting $\omega e_{S_O} = \nu^\top(\omega \lambda_O)$ combines reaction and compatible forcing in the same component-change space. Each component change is then a sum of the form $\sum_p \nu_{pj} r_p$. Rearranging the steady balance and using Equation (3.25) gives

$$c_{out} - c_{in} = \nu^\top r, \quad r = D^{-1}(\rho + \omega \lambda_O). \quad (3.28)$$

The vector r gives one set of process coordinates for the total component change. It is not a unique reconstruction of the biological rates because $\nu^\top$ has rank 15 and 28 columns. It also includes the algebraic representation of oxygen transfer. Individual entries of r therefore should not be interpreted as independently measured reaction extents or microbial activity.

A change satisfying mass conservation need not be described in the original process coordinates. In the toy network, the condition on a change v is $v_A + 2v_B + v_C = 0$. Two independent unit-length columns satisfying it are

$$z_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad z_2 = \frac{1}{\sqrt{30}} \begin{bmatrix} 1 \\ 2 \\ -5 \end{bmatrix}.$$

Their invariant dot products are $(-2 + 2 + 0)/\sqrt{5} = 0$ and $(1 + 4 - 5)/\sqrt{30} = 0$. Their mutual dot product is $(-2 + 2 + 0)/\sqrt{150} = 0$. Their squared lengths are $5/5 = 1$ and $30/30 = 1$. The matrix $Z_{\text{toy}} = [z_1 \ z_2]$ therefore supplies an orthonormal basis of the two-dimensional change space. Every change satisfying mass conservation for the weighted sum has a representation $v = z_1 q_1 + z_2 q_2$.

This change basis differs from the invariant basis. The invariant basis identifies quantities that cannot change. The columns of Z identify directions along which a state can change without changing those quantities. In the full model, five independent invariant rows leave fifteen independent change directions. Let $Z \in \mathbb{R}^{20 \times 15}$ have orthonormal columns spanning $\text{null}(A)$. The fundamental subspace relation gives

$$\text{range}(\nu^\top) = \text{null}(A), \quad AZ = 0, \quad Z^\top Z = I_{15}. \quad (3.29)$$

Every component change consistent with mass conservation can consequently be written as Zq for a unique $q \in \mathbb{R}^{15}$ in the selected orthonormal basis. The coordinates q describe an admissible direction of material redistribution. They do not identify a unique sequence of the 28 kinetic processes.

For one invariant row, the target for sample i is the influent sum $b_{ih} = \sum_j A_{hj} c_{in,ij}$. A feasible effluent must satisfy $\sum_j A_{hj} c_j = b_{ih}$ for each of the five rows and $c_j \geq 0$ for every component. The vertical bar in set notation means that all conditions to its right must hold. For sample i , define the target $b_i = A c_{in,i}$ and the feasible set

$$\mathcal{F}_i = \{c \in \mathbb{R}^{20} \mid Ac = b_i, \ c \geq 0\}. \quad (3.30)$$

The equality describes a 15-dimensional affine space before concentration bounds are applied. An affine space is a linear change space translated by a fixed reference state. Here that reference can be the influent. Non-negativity intersects the affine space with the non-negative orthant, which is the region where every coordinate is non-negative. The intersection is closed because its defining equalities and bounds include their boundary points. It is convex because a weighted average of two feasible states with weights between zero and one retains both their invariant target and non-negative coordinates. A non-negative influent belongs to the set, so the set is nonempty and the influent supplies a feasible anchor.

For the hypothetical three-pool influent $(8, 1, 1)^\top$, the conditions are $c_A + 2c_B + c_C = 11$ and $c_A, c_B, c_C \geq 0$. If only one pool is nonzero, the three boundary states are $(11, 0, 0)^\top$, $(0, 5.5, 0)^\top$, and $(0, 0, 11)^\top$. The feasible region is the triangle between those states in the inventory plane. The state $(6, 1.5, 2)^\top$ lies inside the triangle. A state $(12, -0.5, 0)^\top$ lies in the same inventory plane because $12 - 1 = 11$, but it fails the concentration bound. The equality selects the plane and non-negativity selects the admissible part of it.

Figure 3.2 displays this triangle using c_A and c_B as its axes. The inventory equality supplies the third coordinate through $c_C = 11 - c_A - 2c_B$, so every plotted point represents a complete three-pool state.

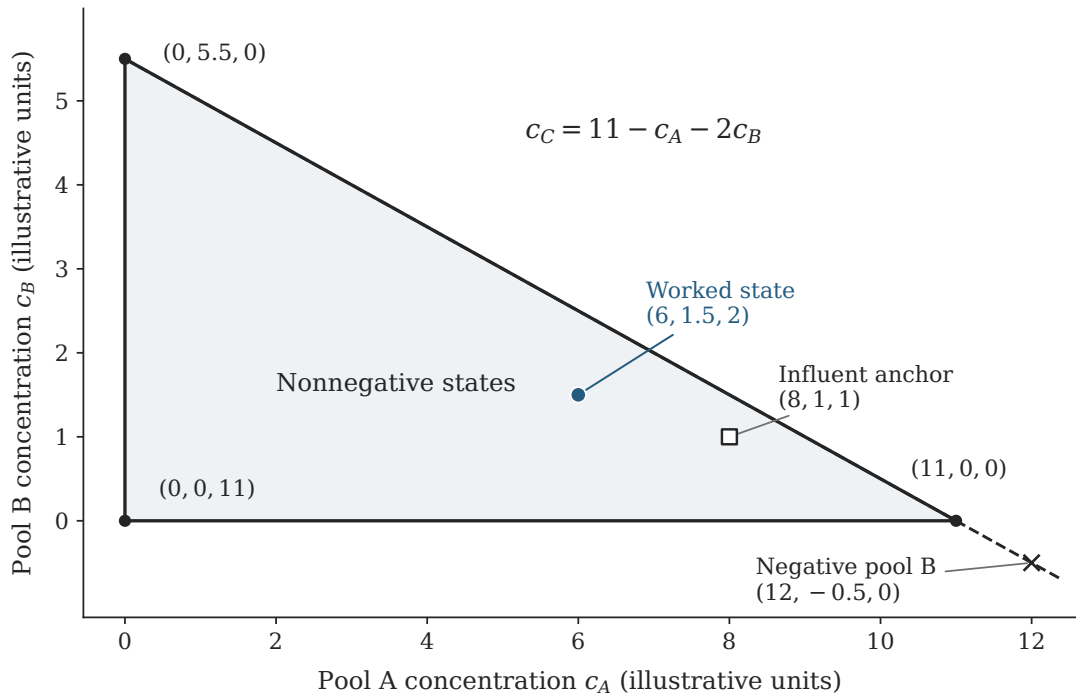


Figure 3.2: Hypothetical non-negative states for the weighted inventory $c_A + 2c_B + c_C = 11$. The shaded triangle includes its boundary. Full state labels retain the third component even though only two coordinates are plotted.

The sloping edge has $c_C = 0$. The two coordinate edges have $c_A = 0$ or $c_B = 0$.

Inside the triangle, all three components are positive and can redistribute the same inventory in different proportions. The cross-marked point preserves the inventory but has negative pool B . The figure therefore shows how concentration bounds reduce the inventory plane to a particular feasible region without selecting a unique kinetic state.

A two-component hypothetical system with sum fixed by mass conservation $c_1 + c_2 = 10$ illustrates the geometry. The equality permits states such as $(-1, 11)$, $(4, 6)$, and $(12, -2)$. Non-negativity retains only the line segment between $(0, 10)$ and $(10, 0)$. Clipping $(-1, 11)$ to $(0, 11)$ sets its negative coordinate to zero but changes the sum to 11. Enforcing the equality alone and enforcing non-negativity alone therefore solve different problems. Simultaneous enforcement requires the intersection.

The geometry also explains why a feasible prediction need not be kinetically correct. Several non-negative states can share all five invariant values. A state with the wrong distribution between ammonium and nitrate may satisfy the invariant equality. A state with an implausible biomass concentration may do the same. The feasible set establishes the declared material accounting and sign restrictions. It does not establish that the 28 process rates balance hydraulics and aeration at that state. This distinction is central to the constraint methods of Kircher and Votsmeier (2025) and to the broader physical-consistency perspective of Valente et al. (2025).

3.7 Reported water quality quantities and hidden component errors

Reported water quality quantities are calculated after a component state has been predicted. Each quantity is first a scalar weighted sum of the components. The coefficients in Table 3.1 give

$$\begin{aligned}
 \text{COD} &= S_F + S_A + S_I + X_I + X_S + X_H + X_{PAO} \\
 &\quad + X_{PHA} + X_{AOB} + X_{NOB}, \\
 \text{TN} &= S_{NH4} + S_{NO2} + S_{NO3} + 0.01S_I + 0.02X_I \\
 &\quad + 0.07(X_H + X_{PAO} + X_{AOB} + X_{NOB}), \\
 \text{TP} &= S_{PO4} + 0.01X_I + 0.02(X_H + X_{PAO} + X_{AOB} + X_{NOB}) \\
 &\quad + X_{PP} + X_{MeP}/4.87, \\
 \text{TSS} &= 0.75(X_I + X_S) + 0.90(X_H + X_{PAO} + X_{AOB} + X_{NOB}) \\
 &\quad + 3.23X_{PP} + 0.60X_{PHA} + X_{MeP} + X_{MeOH}.
 \end{aligned}$$

For example, the factor 0.07 has units of g nitrogen per g biomass COD. Multiplying it by X_H on the biomass COD concentration basis produces a nitrogen concentration. The factor 3.23 has units of g suspended solids per g stored phosphorus. The sum for each reported quantity therefore combines contributions already converted to the same output basis.

A hypothetical reporting state containing only $X_H = 100 \text{ g COD m}^{-3}$ and $X_{PP} = 2 \text{ g P m}^{-3}$ gives

$$\begin{aligned} \text{COD} &= 100, & \text{TN} &= 0.07(100) = 7, \\ \text{TP} &= 0.02(100) + 2 = 4, & \text{TSS} &= 0.90(100) + 3.23(2) = 96.46. \end{aligned}$$

Each numerical output uses its own reported constituent unit. This sparse example teaches the composition calculation and is not asserted to be a reactor steady state. The matrix map records these same four sums by placing the COD coefficients in its first row, the TN coefficients in its second row, and the remaining two coefficient groups in their corresponding rows. With the coefficients in Table 3.1,

$$y = I_{\text{comp}}c, \quad y = (\text{COD}, \text{TN}, \text{TP}, \text{TSS})^\top, \quad I_{\text{comp}} \in \mathbb{R}^{4 \times 20}. \quad (3.31)$$

The four rows convert the component vector to g COD m^{-3} , g N m^{-3} , g P m^{-3} , and g TSS m^{-3} , respectively. All coefficients are non-negative. A non-negative component state therefore gives non-negative reported quantities.

The nitrogen map excludes dissolved dinitrogen. This reporting convention makes TN different from an inventory used for mass conservation that includes nitrogen gas. In a hypothetical redistribution of 3 g N m^{-3} from nitrate to dissolved dinitrogen, the reported TN decreases by 3 g N m^{-3} while the complete nitrogen inventory can remain unchanged. Conversely, a phosphorus inventory that includes every modeled phosphorus pool can coincide with a reporting row. The relationship between a reported total and an invariant must be checked from their coefficients rather than inferred from the word “total.”

The same coefficient table explains constituent conversion. A hypothetical X_H concentration of 100 g COD m^{-3} contributes 100 g COD m^{-3} to COD, 7 g N m^{-3} to TN, 2 g P m^{-3} to TP, and 90 g TSS m^{-3} to TSS under the adopted map. A hypothetical X_{PP} concentration of 2 g P m^{-3} contributes 2 g P m^{-3} to TP and $6.46 \text{ g TSS m}^{-3}$ to TSS. These examples illustrate fixed accounting factors rather than evidence about a treatment outcome.

Aggregation can hide prediction errors. Equal and opposite errors in ammonium and nitrate cancel in reported TN because both have unit nitrogen weights. A negative ammonium value can even be hidden by a sufficiently large nitrate prediction. Likewise, different biomass and substrate errors can compensate in COD while producing distinct errors in TSS because their solids factors differ. Four accurate reported quantities therefore cannot establish an accurate or physically admissible 20-component state.

The reporting map and the invariant operator serve different purposes. The map provides engineering summaries. The invariant operator tests reaction-compatible material accounting at the declared boundary. Neither can recover all of the component information lost by aggregation. Component prediction, composite interpretation, mass conservation, and non-negativity therefore remain separate objects of assessment.

3.8 Mixing and spatial transport as model boundaries

The preceding reactor balance assumes a single mixed concentration for each component in a reactor. This assumption removes spatial coordinates from that reactor state. It is a modeling decision rather than a claim that every treatment unit is uniform. Biofilms, trickling filters, and aerated vessels with incomplete mixing can have concentration gradients and transport limitations (Sadino-Riquelme et al., 2023; Wik, 2003; T. C. Zhang & Bishop, 1994). These effects provide useful context for interpreting the scope of a mixed-reactor surrogate.

A small spatial control volume illustrates the additional accounting. For one component, let V be its fixed volume and $\bar{c}$ its average concentration. Let $F_{\text{adv,in}}$ and $F_{\text{adv,out}}$ be advective mass flow rates. Let $F_{\text{diff,in}}$ and $F_{\text{diff,out}}$ be diffusive mass flow rates. The finite-volume balance is

$$V \frac{d\bar{c}}{dt} = F_{\text{adv,in}} - F_{\text{adv,out}} + F_{\text{diff,in}} - F_{\text{diff,out}} + Vr. \quad (3.32)$$

The symbol r is the net reaction production rate per unit volume. It can be negative when the component is consumed. Every term on the right has units of component mass per time. For a hypothetical volume of 2 m^3 , advective inflow and outflow of 5 and 3 g d^{-1} , diffusive inflow and outflow of 2 and 0 g d^{-1} , and total reaction consumption of 3 g d^{-1} , the accumulation is $5 - 3 + 2 - 0 - 3 = 1 \text{ g d}^{-1}$. Dividing by the volume gives $d\bar{c}/dt = 0.5 \text{ g m}^{-3} \text{ d}^{-1}$. This calculation shows why the reaction term alone cannot determine a local concentration change.

In a one-dimensional domain with coordinate z , the advective flux is vc and the diffusive flux is $-D_z \partial c / \partial z$. Here v is the velocity and D_z is a dispersion or diffusion coefficient. Dividing the balance of a slice by its volume and taking the limit as its thickness decreases gives

$$\frac{\partial c_j}{\partial t} = -\frac{\partial(vc_j)}{\partial z} + \frac{\partial}{\partial z} \left(D_z \frac{\partial c_j}{\partial z} \right) + \sum_{p=1}^{n_p} \nu_{pj} \rho_p + s_j. \quad (3.33)$$

The term s_j represents a specified external source per unit volume. The first two terms express spatial transport. The summation is the same component-by-component reaction accounting used in the mixed reactor. Boundary conditions specify how material enters and leaves the spatial domain. An invariant weighted sum cancels compatible reaction terms, but the transport and external-source terms still remain in its balance.

This spatial formulation explains how the theoretical ideas can extend to other treatment representations. It is not an additional spatial reactor in the present study. Chapter 6 uses mixed reactor stages and a layered clarifier. The stages represent sequential treatment conditions, while the clarifier layers represent solids transport and inventory. Their distinct boundaries must be respected when mass conservation is imposed.

Chapter 4

Learning the Reactor Response and Enforcing Physical Admissibility

4.1 Accuracy and physical admissibility as distinct questions

A surrogate learns the relationship between a reactor input and its steady-state effluent. Its statistical task is to reproduce component concentrations from examples. Its physical task is to return a non-negative component vector with the invariant values appropriate to the supplied influent. These tasks are related because the mechanistic targets satisfy the reactor equations. They remain distinct because an ordinary regression loss does not explicitly restrict every prediction to the sample-specific feasible set (D. Hansen et al., 2024; Kashinath et al., 2021).

For sample i , the two operating values are entered first. They are followed by the twenty influent concentrations in the established component order. Written as a list, the input begins with $(\text{HRT}_i, u_{\text{air},i}, S_{O,\text{in},i}, S_{F,\text{in},i})$ and ends with the remaining influent coordinates through $X_{\text{MeOH},\text{in},i}$. The component list and its units are fixed before any learner is fitted. Transposing the two input blocks into rows permits their horizontal concatenation. The final transpose restores the column convention. The complete input for sample i is

$$x_i = (u_i^\top, c_{\text{in},i}^\top)^\top \in \mathbb{R}^{22}, \quad u_i = (\text{HRT}_i, u_{\text{air},i})^\top. \quad (4.1)$$

A fitted surrogate is a function that takes these twenty-two numbers and returns twenty component values. The subscript m identifies which fitted method is used. For component j , its prediction can be written as $c_{\text{raw},ij}^{(m)} = f_{m,j}(x_{i1}, \dots, x_{i22})$. The superscript (m) identifies the model and is not an exponent. Collecting its twenty scalar output functions, a fitted surrogate f_m produces the raw component prediction

$$c_{\text{raw},i}^{(m)} = f_m(x_i) \in \mathbb{R}^{20}. \quad (4.2)$$

The raw prediction is assessed in its returned physical units. A post prediction projection then produces a second prediction for the same sample. Pairing the two states permits the effect of physical enforcement to be assessed separately from the choice of learner, data partition, and preprocessing.

The comparison includes 13 statistical surrogates. The set spans boosting, randomized ensembles, kernel regression, neighborhood regression, latent-variable regression, sparse multiresponse regression, and neural networks. This breadth addresses the model-selection concern raised by Bhosekar and Ierapetritou (2018). The surveys of Borisov et al. (2024) and Shwartz-Ziv and Armon (2022) further explain why a neural architecture alone should not determine the comparator set for a tabular prediction problem. The purpose of the common protocol is to compare explicitly specified predictors for the same physical state.

4.2 Constructing a mechanistic learning domain

The single-reactor dataset is designed to contain 10,000 accepted steady states. Each candidate varies all 22 input coordinates. A randomized, scrambled, strength-one Latin hypercube places one sample in each stratum of every marginal input distribution. The construction uses seed 42 and no subsequent design optimization. Linear scaling maps the unit-hypercube samples to the bounds in Table 4.1. Latin hypercube sampling distributes coverage across marginal ranges without requiring a full Cartesian grid (McKay et al., 1979). It does not guarantee uniform coverage of every nonlinear response feature or every high-dimensional interaction.

The standalone domain is a declared simulation design. It is not a distribution estimated from plant measurements. In particular, its alkalinity interval is retained in the stated native coordinate. The operating and influent domains of a coupled plant are specified independently. A common component symbol does not establish that two evaluation designs use the same bounds.

For each candidate input, nonlinear least squares solves Equation (3.14) with numerical state bounds $[10^{-8}, 1500]$ in each component's native unit. The cost-change, step-size, and gradient convergence tolerances are 10^{-9} . Each solve allows at most 1200 function evaluations. The generation procedure has an overall attempt cap of 25 times the target retained count. A state is retained only if the solver converges and the maximum absolute component-balance residual is no greater than 10^{-9} on the fixed native component-unit-per-day coordinates. The residual criterion is a numerical acceptance rule rather than a dimensionless physical error measure.

Two physically informed starting states are tried before dynamic relaxation. The first blends the sampled influent with a preceding accepted effluent when one is available. It lowers readily consumed substrate coordinates, retains biomass pools, and estimates dissolved oxygen from dilution and transfer. The second applies stronger sub-

Table 4.1: Design bounds for the standalone reactor learning domain. These intervals specify sampling limits and do not report realized sample extrema.

Input	Meaning	Interval	Unit
HRT	Hydraulic operating input	[6, 36]	h
u_{air}	Dimensionless aeration input	[0.5, 2.5]	1
S_O	Influent dissolved oxygen	[0, 0.5]	g O ₂ m ⁻³
S_F	Influent fermentable substrate	[20, 180]	g COD m ⁻³
S_A	Influent acetate	[5, 80]	g COD m ⁻³
S_{NH_4}	Influent ammonium nitrogen	[12, 55]	g N m ⁻³
S_{NO_2}	Influent nitrite nitrogen	[0, 3]	g N m ⁻³
S_{NO_3}	Influent nitrate nitrogen	[0, 8]	g N m ⁻³
S_{N_2}	Influent dissolved dinitrogen	[0, 2]	g N m ⁻³
S_{PO_4}	Influent soluble phosphate	[2, 18]	g P m ⁻³
S_I	Influent soluble inert organic matter	[10, 90]	g COD m ⁻³
S_{ALK}	Influent alkalinity	[80, 260]	mol m ⁻³
X_I	Influent particulate inert matter	[20, 120]	g COD m ⁻³
X_S	Influent slowly biodegradable substrate	[60, 280]	g COD m ⁻³
X_H	Influent heterotrophic biomass	[15, 100]	g COD m ⁻³
X_{PAO}	Influent phosphate-accumulating biomass	[5, 60]	g COD m ⁻³
X_{PP}	Influent stored polyphosphate	[2, 20]	g P m ⁻³
X_{PHA}	Influent stored polyhydroxyalkanoates	[1, 30]	g COD m ⁻³
X_{AOB}	Influent ammonia-oxidizing biomass	[0.5, 8]	g COD m ⁻³
X_{NOB}	Influent nitrite-oxidizing biomass	[0.5, 8]	g COD m ⁻³
X_{MeP}	Influent precipitated metal phosphate	[0, 12]	g TSS m ⁻³
X_{MeOH}	Influent metal hydroxide	[0, 12]	g TSS m ⁻³

strate depletion and biomass enrichment. Starting vectors are restricted to $[10^{-6}, 1500]$. If both steady solves fail, a stiff backward differentiation formula integrates the dynamic balance for 20 days and supplies a new initial state. The state bounds and starting-state restrictions apply to concentrations. The 28 kinetic process rates are evaluated without an artificial ceiling or clipping.

Solver acceptance, invariant residual, and minimum concentration are recorded separately. A converged bounded least-squares calculation can terminate without meeting the required balance residual. A positive state can also fail mass conservation. Retaining these separate diagnostics prevents one successful numerical check from standing in for all others. The surrogate inputs contain the sampled operating and influent coordinates. Initialization choices and solver history are not predictive features.

4.3 How the thirteen predictors represent the response

4.3.1 Boosting and randomized tree ensembles

XGBoost builds an additive sequence of regression trees. A regression tree applies a sequence of input comparisons and assigns the value associated with the reached terminal region. For one output, an additive tree model has the scalar form

$$f_j(x) = b_j + \beta_1 h_{1j}(x) + \beta_2 h_{2j}(x) + \cdots + \beta_T h_{Tj}(x).$$

Here h_{tj} is the response from tree t , β_t determines its contribution, and b_j is a baseline. The summation notation $b_j + \sum_{t=1}^T \beta_t h_{tj}(x)$ records the same additions. A hypothetical baseline of 10 and two tree contributions of -2 and 1 produce 9. Each new tree addresses the error left by the current ensemble while regularization discourages an unnecessarily complex fit. Its split structure represents changes in response across input regions, and repeated additions permit interactions and nonlinear response shapes. The computational design and regularized objective of Chen and Guestrin (2016) motivate its inclusion as a flexible tabular baseline.

LightGBM also uses gradient-boosted trees, but grows trees through leaf-wise expansion. This strategy can allocate more detail to regions where an additional split offers a larger loss reduction. The efficiency-oriented design of Ke et al. (2017) makes the method relevant when each candidate fit must be repeated over several validation partitions. Its learning rate, leaf count, and minimum leaf support jointly control how closely it follows the training response.

CatBoost uses symmetric boosted trees and a Bayesian bootstrap in the specified regression setting. The work of Prokhorenkova et al. (2018) addresses bias associated with categorical-feature handling and boosting. The reactor benchmark has continuous inputs, so its inclusion evaluates the corresponding regression learner without claiming

a categorical-data advantage. Each level of a symmetric tree uses the same splitting condition across nodes at that level. This structural restriction gives a different approximation class from the leaf-wise trees of LightGBM.

Adaptive boosting (AdaBoost) regression combines shallow trees through adaptive emphasis on observations that are difficult to predict. The base trees have depth three. The estimator count, learning rate, and loss choice control how strongly the ensemble responds to poorly fitted examples. Drucker (1997) develops the regression extension of boosting that supports this comparator. These four boosting methods fit one regressor for each effluent component. All 20 regressors within a method use one shared selected hyperparameter vector.

Random Forest averages randomized regression trees. For component j , the average of T tree responses is $f_j(x) = [h_{1j}(x) + \dots + h_{Tj}(x)]/T$. Hypothetical tree values of 2, 3, and 4 give a mean of 3. Variation in sampled observations and candidate features reduces dependence among trees, so the average is less sensitive to a particular tree fit (Breiman, 2001). Extra Trees adds randomization of split thresholds (Geurts et al., 2006). Both methods fit the 20 outputs jointly in native multiresponse ensembles. Joint output handling lets the same tree partition support several component predictions. It does not impose a stoichiometric relationship among those outputs.

Tree averaging can preserve non-negativity when all leaf responses are non-negative. It cannot generally preserve the balance target for a new influent. For a hypothetical two-component system, two training responses with totals of 10 and 20 have an equally weighted average with total 15. If a new influent requires total 12, the average is positive but inconsistent with mass conservation for that sample. This distinction is caused by the sample-specific target b_i , rather than by whether the predictor averages positive observations.

4.3.2 Kernel, neighborhood, and latent-variable regression

Support vector regression fits a function with an error-insensitive region around each training response. For scalar error e and positive insensitive width ϵ , its error contribution is zero when $|e| \leq \epsilon$ and $|e| - \epsilon$ otherwise. This can be recorded as $\max(0, |e| - \epsilon)$. With a hypothetical width of 0.2, an error of 0.1 contributes zero and an error of -0.5 contributes 0.3. Predictions within that region incur no corresponding error penalty. A linear kernel gives a globally linear input mapping. A radial kernel allows a nonlinear mapping through similarity to training inputs. Its similarity has the form

$$K(z, z') = \exp \left[-\gamma \sum_{\ell=1}^{22} (z_{\ell} - z'_{\ell})^2 \right].$$

Here z and z' contain the standardized input coordinates used by this learner. The positive coefficient γ controls how quickly similarity decreases with squared distance. Identical inputs give similarity one. Greater distance gives a smaller value. The penalty

parameter balances fit and regularity, while the insensitive width sets the tolerated response discrepancy. Smola and Schölkopf (2004) explains these mechanisms and the importance of their parameterization. The benchmark fits one regressor per component with one shared setting vector.

The k -nearest-neighbor predictor finds training observations close to a new input and combines their 20-component responses. For one component and a selected neighborhood $\mathcal{N}(x)$, weighted averaging means

$$f_j(x) = \frac{\sum_{i \in \mathcal{N}(x)} w_i c_{out,ij}}{\sum_{i \in \mathcal{N}(x)} w_i}.$$

The neighborhood size controls the degree of smoothing. Uniform weights give every selected neighbor equal influence, whereas distance weights give closer observations greater influence. Hypothetical neighbor values 2 and 6 give 4 under equal weights. Weights 3 and 1 instead give $[3(2) + 1(6)]/(3 + 1) = 3$. The local behavior analyzed by Mack (1981) explains why input scaling and distance choice matter. An unscaled alkalinity coordinate spanning hundreds of numerical units could dominate a much smaller aeration coordinate even though both influence the reactor. Standardization avoids using native numerical range as an implicit measure of scientific importance.

Partial least squares regression represents the inputs and joint response through latent directions chosen to capture their covariance. One latent score is a weighted input sum $t_h = w_{1h}z_1 + \dots + w_{22h}z_{22}$, where z contains standardized inputs. An output is then a baseline plus a weighted sum of the retained scores. Substituting each score's input sum shows why ordinary partial least squares remains a linear input mapping. It can summarize strongly related input variables with a relatively small number of components. The chemometric formulation described by Wold et al. (2001) provides an inspectable linear baseline. Its wastewater use is illustrated by Woo et al. (2009), who employs a kernel extension for process-variable estimation, and by C. Yang et al. (2021), who combines partial least squares with another predictor for adaptive effluent-quality prediction. The benchmark retains ordinary partial least squares with one to six latent components. The nonlinear extensions in those applications do not make the present baseline nonlinear.

4.3.3 Shared sparse coefficient models

Multi-task Lasso fits one coefficient matrix for all component outputs. A predicted output is a baseline plus a sum of input-coefficient products. For two hypothetical inputs and two outputs, the coefficient rows can be $(3, 4)$ and $(0, 0)$. The first input contributes to both outputs and the second contributes to neither. The Euclidean length of the first row is $\sqrt{3^2 + 4^2} = 5$, and the length of the second row is zero. Adding row lengths gives 5. This operation penalizes an input's complete coefficient group rather than adding the absolute value of every coefficient separately. If row $B_{j\cdot}$,

contains the coefficients of input j across those outputs, the row-group penalty is

$$\|B\|_{2,1} = \sum_j \|B_{j,\cdot}\|_2. \quad (4.3)$$

This penalty can set an entire input row to zero. It therefore promotes a shared selection of inputs rather than 20 unrelated selections. The feature-sharing principle of Argyriou et al. (2008), the multi-task Lasso algorithm of Liu et al. (2009), and the grouped-variable framework of Yuan and Lin (2006) provide the relevant foundations.

Multi-task Elastic Net combines the row-group penalty with a squared Frobenius penalty. The squared Frobenius norm is the sum of the squares of all matrix entries. For the hypothetical coefficient rows just given, it is $3^2 + 4^2 + 0^2 + 0^2 = 25$. With penalty strength $\lambda > 0$ and mixing ratio $r \in (0, 1)$, the combined penalty has the form

$$\lambda r \sum_j \sqrt{\sum_{f=1}^{20} B_{jf}^2} + \frac{\lambda(1-r)}{2} \sum_j \sum_{f=1}^{20} B_{jf}^2.$$

For the two-output illustration only, taking $\lambda = 0.2$ and $r = 0.5$ gives $0.2(0.5)(5) + 0.2(0.5)(25)/2 = 1.75$. The row-group part encourages selection and the squared part shrinks the retained coefficients. This extends the elastic-net principle of Zou and Hastie (2005) to a joint response structure, with multiresponse computation described by Simon et al. (2013). Both methods use standardized inputs and targets. Their global coefficient matrices can be inspected to identify fitted associations. A nonzero row shows that an input contributes to the fitted prediction, but it does not establish a causal effect or a mechanistic reaction rate.

These two predictors and partial least squares are grouped as structurally inspectable surrogates. The grouping identifies their globally available coefficient structures. It is not a measured claim that users understand them more easily or that their coefficients are scientifically correct. It also does not establish physical admissibility. A global linear relation can still predict a negative concentration or violate the influent-specific balance.

4.3.4 Neural predictors

The multilayer perceptron uses three fully connected hidden layers with 128, 64, and 32 units and one joint 20-output layer. One unit first forms a weighted sum, such as $b + w_1 z_1 + w_2 z_2$, and then applies an activation. A rectified linear activation returns the larger of zero and that sum. For hypothetical values $b = 0.2$, $w_1 = 0.5$, $w_2 = -1$, $z_1 = 2$, and $z_2 = 0.3$, the sum is $0.2 + 0.5(2) - 0.3 = 0.9$ and the activated value is 0.9. If the sum were -0.9 , the activated value would be zero. Each hidden layer applies such operations to the values supplied by the preceding layer. Backpropagation updates the weights according to the prediction loss (Rumelhart et al., 1986). The approximation result of Hornik et al. (1989) motivates the expressive capacity of feedforward networks. It

does not guarantee that this finite architecture learns the reactor response from a given sample size. The topology considerations discussed by Curteanu and Cartwright (2011) likewise motivate explicit architectural and training limits rather than an unspecified neural comparator.

TabNet applies sequential attentive feature selection. At a decision step, a feature mask multiplies each input coordinate by its selected attention weight. If two hypothetical standardized inputs are $(2, 1)$ and their mask weights are $(0.8, 0.2)$, the masked inputs are $(1.6, 0.2)$. A different step can use different weights and therefore examine a different input emphasis. The selected features then contribute to a joint output layer. Arik and Pfister (2021) develops this approach for tabular learning. The attention masks provide information about the model’s use of inputs, but they do not impose mass conservation or make the component outputs causal explanations. TabNet remains a neural comparator even though its internal masks can be inspected.

Both neural predictors start from random initialization and use only the simulated training partition available to that fit. No external learned representation is used. The multilayer perceptron uses all available training rows and stops on training-loss convergence or at 250 epochs. Validation-based early stopping is disabled. Its training-loss patience is 15 epochs. TabNet’s epoch count is selected in the inner search and ranges from 20 to 50. Its patience is zero, so a refit runs the selected epoch count without reserving an additional response-based stopping partition.

4.4 Preprocessing and validation without response leakage

The evaluation uses nested subsets at

$$N \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (4.4)$$

A seed-42 permutation determines the nested row order and persistent master fold identifiers before subsets are formed. Every smaller subset is contained in each larger subset. A row retains its outer-fold role at every scale. Five outer folds train on $0.8N$ observations and score the remaining $0.2N$. Thus a design size of 500 gives 400 training rows and 100 scored rows per outer iteration. A design size of 10,000 gives 8000 training rows and 2000 scored rows. The five iterations change which fold is withheld, so each row supplies a scored prediction from a fit that excludes it. These counts explain the design and do not report model performance. The same rows and fold identifiers are used by every surrogate and both prediction types.

Hyperparameters are selected using four inner folds within the largest outer-training partition. The scored outer fold is absent from the search. Within an 8000-row outer-training pool, an inner iteration fits on 6000 rows and scores 2000 different rows from

that pool. Those inner-validation rows are not the 2000 rows reserved for outer assessment. The inner score selects the setting, and the outer score assesses a model fitted with that selected setting. For each outer fold, its selected setting is retained over the smaller nested subsets. Learning curves therefore examine the response to sample size under a fixed fold-specific setting. They do not represent a new optimum obtained through a separate hyperparameter search at every scarce-data scale. The bias–variance discussion of Belkin et al. (2019) cautions against assuming that error must vary monotonically with model capacity or available data. The learning-curve review of Viering and Loog (2023) supports treating the complete curve as an object of assessment.

Every transformation is fitted within the applicable training partition. During inner selection, that partition is the inner-training split. During outer scoring, it is the outer-training subset at the current size. During extrapolation assessment, it is the full in-domain training pool. A training coordinate has one mean and one standard deviation for that fit. If its M training values are $v_1, \dots, v_M$, these quantities are calculated in three steps,

$$\mu = \frac{v_1 + \dots + v_M}{M}, \quad \sigma^2 = \frac{(v_1 - \mu)^2 + \dots + (v_M - \mu)^2}{M}, \quad \sigma = \sqrt{\sigma^2}.$$

The second quantity is the population variance. The population convention divides by the training count M . A z-score subtracts the mean and divides by the standard deviation, giving $z = (v - \mu)/\sigma$. Subtracting the mean centers the coordinate and division puts its numerical variation on a unit-standard-deviation scale. The transformation requires $\sigma > 0$.

For hypothetical training values 2, 2, 6, and 6, the mean is 4. Their centered values are $-2, -2, 2$, and 2 , so the variance is $(4+4+4+4)/4 = 4$ and the standard deviation is 2. A validation value of 6 therefore becomes $(6-4)/2 = 1$. The validation value is not added to the training values before calculating the mean or variance. If this coordinate is an output, a predicted z-score of 0.5 returns to physical units as $4 + 2(0.5) = 5$. These calculations illustrate transformations rather than measured reactor data.

The mean and population standard deviation of a training coordinate define its z-score transformation. Validation inputs are transformed with those training quantities, and target-scaled predictions are restored to physical units before projection or physical diagnostics. Every target has its own inverse transformation $\hat{c}_j = \mu_j + \sigma_j \hat{z}_j$. A single common inverse scale would be incorrect for components with different units or variability.

Tree partitions depend on coordinate ordering, so a positive affine input rescaling preserves candidate threshold partitions. Distance-based, covariance-based, and regularized models require more deliberate scaling because relative numerical magnitudes affect their objectives. Separate-output tree regressors do not combine different component units within one joint fitting loss. The target-only scaling of Extra Trees and

Table 4.2: Output handling and preprocessing for the statistical comparators. Scaling uses only the training partition applicable to each fit.

Surrogates	Output handling	Scaling
XGBoost, LightGBM, Cat-Boost	Separate component regressors	None
AdaBoost	Separate component regressors	Targets only
Random Forest	Joint tree response	None
Extra Trees	Joint tree response	Targets only
Support vector regression	Separate component regressors	Inputs and targets
k -nearest neighbors	Joint neighborhood response	Inputs and targets
Partial least squares, Multi-task Lasso, Multi-task Elastic Net	Joint coefficient structure	Inputs and targets
Multilayer perceptron, TabNet	Joint neural output	Inputs and targets

AdaBoost is retained as a specified benchmark choice. Independent of those training choices, every scored component error is standardized with the designated training reference. Training target scaling and evaluation normalization are distinct operations.

No missing-value imputation, response-based outlier removal, or target filtering is applied. No validation or extrapolation response is used to estimate a scaling parameter. A hypothetical decision to standardize a validation target with the standard deviation of its own fold would change the scoring reference using data outside training. Retaining a training reference instead asks how large an error is relative to the variability the learner was allowed to observe.

4.5 Hyperparameter selection and finite comparison scope

Each surrogate uses a seeded tree-structured Parzen search with 100 trials per study and no pruning or timeout. One trial proposes a complete parameter setting, fits it within the prescribed inner-training partitions, and calculates its validation objective. The objective is an error value to be minimized. It differs from a parameter value such as tree depth or penalty strength. The search mechanism is supported by the optimization framework of Akiba et al. (2019). The objective is the mean squared standardized component error over the inner validation folds. Five studies select the settings for

the five largest-scale outer-training partitions. A sixth, independent four-fold study selects the setting for a final full-domain fit. The design specifies $6(100) = 600$ trials per surrogate and $13(600) = 7800$ across all 13 surrogates. These are allocated search budgets rather than performance findings.

Table 4.3: Search domains and fixed controls for tree ensemble surrogates. Integer limits are inclusive. “Log” denotes a log-uniform search.

Surrogate	Search domain and fixed controls
XGBoost	80–320 trees. Learning rate 0.01–0.2 (log). Depth 3–8. Minimum child weight 0.5–8 (log). Row and feature fractions 0.6–1. Absolute-value penalty 10^{-6} –1 (log). Squared penalty 10^{-3} –10 (log). Histogram or approximate trees. Squared-error objective.
LightGBM	80–320 trees. Learning rate 0.01–0.2 (log). 15–63 leaves. Depth unbounded or in $\{4, 6, 8, 10\}$. Minimum child rows 5–40. Row and feature fractions 0.6–1 with sub-sampling frequency 1. Absolute-value penalty 10^{-6} –1 (log). Squared penalty 10^{-6} –10 (log). Regression objective.
CatBoost	80–320 iterations. Learning rate 0.01–0.2 (log). Depth 4–8. Squared penalty 0.01–20 (log). Bagging temperature 0–5. Random strength 10^{-3} –5 (log). Root mean squared error loss and Bayesian bootstrap.
AdaBoost	50–250 depth-three trees. Learning rate 0.01–1 (log). Linear, squared, or exponential regression loss.
Random Forest	100–400 trees. Depth unbounded or in $\{6, 10, 14, 18\}$. Minimum split size 2–10. Minimum leaf size 1–4. Feature fraction 0.5–1. Bootstrap enabled or disabled.
Extra Trees	100–400 trees. Depth unbounded or in $\{6, 10, 14, 18\}$. Minimum split size 2–10. Minimum leaf size 1–4. Feature fraction 0.5–1. Bootstrap enabled or disabled.

The finite ranges define the actual comparison. A linear or radial support vector model does not represent every possible kernel configuration. A neural predictor limited to 250 epochs does not represent every training budget or architecture. The shared 100-trial allocation does not guarantee equal difficulty of tuning across model families. The study therefore assesses these specified approximation classes under a common protocol rather than claiming universal superiority of one family.

Table 4.4: Search domains and fixed controls for kernel, neighbor, multiresponse, and neural surrogates. Integer limits are inclusive. “Log” denotes a log-uniform search.

Surrogate	Search domain and fixed controls
Support vector regression	Penalty 0.1–10 (log). Error-insensitive width 0.01–0.5 (log). Linear or radial kernel. Radial coefficient set from standardized input variance or inverse input dimension. Tolerance 10^{-3} – 10^{-2} (log). Shrinking enabled. At most 20,000 iterations. Kernel cache of 256 megabytes per output regressor.
k -nearest neighbors	1–15 neighbors. Uniform or distance weights. Automatic, ball-tree, k -dimensional tree, or brute-force search. Leaf size 15–60. Minkowski, Euclidean, or Manhattan distance. Minkowski order in $\{1, 2, 3\}$.
Partial least squares	1–6 latent components. 200–1000 maximum iterations. Tolerance 10^{-8} – 10^{-4} (log). Internal scaling disabled because external training-partition scaling is applied.
Multi-task Elastic Net	Penalty strength 10^{-6} –10 (log). Row-group mixing ratio 0.05–0.95. Cyclic coordinate descent. At most 20,000 iterations and tolerance 10^{-6} .
Multi-task Lasso	Penalty strength 10^{-6} –10 (log). Cyclic coordinate descent. At most 20,000 iterations and tolerance 10^{-6} .
Multilayer perceptron	Hidden widths (128, 64, 32). Rectified linear, hyperbolic tangent, or logistic activation. Adaptive moment optimization. Squared-weight penalty 10^{-6} – 10^{-2} (log). Batch size in $\{64, 128, 256\}$. Learning rate 5×10^{-4} – 10^{-2} (log). Training-loss tolerance 10^{-4} – 10^{-3} (log). Shuffling enabled. Training-loss patience 15 epochs. At most 250 epochs.
TabNet	Equal decision and attention widths in $\{8, 16, 24\}$. Three or four attention steps. Relaxation coefficient 1–2. Sparsity coefficient 10^{-6} – 10^{-2} (log). Adaptive moment learning rate 0.003–0.03 (log). Sparsemax or entmax-1.5 masks. 20–50 selected epochs. Batch size in $\{512, 1024\}$. Virtual batch size 128. Patience zero.

4.6 The Kircher–Votsmeier projection for mass conservation and non-negativity

4.6.1 A positive provisional target and relative-error geometry

Kircher and Votsmeier (2025) develop a projection that combines mass conservation with non-negativity for mechanistic-kinetics surrogates. Its transfer to the reactor prediction problem uses the invariant operator A , the basis Z , and the feasible influent anchor. The projection depends on the completed component prediction rather than on predictor internals. It can therefore be applied to every comparator without changing its training objective or architecture. It still requires the full component vector, the corresponding non-negative influent, and the physical operators.

The first step compares each raw component with its positive floor. If a raw component is above the floor, its provisional value is the raw value. Otherwise, its provisional value is the floor. For component j , the operation is $c_{+,ij}^{(m)} = \max(c_{raw,ij}^{(m)}, \varepsilon_{c,j})$. A hypothetical raw pair $(-0.2, 3)$ becomes $(10^{-8}, 3)$ when both native-unit floors have value 10^{-8} . The corresponding inverse values are 10^8 and $1/3$. This intermediate pair illustrates why a floor is needed before division and why it cannot be treated as a final projection for mass conservation.

A diagonal weighting matrix places each inverse concentration on its diagonal and zero elsewhere. For the hypothetical positive pair $(1, 2)^\top$, its action on a discrepancy column h is

$$W = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}, \quad Wh = \begin{bmatrix} h_1 \\ h_2/2 \end{bmatrix}.$$

No component is mixed with another in this operation. Each component discrepancy is divided by its own provisional concentration, so both weighted coordinates are relative discrepancies. The first step forms a strictly positive provisional target,

$$c_{+,i}^{(m)} = \max(c_{raw,i}^{(m)}, \varepsilon_c), \quad W_i^{(m)} = \text{Diag}(\mathbf{1} \oslash c_{+,i}^{(m)}). \quad (4.5)$$

The maximum and division are componentwise. Each entry of ε_c is 10^{-8} in its component's native unit. The provisional target supplies both a reference state and finite positive weights. It is an intermediate construction, so its clipped coordinates are not the final physical prediction.

For a component change h_j , the weighted discrepancy is $h_j/c_{+,ij}^{(m)}$. The geometry therefore penalizes relative movement. A hypothetical movement of 1 g m^{-3} corresponds to a relative discrepancy of 0.01 at a component value of 100 g m^{-3} and 10 at a component value of 0.1 g m^{-3} . Small coordinates receive strong protection. Large coordinates can absorb larger absolute changes at the same relative cost. This explains why relative-error projection and variance-standardized accuracy need not favor the same redistribution.

4.6.2 Selecting a candidate that satisfies mass conservation

Changes consistent with mass conservation can first be written one component at a time. If Z has fifteen change columns, coordinate j of a change is $Z_{j1}q_1 + \cdots + Z_{j,15}q_{15}$. Adding that change to $c_{in,ij}$ gives the candidate component. Since each column of Z has zero invariant product, every such combination preserves the influent target.

A complete two-component hypothetical calculation shows how these coordinates are chosen. Both components use the same abstract concentration basis. Let the required sum be four and the feasible anchor be $(2, 2)^\top$. The raw pair is $(1, 2)^\top$, which sums to three. The provisional pair is unchanged because both values exceed the floor. A normalized invariant row and a normalized change column are

$$A_{\text{ex}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad Z_{\text{ex}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Their product is $(1 - 1)/2 = 0$. A scalar change amount t gives the candidate $(2 + t, 2 - t)^\top$. In the normalized basis its coordinate is $q = \sqrt{2}t$, because $Z_{\text{ex}}q = (t, -t)^\top$. The sum remains four for every t . All states consistent with mass conservation for sample i have the form

$$c = c_{in,i} + Zq, \quad q \in \mathbb{R}^{15}. \quad (4.6)$$

For the hypothetical pair, the provisional target change from the anchor is $(-1, 0)^\top$. The difference between an allowed change and this target change is $(t + 1, -t)^\top$. Applying the inverse provisional concentrations gives $(t + 1, -t/2)^\top$. Its squared Euclidean norm is

$$E(t) = (t + 1)^2 + \frac{t^2}{4}.$$

At $t = 0$, this value is one. Choosing a negative t moves the first component toward its raw value while requiring an increase in the second to preserve the sum. A derivative describes the change in the objective for a small change in t . The derivative of $(t + 1)^2$ is $2(t + 1)$ and that of $t^2/4$ is $t/2$. Adding them gives the rate of change of the complete objective. The best compromise can be found by differentiating,

$$\frac{dE}{dt} = 2(t + 1) + \frac{t}{2} = \frac{5}{2}t + 2.$$

Setting the derivative to zero gives $t = -4/5$. The second derivative is $5/2 > 0$, so this point is the unique minimum. The resulting candidate is $(6/5, 14/5)^\top = (1.2, 2.8)^\top$. Its relative discrepancies from the provisional target are 0.2 and 0.4, giving squared objective $0.2^2 + 0.4^2 = 0.20$. A candidate $(1.5, 2.5)^\top$ instead gives $0.5^2 + 0.25^2 = 0.3125$. The weighting favors the first candidate because the smaller provisional component receives stronger protection against absolute movement.

Figure 4.1 shows this calculation as a geometric comparison. Curves around the raw pair connect states with the same projection objective. The line defined by mass conservation restricts the candidate to states whose two components sum to four.

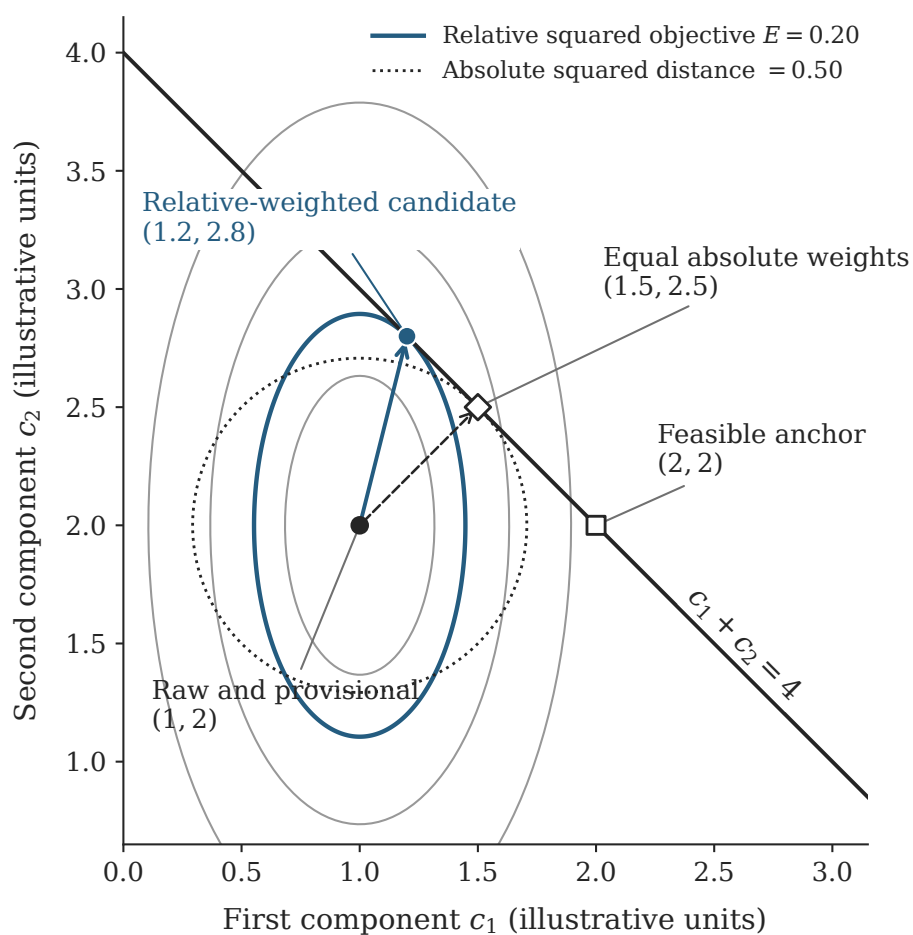


Figure 4.1: Hypothetical projection for mass conservation with relative and equal absolute weights. The highlighted ellipse first touches the line defined by mass conservation at the relative-weighted candidate. The dotted circle touches it at the equal-absolute-weight candidate.

The relative-weighted ellipse is taller than it is wide because movement in the second component is divided by two. A larger absolute change in that component can therefore have the same weighted cost as a smaller change in the first. The circular geometry assigns equal cost to equal absolute changes. Both chosen points satisfy the sum fixed by mass conservation, but the distance definition determines where each geometry touches that line. The feasible anchor locates the reference from which changes satisfying mass conservation are constructed and need not be the nearest candidate.

For the full problem, let $d_j = c_{+,ij}^{(m)} - c_{in,ij}$ denote the provisional change and let $w_j = 1/c_{+,ij}^{(m)}$. The scalar objective adds all twenty squared relative discrepancies,

$$E(q) = \sum_{j=1}^{20} w_j^2 \left(\sum_{k=1}^{15} Z_{jk} q_k - d_j \right)^2.$$

The inner sum builds one component of the change satisfying mass conservation. Subtracting d_j gives its discrepancy from the target change. Multiplication by w_j^2 scales its square. The outer sum combines these component contributions. The notation argmin identifies the coordinate vector at which this objective is smallest, rather than the value of the objective itself. The projection selects the change satisfying mass conservation closest to the provisional target change in the relative-error geometry,

$$q_i^{*,(m)} = \operatorname{argmin}_{q \in \mathbb{R}^{15}} \left\| W_i^{(m)} [Zq - (c_{+,i}^{(m)} - c_{in,i})] \right\|_2^2. \quad (4.7)$$

The squared objective is a quadratic function of q . To minimize it, differentiate with respect to each coordinate q_ℓ . A partial derivative, written with ∂ , varies this one coordinate while holding the others fixed. Differentiating the square and then its inner sum gives

$$\frac{\partial E}{\partial q_\ell} = 2 \sum_{j=1}^{20} w_j^2 Z_{j\ell} \left(\sum_{k=1}^{15} Z_{jk} q_k - d_j \right).$$

Set this derivative to zero, divide by two, and move the target-change contribution to the right-hand side. For every ℓ , the resulting equation is

$$\sum_{k=1}^{15} \left(\sum_{j=1}^{20} w_j^2 Z_{j\ell} Z_{jk} \right) q_k = \sum_{j=1}^{20} w_j^2 Z_{j\ell} d_j.$$

The coefficient of q_k is a weighted dot product of columns ℓ and k of Z . Because $W_i^{(m)}$ is diagonal, its square places w_j^2 on the diagonal. Collecting these fifteen equations gives a 15×15 coefficient matrix and a fifteen-coordinate right-hand side. Differentiation gives

$$Z^\top (W_i^{(m)})^2 Z q_i^{*,(m)} = Z^\top (W_i^{(m)})^2 (c_{+,i}^{(m)} - c_{in,i}). \quad (4.8)$$

For the hypothetical pair, the compact coefficient calculation gives $Z_{\text{ex}}^\top W^2 Z_{\text{ex}} = (1 + 1/4)/2 = 5/8$. Its right-hand side is $-1/\sqrt{2}$. Dividing gives $q^* = -8/(5\sqrt{2}) = -4\sqrt{2}/5$, which equals $\sqrt{2}t$ for the scalar minimum already obtained. The scalar and matrix calculations therefore return the same candidate. For any nonzero vector v , the full

problem's associated quadratic form equals $\|W_i^{(m)}Zv\|_2^2 > 0$. If Zv were zero, multiplying by $Z^\top$ would give $v = Z^\top Zv = 0$, contradicting the choice of v . Positive diagonal weights cannot turn a nonzero column into zero. The squared norm is consequently strictly positive. A positive-definite matrix has a strictly positive such product in every nonzero direction. The matrix is therefore positive definite and the minimizer is unique.

The selected coordinates build the component change through the fifteen sums $\sum_k Z_{jk}q_k^*$. Adding each sum to its matching influent coordinate constructs the state. In the hypothetical pair, this operation is $(2, 2)^\top + (-0.8, 0.8)^\top = (1.2, 2.8)^\top$. Both entries are non-negative and their invariant value is $(1.2 + 2.8)/\sqrt{2} = 4/\sqrt{2}$, equal to the anchor value. This example needs no projection interpolation for non-negativity. The candidate satisfying mass conservation is

$$c_{cand,i}^{(m)} = c_{in,i} + Zq_i^{*,(m)}. \quad (4.9)$$

Since $AZ = 0$, it satisfies $Ac_{cand,i}^{(m)} = Ac_{in,i} = b_i$ in exact arithmetic. If the provisional target is already consistent with mass conservation, its change lies in $\text{null}(A)$. The least-squares residual can then be zero and the candidate reproduces that target to numerical tolerance.

Equation (4.8) provides the derivation rather than the preferred numerical solution. The weighted design $W_i^{(m)}Z$ is solved through an orthogonal-triangular factorization. Singular-value decomposition with relative cutoff 10^{-12} is the fallback. No explicit inverse of the normal-equation matrix is formed. Near-zero provisional targets can produce very unequal weights, so solving the original weighted system avoids the extra conditioning penalty associated with forming a normal-equation inverse.

4.6.3 The largest feasible interpolation toward the candidate

Mass conservation of the candidate does not establish its non-negativity. A different hypothetical three-component example isolates the boundary problem. Its quantity used for mass conservation is the unweighted sum, with feasible influent $(8, 1, 1)^\top$ and candidate satisfying mass conservation $(4, -0.5, 6.5)^\top$. Both sums are ten. Subtracting the influent from the candidate gives the direction $(-4, -1.5, 5.5)^\top$. A fraction α of that direction gives individual concentrations

$$c_1(\alpha) = 8 - 4\alpha, \quad c_2(\alpha) = 1 - 1.5\alpha, \quad c_3(\alpha) = 1 + 5.5\alpha.$$

The first component requires $8 - 4\alpha \geq 0$, so $\alpha \leq 2$. The second requires $1 - 1.5\alpha \geq 0$, so $\alpha \leq 2/3$. The third increases and gives no upper bound for $\alpha \geq 0$. Restricting α to $[0, 1]$ makes $2/3$ the first boundary point. Define

$$\delta_i^{(m)} = c_{cand,i}^{(m)} - c_{in,i} = Zq_i^{*,(m)}. \quad (4.10)$$

The segment $c_{in,i} + \alpha \delta_i^{(m)}$, with $0 \leq \alpha \leq 1$, satisfies mass conservation because $A \delta_i^{(m)} = 0$. For a general decreasing coordinate, start with $c_{in,ij} + \alpha \delta_{ij}^{(m)} \geq 0$. Moving the negative change to the other side gives $c_{in,ij} \geq \alpha(-\delta_{ij}^{(m)})$. The factor $-\delta_{ij}^{(m)}$ is positive, so dividing by it preserves the inequality direction. For every decreasing coordinate, non-negativity requires

$$\alpha \leq \frac{c_{in,ij}}{-\delta_{ij}^{(m)}} \quad \text{when } \delta_{ij}^{(m)} < 0. \quad (4.11)$$

The minimum of these bounds is the first point at which the segment reaches the non-negative-orthant boundary.

If every candidate coordinate is non-negative, the entire direction is allowed and $\alpha = 1$. If at least one candidate coordinate is negative, the smallest decreasing-coordinate ratio supplies a bound below one. Taking that bound reaches the concentration boundary. Multiplication by $1 - \eta$ retreats slightly from the boundary to limit a sign change caused by rounding. With numerical safety margin $\eta = 10^{-12}$, the accepted interpolation factor and final state are

$$\alpha_i^{(m)} = \begin{cases} 1, & c_{cand,i}^{(m)} \geq 0, \\ (1 - \eta) \min_{j \mid \delta_{ij}^{(m)} < 0} \frac{c_{in,ij}}{-\delta_{ij}^{(m)}}, & \text{otherwise,} \end{cases} \quad (4.12)$$

$$\tilde{c}_i^{(m)} = c_{in,i} + \alpha_i^{(m)} \delta_i^{(m)}.$$

If the candidate is negative somewhere, at least one bound lies below one. Non-negativity of the influent makes every bound non-negative. The factor is restricted to $[0, 1]$ against roundoff. Decreasing coordinates satisfy $\tilde{c}_{ij}^{(m)} \geq \eta c_{in,ij} \geq 0$, and increasing coordinates remain non-negative. For invariant row h , the final weighted component sum is $\sum_j A_{hj} c_{in,ij} + \alpha \sum_j A_{hj} \delta_{ij}^{(m)}$. The second sum is zero and the first is b_{ih} . Mass conservation follows from

$$A \tilde{c}_i^{(m)} = A c_{in,i} + \alpha_i^{(m)} A Z q_i^{*,(m)} = b_i. \quad (4.13)$$

The unrestricted boundary factor is the largest feasible interpolation factor. The safety margin makes the returned factor slightly smaller when projection is necessary.

A hypothetical three-component system with sum fixed by mass conservation 10 illustrates this safeguard. Let the feasible influent be $(8, 1, 1)^\top$ and a candidate satisfying mass conservation be $(4, -0.5, 6.5)^\top$. The direction is $(-4, -1.5, 5.5)^\top$. Its decreasing-coordinate bounds are 2 and $2/3$, so the largest feasible factor is $2/3$. Ignoring the numerical safety margin only for this illustration gives $(16/3, 0, 14/3)^\top$. The sum remains 10 and every component is non-negative. The first coordinate changes even though the second coordinate determines the boundary. One scalar shortens the complete direction satisfying mass conservation.

If an influent coordinate is zero and the candidate direction decreases that coordinate, its bound is zero. The final state then returns to the influent anchor. This is a

valid feasible fallback, but it can remove useful information from the raw prediction. Likewise, a component that is absent from the invariant rows can still move during interpolation because the complete direction is shortened. Recording the factor and activation frequency is therefore necessary for interpreting the projection.

4.6.4 Acceptance checks and limits of the guarantee

Numerical acceptance tests the returned state rather than assuming that the exact algebra eliminates rounding error. For row h , form the residual $r_h = \sum_j A_{hj} \tilde{c}_{ij}^{(m)} - b_{ih}$. Square its five values, add them, and take the square root to obtain the mass conservation norm. Separately compare each of the twenty concentrations with the negative of its own tolerance. A hypothetical concentration of -5×10^{-11} passes a 10^{-10} sign tolerance, whereas -2×10^{-10} fails it. These are numerical tolerance examples rather than permitted physical negative concentrations. A returned projection is accepted only when

$$\|A\tilde{c}_i^{(m)} - b_i\|_2 \leq \tau_{mc}, \quad \tilde{c}_{ij}^{(m)} \geq -\tau_{nn,j} \quad \text{for every } j, \quad (4.14)$$

with $\tau_{mc} = 10^{-10}$ in the fixed invariant-coordinate convention and $\tau_{nn,j} = 10^{-10}$ in the native unit of component j . These checks distinguish the exact mathematical construction from its finite-precision realization. A finite raw vector is required. The method does not define an admissible projection of a nonfinite prediction through this algebra.

The final state is feasible for the declared invariants and sign restrictions. It need not minimize the weighted distance over the full intersection $\mathcal{F}_i$. The least-squares stage minimizes distance over an affine space, and the interpolation stage searches one line segment. A general constrained nearest-point problem searches a larger set. The construction also does not minimize the variance-standardized prediction error, whose denominator differs from c_+ . A reduction in mass conservation error therefore does not imply a reduction in statistical error.

No kinetic, equilibrium, stability, or field-performance guarantee follows from the projection. A state satisfying mass conservation and non-negativity can have a large residual in Equation (3.14). The distinction between projection for mass conservation and generalization under a changed input distribution is consistent with Shen et al. (2023) and Valente et al. (2025). The projection also assumes that the influent anchor is feasible for the correct boundary. An unmodeled external source invalidates that assumption even if the numerical projection succeeds.

Figure 4.2 brings the projection steps and their checks into one prediction procedure. The non-negativity decision concerns the candidate satisfying mass conservation. The final numerical audit applies to the state returned by either branch.

The provisional positive target supplies a reference and weights for the solve. It does not bypass the mass conservation calculation. A non-negative candidate can

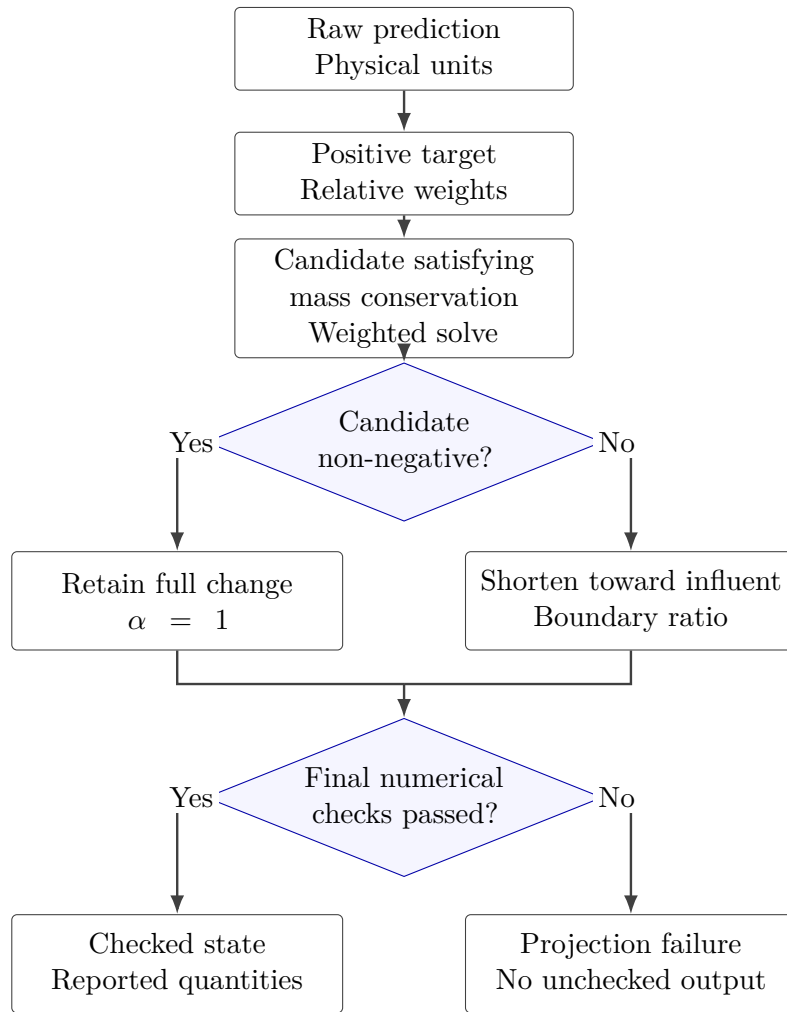


Figure 4.2: Post prediction mass conservation and non-negativity procedure. Positive-target construction, projection for mass conservation, and final-state acceptance have separate roles. The diagram describes the method and contains no empirical branch frequencies.

retain the full change satisfying mass conservation, while a candidate crossing a concentration boundary requires shortening toward the feasible influent. Both branches reach the same final checks. This arrangement prevents either strict positivity of the provisional target or an algebraic mass conservation proof from being mistaken for an independently checked returned state.

Each raw or projected composite is obtained by multiplying the relevant component values by the coefficients already defined in Table 3.1 and adding them. The coefficients do not change between the two prediction types. A paired comparison therefore changes only the component state supplied to the four scalar sums. Raw and projected composites are calculated with the same map,

$$y_{raw,i}^{(m)} = I_{comp} c_{raw,i}^{(m)}, \quad \tilde{y}_i^{(m)} = I_{comp} \tilde{c}_i^{(m)}. \quad (4.15)$$

Feasibility is checked before this aggregation. Composite values cannot reveal every negative coordinate or every inconsistent component redistribution.

4.7 Evaluation of error, feasibility, and projection size

4.7.1 Component errors on a training-defined scale

Let $\mathcal{T}$ denote the applicable training reference and let $\sigma_j^{(\mathcal{T})}$ be the population standard deviation of target component j in that reference. For one assessed concentration, subtract the reference effluent value from the predicted value to obtain its signed physical error. Divide this error by the training standard deviation of the same component. For example, a hypothetical prediction of 3 g N m⁻³ against a reference of 2 g N m⁻³ has signed error 1 g N m⁻³. A training standard deviation of 2 g N m⁻³ gives standardized error 0.5. The constituent units cancel in the division. For prediction type t , which is either raw or projected, define

$$\xi_{ij}^{(m,t)} = \frac{c_{ij}^{(m,t)} - c_{out,ij}}{\sigma_j^{(\mathcal{T})}}. \quad (4.16)$$

The normalized mean squared error (nMSE), normalized root mean squared error (nRMSE), and normalized mean absolute error (nMAE) combine these standardized errors. A two-observation, two-component hypothetical example shows their order of calculation. Suppose the training standard deviations are 2 and 4. Let the reference and predicted values be

Observation	Reference		Prediction	
	Component 1	Component 2	Component 1	Component 2
1	2	4	3	2
2	4	8	2	12

Values within each component use that component's own physical basis. The four physical errors are 1, -2 , -2 , and 4. Dividing each by its corresponding training standard deviation gives

$$\xi = \begin{bmatrix} 1/2 & -2/4 \\ -2/2 & 4/4 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5 \\ -1 & 1 \end{bmatrix}.$$

There are four scored component-observation pairs. Their squared errors sum to $0.25 + 0.25 + 1 + 1 = 2.5$. Their mean squared standardized error is $2.5/4 = 0.625$, and its square root is approximately 0.7906. Their absolute errors sum to $0.5 + 0.5 + 1 + 1 = 3$, so the mean absolute standardized error is $3/4 = 0.75$. The square is taken before averaging for nMSE. The square root is taken after that average for nRMSE. Absolute values are taken before averaging for nMAE.

For the twenty-component assessment, the first summation accumulates errors across observations and the second accumulates them across components. Division by $20n$ gives every scored pair equal weight after standardization. The compact definitions are

$$\text{nMSE}_{m,t} = \frac{1}{20n} \sum_{i=1}^n \sum_{j=1}^{20} (\xi_{ij}^{(m,t)})^2, \quad (4.17)$$

$$\text{nRMSE}_{m,t} = \sqrt{\text{nMSE}_{m,t}}, \quad (4.18)$$

$$\text{nMAE}_{m,t} = \frac{1}{20n} \sum_{i=1}^n \sum_{j=1}^{20} |\xi_{ij}^{(m,t)}|. \quad (4.19)$$

The component state is the primary scoring space because it is the prediction target and the space in which constraints are imposed. The root mean squared score emphasizes larger standardized errors. The absolute score gives a complementary average deviation. A component with zero training standard deviation makes these normalized measures undefined and invalidates the run.

A hypothetical error of 1 g m^{-3} has standardized magnitude 0.1 for a component with training standard deviation 10 g m^{-3} . The same physical error has magnitude 10 for a component with training standard deviation 0.1 g m^{-3} . The score gives equal numerical weight to standardized component variation. It does not claim that every component has equal regulatory importance or equal measurement uncertainty.

The coefficient of determination compares squared prediction error with the error of predicting one constant value for every observation. That constant is the true mean within the scored partition. For component 1 of the hypothetical example, the true mean is $(2 + 4)/2 = 3$. Predicting 3 for both observations would give squared errors $(2 - 3)^2 + (4 - 3)^2 = 2$. The actual predictions give squared errors $(3 - 2)^2 + (2 - 4)^2 = 5$. Dividing 5 by 2 and subtracting from one gives $R^2 = 1 - 5/2 = -1.5$. For component 2, the true mean is 6. The corresponding baseline square sum is $(4 - 6)^2 + (8 - 6)^2 = 8$, and the prediction square sum is $(2 - 4)^2 + (12 - 8)^2 = 20$. Its score is also $1 - 20/8 = -1.5$.

The negative scores mean that the predictions are worse than their scored-partition constant baselines.

The mean used in this comparison is $\bar{c}_{out,j} = (c_{out,1j} + \dots + c_{out,nj})/n$. It belongs to scoring and is not used to fit the model. The denominator of R^2 therefore differs from the training-defined scale of nMSE. For a scored partition with true mean $\bar{c}_{out,j}$, the component coefficient of determination and its equal-weight average are

$$R_{j,m,t}^2 = 1 - \frac{\sum_i (c_{ij}^{(m,t)} - c_{out,ij})^2}{\sum_i (c_{out,ij} - \bar{c}_{out,j})^2}, \quad \bar{R}_{m,t}^2 = \frac{1}{20} \sum_{j=1}^{20} R_{j,m,t}^2. \quad (4.20)$$

An undefined component denominator makes the corresponding macro-average undefined. The coefficient can be negative when the prediction is worse than the scored-partition mean. It complements the standardized error but uses a different response-variability reference.

The macro-average adds the twenty component R^2 values and divides by twenty. It does not pool all concentrations into one baseline denominator. Likewise, pooled nRMSE is the square root of the mean of component squared standardized errors. It is generally different from the arithmetic mean of component root errors. Hypothetical component mean squared standardized errors of 1 and 9 give pooled root error $\sqrt{(1+9)/2} = \sqrt{5}$, whereas the mean of their separate roots is $(1+3)/2 = 2$. This distinction prevents differently constructed aggregate scores from being treated as interchangeable.

Physical-unit errors are retained for individual components. Secondary composite error is calculated separately for COD, TN, TP, and TSS. For a composite, calculate one signed error per observation, square the errors, average them over observations, and take the square root. Dividing that physical-unit root error by the training standard deviation of the composite gives its normalized root error. A hypothetical root error of 5 g COD m⁻³ divided by a training standard deviation of 10 g COD m⁻³ gives 0.5. The normalized root mean squared error of composite q is its physical-unit root mean squared error divided by $\sigma_q^{(\mathcal{T})}$, the training standard deviation of that composite. This is not the primary component-average nRMSE. Composite standard deviations are calculated after applying the common reporting map to the training targets.

4.7.2 Violation frequencies and their magnitudes

The mass conservation residual begins with five scalar errors. For invariant row h , its error is $r_h = \sum_j A_{hj}c_j - b_{ih}$. Its Euclidean magnitude is $\sqrt{r_1^2 + \dots + r_5^2}$. A hypothetical two-row residual of $(3, 4)^\top$ has magnitude five in its fixed invariant-coordinate convention. This example explains the norm and is unrelated to the much smaller acceptance tolerances.

The negative magnitude uses a different operation. First divide each concentration by its training standard deviation. Then retain a negative coordinate if one is

present and replace every non-negative coordinate by zero. Finally add the absolute values of the retained negatives. Hypothetical standardized values $(-2, 1.5, -2)$ become $(-2, 0, -2)$ and have absolute-sum magnitude four. A positive component cannot cancel a negative component in this diagnostic. The mass conservation residual and standardized total negative magnitude of a state c are

$$v_{mc,i}(c) = \|Ac - b_i\|_2, \quad v_{nn,i}^{\text{std}}(c) = \left\| \min(c \oslash \sigma^{(\mathcal{T})}, 0) \right\|_1. \quad (4.21)$$

The minimum is componentwise. The mass conservation residual uses the same full-precision orthonormal basis for every sample. It is a numerical norm of the adopted invariant coordinates rather than a direct mass concentration in one constituent unit. The negative magnitude uses training standard deviations to compare different component bases without summing incompatible physical units.

A frequency counts observations satisfying a stated failure condition and divides that count by the assessed observation count. The indicator $\mathbb{K}[\cdot]$ returns one when its bracketed condition is true and zero otherwise. For a hypothetical four-observation assessment, suppose the mass conservation failure indicators are $(1, 0, 1, 0)$ and the non-negativity failure indicators are $(0, 1, 1, 0)$. The two frequencies are $(1+0+1+0)/4 = 0.5$ and $(0+1+1+0)/4 = 0.5$. An observation with several negative components is still one failing observation in this row-level count. The two row-level violation frequencies are

$$\pi_{mc} = \frac{1}{n} \sum_i \mathbb{K}[v_{mc,i}(c_i) > \tau_{mc}], \quad \pi_{nn} = \frac{1}{n} \sum_i \mathbb{K}[\min_j(c_{ij} + \tau_{nn,j}) < 0]. \quad (4.22)$$

A state is counted as non-negative only when every component passes its tolerance. Component-specific violation frequencies identify which coordinates cross the boundary. In the four-observation example, only the fourth observation passes both conditions, so its joint feasible fraction is one quarter. The third fails both conditions and appears in both marginal failure counts. Adding the two failure frequencies would therefore count it twice. A joint indicator applies both pass conditions to the same observation before averaging. The joint feasible fraction is

$$\pi_{\text{feasible}} = \frac{1}{n} \sum_i \mathbb{K}[v_{mc,i}(c_i) \leq \tau_{mc} \text{ and } \min_j(c_{ij} + \tau_{nn,j}) \geq 0]. \quad (4.23)$$

Mass conservation and non-negativity frequencies are not added to calculate this fraction because the same state can violate both requirements.

Mean, median, and maximum violation magnitudes are retained along with the frequencies. A hypothetical set of many concentrations just below zero can have a high violation frequency but small total negative magnitude. Another set with one strongly negative state can have low frequency and a large maximum magnitude. The two summaries answer different questions. Physical-unit negative magnitudes are also retained separately for each component.

4.7.3 Paired accuracy effects and displacement

The paired accuracy effect subtracts the raw score from the projected score on the same assessment observations. If hypothetical raw and projected nRMSE values are 0.40 and 0.35, the difference is $0.35 - 0.40 = -0.05$. If the projected value were 0.45, the difference would be $+0.05$. This sign convention expresses whether error decreases or increases. It does not identify which components move or whether their mass conservation residual improves. For each outer fold, the effect on the primary error score is

$$\Delta \text{nRMSE}_{m,N,g} = \text{nRMSE}_{m,\text{projected},N,g} - \text{nRMSE}_{m,\text{raw},N,g}. \quad (4.24)$$

A negative difference indicates lower error after projection, and a positive difference indicates higher error. Differences are calculated within paired folds before summary. Component and composite differences are retained because a single aggregate score can conceal opposite changes among outputs.

Displacement measures the difference between the projected prediction and its own raw prediction. It does not subtract the true effluent. Calculate one change per component, divide it by that component's training standard deviation, square it, sum the squares, and take the square root. For the hypothetical weighted-projection pair already calculated, the changes are $1.2 - 1 = 0.2$ and $2.8 - 2 = 0.8$. If its illustrative training standard deviations are 0.1 and 0.4, the standardized changes are 2 and 2. Their Euclidean displacement is $\sqrt{2^2 + 2^2} = 2\sqrt{2}$. The example deliberately uses a training scale different from the provisional concentration scale, as the two operations serve different purposes. The sample-level standardized displacement is

$$d_{c,i}^{\text{std},(m)} = \left\| (\tilde{c}_i^{(m)} - c_{\text{raw},i}^{(m)}) \oslash \boldsymbol{\sigma}^{(\mathcal{T})} \right\|_2. \quad (4.25)$$

This norm measures how far the projection moves the state relative to training variability. It measures movement rather than accuracy. A large displacement can reduce violations of the specified physical constraints while moving away from the true kinetic response. Mean, median, and maximum displacement, componentwise physical-unit movement, the fraction requiring backtracking, and the distribution of α are therefore assessed together.

Learning curves retain the full error trajectory over Equation (4.4). An area summary connects the errors at neighboring sample sizes by straight segments. The area under one segment is its width multiplied by the average of its two endpoint heights. For hypothetical error heights 0.8 and 0.6 at design sizes 500 and 1450, the segment area is $(1450 - 500)(0.8 + 0.6)/2 = 665$. If the next hypothetical height is 0.4 at size 2400, the next area is $(2400 - 1450)(0.6 + 0.4)/2 = 475$. The total area over those two segments is 1140. Dividing by the size interval $2400 - 500 = 1900$ gives an average height of 0.6. These hypothetical heights illustrate integration and are not learning-curve findings.

The full design has ten adjacent intervals. Calculate the trapezoidal area of each interval, add those areas, and divide by $10000 - 500 = 9500$. Division by the total interval converts the area into an average error height. As a descriptive summary, the normalized area under an error curve is calculated by trapezoidal integration,

$$\mathcal{A}_{m,t} = \frac{1}{N_{11} - N_1} \sum_{\ell=1}^{10} \frac{E_{m,t}(N_\ell) + E_{m,t}(N_{\ell+1})}{2} (N_{\ell+1} - N_\ell), \quad (4.26)$$

where $E_{m,t}(N)$ is the mean held-out error at size N . This quantity has the same scale as the selected error measure. It summarizes behavior across the declared size interval and does not replace the individual size scores. Training and validation error at the largest size provide a separate descriptive gap. Neither summary establishes an optimal future sample size without an additional learning-curve model (Viering & Loog, 2023).

Each in-domain metric is summarized as the arithmetic mean and sample standard deviation of five outer-fold values. If these values are $v_1, \dots, v_5$, first calculate $\bar{v} = (v_1 + \dots + v_5)/5$. Then calculate $s = \sqrt{[(v_1 - \bar{v})^2 + \dots + (v_5 - \bar{v})^2]/4}$. The denominator is four because this summary uses the sample standard deviation of five fold values. It differs from the population standard deviation used to standardize a training coordinate. The standard deviation describes partition-to-partition variation. No hypothesis test or inferential model ranking follows from these summaries. They also do not supply calibrated predictive uncertainty for a new state. The ensemble uncertainty approach of Lakshminarayanan et al. (2017) addresses a different task, and no such uncertainty calibration is assumed here.

4.8 Testing inputs beyond the learning domain

Extrapolation assessment uses separately simulated inputs outside the training hyperrectangle. The final setting for each surrogate is selected with four folds on all 10,000 in-domain states. Its preprocessing and predictor are then refitted on that complete pool. No extrapolation response enters selection, fitting, scaling, or calibration. This separation addresses the domain-shift concern discussed in scientific machine learning by Karniadakis et al. (2021) and Shen et al. (2023).

Four inputs can exceed their upper learning bounds. Exceedance is measured relative to the width of the original input interval. For HRT, the width is $36 - 6 = 30$ h. A hypothetical input of 39 h exceeds the upper bound by 3 h, giving exceedance $3/30 = 0.10$. Dividing by the original width allows a comparable severity description for inputs with different units. For input ℓ with bounds $[L_\ell, U_\ell]$, define its normalized exceedance as

$$e_{i\ell} = \frac{x_{i\ell} - U_\ell}{U_\ell - L_\ell}. \quad (4.27)$$

The mild design uses $e_{i\ell} \in [0.10, 0.25]$ and the severe design uses $e_{i\ell} \in [0.35, 0.60]$. Reversing the exceedance calculation gives $x_{i\ell} = U_\ell + e_{i\ell}(U_\ell - L_\ell)$. For HRT, the

mild endpoints are $36 + 0.10(30) = 39$ h and $36 + 0.25(30) = 43.5$ h. The severe endpoints are $36 + 0.35(30) = 46.5$ h and $36 + 0.60(30) = 54$ h. These arithmetic steps derive the declared intervals rather than treatment outcomes. The generator targets a mixture of 70% one-input excursions and 30% two-input excursions. All remaining inputs remain within the original bounds. Every candidate therefore lies outside the training hyperrectangle and consequently outside the training-point convex hull. The hyperrectangle is the set satisfying every original coordinate interval. Every weighted average of training points remains inside those intervals, so a point violating even one interval is also outside their convex hull.

Table 4.5: Fixed exterior intervals for the two extrapolation designs. The intervals specify input conditions rather than prediction outcomes.

Input	Learning interval	Mild interval	Severe interval	Unit
HRT	[6, 36]	[39, 43.5]	[46.5, 54]	h
u_{air}	[0.5, 2.5]	[2.7, 3.0]	[3.2, 3.7]	1
S_{NH4}	[12, 55]	[59.3, 65.8]	[70.1, 80.8]	g N m ⁻³
X_S	[60, 280]	[302, 335]	[357, 412]	g COD m ⁻³

Each severity design targets 600 accepted steady states and retains the same solver, initialization sequence, and balance-residual criterion. Candidate, accepted, failed, valid-but-unselected, and retained counts are distinguished. The realized one-input and two-input proportions are also distinguished from their design targets. The intervals alone do not establish which exterior conditions are hardest to predict because reactor kinetics can respond differently to the four inputs.

Extrapolation errors are standardized with population component standard deviations from all 10,000 in-domain targets. Composite normalization uses the corresponding full-domain composite standard deviations. The nMSE, nRMSE, nMAE, mass conservation and non-negativity diagnostics, displacement, and backtracking diagnostics are calculated on identical raw and projected cases. The coefficient of determination is secondary because the deliberately shifted response distribution changes its denominator.

Each exterior metric is a single descriptive summary over 600 cases from one full-domain refit. It is not an average over the five outer folds and has no fold standard deviation. Satisfaction of the projected constraints is interpreted separately from the predictive accuracy of extrapolation. A state can remain non-negative and consistent with mass conservation while its estimated nitrogen distribution or biomass abundance is inaccurate.

4.9 Computational cost and the meaning of a fast prediction

Computation is partitioned into hyperparameter search, preprocessing and fitting, raw inference, projection alone, and complete projected inference. Search time is reported once for each distinct study because a largest-scale outer setting is reused at all eleven sizes. Summing that same search duration over the nested size rows would count the selection cost repeatedly. Fitting time excludes the search and reflects the preparation required for the specified training partition.

Inference uses a deterministic batch of 512 rows constructed by cycling through the relevant validation inputs. The batch size stays fixed even when fewer than 512 unique validation observations are available. Cycling reuses input rows to make a fixed-sized workload. It does not add independent assessment observations or change the prediction-error denominator. Five warm-up calls precede 30 timed repetitions for each path. Sorting the thirty batch durations puts them in increasing order. Their median is the average of the fifteenth and sixteenth values. The median batch latency is divided by 512 and expressed as milliseconds per sample. For a hypothetical median batch duration of 102.4 ms, the reported per-sample quantity would be $102.4/512 = 0.2$ ms. This is a timing calculation rather than a measured performance value. For in-domain assessment, the five fold medians are summarized by their arithmetic mean and sample standard deviation. Exterior timing uses the same batching and repetitions for one final fit and has no fold uncertainty.

Projection timing includes positive-target construction, relative weighting, the weighted solve, mass conservation and non-negativity checks, and any projection interpolation for non-negativity. Complete projected inference includes both raw prediction and projection. All physical operators, returned component states, preprocessing, and metrics use double precision. TabNet’s internal tensors use single precision before its predictions return to the common physical-coordinate calculations. Pseudorandom generation and stochastic fitting use streams derived from seed 42. Deterministic coordinate-descent settings are retained for the multi-task models.

These timing measures describe the declared batch protocol and computational environment. Dividing a batch duration by its row count does not establish the latency of a single isolated query. Likewise, low projection latency would not establish field readiness. A useful predictor must be assessed through its component accuracy, physical diagnostics, domain of use, preparation cost, and the behavior of downstream calculations. The post prediction projection provides a defined admissibility layer within that assessment.

Chapter 5

Interpretable regression with checked physical deployment

5.1 An inspectable model of component transformations

A component prediction is useful for engineering analysis when its dependence on the inputs can be examined and its material accounting can be checked. These requirements address different questions. A transparent regression equation explains how the predicted state depends on operating conditions and influent loads. A mass conservation test determines whether that state preserves the inventories represented by the reaction network. A non-negativity test determines whether each predicted concentration is physically meaningful. An explicit equation can fail either physical test. A physically projected prediction can also remain difficult to interpret if the underlying regression is opaque.

Invariant-constrained second-order regression (ICSOR) combines an explicit polynomial driver, learned dependence among effluent components, and checked deployment. The driver retains separate terms for operating conditions, influent composition, curvature, and interactions. The coupling relation expresses how the driver contributes to the predicted component state. Deployment then checks that state and applies a constrained projection when necessary. The interpretation of the regression and the guarantee provided by deployment therefore remain distinct.

The preference for an inspectable regression follows the argument of Rudin (2019) that transparent model structures can support decisions that require scrutiny. It also follows the broader development of models that combine learned relations with physical structure discussed by Shen et al. (2023). Transparency does not establish that a fitted coefficient is a biological mechanism. It makes the relation available for examination, comparison with process knowledge, and testing across operating conditions.

Activated sludge nutrient removal provides a strong reason to retain interactions.

Carbon availability influences denitrification and the activity of organisms that store phosphorus. Oxygen availability influences both carbon oxidation and nitrification. Wentzel et al. (1992) describe these connected pathways, and Guerrero et al. (2011) examine the role of carbon source in the competition between phosphorus-storing organisms and denitrifiers. A regression that represents an aeration effect independently of loading can overlook such dependencies. An interaction term can describe an association between an operating action and a loading condition while keeping that association visible.

The prediction target is the effluent component vector $c_{out} \in \mathbb{R}^F$. The adopted process basis has $F = 20$ components and $P = 28$ reactions. The operating vector $u \in \mathbb{R}^{M_{op}}$ has $M_{op} = 2$ entries for HRT and aeration level. The full influent vector is $c_{in} \in \mathbb{R}^F$. Component concentrations retain their physical units. The four reported composites are obtained from a fixed map $I_{comp} \in \mathbb{R}^{4 \times F}$. That map is applied after component prediction.

5.2 Mass conservation in the adopted state space

The link between a component balance and a regression constraint can first be seen in a hypothetical conversion of one component into another. Suppose that one unit of component 1 is converted into one unit of component 2. The two components are expressed in a common inventory unit for this illustration. Their changes satisfy

$$c_{out,1} - c_{in,1} = -\xi, \quad c_{out,2} - c_{in,2} = \xi. \quad (5.1)$$

Adding these equations eliminates the unknown conversion amount. The relation available to the regression is $c_{out,1} + c_{out,2} = c_{in,1} + c_{in,2}$. With an illustrative influent $(6, 4)^\top$ and conversion amount 2, the effluent is $(4, 6)^\top$. Both totals are 10. Many other non-negative pairs also have that total. The balance therefore limits the possible regression output without identifying the particular state produced by the kinetics.

The same calculation can be written using the small matrices

$$\nu = \begin{bmatrix} -1 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad A\nu^\top = 1(-1) + 1(1) = 0. \quad (5.2)$$

These matrices describe the reduced illustration, rather than the 20-component process. All numerical teaching examples use hypothetical coefficients and states. Their dimensions are reduced to make each operation visible, and they report no process observations. The adopted reactor retains its stated component basis, reaction matrix, and operating domain.

The stoichiometric matrix $\nu \in \mathbb{R}^{P \times F}$ records reactions as rows and components as columns. Let $V_0 \in \mathbb{R}^{F \times K}$ span its right null space, where $K = 5$. The invariant operator is

$$A = V_0^\top \in \mathbb{R}^{K \times F}, \quad \nu V_0 = 0, \quad A\nu^\top = 0. \quad (5.3)$$

The columns of V_0 are independent, so A has full row rank. With an orthonormal null-space basis, $AA^\top = I_K$. A different full-rank basis describes the same mass conservation conditions, although its row scaling changes the numerical size of a soft invariant penalty.

For the control volume and boundary accounting represented by the adopted process model, a reaction-induced change has the form

$$c_{out} - c_{in} = \nu^\top \xi, \quad \xi \in \mathbb{R}^P. \quad (5.4)$$

The reaction-progress vector ξ expresses the net contribution of the reactions in concentration-equivalent units. Multiplication by A removes that unknown contribution and gives

$$A(c_{out} - c_{in}) = 0. \quad (5.5)$$

This construction uses the mass conservation logic of the photochemical surrogate framework of Sturm and Wexler (2020). Sturm and Wexler (2022) further show how a stoichiometric transformation can connect learned reaction information to state satisfying mass conservation changes. The same linear algebra applies here, but the component combinations used for mass conservation are those of the activated sludge reaction network.

The equality in Equation (5.5) applies only when the modeled boundary supports the change relation in Equation (5.4). External gas transfer, chemical dosing, solids withdrawal, or another boundary flux must be included in the balance where it occurs. If an explicit boundary contribution s is present, the relevant relation becomes $A(c_{out} - c_{in} - s) = 0$. Using $Ac_{out} = Ac_{in}$ while omitting a boundary contribution would impose the wrong constraint. The invariant operator does not make an incomplete material balance correct.

For a new influent state, the component set used by checked deployment is

$$\mathcal{F}(c_{in}) = \{c \in \mathbb{R}^F \mid Ac = Ac_{in}, c \geq \mathbf{0}\}. \quad (5.6)$$

When $c_{in} \geq \mathbf{0}$, this set contains c_{in} and is therefore nonempty. Its members satisfy the modeled linear invariants and component non-negativity. Membership does not show that a state solves the nonlinear kinetic equations, is dynamically stable, or is reachable at the stated operating point. The set gives necessary physical conditions within the declared boundary.

The reported composites provide another reason to work in component space. Distinct component states can have the same COD, TN, TP, and TSS values. A negative component can also be hidden by positive contributions to an aggregate. When I_{comp} is entrywise non-negative, a non-negative component state produces non-negative composites. The reverse implication does not generally hold. Mass conservation and non-negativity are therefore tested before the component state is collapsed into the four reported quantities.

5.3 A partitioned second-order driver

5.3.1 Separating operation from loading

Operating conditions and influent composition play different engineering roles. HRT and aeration define aspects of the reactor environment. Influent components define the material presented to that environment. This distinction is consistent with the operating-sensitivity analysis of de Araujo et al. (2013) and the separation of operational and influent uncertainty examined by Machado et al. (2014).

Start with two hypothetical operating entries u_1, u_2 and two influent entries c_1, c_2 . A second-order model uses the individual entries, their squares, and their pairwise products. The operating products are $u_1^2, u_1u_2, u_2u_1, u_2^2$. Keeping both orders is an accounting convention for the feature vector. It does not make u_1u_2 different from u_2u_1 .

For two column vectors, the Kronecker operation multiplies the entire second vector by the first entry of the first vector, then by its second entry. Thus

$$\begin{aligned} u \otimes u &= \begin{bmatrix} u_1u_1 \\ u_1u_2 \\ u_2u_1 \\ u_2u_2 \end{bmatrix}, \quad c \otimes c = \begin{bmatrix} c_1c_1 \\ c_1c_2 \\ c_2c_1 \\ c_2c_2 \end{bmatrix}, \\ u \otimes c &= \begin{bmatrix} u_1c_1 \\ u_1c_2 \\ u_2c_1 \\ u_2c_2 \end{bmatrix}. \end{aligned} \tag{5.7}$$

For the illustrative values $u = (2, 3)^\top$ and $c = (4, 5)^\top$, these three blocks are respectively $(4, 6, 6, 9)^\top$, $(16, 20, 20, 25)^\top$, and $(8, 10, 12, 15)^\top$. Including the baseline 1 and the four first-order entries gives 17 ordered features in this reduced example. The same construction applies to the larger adopted input vector.

The feature vector retains both roles through the ordered expansion

$$\phi(u, c_{in}) = \begin{bmatrix} 1 \\ u \\ c_{in} \\ u \otimes u \\ c_{in} \otimes c_{in} \\ u \otimes c_{in} \end{bmatrix} \in \mathbb{R}^D, \quad D = 1 + M_{op} + F + M_{op}^2 + F^2 + M_{op}F. \tag{5.8}$$

The symbol $\otimes$ denotes the Kronecker product. The adopted dimensions give $D = 467$. The ordered expansion includes both u_iu_j and u_ju_i , and both $c_{in,i}c_{in,j}$ and $c_{in,j}c_{in,i}$. These pairs are equal as numerical features. Their interpretation requires the symmetry convention developed below.

Multiplication by regression coefficients can also be read one entry at a time. For an illustrative first driver, take the baseline coefficient as 1, operating coefficients as $(2, -1)$, and influent coefficients as $(0.5, 0)$. Let its operating quadratic row be $(0.1, 0.1, 0.1, 0)$, influent quadratic row be $(0, 0, 0, -0.05)$, and operating-load row be $(0, 0, 0.25, 0)$. Applying these rows to the numerical feature blocks gives

$$\begin{aligned}
 r_1 &= 1 + \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\
 &\quad + \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \\ 6 \\ 9 \end{bmatrix} \\
 &\quad + \begin{bmatrix} 0 & 0 & 0 & -0.05 \end{bmatrix} \begin{bmatrix} 16 \\ 20 \\ 20 \\ 25 \end{bmatrix} \\
 &\quad + \begin{bmatrix} 0 & 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 8 \\ 10 \\ 12 \\ 15 \end{bmatrix} \\
 &= 1 + 1 + 2 + 1.6 - 1.25 + 3 = 7.35.
 \end{aligned} \tag{5.9}$$

Each row-vector multiplication is a sum of coefficient times feature. It is the same operation as an ordinary linear regression, but the supplied features include nonlinear products. A second hypothetical driver $r_2 = -u_1 + u_2 + 0.2c_2$ would equal 2 at the same inputs. The two driver values need not be predicted concentrations until the coupled relation is solved.

Before grouping the coefficients into matrices, the first driver in this illustration can be written as

$$r_1 = 1 + 2u_1 - u_2 + 0.5c_1 + 0.1u_1^2 + 0.2u_1u_2 - 0.05c_2^2 + 0.25u_2c_1. \tag{5.10}$$

The coefficient 0.2 of u_1u_2 is the sum of the two ordered coefficients 0.1. In a general row, the scalar calculation is $r_f = \sum_{\ell=1}^D B_{f\ell}\phi_\ell$. Carrying out this sum for all components gives the driver $r \in \mathbb{R}^F$ through

$$r(u, c_{in}) = B\phi(u, c_{in}), \quad B \in \mathbb{R}^{F \times D}. \tag{5.11}$$

The driver is an intermediate regression quantity. It is not a vector of mechanistic reaction rates and is not the final deployed state. Its coefficients have no sign restriction, so the regression can represent both increasing and decreasing effects. Non-negativity

is imposed on component states where it has a physical meaning. The driver block form is

$$\begin{aligned} r(u, c_{in}) = & b + W_u u + W_{in} c_{in} + \Theta_{uu}(u \otimes u) \\ & + \Theta_{cc}(c_{in} \otimes c_{in}) + \Theta_{uc}(u \otimes c_{in}). \end{aligned} \quad (5.12)$$

Here $b \in \mathbb{R}^F$ is a baseline term. The matrices $W_u \in \mathbb{R}^{F \times M_{op}}$ and $W_{in} \in \mathbb{R}^{F \times F}$ contain first-order operating and influent terms. The matrices $\Theta_{uu} \in \mathbb{R}^{F \times M_{op}^2}$ and $\Theta_{cc} \in \mathbb{R}^{F \times F^2}$ contain curvature within each input group. The matrix $\Theta_{uc} \in \mathbb{R}^{F \times M_{op} F}$ contains operating-load interactions.

5.3.2 Illustrative operating and loading effects

Consider an illustrative driver for one component with aeration level a and influent ammonium concentration n . A term of the form

$$r_f = \beta_0 + \beta_a a + \beta_n n + \beta_{aa} a^2 + \beta_{an} a n \quad (5.13)$$

has an aeration derivative $\partial r_f / \partial a = \beta_a + 2\beta_{aa} a + \beta_{an} n$. The effect described by this derivative changes with both aeration level and ammonium loading. This example uses a hypothetical polynomial and reports no fitted relation. It explains why a single linear aeration coefficient cannot summarize the response represented by the entire polynomial.

The same reasoning applies to simultaneous influent loads. A carbon-ammonium interaction allows the modeled association with carbon availability to depend on nitrogen loading. A phosphorus-storage interaction allows the modeled association with dissolved phosphate to depend on a particulate storage pool. These terms provide a transparent approximation to a nonlinear response. They do not replace the microbial pathways described by Henze et al. (1999), and their signs do not establish causation.

The choice of a second-order library limits the model deliberately. It can represent smooth local curvature and pairwise input interactions. It cannot reproduce every threshold, saturation relation, or higher-order interaction exactly. Explicit nonlinear libraries are useful for inspecting candidate structure, as illustrated by Brunton et al. (2016). Their usefulness depends on the sampling domain, the adequacy of the library, and the stability of the fitted terms.

5.3.3 Symmetry and the meaning of cross terms

For a two-entry vector, direct multiplication gives

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = q_{11} x_1^2 + (q_{12} + q_{21}) x_1 x_2 + q_{22} x_2^2. \quad (5.14)$$

Only the sum of the two off-diagonal coefficients enters this polynomial. Replacing them by their average leaves that sum unchanged. For example,

$$Q = \begin{bmatrix} 0.1 & 0.3 \\ -0.1 & 0 \end{bmatrix}, \quad \frac{Q + Q^\top}{2} = \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0 \end{bmatrix}. \quad (5.15)$$

At $x = (2, 3)^\top$, the first matrix gives $0.4 + 1.8 - 0.6 = 1.6$. The symmetric matrix gives $0.4 + 0.6 + 0.6 = 1.6$. The asymmetric coefficients cannot be distinguished from these predictions. Their squared sums are different. The original matrix has squared sum $0.01 + 0.09 + 0.01 = 0.11$, while its symmetric part has squared sum $0.01 + 0.01 + 0.01 = 0.03$. The removed part has squared sum 0.08 and contributes no prediction. This is why a positive coefficient penalty favors the symmetric representation.

For component f , reshape the corresponding rows of Θ_{uu} and Θ_{cc} into square matrices $Q_{uu}^{(f)}$ and $Q_{cc}^{(f)}$. A quadratic term can then be written as $x^\top Qx$. For any real square matrix,

$$x^\top Qx = x^\top \text{sym}(Q)x, \quad \text{sym}(Q) = \frac{Q + Q^\top}{2}. \quad (5.16)$$

The skew-symmetric part contributes nothing to the prediction. It can nevertheless increase the coefficient norm because

$$\|Q\|_F^2 = \|\text{sym}(Q)\|_F^2 + \|\text{skew}(Q)\|_F^2. \quad (5.17)$$

Replacing each square slice by its symmetric part preserves every fitted value and cannot increase a positive ridge penalty. The projection is applied after each driver update. With a strictly positive driver ridge weight, the exact minimum-norm solution has no skew-symmetric part.

Symmetry also determines how a coefficient display is read. If $q_{ij} = q_{ji}$, the two displayed off-diagonal cells together contribute $2q_{ij}x_ix_j$. A displayed cell is therefore half the coefficient of the corresponding unordered cross term. For an illustrative matrix with $q_{12} = q_{21} = 0.3$, the polynomial contribution is $0.6x_1x_2$. Reading either cell as the entire interaction coefficient would halve its contribution.

Each component has $M_{op}(M_{op} + 1)/2 = 3$ distinct operating quadratic terms and $F(F + 1)/2 = 210$ distinct influent quadratic terms. Together with the baseline, linear terms, and operating-load interactions, the library has 276 distinct polynomial terms per component. These are structural dimensions of the model. They do not describe how many coefficients are statistically important or biologically meaningful.

5.4 Learned coupling among effluent components

Consider two hypothetical component equations in which each channel has a coupling coefficient 0.2 to the other channel. With driver values 4 and 2, their equations are

$$c_1 = 4 + 0.2c_2, \quad c_2 = 2 + 0.2c_1. \quad (5.18)$$

Substituting the second equation into the first gives $c_1 = 4 + 0.2(2 + 0.2c_1)$. Moving the remaining c_1 term to the left gives $0.96c_1 = 4.4$. Therefore $c_1 = 55/12$ and $c_2 = 35/12$. The solution is approximately $(4.5833, 2.9167)^\top$. These arithmetic values demonstrate a simultaneous regression relation. They do not describe a biochemical conversion.

Moving the coupled terms to the left makes the same calculation a matrix solve,

$$\Gamma = \begin{bmatrix} 0 & 0.2 \\ 0.2 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & -0.2 \\ -0.2 & 1 \end{bmatrix}, \quad R \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}. \quad (5.19)$$

The determinant of R is $1 - 0.2^2 = 0.96$. Its inverse is

$$R^{-1} = \frac{1}{0.96} \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix}. \quad (5.20)$$

Multiplying this inverse by the driver gives the same two scalar solutions. The inverse also explains why a change in one driver can change both predicted components.

The regression includes a matrix $\Gamma \in \mathbb{R}^{F \times F}$ that describes fitted dependence among the effluent components. Define

$$R = I_F - \Gamma, \quad Rc = r(u, c_{in}). \quad (5.21)$$

Equivalently, the equation for component f is $c_f = r_f + \sum_{j \neq f} \Gamma_{fj} c_j$. This form separates the input-dependent driver from dependence among the predicted component channels. The diagonal of Γ is fixed to zero to avoid a redundant self-coupling coefficient.

The sign of Γ_{fj} describes reinforcement or compensation within this fitted simultaneous relation. It does not describe a one-way biochemical reaction from component j to component f . Activated sludge reactions involve several components at once, and the stoichiometric matrix already specifies those transformations. The coupling matrix is learned from state patterns. Its entries can reflect correlated transformations, loading patterns, and the allocation of effects between the driver and coupling blocks.

When R is nonsingular, the raw component prediction is $R^{-1}r$. The dependence on R^{-1} means that the sign of a driver coefficient alone does not determine the response of the raw component prediction. Coupling can redistribute or amplify that response. For interpretation, the driver, coupling matrix, and complete raw mapping are examined together.

The adopted coupling constraints require

$$\Gamma_{ff} = 0, \quad |\Gamma_{fj}| \leq \gamma_{abs} \quad (f \neq j), \quad \kappa_2(I_F - \Gamma) \leq \kappa_{max}. \quad (5.22)$$

The condition number is the ratio of the largest to smallest singular value of the coupled-system matrix. A nearly singular matrix can amplify small driver perturbations and numerical errors. Bounding individual entries does not by itself guarantee a well-conditioned matrix, so the condition check is separate from the box constraints.

The small symmetric matrix in Equation (5.19) illustrates this measure without a large singular-value calculation. In the direction $(1, 1)^\top$, multiplication by R scales the vector by 0.8. In the direction $(1, -1)^\top$, it scales the vector by 1.2. These orthogonal directions give singular values 0.8 and 1.2, so the condition number is $1.2/0.8 = 1.5$.

If the two illustrative couplings were 0.99, the same directions would have scales 0.01 and 1.99. The condition number would be 199. A driver error of magnitude ϵ in each component along the first direction would produce a state error of 100ϵ in each component. This hypothetical choice is outside the adopted coefficient bound and serves only to show the mechanism of amplification.

Even a small entry bound can permit singularity in a larger matrix. In a hypothetical four-component matrix with every off-diagonal entry equal to $1/3$, every row sum of Γ is one. Hence $\Gamma \mathbf{1} = \mathbf{1}$ and $(I - \Gamma)\mathbf{1} = 0$. Each entry is smaller than the adopted bound 0.368, but the coupled-system matrix is singular. This example explains why entry bounds and the condition-number test perform separate tasks.

Shrinking a candidate coupling matrix toward zero can provide a safe candidate because $I_F - 0 = I_F$. A zero coupling matrix gives a condition number of one. A prescribed $\kappa_{max} > 1$ therefore permits this fallback. The admissible coupling set contains a nonlinear condition-number requirement. It is not generally a convex set, even though the box-constrained estimation problem used to propose a coupling update is convex.

5.5 Fitting in prediction space

5.5.1 The coupled training objective

The squared error of one prediction is the square of its difference from the reference. For several components and observations, fitting adds these squared differences. A compact matrix norm represents this sum. For any matrix M with entries M_{ij} ,

$$\|M\|_F^2 = \sum_i \sum_j M_{ij}^2. \quad (5.23)$$

For example, a residual matrix $\begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$ has squared norm $1^2 + (-2)^2 + 3^2 + 0^2 = 14$. The norm is $\sqrt{14}$, while a least-squares objective uses 14. Squaring prevents positive and negative errors from canceling when they are added.

Let i identify an observation, f a component, k an invariant, and ℓ a feature. The entry $\hat{c}_{if}$ is the fitted concentration, while $\phi_{i\ell}$ is the corresponding feature entry. Before

grouping the five training contributions into norms, their scalar form is

$$\begin{aligned}
J = & \sum_{i=1}^N \sum_{f=1}^F (c_{out,if} - \hat{c}_{if})^2 \\
& + \lambda_{inv} \sum_{i=1}^N \sum_{k=1}^K \left[\sum_{f=1}^F A_{kf} (\hat{c}_{if} - c_{in,if}) \right]^2 \\
& + \lambda_{sys} \sum_{i=1}^N \sum_{f=1}^F \left[\hat{c}_{if} - \sum_{j=1}^F \Gamma_{fj} \hat{c}_{ij} - \sum_{\ell=1}^D B_{f\ell} \phi_{i\ell} \right]^2 \\
& + \lambda_B \sum_{f=1}^F \sum_{\ell=1}^D B_{f\ell}^2 + \lambda_{\Gamma} \sum_{f=1}^F \sum_{j=1}^F \Gamma_{fj}^2.
\end{aligned} \tag{5.24}$$

The second contribution first computes an inventory residual by adding component changes with their invariant weights. It then squares that residual. It does not square each component change before the invariant calculation. The third contribution compares the fitted component state, after coupling, with the driver generated by the inputs. The last two sums penalize coefficient magnitude rather than prediction residuals.

The dimensions of the matrices show how these sums are assembled. The entry in row i , column k of $(\hat{C} - C_{in})A^\top$ is the inventory residual in the second sum. The entry in row i , column f of $\hat{C}R^\top$ is $\hat{c}_{if} - \sum_j \Gamma_{fj} \hat{c}_{ij}$. The transpose appears because observations are stored as rows, while the single-observation equation $Rc = r$ uses column vectors.

For N training observations, $\Phi \in \mathbb{R}^{N \times D}$ contains feature vectors as rows. The matrices C_{in} and C_{out} contain influent and reference effluent states in the same row convention. An auxiliary fitted-state matrix $\hat{C} \in \mathbb{R}_+^{N \times F}$ separates the predicted training states from the regression coefficients. The training criterion is

$$\begin{aligned}
J(B, \Gamma, \hat{C}) = & \|C_{out} - \hat{C}\|_F^2 + \lambda_{inv} \|(\hat{C} - C_{in})A^\top\|_F^2 \\
& + \lambda_{sys} \|\hat{C}(I_F - \Gamma)^\top - \Phi B^\top\|_F^2 \\
& + \lambda_B \|B\|_F^2 + \lambda_{\Gamma} \|\Gamma\|_F^2.
\end{aligned} \tag{5.25}$$

The estimation problem minimizes this criterion subject to non-negative fitted training states, zero coupling diagonal, coupling box bounds, and the conditioning safeguard. All penalty weights are non-negative. The adopted ridge weights and coupled-system weight are strictly positive.

The first term measures component prediction error. The second term discourages disagreement with the modeled invariants. The third term encourages agreement between the fitted states and the coupled regression equation. The final two terms limit coefficient magnitude. A non-negative fitted training state can therefore balance prediction error, invariant mismatch, and coupled-system mismatch. It need not satisfy either equality exactly.

An illustrative one-observation calculation makes the five contributions concrete. Take $c_{out} = (3, 1)^\top$, $c_{in} = (2, 2)^\top$, $\hat{c} = (2.5, 1.2)^\top$, and $A = [1 \ 1]$. Use the hypothetical

coupling matrix in Equation (5.19). For a single baseline feature, take $B = (2, 1)^\top$, so the driver is $(2, 1)^\top$. The five unweighted contributions are

$$\begin{aligned}
 \text{prediction error} &= (3 - 2.5)^2 + (1 - 1.2)^2 = 0.29, \\
 \text{inventory error} &= (2.5 + 1.2 - 2 - 2)^2 = 0.09, \\
 \text{system error} &= (2.5 - 0.2(1.2) - 2)^2 \\
 &\quad + (1.2 - 0.2(2.5) - 1)^2 = 0.1576, \\
 \text{driver squared sum} &= 2^2 + 1^2 = 5, \\
 \text{coupling squared sum} &= 0.2^2 + 0.2^2 = 0.08.
 \end{aligned} \tag{5.26}$$

With illustrative weights $\lambda_{inv} = 2$, $\lambda_{sys} = 3$, $\lambda_B = 0.1$, and $\lambda_\Gamma = 0.5$, the objective is $0.29 + 2(0.09) + 3(0.1576) + 0.1(5) + 0.5(0.08) = 1.4828$. These weights are chosen only for arithmetic teaching. The adopted numerical fitting weights remain those in Table 5.1. The example shows why changing a weight changes the relative cost of a discrepancy rather than making that discrepancy an exact equality.

This distinction matters for a new input. Neither a non-negative $\hat{C}$ nor a finite invariant penalty guarantees that $R^{-1}B\phi$ is non-negative or consistent with mass conservation. The auxiliary state is a variable in the fitting problem, while the raw prediction is obtained by solving the learned coupled relation. A soft residual penalty encourages agreement but does not impose equality. Physics-informed residual training uses this general penalty principle, as developed by Raissi et al. (2019). Exact feasibility requires the separate deployment conditions below.

The objective is expressed in physical component coordinates. Components have different units and numerical scales, so the magnitudes of their residuals and coefficients reflect those conventions. The four penalty weights also depend on the fixed feature and invariant scaling. They cannot be transferred unchanged to arbitrary unit systems. Any rescaling must transform the corresponding features, coefficients, invariant operator, and reporting map consistently. Pinheiro et al. (2025) discuss the influence of feature scaling on learning. For physical interpretation, unit consistency is as necessary as numerical conditioning.

5.5.2 Deterministic block updates

The term $\hat{C}\Gamma^\top$ makes the joint objective nonconvex. Holding two variable blocks fixed gives a simpler problem in the remaining block. The fitting procedure updates the driver, coupling matrix, and fitted states in that order within each cycle.

This distinction can be checked with the reduced residual $(1 - \gamma z)^2$, where γ is a coupling and z a fitted component. If z is fixed, this is a quadratic function of γ . If γ is fixed, it is a quadratic function of z . It is not jointly convex. At $(\gamma, z) = (0.2, 5)$ and $(0.1, 10)$, the residual is zero. At the midpoint $(0.15, 7.5)$, it is $(1 - 1.125)^2 = 1/64$. A convex function could not exceed the average of the two zero endpoint values at

their midpoint. This small example identifies the product responsible for the lack of a general joint-convexity guarantee.

For fixed Γ and $\hat{C}$, the driver update is a ridge-regularized linear least-squares problem. Define $T = \hat{C}R^\top$. For one output row, write its coefficient column as β and its target column from T as t . The part of the objective that depends on this row is

$$\lambda_{sys} \sum_{i=1}^N \left(t_i - \sum_{\ell=1}^D \phi_{i\ell} \beta_\ell \right)^2 + \lambda_B \sum_{\ell=1}^D \beta_\ell^2. \quad (5.27)$$

Differentiate with respect to one coefficient β_h . The derivative of the squared residual contributes its residual times the feature ϕ_{ih} , while the ridge term contributes $2\lambda_B\beta_h$. Setting the derivative to zero gives

$$\lambda_{sys} \sum_{i=1}^N \phi_{ih} \left(\sum_{\ell=1}^D \phi_{i\ell} \beta_\ell - t_i \right) + \lambda_B \beta_h = 0. \quad (5.28)$$

Writing this equation for every coefficient gives

$$(\lambda_{sys} \Phi^\top \Phi + \lambda_B I_D) \beta = \lambda_{sys} \Phi^\top t. \quad (5.29)$$

The matrix $\Phi^\top \Phi$ contains sums of products of feature columns. The vector $\Phi^\top t$ contains sums of feature times target. Adding the positive ridge weight to the diagonal prevents a zero-curvature direction when feature columns are duplicated.

For a hypothetical calculation with one coefficient and no intercept, let $\Phi = (1, 2)^\top$, $t = (2, 3)^\top$, and both weights equal one. The objective becomes

$$(2 - \beta)^2 + (3 - 2\beta)^2 + \beta^2 = 13 - 16\beta + 6\beta^2. \quad (5.30)$$

Its derivative is $-16 + 12\beta$, so the minimizer is $\beta = 4/3$. The matrix calculation gives $\Phi^\top \Phi = 1^2 + 2^2 = 5$, $\Phi^\top t = 1(2) + 2(3) = 8$, and $(5 + 1)\beta = 8$. Both calculations agree. Without the penalty, the coefficient would be $8/5$. The reduction illustrates ridge shrinkage for one coefficient. The adopted driver still includes its baseline and all stated features.

Assembling the normal equations for all output rows gives the exact solution

$$B^\top = (\lambda_{sys} \Phi^\top \Phi + \lambda_B I_D)^{-1} \lambda_{sys} \Phi^\top \hat{C} R^\top. \quad (5.31)$$

The positive driver ridge weight makes the matrix in this expression positive definite, even when ordered quadratic columns are duplicated. In computation, the corresponding linear system is solved. An explicit matrix inverse is unnecessary. The square quadratic slices are then symmetrized without changing their predictions.

For fixed B and $\hat{C}$, define $E = \hat{C} - \Phi B^\top$.

The coupling calculation can be read one output row at a time. For row f , the residual at observation i is $E_{if} - \sum_{j \neq f} \hat{c}_{ij} \Gamma_{fj}$. Its minimization uses the other fitted component columns as predictors of E_{if} , with a ridge penalty and box bounds. In a

hypothetical row with one permitted coupling γ , two predictor values 1 and 2, and residual targets 0.4 and 0.8, equal unit weights give

$$(0.4 - \gamma)^2 + (0.8 - 2\gamma)^2 + \gamma^2 = 0.8 - 4\gamma + 6\gamma^2. \quad (5.32)$$

The derivative $-4 + 12\gamma$ vanishes at $\gamma = 1/3$. This value lies below the adopted absolute entry bound 0.368. If the illustrative box bound were instead 0.25, the convex scalar problem would attain its minimum at 0.25. The conditioning test would still be applied to the complete coupling matrix. A row-wise scalar minimizer cannot certify the conditioning of that matrix.

Collecting these scalar residuals across all output rows gives the coupling proposal

$$\min_{\Gamma} \lambda_{sys} \|E - \hat{C}\Gamma^\top\|_F^2 + \lambda_\Gamma \|\Gamma\|_F^2 \quad (5.33)$$

subject to its zero diagonal and box bounds. This is a convex quadratic program. Its candidate is subsequently screened using Equation (5.22). A candidate that fails the condition check is shrunk toward zero or replaced by an admissible fallback. Screening preserves numerical admissibility, but it need not preserve the minimum of the unrestricted convex subproblem.

For fixed B and Γ , the fitted-state update separates over the training rows. For sample i , write $\hat{c}_i$ as a column vector.

Expand the three residual sums for one observation before collecting their coefficients. The data term contributes $\hat{c}^\top \hat{c} - 2c_{out}^\top \hat{c}$. The invariant term contributes $\lambda_{inv} \hat{c}^\top A^\top A \hat{c} - 2\lambda_{inv} c_{in}^\top A^\top A \hat{c}$. The system term contributes $\lambda_{sys} \hat{c}^\top R^\top R \hat{c} - 2\lambda_{sys} (B\phi)^\top R \hat{c}$. Terms independent of $\hat{c}$ do not affect its minimizer. Collecting its quadratic terms gives H_C , and collecting its linear terms gives h_i .

Write the expanded quadratic as $\hat{c}_i^\top H_C \hat{c}_i - 2h_i^\top \hat{c}_i$. The entry $(H_C)_{fg}$ multiplies the ordered product $\hat{c}_{if} \hat{c}_{ig}$, and $(h_i)_f$ supplies its linear term. For distinct components, the two mirrored entries contribute together, following the quadratic symmetry convention. Let δ_{fg} equal one when $f = g$ and zero otherwise. Reading the coefficients entry by entry gives

$$\begin{aligned} (H_C)_{fg} &= \delta_{fg} + \lambda_{inv} \sum_{k=1}^K A_{kf} A_{kg} + \lambda_{sys} \sum_{j=1}^F R_{jf} R_{jg}, \\ (h_i)_f &= c_{out,if} + \lambda_{inv} \sum_{k=1}^K A_{kf} \sum_{g=1}^F A_{kg} c_{in,ig} \\ &\quad + \lambda_{sys} \sum_{j=1}^F R_{jf} \sum_{\ell=1}^D B_{j\ell} \phi_{i\ell}. \end{aligned} \quad (5.34)$$

For example, the invariant contribution to $(H_C)_{12}$ adds the product of the first and second component weights over the invariant rows. The system contribution adds the product of columns 1 and 2 of R . These products explain why correcting one fitted component can change the preferred value of another.

Completing the square explains what the non-negative solve is adjusting. With $c_0 = H_C^{-1}h_i$,

$$\hat{c}^\top H_C \hat{c} - 2h_i^\top \hat{c} = (\hat{c} - c_0)^\top H_C (\hat{c} - c_0) - h_i^\top H_C^{-1} h_i. \quad (5.35)$$

The unrestricted minimizer is c_0 . When it contains a negative entry, the constrained problem seeks the closest non-negative state in the quadratic geometry defined by H_C . Off-diagonal entries tilt that geometry, so ordinary coordinate clipping need not give the closest state.

For example, take hypothetical $H_C = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ and $h_i = (-1, 2)^\top$. These coefficients can arise from unit data, invariant, and system weights with $A = [1 \ 1]$ and $R = I$. A signed driver can make the first entry of h_i negative. The unrestricted solution is $c_0 = (-5/8, 7/8)^\top$. Clipping gives $(0, 7/8)^\top$. Instead, set the constrained first component to zero and minimize the remaining expression $3c_2^2 - 4c_2$. Its derivative $6c_2 - 4$ gives $c_2 = 2/3$.

At $(0, 2/3)^\top$, the derivative in the first coordinate is $2(c_2 + 1) = 10/3$, which is positive. Increasing that zero component would increase the objective, so the boundary value is appropriate. The objective is $-4/3$ at this state and $-77/64$ at the clipped state. The constrained solution has the smaller objective. The negative objective values are possible because the constants removed during expansion are absent. The complete residual-based training criterion remains non-negative.

The general coefficient arrays derived above are assembled in the compact form

$$\begin{aligned} H_C &= I_F + \lambda_{inv} A^\top A + \lambda_{sys} R^\top R, \\ h_i &= c_{out,i} + \lambda_{inv} A^\top A c_{in,i} + \lambda_{sys} R^\top B \phi_i. \end{aligned} \quad (5.36)$$

Each row minimizes $\hat{c}_i^\top H_C \hat{c}_i - 2h_i^\top \hat{c}_i$ over $\hat{c}_i \geq \mathbf{0}$. The identity term makes H_C positive definite, so the row minimizer is unique. Clipping the unconstrained solution componentwise generally does not solve this problem because the off-diagonal terms in H_C connect component adjustments.

The quadratic-program method of Stellato et al. (2018) provides an algorithmic foundation for the constrained coupling and fitted-state blocks. These blocks use their quadratic structure directly. A generic stochastic first-order method could also optimize a related criterion, but it would require separate mechanisms to control non-negativity, coupling bounds, and conditioning. The block formulation keeps each requirement attached to an explicit subproblem.

5.5.3 Descent, stopping, and numerical safeguards

Exact minimization over a block cannot increase the objective when its admissible set contains the previous block value. Since $J \geq 0$, an objective sequence that is monotone non-increasing converges in value. This argument does not establish a global minimum of the joint problem. The convergence analysis of Razaviyayn et al. (2013)

specifies additional conditions needed to relate blockwise minimization to stationary solutions.

The conditioning screen requires particular care. Shrinking a coupling proposal can increase the training criterion relative to the previous admissible coupling matrix. The procedure therefore evaluates the objective after screening and retains the previous admissible matrix when a screened candidate cannot provide an acceptable non-increasing update. Solver tolerances also require an explicit allowance for numerical error. A claim of monotone descent applies to accepted objective-decreasing updates. It does not follow from box-constrained minimization alone.

Each restart begins with an admissible coupling matrix and non-negative fitted states. A ridge solve supplies the initial driver. Full cycles continue until an iteration limit is reached or the slope of the trailing objective history is sufficiently small. A running minimum records the best admissible state reached. When several restarts are used, the admissible state with the smallest recorded objective is retained. A nearly flat objective indicates limited further progress under that stopping rule. It does not certify parameter uniqueness or global optimality.

Figure 5.1 separates the three fitting updates from the checks that make an update acceptable. The scalar calculations in Equations (5.30) and (5.32) illustrate the first and third computational boxes in the left column. The non-negative-state example explains the return path in the right column.

The cycle returns to the driver because each block changes the conditions faced by the others. The conditioning screen acts on the coupling proposal before the fitted-state update. The full objective is then recorded after the cycle. The restart decision concerns the prescribed search procedure, while the final selection compares admissible objective values. These steps explain why a sequence of convex block solves does not establish a jointly global optimum.

Table 5.1: Specified fitting settings for the structured regression.

Setting	Value
Baseline feature	Included
Invariant penalty λ_{inv}	8.68×10^{-3}
Coupled-system penalty λ_{sys}	1.27×10^{-3}
Driver ridge penalty λ_B	6.09×10^{-5}
Coupling ridge penalty λ_Γ	6.55×10^{-4}
Off-diagonal bound γ_{abs}	0.368
Maximum full fitting cycles	60
Restarts	1
Trailing objective window	12 cycles
Objective slope tolerance	2.85×10^{-7}

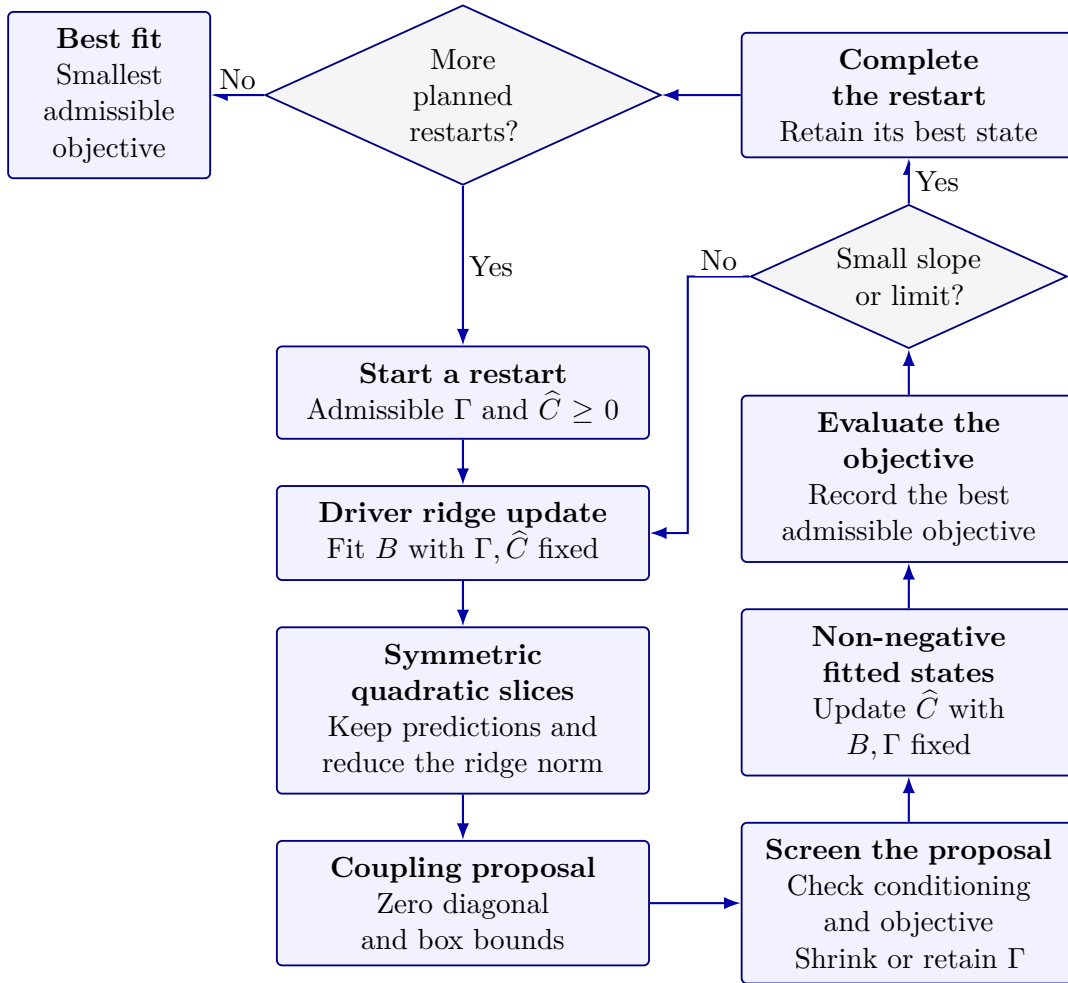


Figure 5.1: Conceptual fitting cycle with explicit symmetry, conditioning, descent, and non-negativity checks. The prescribed restart count can be one.

Table 5.1 specifies numerical fitting choices. The condition threshold κ_{max} remains an additional prescribed numerical safeguard and is checked independently of the coefficient bound. Hyperparameter assessment uses validation observations within the training pool. Held-out test observations do not determine penalty weights or stopping choices. Fitting settings remain fixed within a sample-size comparison so that changes in the learning curve reflect changing sample availability under a common model specification.

5.6 Checked deployment

5.6.1 The raw coupled state

For a new pair $(u_*, c_{in,*})$, the fitted parameters give

$$\hat{R} = I_F - \hat{\Gamma}, \quad d_* = \hat{B}\phi(u_*, c_{in,*}), \quad \hat{R}c_{raw} = d_*. \quad (5.37)$$

The notation $c_{raw} = \hat{R}^{-1}d_*$ describes the mapping mathematically. A linear-system solve evaluates it numerically. The accepted conditioning bound limits the sensitivity of that solve. The raw state is returned only when it satisfies both the invariant and non-negativity checks.

For a vector, the Euclidean norm is the square root of the sum of squared entries. Thus an illustrative invariant-residual vector $(3, 4)^\top$ has norm $\sqrt{3^2 + 4^2} = 5$. The absolute norm adds the absolute entries instead. For an illustrative concentration vector $(-0.3, 0.1, -0.4)^\top$, retaining only negative entries gives $(-0.3, 0, -0.4)^\top$, whose absolute norm is 0.7. The first calculation measures the size of a mass conservation discrepancy. The second measures total negative concentration in the stated numerical coordinates.

For any candidate c , define the diagnostic residuals

$$v_{mc}(c) = \|A(c - c_{in,*})\|_2, \quad v_{nn}(c) = \|\min(c, \mathbf{0})\|_1. \quad (5.38)$$

The minimum is taken componentwise. The adopted diagnostic tolerance is 10^{-10} . Algebraic equality and non-negativity define the mathematical requirement. Residuals below the numerical tolerance define its computational assessment. Invariant residual magnitudes are meaningful only under the stated scaling of A .

5.6.2 Affine projection for mass conservation

The nearest state satisfying mass conservation can first be derived for the hypothetical two-component inventory. Write the raw prediction as $(p_1, p_2)^\top$ and the required total as T . The two unknown projected concentrations minimize $(c_1 - p_1)^2 + (c_2 - p_2)^2$ while satisfying $c_1 + c_2 = T$. Introduce a multiplier η for that equality and write

$$\mathcal{L} = \frac{1}{2}(c_1 - p_1)^2 + \frac{1}{2}(c_2 - p_2)^2 + \eta(c_1 + c_2 - T). \quad (5.39)$$

The factor one-half simplifies differentiation without changing the minimizer. Differentiating with respect to the two concentrations and the multiplier gives

$$c_1 - p_1 + \eta = 0, \quad c_2 - p_2 + \eta = 0, \quad c_1 + c_2 = T. \quad (5.40)$$

The first two equations give $c_1 = p_1 - \eta$ and $c_2 = p_2 - \eta$. Substituting them into the inventory equality gives $p_1 + p_2 - 2\eta = T$. Thus

$$\begin{aligned} \eta &= \frac{p_1 + p_2 - T}{2}, \\ c_1 &= p_1 - \frac{p_1 + p_2 - T}{2}, \\ c_2 &= p_2 - \frac{p_1 + p_2 - T}{2}. \end{aligned} \quad (5.41)$$

The unweighted distance shares the inventory discrepancy equally between these two coordinates. With $p_1 = 12$, $p_2 = 1$, and $T = 10$, the discrepancy is 3 and the multiplier is 1.5. The projected state is $(10.5, -0.5)^\top$. The equality is satisfied even though the second component is negative. This example identifies the inequality that is absent from an equality-only projection.

When the raw state fails its checks, the first projection solves

$$c_{aff} = \arg \min_{c \in \mathbb{R}^F} \|c - c_{raw}\|_2^2 \quad \text{subject to} \quad Ac = Ac_{in,*}. \quad (5.42)$$

Introduce a Lagrange multiplier $\eta \in \mathbb{R}^K$. Stationarity gives $c - c_{raw} + A^\top \eta = 0$. Substitution into the equality gives $AA^\top \eta = A(c_{raw} - c_{in,*})$. The full row rank of A therefore gives the unique solution

$$c_{aff} = c_{raw} - A^\top (AA^\top)^{-1} A(c_{raw} - c_{in,*}). \quad (5.43)$$

The fixed matrix $A^\top (AA^\top)^{-1} A$ can be prepared once. The affine projection is then a matrix-vector operation for each new prediction.

Each matrix step follows the same scalar derivation. The multiplier column contains one multiplier per independent inventory. The vector $A^\top \eta$ converts these inventory projections into component adjustments. Solving $AA^\top \eta = A(c_{raw} - c_{in,*})$ determines the adjustments needed to remove the inventory residual. For the two-component illustration,

$$AA^\top = 2, \quad A^\top (AA^\top)^{-1} A = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}. \quad (5.44)$$

Taking the illustrative influent as $(6, 4)^\top$, the raw-minus-influent difference is $(6, -3)^\top$. Multiplying it by this projection matrix gives $(1.5, 1.5)^\top$, which is subtracted from the raw state. This recovers the scalar calculation without guessing how much either component should change.

This projection is an orthogonal projection onto the affine set defined by mass conservation in the chosen numerical component coordinates. It removes the part

of the prediction error normal to that set. It retains the part tangent to the set, which can still be large. The projection therefore enforces mass conservation without establishing predictive accuracy. Its distance also depends on the component unit convention. Sturm and Silva (2025) develop related projections for mass conservation and explain why species-dependent weights matter when concentration scales differ.

The affine state is checked for both mass conservation residuals and non-negativity before deployment. The equality follows analytically from Equation (5.43), but checking its residual catches numerical or operator inconsistency. An affine state that satisfies mass conservation can still contain a negative concentration. That possibility requires a projection with inequality constraints.

5.6.3 Joint mass conservation and non-negativity

The absolute-distance projection can also be written before introducing compact vector inequalities. For the state that satisfies mass conservation but violates non-negativity $(10.5, -0.5)^\top$, require $c_1 + c_2 = 10$ and $c_1, c_2 \geq 0$. Introduce non-negative adjustment bounds ρ_1, ρ_2 . The constraints

$$\begin{aligned} \rho_1 &\geq c_1 - 10.5, & \rho_1 &\geq 10.5 - c_1, \\ \rho_2 &\geq c_2 + 0.5, & \rho_2 &\geq -0.5 - c_2 \end{aligned} \tag{5.45}$$

make ρ_1 at least the absolute first adjustment and ρ_2 at least the absolute second adjustment. Minimizing $w_1\rho_1 + w_2\rho_2$ with positive weights makes each bound equal to the corresponding absolute adjustment. No absolute-value function remains in the optimization constraints.

Set $c_2 = t$ to impose the inventory equality, so $c_1 = 10 - t$ and $0 \leq t \leq 10$. Both absolute adjustments then equal $t + 0.5$. The objective is $(w_1 + w_2)(t + 0.5)$, which increases with t . Its minimum is at $t = 0$, giving $(10, 0)^\top$. For equal unit weights, its cost is 1. At $t = 1$, the state would be $(9, 1)^\top$, with cost 3. This direct comparison explains the optimal projection in the reduced problem.

When the affine state fails the non-negativity check, the final projection solves

$$\begin{aligned} \hat{c} &= \arg \min_{c, \rho \in \mathbb{R}^F} w_c^\top \rho \\ \text{subject to} \quad &Ac = Ac_{in,*}, \\ &c \geq \mathbf{0}, \\ &-\rho \leq c - c_{aff} \leq \rho, \\ &\rho \geq \mathbf{0}. \end{aligned} \tag{5.46}$$

For strictly positive fixed weights w_c , minimization makes $\rho_f = |c_f - c_{aff,f}|$ at an optimum. The objective is therefore the weighted absolute adjustment $\sum_f w_{c,f} |c_f - c_{aff,f}|$. The reference is the affine state. The program does not minimize distance to the original raw state or re-estimate the regression coefficients.

The program is linear because the absolute values are represented by auxiliary variables and linear inequalities. The revised-simplex method described by Huangfu and Hall (2018) provides an algorithmic foundation for solving this type of projection. The mathematical feasible set is nonempty whenever the supplied influent state is non-negative under the adopted boundary relation. The positive weights make the adjustment objective bounded below and discourage unconstrained movement of any component. A minimizer exists, although several minimizing states can exist when the absolute-distance objective has a flat face.

Large weights make changes to a component more costly. Small weights permit larger changes when those changes help satisfy the specified constraints. Weights are fixed using the intended component scaling and scientific priorities. They are not chosen after seeing the reference target for a deployed observation. A sensitivity analysis of the weights can examine how much the returned state depends on that modeling choice. Equal numerical weights also embody a choice because component units and typical magnitudes differ.

Three hypothetical components show how weights can change an admissible allocation. Let the inventory row be $[1 \ 1 \ 1]$ and the affine reference satisfying mass conservation be $(-1, 4, 7)^\top$, whose total is 10. Restoring its first component to zero requires removal of one unit from another component. Two feasible candidates are $(0, 3, 7)^\top$ and $(0, 4, 6)^\top$. For weights $(1, 10, 1)^\top$, their costs are respectively $1(1) + 10(1) = 11$ and $1(1) + 1(1) = 2$. The second component is more costly to change, so removing material from the third is preferable. With weights $(1, 1, 10)^\top$, the two costs become 2 and 11. The preferred allocation reverses. Both allocations preserve the same inventory, and neither preference is supplied by mass conservation alone.

Kircher and Votsmeier (2025) show why a projection for mass conservation needs simultaneous non-negativity enforcement in reaction-kinetics surrogates. Their scaled-space treatment protects low-concentration species during projection. Equation (5.46) addresses the same joint requirement through a weighted absolute-distance problem for activated sludge components. It imposes the adopted linear invariants and non-negativity on the returned state.

5.6.4 Illustrative projection and the failure of clipping

Consider an illustrative two-component system with $A = [1 \ 1]$, a total influent inventory of 10, and a raw prediction $c_{raw} = (12, 1)^\top$. The raw total is 13. The affine formula subtracts 1.5 from each component and gives $c_{aff} = (10.5, -0.5)^\top$. The total is now 10, but the second component is negative.

Clipping that negative value to zero produces $(10.5, 0)^\top$, which violates the total inventory. With equal projection weights, Equation (5.46) instead returns $(10, 0)^\top$. Both constraints hold. These values are a hypothetical illustration of the projection geometry. They are not activated sludge predictions or experimental findings.

The deployment order uses this geometry directly. A feasible raw state needs no adjustment. An infeasible raw state is first projected onto the mass conservation set. A feasible affine state can then be returned. The linear program is required only when the affine state still fails non-negativity. Every returned candidate is checked under the declared numerical tolerances. An unsuccessful numerical solve does not authorize an unchecked prediction.

Figure 5.2 makes the branch conditions explicit. Both physical checks appear at the raw and affine stages. The last projection branch also requires successful optimization and an independent admissibility check.

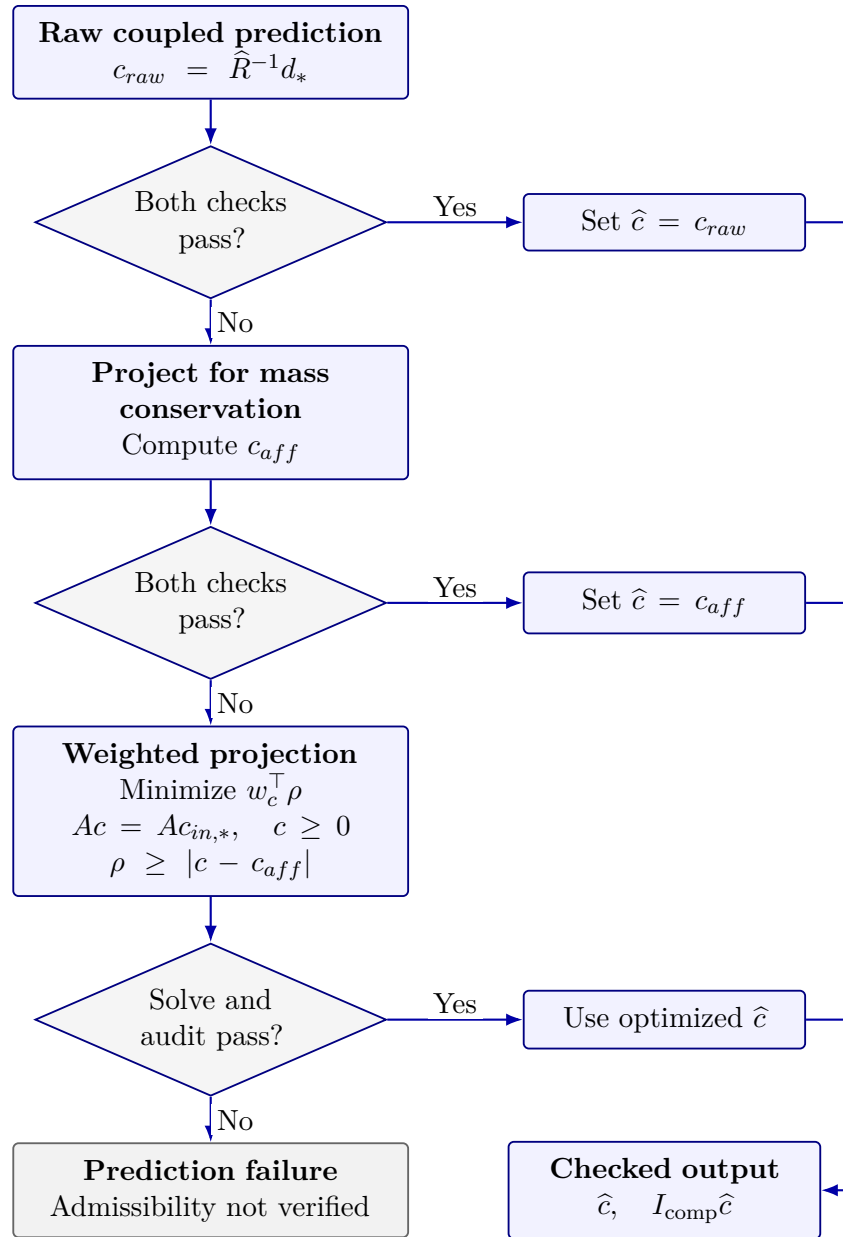


Figure 5.2: Checked deployment of one predicted component state. Accepted branches merge before the common reporting map is applied. A failed final audit produces a reported prediction failure.

The hypothetical state $(12, 1)^\top$ follows the two No branches because its total is incorrect and its affine projection contains a negative component. The weighted projection supplies $(10, 0)^\top$, which reaches the return branch only after the stated checks pass. A state that satisfies both conditions at an earlier branch reaches the same reporting step. The different paths therefore change the projection effort while retaining the same acceptance requirements.

The component state returned by this rule is the deployed output. Its reported quantities are

$$\hat{y}_{\text{comp}} = I_{\text{comp}} \hat{c}. \quad (5.47)$$

The deployment constraint is responsible for admissibility. The soft training penalties can reduce the need for projection, but that possibility is an evaluation question rather than a mathematical guarantee.

5.7 Interpreting the complete prediction map

5.7.1 Driver coefficients and back projection

The transformation from driver coefficients to output coefficients can be computed with small matrices. Use the hypothetical coupling matrix in Equation (5.19), a reduced feature vector $(1, a)^\top$, and driver coefficients

$$B = \begin{bmatrix} 4 & 1 \\ 2 & 0 \end{bmatrix}, \quad r = \begin{bmatrix} 4 + a \\ 2 \end{bmatrix}. \quad (5.48)$$

The first coefficient column describes the baseline, and the second describes dependence on the hypothetical scalar input a . Multiplication through the coupling inverse gives

$$\begin{aligned} R^{-1}B &= \frac{1}{0.96} \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 2 & 0 \end{bmatrix} \\ &= \frac{1}{0.96} \begin{bmatrix} 4.4 & 1 \\ 2.8 & 0.2 \end{bmatrix} = \begin{bmatrix} 55/12 & 25/24 \\ 35/12 & 5/24 \end{bmatrix}. \end{aligned} \quad (5.49)$$

The second component now depends on a , even though its driver coefficient for a was zero. This dependence is introduced by the simultaneous coupling solve.

For a hypothetical reported quantity $y = c_1 + 2c_2$, its reporting row is $[1 \ 2]$. Its transformed coefficient row is

$$[1 \ 2] R^{-1}B = [125/12 \ 35/24]. \quad (5.50)$$

At $a = 2$, the driver is $(6, 2)^\top$, the raw component state is $(20/3, 10/3)^\top$, and the reported quantity is $40/3$. The transformed row gives the same value through $125/12 + (35/24)2 = 40/3$. This reporting row is chosen for teaching and is not an adopted COD or nutrient composition row.

The rows of $\hat{B}$ refer to component drivers. They cannot be labeled directly as COD, TN, TP, or TSS coefficients. The raw component coefficient matrix is

$$B_{\text{raw}} = \hat{R}^{-1} \hat{B}, \quad c_{\text{raw}} = B_{\text{raw}} \phi. \quad (5.51)$$

The corresponding map in reported-composite space is

$$B_{\text{raw,comp}} = I_{\text{comp}} \hat{R}^{-1} \hat{B}, \quad y_{\text{raw,comp}} = B_{\text{raw,comp}} \phi. \quad (5.52)$$

This transformation carries the driver coefficients through the learned coupling and then through the reporting operator. Inspection of a COD quadratic block uses its row in $B_{\text{raw,comp}}$, rather than an arbitrary row of $\hat{B}$.

Figure 5.3 shows the complete raw prediction chain using the named feature groups. It keeps the component driver, simultaneous coupling solve, and reporting operation visible as separate mathematical steps.

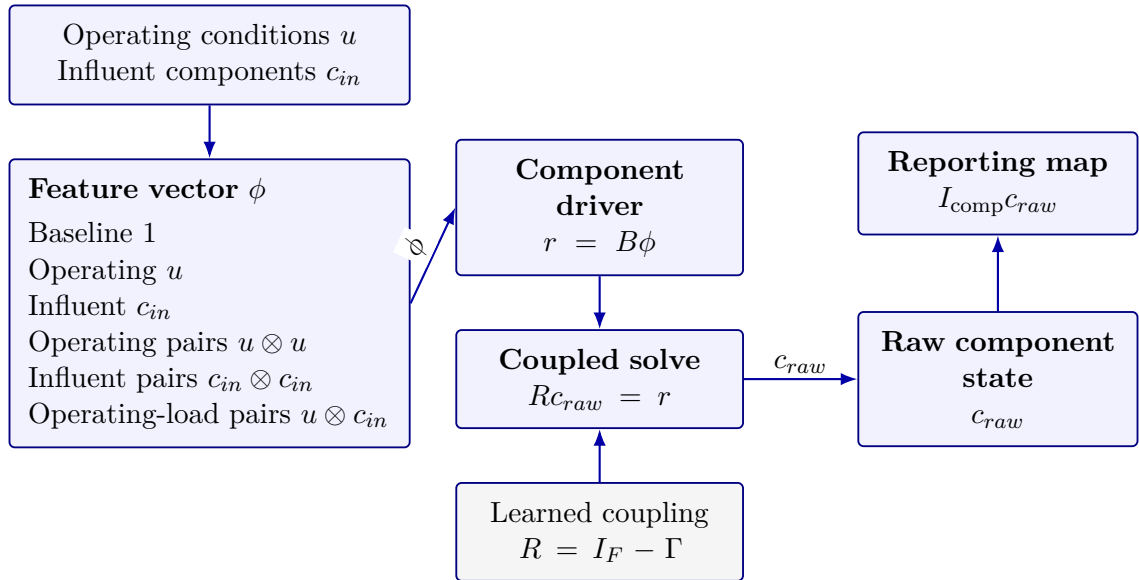


Figure 5.3: Named feature blocks and the transformation from driver to raw component and reported-composite predictions. Composing the three operations gives the raw reported coefficient map $I_{\text{comp}} R^{-1} B$.

In the two-feature example, the second driver has no direct coefficient for the scalar input, but the coupling path gives its raw component a nonzero input coefficient. The reporting path combines both component coefficients. Reading the paths in this order explains the back projection in Equations (5.49) and (5.50). Checked deployment acts on the raw component state before the final reported quantities are calculated.

The named blocks can be displayed as coefficient maps. Operating terms are grouped separately from influent terms. Symmetric quadratic maps display both ordered cells while reporting unique cross terms consistently. Operating-load maps show which operating variables interact with which influent components. The coupling matrix is displayed separately because it has a different interpretation from the input-driver terms.

A coefficient-display threshold is used only to simplify reporting after fitting. For each reported-composite raw-head map, the threshold is 10^{-4} times its largest absolute coefficient. For the coupling matrix, the threshold is 10^{-4} times its largest absolute off-diagonal entry. The threshold does not modify the fitted mapping, select hyperparameters, or impose sparsity during training. Symmetric quadratic summaries count unique upper-triangular terms. A small coefficient can still have a large contribution when multiplied by a large feature. Interpretation therefore examines coefficient magnitude, input scale, and the contribution of the corresponding term over the declared domain.

5.7.2 Analytical jacobians of the raw head

Differentiation of a scalar polynomial shows why a coefficient and a marginal response differ. For the illustrative driver in Equation (5.10),

$$\begin{aligned}\frac{\partial r_1}{\partial u_1} &= 2 + 0.2u_1 + 0.2u_2, \\ \frac{\partial r_1}{\partial u_2} &= -1 + 0.2u_1 + 0.25c_1, \\ \frac{\partial r_1}{\partial c_1} &= 0.5 + 0.25u_2, \quad \frac{\partial r_1}{\partial c_2} = -0.1c_2.\end{aligned}\tag{5.53}$$

At $u = (2, 3)^\top$ and $c = (4, 5)^\top$, these derivatives are 3, 0.4, 1.25, and -0.5 . In particular, the first derivative is 3 although the linear coefficient of u_1 is 2. The operating quadratic term and interaction with u_2 supply the remaining contribution.

The derivative of the second illustrative driver $r_2 = -u_1 + u_2 + 0.2c_2$ with respect to u_1 is -1 . Differentiate the coupled equations with the coupling held fixed. Their derivative equations are $\partial c_1/\partial u_1 - 0.2\partial c_2/\partial u_1 = 3$ and $-0.2\partial c_1/\partial u_1 + \partial c_2/\partial u_1 = -1$. Thus the same inverse used for prediction gives

$$\begin{bmatrix} \partial c_1/\partial u_1 \\ \partial c_2/\partial u_1 \end{bmatrix} = \frac{1}{0.96} \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 35/12 \\ -5/12 \end{bmatrix}.\tag{5.54}$$

For the hypothetical reporting row $[1 \ 2]$, the reported derivative is $35/12 + 2(-5/12) = 25/12$. Differentiating the driver, solving through the coupling, and applying the reporting weights are three separate operations.

Putting several derivatives in one matrix gives a Jacobian. Its rows identify the output being differentiated, and its columns identify the input being varied. At the illustrative inputs, the operating-driver Jacobian and its coupled transformation are

$$J_{r,u} = \begin{bmatrix} 3 & 0.4 \\ -1 & 1 \end{bmatrix}, \quad R^{-1}J_{r,u} = \begin{bmatrix} 35/12 & 5/8 \\ -5/12 & 9/8 \end{bmatrix}.\tag{5.55}$$

The first column is the derivative calculation above. The second follows by solving the same coupled system for the derivative column $(0.4, 1)^\top$. A Jacobian therefore

collects several scalar sensitivity calculations rather than introducing a different kind of response.

A coefficient describes a term in the polynomial. A Jacobian describes the change in the whole mapping at an operating point. For component f , let $H_{uc}^{(f)} \in \mathbb{R}^{M_{op} \times F}$ reshape its interaction coefficients so that the cross term is $u^\top H_{uc}^{(f)} c_{in}$. The driver can then be expressed as

$$\begin{aligned} r_f = & b_f + (W_u)_{f,:} u + (W_{in})_{f,:} c_{in} + u^\top Q_{uu}^{(f)} u \\ & + c_{in}^\top Q_{cc}^{(f)} c_{in} + u^\top H_{uc}^{(f)} c_{in}. \end{aligned} \quad (5.56)$$

With symmetric square slices, its gradients are

$$\begin{aligned} \nabla_u r_f &= (W_u)_{f,:}^\top + 2Q_{uu}^{(f)} u + H_{uc}^{(f)} c_{in}, \\ \nabla_{c_{in}} r_f &= (W_{in})_{f,:}^\top + 2Q_{cc}^{(f)} c_{in} + (H_{uc}^{(f)})^\top u. \end{aligned} \quad (5.57)$$

The quadratic gradients in Equation (5.57) follow by differentiating each scalar product. For $x^\top Q x = \sum_i \sum_j q_{ij} x_i x_j$, differentiation with respect to x_h collects the terms with $i = h$ and those with $j = h$. Their sum is $\sum_j (q_{hj} + q_{jh}) x_j$. It becomes the h th entry of $2Qx$ when Q is symmetric. For $u^\top H c = \sum_i \sum_j h_{ij} u_i c_j$, differentiation with respect to u_i gives $\sum_j h_{ij} c_j$, while differentiation with respect to c_j gives $\sum_i h_{ij} u_i$. These are the entries of Hc and $H^\top u$.

Stacking the transposed gradients gives $J_{r,u} \in \mathbb{R}^{F \times M_{op}}$ and $J_{r,c} \in \mathbb{R}^{F \times F}$. The raw-state and raw-composite Jacobians follow through the same back projection as the coefficients,

$$\begin{aligned} J_{raw,u} &= \hat{R}^{-1} J_{r,u}, & J_{raw,c} &= \hat{R}^{-1} J_{r,c}, \\ J_{raw,comp,u} &= I_{comp} J_{raw,u}, & J_{raw,comp,c} &= I_{comp} J_{raw,c}. \end{aligned} \quad (5.58)$$

These derivatives describe the complete fitted raw head at the specified inputs. They include curvature and interactions. They do not include the projection applied during deployment.

An illustrative aeration sensitivity can be inspected at low and high ammonium loading while holding the remaining inputs fixed. A difference in the two derivatives identifies a loading-dependent association within the fitted model. Oxygen-transfer limitations and heterotrophic competition provide process context for assessing such a pattern, as discussed by Stenstrom and Song (1991). Agreement with that context is a plausibility check. It is not evidence that changing aeration in a plant will cause exactly the fitted response.

5.7.3 Sensitivity after the affine projection

Return to the two-feature example in Equation (5.49). Its raw derivative with respect to a is $(25/24, 5/24)^\top$, whose total is $5/4$. This raw response changes the total

component inventory. At fixed influent inventory, an affine projection must remove that total change. For the two-component invariant,

$$P_A = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad (5.59)$$

$$\frac{\partial c_{aff}}{\partial a} = P_A \begin{bmatrix} 25/24 \\ 5/24 \end{bmatrix} = \begin{bmatrix} 5/12 \\ -5/12 \end{bmatrix}.$$

The two entries now sum to zero. For the reporting row $[1 \ 2]$, the derivative is $5/12 + 2(-5/12) = -5/12$. The corresponding raw derivative was $35/24$. The projection changes both its magnitude and sign because it enforces a different material-accounting condition on the complete returned state.

The influent reference also changes when an influent component is perturbed. In this reduced example, the driver contains no influent term. Increasing either influent component by δ increases the required total by δ . The unweighted affine projection then adds $\delta/2$ to each predicted component. Its influent derivative is the matrix $\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, even though the raw influent derivative is zero. This contribution comes from the right-hand side of the mass conservation equality.

Define the complementary projection matrices

$$Q_A = A^\top (AA^\top)^{-1} A, \quad P_A = I_F - Q_A. \quad (5.60)$$

Then $c_{aff} = P_A c_{raw} + Q_A c_{in}$. With the fitted coefficients and invariant operator held fixed, its Jacobians are

$$J_{aff,u} = P_A J_{raw,u}, \quad J_{aff,c} = P_A J_{raw,c} + Q_A. \quad (5.61)$$

The second expression includes the change in the mass conservation reference when the influent changes. Omitting Q_A would treat that reference as fixed and give the wrong influent sensitivity.

Multiplication by A gives $AJ_{aff,u} = 0$ and $AJ_{aff,c} = A$. An operating perturbation changes the affine prediction only along directions that preserve its fixed influent inventories. An influent perturbation changes those inventories consistently with the influent reference. These relations provide analytical checks on the sensitivity calculation.

When the affine branch is used over a neighborhood of inputs, its coefficients and Jacobians describe that branch. The complete deployment rule still includes transitions between raw, affine, and linear-program branches. A single coefficient map does not describe every branch at once.

5.7.4 Active constraints in the final projection

An explicit boundary in the same hypothetical model shows the change in sensitivity caused by non-negativity. Keep the total influent inventory equal to 10. The affine

components obtained from Equation (5.49) are

$$c_{aff,1} = \frac{35}{6} + \frac{5a}{12}, \quad c_{aff,2} = \frac{25}{6} - \frac{5a}{12}. \quad (5.62)$$

Their sum is 10. Near $a = 10$, the first component is positive, while the second reaches zero at $a = 10$. Immediately below this boundary the affine state is feasible and its derivative is $(5/12, -5/12)^\top$. Above the boundary the affine second component is negative. The positive-weight absolute projection returns $(10, 0)^\top$, so its derivative with respect to a is $(0, 0)^\top$. The final projected state therefore has a different derivative on each side of the boundary. A single derivative is not available at the boundary itself.

In the projected branch, the binding equalities are $c_1 + c_2 = 10$ and $c_2 = 0$. Differentiating them gives $\partial c_1/\partial a + \partial c_2/\partial a = 0$ and $\partial c_2/\partial a = 0$. These two equations directly give the zero derivative. The general active-set calculation uses the same idea with all binding inventory and adjustment constraints.

The linear program can make a component concentration equal to zero or bind one of the absolute-adjustment inequalities. These binding constraints define an active set. If the optimal solution is unique and its nondegenerate optimal basis remains unchanged over a local neighborhood, the solution varies affinely with the relevant right-hand sides in that neighborhood. A branch-specific sensitivity can then be derived from that fixed basis.

At an active-set transition, the derivative can change abruptly. When several optimal projected states exist, the returned solution can also depend on the rule used to select among them. A global raw-head coefficient display cannot resolve these effects. Local deployment sensitivity is assessed by differentiating a verified fixed basis where appropriate or by perturbing admissible inputs and resolving the entire deployment problem. The perturbation size and any branch changes are recorded as part of that sensitivity assessment.

This distinction prevents three different quantities from being conflated. A driver coefficient belongs to $\hat{B}$. A raw-state sensitivity passes through $\hat{R}^{-1}$. A deployed-state sensitivity also depends on the mass conservation reference and active projection constraints. Each quantity answers a different question about the model.

5.8 Assessment design for the structured regression

5.8.1 Common inputs and comparator models

The structured-regression assessment uses a design collection of 10,000 accepted steady-state reactor cases. Every method receives the same 22 physical inputs and predicts the same 20 effluent components. Its reported composites are obtained from the common composition matrix. The input domain, reaction model, simulation bounds, and acceptance criterion remain those of the declared reactor benchmark. The compar-

ison therefore changes the predictor while holding the modeled process and prediction task fixed.

The nine unconstrained comparators are XGBoost, LightGBM, CatBoost, Adaboost, random forest, support vector regression, k -nearest neighbors, partial least squares, and a fully connected multilayer perceptron. These methods represent boosted and randomized tree ensembles, kernel regression, local neighbor averaging, latent linear regression, and nonlinear neural regression. Their foundations include the ensemble construction of Breiman (2001), the boosting formulations of Chen and Guestrin (2016); Drucker (1997); Ke et al. (2017); Prokhorenkova et al. (2018), and the latent regression of Wold et al. (2001). The comparators have no invariant penalty, auxiliary non-negative fitted-state matrix, or learned component-coupling equation during fitting.

The neural comparator has hidden layers with 128, 64, and 32 units and logistic activation. Dense connections map the 22 inputs through these layers to 20 component outputs. Its capacity is evaluated under the same samples and reported-composite map as the other predictors. A transparent polynomial and a neural network can therefore differ in their internal structure without changing the supervised target.

5.8.2 Training partitions and hyperparameter selection

The primary assessment uses a row-wise 80% training and 20% held-out split with random seed 42. The specified collection gives a training pool of 8000 rows and a held-out set of 2000 rows. Hyperparameter selection uses only the training pool. Half of that pool forms a 4000-row tuning subset. A further 80% and 20% split gives 3200 tuning-training rows and 800 tuning-validation rows. This is one internal validation split rather than a cross-validation design.

Each predictor receives 100 hyperparameter trials under a seeded tree-structured Parzen estimator with seed 42. No trial pruning or wall-clock timeout is imposed. Bergstra et al. (2011) develop this search approach, while Akiba et al. (2019) provide the associated hyperparameter optimization framework. The selection criterion is mean squared error in the shared physical component space on the internal validation set. Selected settings are held fixed for the primary assessment and the repeated sample-size design. The final primary fit uses the full 8000-row training pool.

For the unconstrained comparators, input and output standardization parameters are estimated from the relevant training partition. Predictions are transformed back to physical component coordinates before any invariant check, reporting map, or error assessment. ICSOR remains in physical coordinates so that its driver, invariant operator, and deployed constraints use the same component convention. Internal-validation transformations are estimated from tuning-training rows only. Held-out observations do not determine the means, standard deviations, hyperparameters, or projection weights.

Table 5.2 gives the specified numerical configurations. These values define fitted

model settings rather than measured predictive performance. The internal search compares candidate configurations according to the stated validation objective.

Table 5.2: Specified configurations of the nine unconstrained comparators.

Comparator	Configuration
XGBoost	309 trees, learning rate 0.125, maximum depth 3, minimum child weight 6.84, row fraction 0.915, feature fraction 0.828, absolute penalty 1.18×10^{-3} , and squared penalty 0.382.
LightGBM	302 trees, learning rate 0.0809, 16 leaves, maximum depth 8, minimum leaf count 14, row fraction 0.845, feature fraction 0.869, absolute penalty 3.17×10^{-3} , and squared penalty 3.36×10^{-4} .
CatBoost	236 boosting iterations, learning rate 0.172, depth 4, leaf squared penalty 0.0266, bagging temperature 0.0277, and random strength 0.0657.
AdaBoost	236 estimators, learning rate 0.999, and squared regression loss.
Random forest	390 trees, maximum depth 14, minimum split count 3, minimum leaf count 1, feature fraction 0.500, and no bootstrap resampling.
Support vector regression	Polynomial kernel of degree 4, regularization parameter 0.118, insensitive-error width 0.0458, constant offset 0.889, and solver tolerance 8.96×10^{-4} . The kernel scale is the reciprocal of the product of input dimension and training input variance.
k -nearest neighbors	15 neighbors, inverse-distance weighting, Manhattan distance, and a k -dimensional search tree with leaf size 40.
Partial least squares	6 latent components, maximum 887 fitting iterations, and tolerance 9.91×10^{-5} .
Multilayer perceptron	Hidden widths (128, 64, 32), logistic activation, adaptive moment estimation, squared penalty 6.33×10^{-4} , batch size 64, initial learning rate 1.08×10^{-3} , maximum 500 iterations, and tolerance 1.63×10^{-5} . Training rows are not shuffled, and early stopping is disabled.

5.8.3 Scoring states and physical diagnostics

The scored ICSOR output is the final checked component state. For comparator accuracy assessment, the raw component prediction receives the same affine projection for mass conservation in Equation (5.43). The composition map is then applied to the projected state. This comparison describes the accuracy of a checked structured prediction and comparator predictions after affine projection. The comparator projection imposes neither component non-negativity nor the weighted linear projection.

Physical diagnostics use the direct raw states of the comparators. For ICSOR, diagnostics distinguish the raw coupled state, affine state, and final deployed state. This stage separation prevents affine projection for mass conservation used during

scoring from being attributed to the learner. At each stage, the violation frequency for diagnostic v is

$$f_v = \frac{100}{n} \sum_{i=1}^n \mathbf{1}\{v(c_i) > 10^{-10}\}, \quad (5.63)$$

where n is the number of assessed observations. The diagnostics v_{mc} and v_{nn} are defined in Equation (5.38). Frequencies, residual magnitudes, adjustment distances, and deployment-branch use describe different aspects of the same prediction procedure. None is treated as a substitute for predictive error.

5.8.4 Composite error measures

The differences among the five measures can first be computed for two hypothetical observations of one reported quantity. Take reference values (2, 4) and predictions (3, 2). Their residuals are (1, -2). Squaring them gives (1, 4), while taking absolute values gives (1, 2). The mean squared error is $(1+4)/2 = 2.5$, its square root is approximately 1.5811, and the mean absolute error is $(1+2)/2 = 1.5$. These measures describe different summaries of the same two errors.

The absolute relative errors are $1/2$ and $2/4$, so their mean is 0.5. The reference mean is 3. Its total squared variation is $(2-3)^2 + (4-3)^2 = 2$. The prediction residual squared sum is 5. Therefore the coefficient of determination is $1 - 5/2 = -1.5$. Its negative value says that the hypothetical predictions have greater squared error than assigning the reference mean to both observations. None of these example values is a performance finding.

For reported quantity $q \in \{1, 2, 3, 4\}$, define the prediction residual $e_{iq} = \hat{y}_{iq} - y_{iq}$. The five measures are mean squared error, root mean squared error, mean absolute error, mean absolute relative error, and the coefficient of determination. They are computed per reported quantity and summarized across the four quantities. The squared and absolute measures are

$$E_{2,q} = \frac{1}{n} \sum_i e_{iq}^2, \quad E_{root,q} = \sqrt{E_{2,q}}, \quad E_{1,q} = \frac{1}{n} \sum_i |e_{iq}|. \quad (5.64)$$

For strictly nonzero reference values, the relative measure is

$$E_{rel,q} = \frac{1}{n} \sum_i \frac{|e_{iq}|}{|y_{iq}|}. \quad (5.65)$$

It is reported as a ratio. Multiplication by 100 converts it to a percentage. A zero reference value makes this expression undefined, and a very small value can dominate it. Any zero-reference convention is declared explicitly before comparing predictors. A finite relative-error summary is not assigned to a target for which the unmodified expression is undefined.

The per-target coefficient of determination is

$$R_q^2 = 1 - \frac{\sum_i e_{iq}^2}{\sum_i (y_{iq} - \bar{y}_q)^2}. \quad (5.66)$$

A target with zero reference variance makes this expression undefined. A negative value is possible when the predictor has a greater squared error than the reference-mean predictor. This coefficient measures relative prediction error and is not a correlation coefficient.

The aggregate squared error pools the four composite residuals. Its root is

$$E_{root} = \left(\frac{1}{4n} \sum_{q=1}^4 \sum_{i=1}^n e_{iq}^2 \right)^{1/2}. \quad (5.67)$$

Aggregate absolute and relative errors use the corresponding mean over observations and composites. Aggregate R^2 uses the mean of the four target-specific coefficients. The composite coordinates retain their own physical units. Their pooled error is a numerical benchmark summary under that coordinate convention, rather than a single common material quantity. Per-target scores remain necessary to show which reported quantities contribute to an aggregate comparison.

Pooling before taking the square root also matters. If four hypothetical target mean squared errors are 1, 9, 1, and 9, the pooled mean squared error is 5 and its root is $\sqrt{5}$, approximately 2.2361. The arithmetic mean of the four target root errors is instead $(1 + 3 + 1 + 3)/4 = 2$. These are different summaries. Equation (5.67) specifies the first calculation.

The primary ordering measure is aggregate root mean squared error. The other measures provide complementary assessments. Squared error emphasizes larger residuals, absolute error gives average deviation, relative error emphasizes deviations in relation to the reference magnitude, and R^2 compares the squared residual with reference variation. A predictor need not receive the same ordering under all these measures.

5.8.5 Repeated sample-size assessment

The sample-size design uses

$$n \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (5.68)$$

At each size, ten subsamples are drawn without replacement from the design collection. Deterministic seeds are derived from seed 42. Each run uses a row-wise 80% training and 20% held-out split. All ten predictors receive the same sampled rows and partitions. The design specifies 110 fits per predictor across the size schedule. This is repeated random splitting rather than ten-fold cross-validation.

The fixed hyperparameters isolate sensitivity to available observations under one model specification. The mean and spread of each held-out score are summarized across the ten runs. Learning curves are assessed over the entire schedule because behavior at one sample size does not establish behavior at other sizes. Geman et al. (1992) explain the relation between flexibility and the bias-variance problem, and Viering and Loog (2023) review the varied shapes of learning curves.

The learning-curve area can first be calculated over a reduced hypothetical schedule. Suppose an error measure has values 6, 4, and 3 at sample sizes 100, 200, and 400. Between the first two sizes, a trapezoid has width 100 and mean height $(6 + 4)/2 = 5$, giving area 500. Between the next two, its width is 200 and mean height $(4 + 3)/2 = 3.5$, giving area 700. The total area is 1200. Dividing by the total width $400 - 100 = 300$ gives a normalized area of 4. A larger interval contributes more to this average than a smaller interval, even when each has one pair of endpoints. This schedule is a teaching illustration and does not replace the stated assessment sizes.

For successive retained sizes $n_1, \dots, n_S$, write the mean error of predictor m under measure j as $\overline{M}_{m,j}(n_s)$. The general numerical calculation is

$$\frac{1}{n_{\max} - n_{\min}} \sum_{s=1}^{S-1} (n_{s+1} - n_s) \frac{\overline{M}_{m,j}(n_s) + \overline{M}_{m,j}(n_{s+1})}{2}. \quad (5.69)$$

Let $\overline{M}_{m,j}(n)$ denote the mean held-out measure j for predictor m at sample size n . The normalized area under its learning curve is

$$L_{m,j} = \frac{1}{n_{\max} - n_{\min}} \int_{n_{\min}}^{n_{\max}} \overline{M}_{m,j}(n) \, dn. \quad (5.70)$$

The trapezoidal rule evaluates this integral over the stated sample sizes. Normalization by the sample-size interval gives an average height of the learning curve and preserves the units of measure j . It does not normalize the prediction error by a target standard deviation. Cawley (2011); Guyon et al. (2011) provide related trajectory summaries in active learning. The present design changes the number of sampled observations without choosing observations through an active acquisition rule.

A rank discards the numerical distance between scores and keeps their order. For four hypothetical predictor errors 3.0, 3.0, 4.5, and 6.0, the dense ranks are 1, 1, 2, and 3. The tied first two predictors receive the same rank. There is no skipped rank after the tie. The difference between the third and fourth errors does not influence the difference between their ranks. If one predictor receives ranks 1, 2, and 2 in three illustrative comparisons, its mean rank is $(1 + 2 + 2)/3 = 5/3$. The actual assessment averages over the stated combinations of target, measure, and size.

For each measure j , reported quantity q , and retained size n_s , predictors receive a dense rank $r_{m,j,q}(n_s)$ from their mean held-out score. Lower squared, root, absolute, and relative errors receive better ranks. Larger R^2 values receive better ranks. Equal scores receive equal ranks, and the next distinct score receives the next consecutive rank. With $S = 11$ sizes and five measures, the summaries are

$$\bar{r}_{m,j} = \frac{1}{4S} \sum_{q=1}^4 \sum_{s=1}^S r_{m,j,q}(n_s), \quad \bar{r}_m = \frac{1}{5} \sum_{j=1}^5 \bar{r}_{m,j}. \quad (5.71)$$

These are descriptive summaries of relative ordering. They do not establish statistical significance. Demšar (2006) develops broader rank-based model comparisons, while

Shevchenko et al. (2024) emphasize variability in benchmarking. Their contributions support careful comparative assessment without equating a mean rank with an absolute error or a significance test.

At the largest sample size, the generalization gap is

$$\Delta E_{root,m} = E_{root,m}^{held-out} - E_{root,m}^{training}. \quad (5.72)$$

A positive gap indicates greater held-out error. A small gap can accompany either accurate prediction or high-error underfitting, so it is interpreted with the corresponding absolute scores. Training time is recorded for every repeated fit and summarized as a distribution across runs. Deployment effort is assessed separately through prediction time, projection-branch use, and constrained-solve effort. This separation makes the cost of learning distinct from the cost of returning a checked state.

For a hypothetical predictor with training error 0.5 and held-out error 0.8, the gap is 0.3. Another with errors 5.0 and 5.1 has a gap of only 0.1. The second has the smaller gap but much larger absolute errors. This arithmetic explains why the gap cannot by itself identify the more useful surrogate.

5.9 Interpretability assessment and modeling limits

The interpretability assessment examines the six driver blocks, the coupling matrix, and the corresponding component and composite raw-head maps. It evaluates whether important terms remain stable across repeated training partitions and whether the associated response patterns agree with the declared process domain. Stability assessment is particularly relevant because an explicit polynomial library does not remove identifiability problems. Brunton et al. (2016) connect interpretable nonlinear terms to the problem of recovering structure from data, while Machado et al. (2014) show how operational and influent uncertainty complicate parameter identification in activated sludge models.

The driver and coupling parameters can trade off in the relation $Rc = B\phi$. Different joint stationary solutions can therefore predict similar component states with different coefficients. Symmetry removes the redundancy of ordered self-quadratic pairs. It does not establish uniqueness of B and Γ together. Coefficient signs and magnitudes are interpreted as properties of a specified fitted representation. Repeated stability checks and response comparisons provide stronger evidence than an isolated coefficient value.

Prediction error and physical diagnostics are assessed separately for the raw head, the affine state, and the final deployed state. The comparison distinguishes what the regression learns from what the projection enforces. The magnitude of the projection is also examined because a feasible deployed state can lie far from its raw prediction. Computational assessment records fitting effort and deployment effort separately.

These measures establish the cost of the structured regression and the cost of enforcing its physical conditions without assuming an accuracy benefit.

The scope remains steady-state component prediction within the stated input domain and control volume. The projected state preserves the adopted invariants and non-negativity when the optimization and numerical checks succeed. It does not inherit the kinetic equations, dynamic behavior, or thermodynamic conditions of the full process model. The raw head supplies an inspectable approximation. The deployment constraints supply defined material-accounting conditions. Their combined use is evaluated as an engineering surrogate under these limits.

Chapter 6

Connected plant surrogates and operating optimization

6.1 From an admissible reactor prediction to a plant decision

An activated sludge plant repeatedly moves material between biological reactors and a secondary clarifier. Settled biomass returns to the biological treatment train. Mixed liquor may return to an upstream stage to support nitrogen removal. A separate waste stream removes accumulated solids. Aeration, recycle, and wasting therefore affect more than one treatment unit. The account of biological treatment given by Henze et al. (2008) explains why these decisions need a common plant balance. The clarification and thickening model of Härtel and Pöpel (1992) adds the storage and separation processes that determine which solids return to the reactors.

The operating question is how to choose these settings while balancing treated-water quality and resource use. de Araujo et al. (2013) formulate activated sludge operation as a constrained economic problem and examine the sensitivity of the resulting decisions. Padrón-Páez et al. (2020) examine wastewater treatment design through multiple objectives. Aboagye et al. (2021) organize treatment-network design and sustainability assessment within mathematical programming. These studies make a common point. A preferred decision depends on the objective, the feasible operating region, and the quantities used to measure treatment and resource use.

A process surrogate can reduce the cost of evaluating the same plant at many candidate settings. McBride and Sundmacher (2019) describe surrogate construction from experiments and process simulations. Bhosekar and Ierapetritou (2018) connect surrogate prediction to feasibility analysis and optimization. Applications include the Kriging treatment model of He et al. (2023), the machine learning process model within predictive control developed by Bernardelli et al. (2020), and the unit surrogates used for wastewater resource-recovery optimization by Durkin et al. (2024). Each application

links prediction to a decision problem. Its objective and constraints determine which prediction errors matter most.

A physically admissible prediction for one reactor does not by itself establish a consistent plant prediction. The predicted clarifier underflow may disagree with the predicted return-sludge contribution at the mixer. Separate predictions may also assign different amounts of a inventory used for mass conservation to connected streams. ICSOR supplies a useful starting point because it separates second-order statistical prediction from enforcement of declared physical relations (E. F. Selerio Jr., 2026). Extending this principle to a plant requires a joint response and a joint projection. The projection must account for the mixer, all reactors, both clarifier outlets, and the solids stored in the clarifier.

Three questions remain distinct. A network prediction can satisfy the imposed balances. It can also approximate a mechanistic reference accurately. Finally, it can lead an optimizer to a useful operating decision. None of these properties is a substitute for the others. The surrogate comparisons studied by Pedrozo et al. (2025) make optimization performance a separate endpoint from predictive accuracy. The gray-box methods of Hameed et al. (2026) also retain checks against the original model when managing approximation error. The connected formulation developed here therefore includes independent mechanistic evaluation of selected controls.

6.2 Hydraulic connections and reactor inventories

6.2.1 Flow through the treatment train

Let Q_0 denote the fresh-influent flow in $\text{m}^3 \text{d}^{-1}$ and let $c_{\text{in}} \in \mathbb{R}_+^F$ denote its component concentrations. The plant has n_r completely mixed reactors and an L -layer clarifier. The reactor concentrations are $c_1, \dots, c_{n_r}$. The mixer concentration is m . Clarifier overflow and underflow concentrations are c_E and c_U . The overflow is the final plant outlet, so $c_{\text{out}} = c_E$.

The dimensionless ratios r_I , r_R , and w describe mixed-liquor recycle, return activated sludge, and waste activated sludge flow relative to Q_0 . Mixed-liquor recycle leaves the final reactor before the clarifier. Return and waste sludge share the clarifier underflow concentration. Each ratio is an actual flow divided by Q_0 . Denote the train, clarifier-feed, underflow, and overflow flows by Q_P, Q_C, Q_U, Q_E . The mixer, reactor splitter, underflow splitter, and external water balance respectively give

$$Q_P = Q_0 + r_I Q_0 + r_R Q_0, \quad (6.1)$$

$$Q_C = Q_P - r_I Q_0, \quad (6.2)$$

$$Q_U = r_R Q_0 + w Q_0, \quad (6.3)$$

$$Q_E = Q_0 - w Q_0. \quad (6.4)$$

The second equation removes the mixed-liquor recycle before clarification. The third includes both underflow destinations. Dividing each hydraulic balance by Q_0 gives

$$q_P = 1 + r_I + r_R, \quad q_C = 1 + r_R, \quad q_U = r_R + w, \quad q_E = 1 - w. \quad (6.5)$$

Here q_P is the reactor-train flow ratio and q_C is the clarifier-feed flow ratio. The identity $q_C = q_E + q_U$ confirms the clarifier water balance. The adopted domain requires $r_I \geq 0$, $r_R \geq 0$, $0 \leq w < 1$, and $q_U > 0$. The outlet concentrations are then defined by positive outlet flows.

Figure 6.1 places these flow balances on the physical connections of the five-reactor plant. The two recycle routes return material to the mixer, while treated water and waste sludge cross the external boundary.

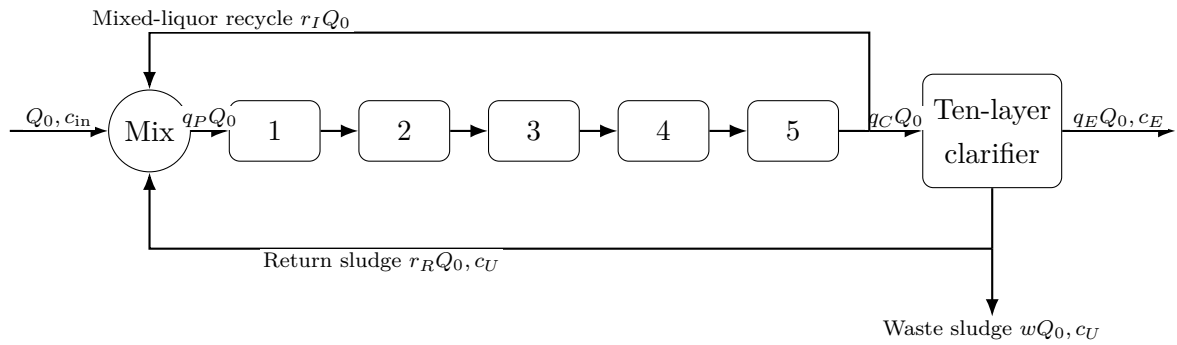


Figure 6.1: Connected plant configuration used for the operating study. Reactors 1 and 2 are unaerated. Reactors 3–5 have independent aeration controls.

The diagram shows why the reactor-train and clarifier-feed flows differ. Mixed-liquor recycle leaves before the clarifier, while return sludge leaves the clarifier underflow. Their concentrations consequently come from different locations. The mixer calculation uses each stream's own flow and concentration rather than assigning one shared recycle concentration.

The mixer balance is a component balance rather than a balance of a single water-quality total. For component f , its incoming and outgoing mass rates satisfy

$$Q_P m_f = Q_0 c_{in,f} + r_I Q_0 c_{n_r,f} + r_R Q_0 c_{U,f}. \quad (6.6)$$

Each product has units of g d^{-1} when the component basis is g m^{-3} . Divide by Q_0 and then by q_P to obtain

$$m_f = \frac{c_{in,f} + r_I c_{n_r,f} + r_R c_{U,f}}{1 + r_I + r_R}. \quad (6.7)$$

For two components, the same calculation is performed independently in each row,

$$\begin{bmatrix} q_P & 0 \\ 0 & q_P \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} = \begin{bmatrix} c_{in,1} + r_I c_{n_r,1} + r_R c_{U,1} \\ c_{in,2} + r_I c_{n_r,2} + r_R c_{U,2} \end{bmatrix}. \quad (6.8)$$

The complete concentration vector therefore satisfies

$$q_P m = c_{\text{in}} + r_I c_{n_r} + r_R c_U. \quad (6.9)$$

A high mixer concentration can therefore result from internal circulation. It does not imply that an external pollutant load has been created. The component-resolved framework of X. Zhang et al. (2026) treats these recirculation and discharge streams explicitly.

A hypothetical mixing calculation makes the units and recycle effect explicit. Let $Q_0 = 10,000 \text{ m}^3 \text{ d}^{-1}$, $r_I = 2$, $r_R = 0.5$, and $w = 0.02$. The four physical flows are 35,000, 15,000, 5,200, and 9,800 $\text{m}^3 \text{ d}^{-1}$ in the order Q_P, Q_C, Q_U, Q_E . For one particulate component, specify fresh, final-reactor, and underflow concentrations of 200, 1,000, and 2,000 g m^{-3} . The mixer calculation is

$$q_P m_f = 200 + 2(1,000) + 0.5(2,000) = 3,200, \quad (6.10)$$

$$m_f = 3,200/3.5 = 914.29 \text{ g m}^{-3}. \quad (6.11)$$

The physical incoming mass rates are 2,000, 20,000, and 10,000 kg d^{-1} . Their sum is the 32,000 kg d^{-1} leaving the mixer. The mixer concentration exceeds the fresh concentration because the two return streams contain concentrated material. These specified streams illustrate mixing and do not constitute a complete calculated plant state.

The nominal stage HRT is measured on the fresh-flow basis. A volume divided by a flow gives time. Thus V_i/Q_0 has units of days, and multiplying by 24 converts it to hours. The actual flow through the reactor is $Q_P = Q_0 q_P$, so its actual residence time is V_i/Q_P . The inverse of that actual time is the dilution rate. With reactor volume V_i ,

$$h_i = \frac{24V_i}{Q_0}, \quad H = \sum_{i=1}^{n_r} h_i, \quad d_i = \frac{Q_0 q_P}{V_i} = \frac{24q_P}{h_i}. \quad (6.12)$$

The factors h_i and H are in hours. The dilution rate d_i is in d^{-1} . The actual residence time during one passage through stage i is h_i/q_P . For a hypothetical plant with $H = 24$ h, $r_I = 2$, and $r_R = 0.5$, the reactor-train flow ratio is 3.5. Its train residence time per passage is about 6.86 h. Equal tanks then have a residence time per passage of about 1.37 h. These illustrative numbers explain why nominal HRT and recycle flow must be interpreted together.

Fixed stage fractions can be represented by $h_i = f_i^V H$, where $f_i^V > 0$ and $\sum_i f_i^V = 1$. The case study uses equal fractions. Varying H at fixed Q_0 varies biological reactor volume. The resulting problem combines capacity selection and operating decisions. For an existing plant with installed tanks, the volumes would be fixed and H would be removed from the free-control vector.

Let ϑ collect only settings that vary freely. A constant setting remains a parameter. Treating it as a varying regression coordinate would create a zero-variance feature. A

carbon or chemical addition may also enter the control vector. An addition represented only by a source term has negligible liquid carrier flow. An appreciable carrier flow requires a new hydraulic stream and revised downstream dilution rates.

6.2.2 Biological conversion and non-negativity

The reaction network uses the component and process representation of Chapter 3. Its stoichiometric matrix is $\nu \in \mathbb{R}^{P \times F}$ and its process-rate vector in stage i is $\rho_i \in \mathbb{R}_+^P$. For component f , the dynamic component equation is

$$V_i \frac{dc_{i,f}}{dt} = Q_P c_{i-1,f} - Q_P c_{i,f} + V_i \sum_{p=1}^P \nu_{pf} \rho_{i,p} + V_i \omega_i(e_{S_O})_f + V_i \xi_{i,f}. \quad (6.13)$$

The reaction sum combines the change caused by each process. A negative ν_{pf} consumes component f , while a positive entry produces it. The oxygen selector is one for the oxygen component and zero elsewhere. At steady state, the accumulation term is zero. Dividing the remaining equation by V_i gives

$$0 = d_i(c_{i-1,f} - c_{i,f}) + \sum_{p=1}^P \nu_{pf} \rho_{i,p} + \omega_i(e_{S_O})_f + \xi_{i,f}. \quad (6.14)$$

Putting all F component equations into one vector gives the steady reactor equation, with $c_0 = m$,

$$0 = d_i(c_{i-1} - c_i) + \nu^\top \rho_i(c_i, \vartheta) + k_L a_i(a_i)(S_O^* - c_{i,S_O})e_{S_O} + \xi_i(\vartheta). \quad (6.15)$$

The vector e_{S_O} selects dissolved oxygen. The parameter S_O^* is its saturation concentration. The vector ξ_i represents declared stage additions. Each term has units of component concentration per day. The treatment-model framework of Henze et al. (2015) separates hydraulic transport, stoichiometric conversion, and external sources in this way.

At a zero component concentration, the combined reaction and external-source terms must not direct that component below zero. With non-negative inflow and a locally Lipschitz balance function defined on the boundary, this condition preserves the non-negative state space of the dynamic model (Haddad et al., 2010). A numerical integrator does not inherit this property automatically. Acceptance therefore checks concentrations and reaction rates independently of the solver's stopping message.

The same distinction matters for a surrogate. A fitted quadratic equation can predict a negative concentration even when every training state is non-negative. Polynomial approximation imposes no sign restriction. The work of Kircher and Votsmeier (2025) addresses the simultaneous requirements of mass conservation and non-negativity in kinetic surrogates. Both requirements need explicit treatment.

6.2.3 Stage invariants and the external plant balance

Consider a hypothetical reaction that transfers material from component 1 to component 2 without changing their combined amount. Its stoichiometric and inventory rows are

$$\nu = \begin{bmatrix} -1 & 1 \end{bmatrix}, \quad A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}. \quad (6.16)$$

The factor $1/\sqrt{2}$ gives the inventory row unit length. Matrix multiplication shows the cancellation directly,

$$A\nu^\top = \frac{1}{\sqrt{2}} [1(-1) + 1(1)] = 0. \quad (6.17)$$

At an illustrative dilution rate of 2 d^{-1} , take upstream concentrations $(12, 3)$ and downstream concentrations $(8, 7) \text{ g m}^{-3}$, with a reaction rate of $8 \text{ g m}^{-3} \text{ d}^{-1}$. The component equations are

$$0 = 2(12 - 8) - 8, \quad (6.18)$$

$$0 = 2(3 - 7) + 8. \quad (6.19)$$

Adding them removes the unknown reaction rate. Both upstream and downstream totals are 15 g m^{-3} . The normalized invariant is that total divided by $\sqrt{2}$. The full reactor model applies the same cancellation to more components and more processes.

Let $A \in \mathbb{R}^{K \times F}$ contain independent inventory rows such that

$$A\nu^\top = 0, \quad Ae_{S_O} = 0, \quad \text{rank}(A) = K. \quad (6.20)$$

The rows are normalized to unit Euclidean length. Unit row normalization does not make them orthogonal. Physically named inventories and an orthonormal basis can span the same subspace defined by mass conservation while using different coordinates. The stoichiometric fixed-layer formulation of Sturm and Wexler (2022) illustrates the value of separating quantities used for mass conservation from the rates that redistribute them.

For inventory row k , multiply component equation f by A_{kf} and sum over f . The reaction contribution becomes

$$\sum_{f=1}^F A_{kf} \sum_{p=1}^P \nu_{pf} \rho_{i,p} = \sum_{p=1}^P \left(\sum_{f=1}^F A_{kf} \nu_{pf} \right) \rho_{i,p} = 0. \quad (6.21)$$

The inner sum is entry (k, p) of $A\nu^\top$. Oxygen transfer also disappears because $Ae_{S_O} = 0$. The remaining scalar equation is

$$d_i \left(\sum_f A_{kf} c_{i-1,f} - \sum_f A_{kf} c_{i,f} \right) + \sum_f A_{kf} \xi_{i,f} = 0. \quad (6.22)$$

Moving the transport difference to the other side and dividing by the positive d_i gives the remaining stage relation,

$$A(c_i - c_{i-1}) = d_i^{-1} A\xi_i(\vartheta). \quad (6.23)$$

If there is no stage addition, $Ac_i = Ac_{i-1}$. For example, nitrification can transfer nitrogen from ammonium to oxidized forms while denitrification transfers it to dissolved nitrogen gas. These changes do not remove nitrogen from the model's complete nitrogen inventory. A reporting quantity that excludes dissolved nitrogen gas can decrease while the inventory used for mass conservation remains unchanged. Nutrient-form interpretation in U.S. Environmental Protection Agency (2009) supports the need to inspect individual species as well as reported totals.

In the two-component example, adding component 1 at $2 \text{ g m}^{-3} \text{ d}^{-1}$ changes the downstream concentrations to (9, 7) when the same reaction rate is used. The two balances are $2(12 - 9) - 8 + 2 = 0$ and $2(3 - 7) + 8 = 0$. Their total changes by $2/2 = 1 \text{ g m}^{-3}$. Thus the invariant transition must include a dose whenever that dose carries the quantity used for mass conservation. A zero invariant difference would incorrectly remove the added material.

Internal recycle streams also cancel from an external invariant balance. First add the stage transitions. Every intermediate Ac_i appears once with a plus sign and once with a minus sign. This gives

$$A(c_{n_r} - m) = \sum_i \frac{V_i}{Q_0 q_P} A\xi_i, \quad (6.24)$$

$$q_P Ac_{n_r} - q_P Am = D_{\text{add}}, \quad D_{\text{add}} = \sum_i \frac{V_i}{Q_0} A\xi_i. \quad (6.25)$$

Next multiply the mixer balance by A and substitute it for $q_P Am$,

$$q_P Ac_{n_r} = Ac_{\text{in}} + r_I Ac_{n_r} + r_R Ac_U + D_{\text{add}}, \quad (6.26)$$

$$(q_P - r_I) Ac_{n_r} = Ac_{\text{in}} + r_R Ac_U + D_{\text{add}}. \quad (6.27)$$

Since $q_P - r_I = q_C$, the left-hand side is the clarifier-feed inventory flow. The clarifier balance gives a second expression for that same quantity,

$$q_C Ac_{n_r} = q_E Ac_E + (r_R + w) Ac_U. \quad (6.28)$$

Equating the expressions and subtracting $r_R Ac_U$ cancels the return-sludge flow. The mixed-liquor recycle already canceled in $q_P - r_I$. The remaining external relation is

$$Ac_{\text{in}} + \sum_{i=1}^{n_r} \frac{V_i}{Q_0} A\xi_i = q_E Ac_E + w Ac_U. \quad (6.29)$$

Only fresh feed, stage additions, effluent, and waste sludge remain. The recycle ratios still influence the state and treatment processes. Their cancellation means that they are internal transfers rather than external sources or sinks.

For ASM2d-TSN, $F = 20$, $P = 28$, and $K = 5$. The five inventories concern soluble inert material, total nitrogen including dissolved nitrogen gas, total phosphorus, an alkalinity-related combination, and a metal-related combination. The last two

unnormalized combinations are

$$I_{\text{alk}} = S_{\text{ALK}} - \frac{S_{\text{NH}_4}}{14} + \frac{S_{\text{NO}_2}}{14} + \frac{S_{\text{NO}_3}}{14} - \frac{S_{\text{PO}_4}}{31}, \quad (6.30)$$

$$I_{\text{metal}} = 3.45X_{\text{MeP}} + 4.87X_{\text{MeOH}}. \quad (6.31)$$

The nitrogen and phosphorus rows use the constituent weights in Table 3.1. The nitrogen inventory also gives dissolved nitrogen gas unit weight. The stoichiometric rank is 15 under the stated parameterization. The inventory identities are audited rather than inferred from a reporting label such as TN.

6.3 Clarification, component separation, and stored solids

6.3.1 A nonreactive separator with two outlet flows

The clarifier uses a nonreactive settling representation. Soluble components follow the liquid. Particulate components can settle and concentrate in the underflow. Takács et al. (1991) provide the settling and thickening formulation used for the aggregate solids profile. Jeppsson and Diehl (1996) explain why component propagation through a clarifier requires attention beyond a single aggregate solids concentration.

Selection means taking designated entries of a vector without changing them. For the illustrative order $(S_A, S_{\text{NH}_4}, X_H, X_I)$, the two selector matrices are

$$E_S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad E_X = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (6.32)$$

Thus $E_{SC} = (S_A, S_{\text{NH}_4})^\top$ and $E_X c = (X_H, X_I)^\top$. Transposing a selector puts selected entries back into their original positions and fills the other positions with zeros. In this example,

$$E_S^\top E_S = \text{diag}(1, 1, 0, 0), \quad E_X^\top E_X = \text{diag}(0, 0, 1, 1). \quad (6.33)$$

Their sum is the identity matrix because every component belongs to exactly one group. Their cross-product is zero because the groups do not overlap. For the complete ordered model, $E_S = [I_{10} \ 0]$ and $E_X = [0 \ I_{10}]$. In general, let $E_S \in \mathbb{R}^{F_S \times F}$ and $E_X \in \mathbb{R}^{F_X \times F}$ select soluble and particulate components. They satisfy

$$E_S^\top E_S + E_X^\top E_X = I_F, \quad E_S E_X^\top = 0. \quad (6.34)$$

The column t_X maps component concentrations to TSS. Its entries are non-negative and its soluble entries are zero.

For the four-component order above, $t_X = (0, 0, 0.90, 0.75)^\top$. If $X_H = 100$ and $X_I = 40$ g COD m⁻³, the TSS calculation is $0.90(100) + 0.75(40) = 120$ g TSS m⁻³. The two factors convert component COD bases to suspended-solids mass. Adding the

native numerical concentrations without these conversion factors would give a different quantity.

For one component, mass conservation across the clarifier gives

$$Q_C c_{nr,f} = Q_E c_{E,f} + Q_U c_{U,f}. \quad (6.35)$$

The statistical response uses component flows divided by Q_0 ,

$$g_E = q_E c_E, \quad g_U = q_U c_U. \quad (6.36)$$

Their units are the same as concentration because the flow ratios are dimensionless. The actual component discharge rates are $Q_0 g_E$ and $Q_0 g_U$. Writing balances in these coordinates avoids dividing by a small underflow ratio inside the equality system. The component balance and soluble pass-through are

$$g_E + g_U = q_C c_{nr}, \quad E_S g_U = q_U E_S c_{nr}. \quad (6.37)$$

The two relations imply $E_S g_E = q_E E_S c_{nr}$. Settling therefore cannot remove a soluble nutrient directly in this model.

For a soluble component f , pass-through fixes $c_{U,f} = c_{nr,f}$. Substitute $g_{U,f} = q_U c_{nr,f}$ into the total balance to obtain

$$g_{E,f} = (q_C - q_U) c_{nr,f} = q_E c_{nr,f}, \quad (6.38)$$

$$c_{E,f} = g_{E,f} / q_E = c_{nr,f}. \quad (6.39)$$

For a hypothetical feed concentration of 20 g m^{-3} and flow ratios $q_C = 1.5$, $q_U = 0.52$, and $q_E = 0.98$, the feed, underflow, and overflow component flows are 30, 10.4, and 19.6 g m^{-3} on the fresh-flow basis. Both outlet concentrations remain 20 g m^{-3} .

The particulate densification requirement is

$$E_X g_U \geq q_U E_X c_{nr}. \quad (6.40)$$

After division by q_U , each underflow particulate concentration is at least its feed concentration. Non-negative overflow and the total component balance also prevent the underflow from recovering more than the incoming particulate flow.

For an illustrative particulate component, take $q_C = 1.5$, $q_U = 0.52$, and feed concentration 100 g m^{-3} . Densification requires $g_U \geq 52 \text{ g m}^{-3}$ on the fresh-flow basis. Mass conservation and non-negative overflow require $g_U \leq 150 \text{ g m}^{-3}$. Thus the underflow recovery fraction lies between $0.52/1.5$ and one. A soluble component with the same feed concentration instead has $g_U = 52$ and unchanged soluble outlet concentrations. These examples concern the constraints alone and do not predict a settling-model outcome.

An admissible illustrative choice within that particulate interval is $g_U = 140$. Mass conservation then gives $g_E = 150 - 140 = 10$. Dividing by the respective flow ratios

gives $c_U = 140/0.52 = 269.23$ and $c_E = 10/0.98 = 10.20 \text{ g m}^{-3}$. The underflow has recovered $140/150 = 93.33\%$ of the incoming component flow. The concentrations alone do not give that recovery fraction because the outlet water flows differ.

For any particulate component with positive feed concentration, the same reasoning gives

$$\frac{q_U}{q_C} \leq \frac{g_{U,f}}{q_C c_{n_r,f}} \leq 1. \quad (6.41)$$

Multiplication by the non-negative TSS weights preserves inequalities. Hence $X_U \geq X_N$. The solids balance $q_C X_N = q_E X_E + q_U X_U$ then gives

$$q_E X_E \leq (q_C - q_U) X_N = q_E X_N. \quad (6.42)$$

Division by $q_E > 0$ establishes the other endpoint ordering.

Applying t_X gives feed and outlet solids concentrations

$$X_N = t_X^\top c_{n_r}, \quad X_E = \frac{t_X^\top g_E}{q_E}, \quad X_U = \frac{t_X^\top g_U}{q_U}. \quad (6.43)$$

The declared constraints imply $X_E \leq X_N \leq X_U$. They bound separation but do not determine its magnitude. The nonlinear settling calculation and an empirical endpoint model supply additional information through different routes.

6.3.2 The layer-resolved mechanistic model

Clarifier layers are numbered from the surface to the hopper. Their TSS concentrations are $s_1, \dots, s_L$, with $s_1 = X_E$ and $s_L = X_U$. The exact settling velocity at concentration s is

$$v(s, X_N) = \max \left\{ 0, \min \left[v_{\max}, v_0 \left(e^{-r_h(s - f_{\text{ns}} X_N)} - e^{-r_p(s - f_{\text{ns}} X_N)} \right) \right] \right\}. \quad (6.44)$$

The gravitational solids flux is $sv(s, X_N)$. The nonsettleable fraction f_{ns} makes the settling offset depend on the feed concentration. At an interface, a receiver concentration above X_t activates a limitation by the smaller adjacent gravitational flux. Otherwise, the upper-layer gravitational flux passes to the receiver. Hydraulic transport uses the overflow velocity above the feed layer and the underflow velocity below it.

Each finite-volume layer equation is its incoming solids flux minus its outgoing solids flux, multiplied by $A_{\text{cl}}/V_{\text{cl},\ell}$. The feed layer also receives $Q_0 q_C X_N / V_{\text{cl},\ell_F}$. Internal interface fluxes cancel when these equations are summed. At steady state, the resulting total solids balance agrees with Equation (6.37) after application of t_X .

A three-layer illustration shows that cancellation without hiding it inside a summation. Place the feed in layer 2. Let B_{12} be the net solids rate from layer 2 toward layer 1 and B_{23} the net rate from layer 2 toward layer 3. These signed rates combine

hydraulic and gravitational transport in g d^{-1} . The layer mass equations are

$$V_1 \dot{s}_1 = B_{12} - Q_E s_1, \quad (6.45)$$

$$V_2 \dot{s}_2 = Q_C X_N - B_{12} - B_{23}, \quad (6.46)$$

$$V_3 \dot{s}_3 = B_{23} - Q_U s_3. \quad (6.47)$$

Adding them removes both interface rates. At steady state, the remaining equation is $Q_C X_N = Q_E X_E + Q_U X_U$. This cancellation holds even though the interface rates depend nonlinearly on the neighboring concentrations. Mass conservation of the whole clarifier therefore does not determine its internal fluxes or profile.

The mechanistic state retains one solids concentration per layer. It does not retain a separate transient concentration for every particulate species in every layer. At a steady state, particulate outlet components are reconstructed with the final-reactor particulate composition,

$$E_X c_E = \frac{s_1}{X_N} E_X c_{nr}, \quad E_X c_U = \frac{s_L}{X_N} E_X c_{nr}. \quad (6.48)$$

Soluble outlet components equal their feed concentrations. This reduced representation defines the physical scope of the mechanistic reference. Its steady component mapping is well defined when $X_N > 0$. It does not establish the dynamic propagation of each particulate species considered in more detailed component-transport models (Jeppsson & Diehl, 1996).

6.3.3 An exact interval for aggregate inventory under the endpoint envelope

Inventory is concentration multiplied by volume, summed over the vessel. For an illustrative four-layer clarifier, it is

$$M = V_1 X_E + V_2 s_2 + V_3 s_3 + V_4 X_U. \quad (6.49)$$

For example, four 100-m^3 layers with concentrations $(10, 200, 600, 1,000)$ g m^{-3} store $100(10 + 200 + 600 + 1,000) = 181,000$ g, or 181 kg. The general clarifier's stored solids mass is

$$M_{\text{cl}} = \sum_{\ell=1}^L V_{\text{cl},\ell} s_{\ell}, \quad V_{\text{cl}} = \sum_{\ell=1}^L V_{\text{cl},\ell}. \quad (6.50)$$

An outlet concentration cannot determine this mass. Two clarifiers can have the same endpoints and different internal storage. Whole-plant solids retention time therefore requires the inventory as well as the two outlet solids flows. The benchmark formulation of Alex et al. (2008) includes reactor and settler inventories in its sludge-age calculation.

The mechanistic acceptance envelope requires

$$X_E \leq s_{\ell} \leq X_U, \quad \ell = 2, \dots, L-1. \quad (6.51)$$

This is an endpoint envelope. It does not require monotonic concentration from one layer to the next. With positive layer volumes, minimizing the inventory assigns every internal layer X_E . Maximizing it assigns every internal layer X_U .

For four layers, the envelope bounds each of the two unknown terms separately,

$$V_2X_E + V_3X_E \leq V_2s_2 + V_3s_3 \quad (6.52)$$

$$\leq V_2X_U + V_3X_U. \quad (6.53)$$

Adding the known endpoint terms gives

$$(V_1 + V_2 + V_3)X_E + V_4X_U \leq M, \quad (6.54)$$

$$M \leq V_1X_E + (V_2 + V_3 + V_4)X_U. \quad (6.55)$$

For the hypothetical four equal layers above, these values are 103,000 and 301,000 g. The 181,000-g inventory lies inside the interval. If both internal concentrations were instead 400 g m^{-3} , the mass would still be 181,000 g. Thus matching endpoints and total inventory does not identify the internal distribution.

Figure 6.2 shows the same four-layer arithmetic as concentration bars. Changing the two internal layers changes stored mass while the surface and hopper concentrations remain fixed. The left and right panels attain the lower and upper interval endpoints derived from the envelope.

The internal layers in the middle panel need not resemble those of another profile with the same inventory. Their total mass is the retained scalar information. The nonlinear layer equations supply the additional information needed to determine a mechanistic profile.

With L layers, the identical argument replaces $V_2 + V_3$ by $\sum_{\ell=2}^{L-1} V_{\text{cl},\ell}$. That internal-volume sum is $V_{\text{cl}} - V_{\text{cl},1} - V_{\text{cl},L}$. Collecting coefficients of each endpoint gives the general interval,

$$M_{\text{cl}}^- = (V_{\text{cl}} - V_{\text{cl},L})X_E + V_{\text{cl},L}X_U, \quad (6.56)$$

$$M_{\text{cl}}^+ = V_{\text{cl},1}X_E + (V_{\text{cl}} - V_{\text{cl},1})X_U, \quad (6.57)$$

$$M_{\text{cl}}^- \leq M_{\text{cl}} \leq M_{\text{cl}}^+. \quad (6.58)$$

These bounds are necessary and sufficient for a profile with the prescribed endpoints and envelope to have the stated inventory. To see sufficiency, give every internal layer a common concentration s_* between X_E and X_U . The inventory is then

$$V_{\text{cl},1}X_E + V_{\text{cl},L}X_U + (V_{\text{cl}} - V_{\text{cl},1} - V_{\text{cl},L})s_*. \quad (6.59)$$

This expression moves continuously through the full interval as s_* varies. For equal-volume layers, it also gives a simple constructive profile for any admitted inventory.

As an illustrative calculation, take ten layers of 600 m^3 , $X_E = 10 \text{ g m}^{-3}$, and $X_U = 5,000 \text{ g m}^{-3}$. The allowable inventory interval is 3,054,000–27,006,000 g. The

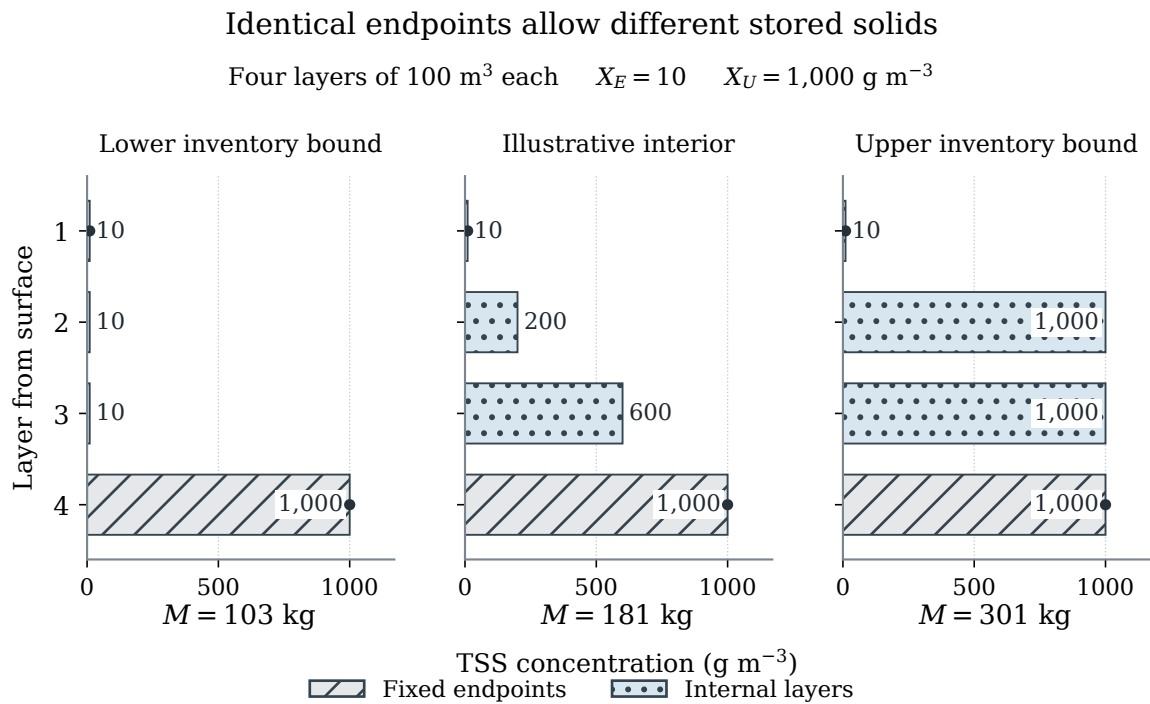


Figure 6.2: Hypothetical four-layer profiles with identical overflow and underflow concentrations. Each layer has volume 100 m^3 . The lower-bound, illustrative interior, and upper-bound profiles store 103, 181, and 301 kg of solids. Hatching identifies the fixed endpoints, while dotted fills identify internal layers. The diagram represents the adopted endpoint envelope and does not assert satisfaction of nonlinear settling equations.

broad interval reflects how little the endpoints alone say about internal storage. An inventory inside it has an envelope-compatible profile. It may still have no profile that satisfies the nonlinear settling equations. Joint projection enforces the interval, while the layer-resolved mechanistic evaluation checks the flux equations.

For these ten equal layers, the calculation can be written out as

$$M_{\text{cl}} = 600(X_E + s_2 + s_3 + \cdots + s_9 + X_U), \quad (6.60)$$

$$M_{\text{cl}}^- = 600(9X_E + X_U), \quad (6.61)$$

$$M_{\text{cl}}^+ = 600(X_E + 9X_U). \quad (6.62)$$

The lower endpoint has nine layers at X_E and one at X_U . The upper endpoint has one layer at X_E and nine at X_U . This explains both coefficients in the compact interval rather than treating them as fitted constants.

6.4 Statistical prediction of the connected response

6.4.1 A response that preserves network and storage information

For a teaching example with one reactor and two components, denote a soluble component by S and a particulate component by X . The stored response is the explicit nine-entry column

$$\chi_{\text{toy}} = (m_S, m_X, c_{1,S}, c_{1,X}, g_{E,S}, g_{E,X}, g_{U,S}, g_{U,X}, M_{\text{cl}})^\top. \quad (6.63)$$

Adding a second reactor inserts $c_{2,S}$ and $c_{2,X}$ before the outlet coordinates and increases the length to eleven. Every reactor contributes F entries. The mixer and two outlets each contribute another F , while the clarifier inventory contributes one. The connected surrogate therefore predicts

$$\chi = \text{col}(m, c_1, \dots, c_{n_r}, g_E, g_U, M_{\text{cl}}) \in \mathbb{R}^{n_\chi}, \quad n_\chi = (n_r + 3)F + 1. \quad (6.64)$$

Here col stacks its arguments into one column. The response preserves concentrations needed for stage inventories, outlet flows needed for material balances, and clarifier storage needed for solids retention time. It omits the internal layer profile. The full mechanistic state is

$$y = \text{col}(c_1, \dots, c_{n_r}, s_1, \dots, s_L) \in \mathbb{R}_+^{n_y}, \quad n_y = n_r F + L. \quad (6.65)$$

The mapping from y to χ reconstructs the mixer, both outlets, and inventory deterministically. This mapping lets two different response dimensions support the same engineering calculations.

For the five-reactor case, the response count is $20 + 5(20) + 20 + 20 + 1 = 161$. The full-state count is $5(20) + 10 = 110$. Table 6.1 makes the response ordering explicit. A position identifies a coordinate and does not imply that neighboring entries have the same physical unit.

Table 6.1: Blocks in the 161-coordinate connected response.

Block	Positions	Coordinates
Mixer	1–20	$m_1, \dots, m_{20}$
Reactor 1	21–40	$c_{1,1}, \dots, c_{1,20}$
Reactor 2	41–60	$c_{2,1}, \dots, c_{2,20}$
Reactor 3	61–80	$c_{3,1}, \dots, c_{3,20}$
Reactor 4	81–100	$c_{4,1}, \dots, c_{4,20}$
Reactor 5	101–120	$c_{5,1}, \dots, c_{5,20}$
Overflow flows	121–140	$g_{E,1}, \dots, g_{E,20}$
Underflow flows	141–160	$g_{U,1}, \dots, g_{U,20}$
Clarifier inventory	161	M_{cl}

6.4.2 Complete second-order features and ridge estimation

The statistical input includes every free control and every fresh-influent component. Standardization first subtracts a fitting-set mean and then divides by a fitting-set standard deviation. For one operating coordinate and one influent coordinate,

$$z_1 = \frac{\vartheta_1 - \bar{\vartheta}_1}{s_{\vartheta_1}}, \quad z_2 = \frac{c_{in,1} - \bar{c}_{in,1}}{s_{in,1}}. \quad (6.66)$$

These are dimensionless coordinates. For hypothetical aeration 0.8, fitting mean 0.5, and fitting scale 0.1, the standardized value is 3. For hypothetical ammonium concentration 40 g m^{-3} with mean 30 and scale 10 g m^{-3} , the value is 1. The different native units no longer determine the regression penalty. For fitting-set centers and positive diagonal scales, define

$$z = \text{col} \left[D_{\vartheta, \text{fit}}^{-1}(\vartheta - \bar{\vartheta}), D_{in, \text{fit}}^{-1}(c_{in} - \bar{c}_{in}) \right] \in \mathbb{R}^{d_z}, \quad d_z = d_\vartheta + F. \quad (6.67)$$

For two standardized inputs, their symmetric product matrix is

$$zz^\top = \begin{bmatrix} z_1^2 & z_1 z_2 \\ z_2 z_1 & z_2^2 \end{bmatrix}. \quad (6.68)$$

Because multiplication commutes, $z_1 z_2$ and $z_2 z_1$ are the same feature. One triangular copy therefore gives the unique quadratic entries $(z_1^2, z_1 z_2, z_2^2)$. Including the linear entries and intercept produces six terms,

$$(1, z_1, z_2, z_1^2, z_1 z_2, z_2^2)^\top. \quad (6.69)$$

Three inputs would give an intercept, three linear terms, three squares, and three distinct cross-products, for ten features. In general there are d_z squares and $d_z(d_z - 1)/2$ cross-products. Their sum is $d_z(d_z + 1)/2$. The complete quadratic feature family is

$$r(z) = \text{col} \left[z, \text{vech}(zz^\top) \right], \quad (6.70)$$

$$\phi(\vartheta, c_{\text{in}}) = \text{col} \left[1, D_{r, \text{fit}}^{-1}(r(z) - \bar{r}) \right], \quad (6.71)$$

$$n_\phi = 1 + d_z + \frac{d_z(d_z + 1)}{2}. \quad (6.72)$$

The operator vech retains one triangular copy of the symmetric matrix. It includes every square and every pairwise product once. A product between ammonium loading and aeration allows their joint effect to differ from the sum of their separate effects. A square of the recycle coordinate permits local curvature. These terms are statistical approximations. They are not a replacement for kinetic rate laws.

Quadratic features undergo their own fitting-set centering and scaling after they are constructed. For the hypothetical standardized inputs $(3, 1)$, the unscaled nonconstant terms are $(3, 1, 9, 3, 1)$. If the first squared feature has fitting mean 1 and standard deviation 2, its final value is $(9 - 1)/2 = 4$. If the cross-product has fitting mean 0.2 and standard deviation 0.5, its final value is $(3 - 0.2)/0.5 = 5.6$. This second standardization prevents squares with large variance from receiving a weaker effective ridge penalty merely because they are numerically larger.

The unique-monomial representation spans the same quadratic functions as an ordered representation that retains both $z_1 z_2$ and $z_2 z_1$. Their two coefficients enter prediction only through their sum. For an effective cross-product coefficient b , an ordered representation can assign $b/2$ to each duplicate. The squared coefficient penalty is then $b^2/2$, compared with b^2 for one unique coefficient. Thus removing duplicates preserves the polynomial family but changes the regularization convention unless the penalty is adjusted. The connected fit selects its penalty for the stated unique-feature representation. For 27 inputs, the count is $1 + 27 + 378 = 406$.

For response coordinate k , write its standardized target as $\tilde{\chi}_{ik} = (\chi_{ik} - \bar{\chi}_k)/s_{\chi,k}$. Its fitted value is $C_{k0} + \sum_{j=1}^{n_\phi-1} C_{kj} \phi_{ij}$. The scalar ridge objective for all responses is

$$\frac{1}{nn_\chi} \sum_{i=1}^n \sum_{k=1}^{n_\chi} \left(\tilde{\chi}_{ik} - \sum_{j=0}^{n_\phi-1} C_{kj} \phi_{ij} \right)^2 + \frac{\gamma}{n_\chi} \sum_{k=1}^{n_\chi} \sum_{j=1}^{n_\phi-1} C_{kj}^2. \quad (6.73)$$

The index $j = 0$ is the intercept, with feature value one. It appears in prediction and does not appear in the penalty sum. Dividing a physical prediction error by $s_{\chi,k}$ is equivalent to weighting its squared error by $1/s_{\chi,k}^2$. Response scaling therefore gives each coordinate comparable importance in the joint statistical score.

For example, an illustrative concentration error of 10 g m^{-3} with a response scale of 20 g m^{-3} contributes $0.5^2 = 0.25$ before averaging. An inventory error of $10,000 \text{ g}$ with a scale of $200,000 \text{ g}$ contributes $0.05^2 = 0.0025$. Comparing the unscaled numbers would give the opposite impression. The scales define what counts as a large error.

Let $\Phi_{\mathcal{I}}$ contain the feature rows for fitting indices $\mathcal{I}$ and let $\tilde{Y}_{\mathcal{I}}$ contain standardized joint responses. With $R_{\phi} = \text{diag}(0, 1, \dots, 1)$, the coefficient matrix solves

$$\hat{C}_{\gamma} = \arg \min_C \left\{ \frac{\|\tilde{Y}_{\mathcal{I}} - \Phi_{\mathcal{I}} C^{\top}\|_F^2}{|\mathcal{I}|n_{\chi}} + \frac{\gamma}{n_{\chi}} \|C R_{\phi}\|_F^2 \right\}. \quad (6.74)$$

The intercept is unpenalized. A positive γ regularizes correlated quadratic features. Hastie et al. (2009) explain this stabilization and the corresponding trade-off between model flexibility and coefficient shrinkage. The raw prediction is

$$\chi_{\text{raw}} = \bar{\chi} + D_{\chi} \hat{C}_{\gamma} \phi(\vartheta, c_{\text{in}}). \quad (6.75)$$

The response scale D_{χ} is fixed from fitting data. It places the inventory, component concentrations, and outlet flows on comparable statistical scales. The raw prediction still has no enforced physical constraints.

Differentiating the scalar ridge objective for one response column with coefficient vector β gives its normal equations,

$$(\Phi^{\top} \Phi + n\gamma R_{\phi}) \hat{\beta} = \Phi^{\top} \tilde{\chi}. \quad (6.76)$$

The term $n\gamma R_{\phi}$ adds positive curvature to the nonintercept coefficient directions. In a hypothetical one-feature teaching problem, take three feature and target values both equal to $(-\sqrt{3/2}, 0, \sqrt{3/2})$. Their means are zero and their population standard deviations are one. Symmetry gives a zero intercept. With slope b , the objective becomes $(1-b)^2 + \gamma b^2$. Differentiation gives $2(b-1) + 2\gamma b = 0$, so $\hat{b} = 1/(1+\gamma)$. For $\gamma = 1$, the fitted slope is 0.5 rather than the unpenalized value of one. This simple calculation shows coefficient shrinkage without claiming that a plant response follows one feature.

Finally, the physical prediction is recovered coordinate by coordinate as $\chi_{\text{raw},k} = \bar{\chi}_k + s_{\chi,k} \sum_j \hat{C}_{kj} \phi_j$. A hypothetical mean of 100 g m^{-3} , scale of 20 g m^{-3} , and standardized prediction of 0.5 give 110 g m^{-3} . This return to physical units is required before imposing component balances.

The quadratic feature family is retained from ICSOR (E. F. Selerio Jr., 2026). Its coefficients describe the raw statistical relation. If projection is active, a raw coefficient alone does not give the derivative of the final projected response. That derivative also depends on the constraints, their control dependence, and the active inequalities.

6.4.3 A separate positive model for overflow solids

Low overflow TSS calls for a different error scale from large reactor solids concentrations. An absolute error of 1 g m^{-3} is 1% of 100 g m^{-3} and 100% of 1 g m^{-3} . A joint standardized response score may not adequately describe this distinction at the discharge point. The relative-error argument of Tofallis (2015) motivates a separate logarithmic model.

For positive mechanistic targets, define

$$\ell_E = \log(X_E/X_\star), \quad X_\star = 1 \text{ mg L}^{-1}. \quad (6.77)$$

The ratio inside the logarithm is dimensionless. The scalar ridge fit uses the same complete quadratic features,

$$\hat{\beta}_{E,\gamma_E} = \arg \min_{\beta} \left\{ \frac{\|\ell_{E,\mathcal{I}} - \Phi_{\mathcal{I}}\beta\|_2^2}{|\mathcal{I}|} + \gamma_E \|R_{\phi}\beta\|_2^2 \right\}. \quad (6.78)$$

Its concentration-scale prediction is

$$\widehat{X}_{E,\log} = X_\star \exp(\hat{\beta}_{E,\gamma_E}^\top \phi) > 0. \quad (6.79)$$

A twofold concentration error has the same logarithmic discrepancy at low and high concentrations. Exponentiation guarantees a positive prediction. It does not guarantee an accurate concentration, a mechanistic minimum, or a effluent estimate consistent with mass conservation.

For hypothetical targets of 2 and 8 mg L⁻¹, the dimensionless logarithms are log 2 and log 8. If a single fitted log value represents both equally, squared log error is minimized at

$$\frac{\log 2 + \log 8}{2} = \log 4. \quad (6.80)$$

Back-transformation gives $1 \times \exp(\log 4) = 4 \text{ mg L}^{-1}$. The arithmetic mean is 5 mg L⁻¹, which explains why exponentiation is not an unbiased concentration-scale mean adjustment. A fitted log value of -2 instead gives 0.1353 mg L⁻¹. It remains positive even though it is below the reference concentration $X_\star$. The reference establishes units and does not impose a lower bound.

Under squared logarithmic error, direct exponentiation represents a conditional geometric center rather than an unbiased concentration-scale mean. No concentration-scale bias adjustment is applied. The separate model also avoids imposing a constant solids-removal fraction across all influents and controls. Such a fraction would disregard variation in feed composition, hydraulics, recycle, and biological state. Its fitted endpoint remains an empirical closure rather than a mass conservation law.

6.5 Joint projection onto the declared plant constraints

6.5.1 Equalities and inequalities at fixed controls

The projection adjusts every connected response block together. The required matrix rows can first be assembled for a hypothetical one-reactor, two-component plant. Use the nine-coordinate order already given for χ_{toy} . Let $r_I = 1$, $r_R = 0.5$, and $w = 0.1$,

so $q_P = 2.5$, $q_C = 1.5$, $q_U = 0.6$, and $q_E = 0.9$. These values are teaching inputs and do not define the five-reactor study domain. The soluble component is S and the particulate component X has unit TSS weight. Suppose their sum is fixed by mass conservation in the reactor and the fresh concentrations are $(30, 20) \text{ g m}^{-3}$.

The two mixer rows, one inventory row, two clarifier rows, soluble pass-through, and fitted endpoint are then

$$2.5m_S - c_{1,S} - \frac{5}{6}g_{U,S} = 30, \quad (6.81)$$

$$2.5m_X - c_{1,X} - \frac{5}{6}g_{U,X} = 20, \quad (6.82)$$

$$c_{1,S} + c_{1,X} - m_S - m_X = 0, \quad (6.83)$$

$$g_{E,S} + g_{U,S} - 1.5c_{1,S} = 0, \quad (6.84)$$

$$g_{E,X} + g_{U,X} - 1.5c_{1,X} = 0, \quad (6.85)$$

$$g_{U,S} - 0.6c_{1,S} = 0, \quad (6.86)$$

$$g_{E,X} = 0.9(10) = 9. \quad (6.87)$$

The last row illustrates a fitted overflow concentration of 10 g m^{-3} . The inventory row is written without its harmless unit-length factor to keep the numbers clear. Multiplying a zero equality by a positive factor does not change its solutions. In the same coordinate order, the explicit equality matrix is

$$\begin{bmatrix} 2.5 & 0 & -1 & 0 & 0 & 0 & -5/6 & 0 & 0 \\ 0 & 2.5 & 0 & -1 & 0 & 0 & 0 & -5/6 & 0 \\ -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1.5 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1.5 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & -0.6 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \chi_{\text{toy}} = \begin{bmatrix} 30 \\ 20 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 9 \end{bmatrix}. \quad (6.88)$$

Every row expresses one previously derived relation. A zero entry means that the coordinate is absent from that relation. The inventory coordinate has a zero column because these equalities do not determine stored solids.

The illustrative response

$$\chi_{\text{toy}} = (30, 50, 30, 50, 27, 9, 18, 66, 18,000)^\top \quad (6.89)$$

satisfies all seven rows. For example, its particulate mixer row is $2.5(50) - 50 - (5/6)(66) = 20$. Its clarifier particulate row is $9 + 66 = 1.5(50)$. Underflow concentration is $66/0.6 = 110 \text{ g m}^{-3}$. These are direct arithmetic checks of the linear constraints and do not verify reaction kinetics.

The full physical equality operator $\mathcal{H}(\vartheta)\chi = b(\vartheta, c_{\text{in}})$ collects

$$\text{col} \begin{bmatrix} q_P m - r_I c_{n_r} - (r_R/q_U)g_U - c_{\text{in}} \\ \{A(c_i - c_{i-1}) - d_i^{-1}A\xi_i\}_{i=1}^{n_r} \\ g_E + g_U - q_C c_{n_r} \\ E_S g_U - q_U E_S c_{n_r} \end{bmatrix} = 0. \quad (6.90)$$

The empirical overflow relation is appended as

$$t_X^\top g_E = q_E \widehat{X}_{E,\log}(\vartheta, c_{\text{in}}). \quad (6.91)$$

The complete operator is denoted $\mathcal{H}_{\log}$ with right-hand side $b_{\log}$. It has $2F + n_r K + F_S + 1$ rows. The last row fixes total overflow TSS. It does not prescribe its particulate composition.

The remaining teaching-plant requirements are also individual matrix rows. With three 100-m³ layers, $V_{\text{cl}} = 300 \text{ m}^3$ and $V_1 = V_3 = 100 \text{ m}^3$. Densification is $0.6c_{1,X} - g_{U,X} \leq 0$. The inventory inequalities follow by substituting $X_E = g_{E,X}/0.9$ and $X_U = g_{U,X}/0.6$ into the endpoint bounds and multiplying by $0.9(0.6) = 0.54$,

$$120g_{E,X} + 90g_{U,X} - 0.54M_{\text{cl}} \leq 0, \quad (6.92)$$

$$-60g_{E,X} - 180g_{U,X} + 0.54M_{\text{cl}} \leq 0. \quad (6.93)$$

The lower row has coefficient $120 = 0.6(300 - 100)$ and coefficient $90 = 0.9(100)$. These are hydraulic and volume factors. They are not regression coefficients. At the illustrative response above, densification gives $30 - 66 = -36$, while both inventory rows give $-2,700$. The corresponding inventory interval is 13,000–23,000 g, which contains 18,000 g.

The three inequality rows can also be displayed in the response order,

$$\begin{bmatrix} 0 & 0 & 0 & 0.6 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 120 & 0 & 90 & -0.54 \\ 0 & 0 & 0 & 0 & 0 & -60 & 0 & -180 & 0.54 \end{bmatrix} \chi_{\text{toy}} \leq 0. \quad (6.94)$$

Non-negativity would add one row $-\chi_{\text{toy},k} \leq 0$ for each of the nine coordinates. The inequality matrix has the same column order as the equality matrix. Changing that order in one operator would constrain the wrong physical quantities.

For any plant size, collecting the same row patterns gives the physical inequality operator,

$$\mathcal{G}(\vartheta)\chi = \text{col} \begin{bmatrix} q_U E_X c_{n_r} - E_X g_U \\ q_U(V_{\text{cl}} - V_{\text{cl},L})t_X^\top g_E + q_E V_{\text{cl},L}t_X^\top g_U - q_E q_U M_{\text{cl}} \\ q_E q_U M_{\text{cl}} - q_U V_{\text{cl},1}t_X^\top g_E - q_E(V_{\text{cl}} - V_{\text{cl},1})t_X^\top g_U \end{bmatrix} \leq 0. \quad (6.95)$$

The first block enforces particulate densification. The last two rows are the inventory bounds multiplied by the positive product $q_E q_U$. This multiplication removes divisions without changing feasibility. Non-negativity requires $\chi \geq 0$ separately.

At fixed $(\vartheta, c_{\text{in}})$, every constraint is affine in χ . This remains true for the overflow closure because its fitted right-hand side is then a known scalar. The endpoint-envelope inequalities also remain affine even though their coefficients depend on the controls. The distinction between fixed-control linearity and control dependence is central to the numerical formulation.

6.5.2 The scaled nearest-response problem

Projection size is first defined for an individual response coordinate. If the raw value is $\chi_{\text{raw},k}$ and its positive scale is s_k , then

$$u_k = \frac{\chi_k - \chi_{\text{raw},k}}{s_k}. \quad (6.96)$$

The scalar projection objective is $\frac{1}{2} \sum_{k=1}^{n_\chi} u_k^2$. Thus a change of two scale units in one component costs $\frac{1}{2}(2)^2 = 2$, while a change of half a scale unit costs $\frac{1}{2}(0.5)^2 = 0.125$. With $D_\chi = \text{diag}(s_1, \dots, s_{n_\chi})$, the response relation becomes $\chi = \chi_{\text{raw}} + D_\chi u$.

For a mixer component row, the substitution is explicit,

$$q_P(m_{\text{raw},f} + s_{m,f}u_{m,f}) - r_I(c_{\text{raw},n_r,f} + s_{n_r,f}u_{n_r,f}) \quad (6.97)$$

$$- \frac{r_R}{q_U}(g_{\text{raw},U,f} + s_{U,f}u_{U,f}) = c_{\text{in},f}. \quad (6.98)$$

The scales multiply unknown projections, while all raw values are known at the current inputs. Moving the raw terms to the right gives one linear equation in u . The other balances are transformed in the same way. Let D_b and D_g be positive diagonal row scales. The projected standardized displacement solves

$$\begin{aligned} \hat{u} &= \arg \min_{u \in \mathbb{R}^{n_\chi}} \frac{1}{2} u^\top u, \\ \text{subject to } & D_b^{-1} \{ \mathcal{H}_{\log}(\vartheta)(\chi_{\text{raw}} + D_\chi u) - b_{\log}(\vartheta, c_{\text{in}}) \} = 0, \\ & \chi_{\text{raw}} + D_\chi u \geq 0, \\ & D_g^{-1} \mathcal{G}(\vartheta)(\chi_{\text{raw}} + D_\chi u) \leq 0, \\ & \hat{\chi} = \chi_{\text{raw}} + D_\chi \hat{u}. \end{aligned} \quad (6.99)$$

The identity Hessian defines a strictly convex quadratic program. In physical coordinates, its metric is $D_\chi^{-\top} D_\chi^{-1}$. Response scaling determines which physical changes count as large projections. Positive constraint-row scaling changes numerical conditioning while leaving the feasible set unchanged. Sturm and Silva (2025) motivate species-sensitive weighting for projection. Boyd and Vandenberghe (2004) supply the convex optimization principles used for feasibility, uniqueness, and optimality.

The physically constrained surrogate combines the original ICSOR separation of prediction and projection with a connected plant response (E. F. Selerio Jr., 2026). The manifold projection developed by Valente et al. (2025) provides related output-projection context. Analytic neural-output constraints in Beucler et al. (2021), flux learning for mass conservation in Sturm and Wexler (2020), and constrained training in Ruckstuhl et al. (2021) provide alternative locations for physical enforcement. H. Li et al. (2024) include governing-equation residuals in physics-informed unit-operation models, including an activated sludge reactor. A residual penalty makes disagreement costly during fitting. It differs from an equality that each admitted prediction must

satisfy. These studies do not supply the present plant balances or inventory interval. Here those constraints follow from the declared process model.

For a nonempty feasible set, closedness, coercivity, and strict convexity give a unique projected response. Nonemptiness is an assumption that must be checked. Positive outlet flows and a full-rank inventory operator alone do not guarantee it. The fitted endpoint may conflict with mass conservation, non-negativity, or densification. For instance, demanding overflow solids above every feed solids concentration allowed by the other balances contradicts $X_E \leq X_N$. The empirical closure is therefore capable of making an otherwise feasible physical polyhedron empty. A failed projection is retained as a failure. It is not changed by clipping the endpoint or relaxing a balance.

Figure 6.3 distinguishes the three inputs to the joint projection. The raw response supplies the starting point for projection. Derived physical relations define balance and envelope requirements. The separate logarithmic prediction supplies a fitted endpoint equality. Its empirical origin remains relevant even though it enters the same equality system as the physical balances.

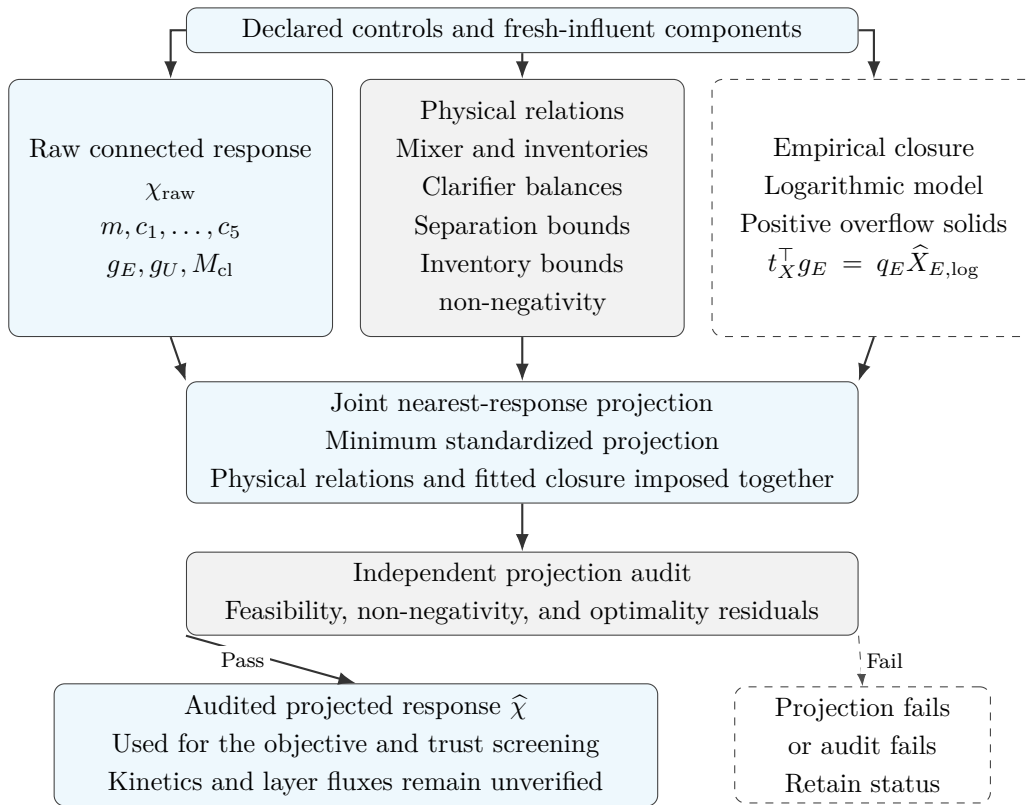


Figure 6.3: Conceptual information flow in the connected projection. The dashed closure block denotes an empirically fitted relation rather than a physical identity. Projection corrects all response blocks jointly. An audited solution satisfies the imposed relations to their numerical tolerances, while original-model evaluation remains necessary for complete kinetic and settling verification.

The projected response is therefore a constrained prediction, not a solved layer

profile. Its smallest standardized projection is meaningful only for the stated response scales and feasible set. Keeping the empirical closure visible also explains why an original mechanistic target can lie outside that set.

6.5.3 An independently auditable optimality certificate

Consider a separate hypothetical two-component projection. The raw prediction is $(-2, 9)$, its coordinate scales are $(1, 2)$, and the required total fixed by mass conservation is 10. The physical-coordinate problem is

$$\min_{c_1, c_2} \frac{1}{2}(c_1 + 2)^2 + \frac{1}{8}(c_2 - 9)^2, \quad c_1 + c_2 = 10, \quad c_1 \geq 0, \quad c_2 \geq 0. \quad (6.101)$$

The second coefficient is $1/(2 \times 2^2)$. A change of one physical unit in component 2 costs less than the same change in component 1 because its scale is larger.

First omit non-negativity and introduce an equality multiplier λ . Stationarity and the total fixed by mass conservation give

$$c_1 + 2 + \lambda = 0, \quad (6.102)$$

$$\frac{1}{4}(c_2 - 9) + \lambda = 0, \quad (6.103)$$

$$c_1 + c_2 = 10. \quad (6.104)$$

The first two equations imply $c_1 = -2 - \lambda$ and $c_2 = 9 - 4\lambda$. Their sum gives $7 - 5\lambda = 10$, so $\lambda = -0.6$ and $(c_1, c_2) = (-1.4, 11.4)$. mass conservation alone leaves a negative component.

Now impose the active boundary $c_1 = 0$, which forces $c_2 = 10$. Use multiplier $\mu_1 \geq 0$ for $-c_1 \leq 0$ and $\mu_2 \geq 0$ for $-c_2 \leq 0$. The Lagrangian is the objective plus $\lambda(c_1 + c_2 - 10) - \mu_1 c_1 - \mu_2 c_2$. Since component 2 is positive, its inequality is inactive and $\mu_2 = 0$. Its stationarity equation gives $\lambda = -0.25$. Component 1 then gives $2 - 0.25 - \mu_1 = 0$, so $\mu_1 = 1.75$. Both inequality multipliers are non-negative, and both products $\mu_j c_j$ vanish. These checks certify the boundary solution rather than merely declaring it plausible.

In standardized coordinates, $u = (2, 0.5)^\top$. The same problem has

$$E = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad e = 3, \quad G_u = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad g = \begin{bmatrix} -2 \\ 9 \end{bmatrix}. \quad (6.105)$$

The equality is $u_1 + 2u_2 = 3$. The inequality rows are the two physical non-negativity requirements after substitution. This example explains the standardized form before its use for the larger plant.

At fixed inputs, write the standardized problem as

$$\min_u \frac{1}{2}u^\top u, \quad Eu = e, \quad G_u u \leq g. \quad (6.106)$$

Let $\lambda^=$ be unrestricted equality multipliers and $\lambda^\leq \geq 0$ inequality multipliers. The optimality conditions are primal feasibility, dual feasibility, stationarity, and complementarity,

$$u + E^\top \lambda^= + G_u^\top \lambda^\leq = 0, \quad (6.107)$$

$$\lambda_j^\leq (g_j - (G_u u)_j) = 0 \quad \text{for every } j. \quad (6.108)$$

To derive the dual value, collect the linear terms in the Lagrangian. Define $v = E^\top \lambda^= + G_u^\top \lambda^\leq$. Completing the square gives

$$\frac{1}{2} u^\top u + v^\top u - e^\top \lambda^= - g^\top \lambda^\leq = \frac{1}{2} \|u + v\|_2^2 - \frac{1}{2} \|v\|_2^2 \quad (6.109)$$

$$- e^\top \lambda^= - g^\top \lambda^\leq. \quad (6.110)$$

The first term has minimum zero at $u = -v$. Minimizing over u therefore gives the dual value,

$$d(\lambda^=, \lambda^\leq) = -\frac{1}{2} \|E^\top \lambda^= + G_u^\top \lambda^\leq\|_2^2 - e^\top \lambda^= - g^\top \lambda^\leq. \quad (6.111)$$

For primal- and dual-feasible quantities, the gap is

$$\Gamma = \frac{1}{2} \|u + E^\top \lambda^= + G_u^\top \lambda^\leq\|_2^2 + (\lambda^\leq)^\top (g - G_u u) \geq 0. \quad (6.112)$$

It vanishes exactly when stationarity and complementarity hold. This gives a mathematical certificate of the unique projection (Boyd & Vandenberghe, 2004). Numerical acceptance checks the individual residuals independently. A small reported gap does not replace a non-negativity or equality check.

The hypothetical two-component example permits every certificate number to be checked. With $\lambda^= = -0.25$ and $\lambda^\leq = (1.75, 0)^\top$,

$$E^\top \lambda^= + G_u^\top \lambda^\leq = (-2, -0.5)^\top = -u, \quad (6.113)$$

$$g - G_u u = (0, 10)^\top, \quad (6.114)$$

$$\frac{1}{2} u^\top u = \frac{1}{2} (4 + 0.25) = 2.125, \quad (6.115)$$

$$d = -2.125 - 3(-0.25) - [-2(1.75)] = 2.125. \quad (6.116)$$

The primal and dual values agree. Stationarity is zero coordinate by coordinate, and the complementarity sum is $1.75(0) + 0(10) = 0$. The gap is consequently zero. A multiplier with the wrong sign would fail dual feasibility even if some other residual happened to be small.

Every distinct trial receives a projection solve from an independent initialization. An operator-splitting quadratic solver of the type studied by Stellato et al. (2020) supplies a numerical candidate. Multipliers are reconstructed independently through bounded-variable least squares for the audit. Numerical retries change the solver settings while preserving the mathematical problem.

6.5.4 The conditional error guarantee

Let $\mathcal{C}$ be the fixed-input feasible set and use the norm $\|v\|_{D_\chi^{-1}} = \|D_\chi^{-1}v\|_2$. This norm has the scalar meaning

$$\|v\|_{D_\chi^{-1}}^2 = \sum_{k=1}^{n_\chi} \frac{v_k^2}{s_k^2}. \quad (6.117)$$

To see the error argument, temporarily work in standardized coordinates. Write $a_k = \chi_{\text{raw},k}/s_k$, $q_k = \hat{\chi}_k/s_k$, and $t_k = \chi_{\text{ref},k}/s_k$. The transformed feasible set is convex because the scales define an invertible linear change of coordinates.

For any feasible comparison point z , the segment $q + \tau(z - q)$ remains feasible for $0 \leq \tau \leq 1$. The projection objective cannot decrease when leaving its minimizer along that segment. Its derivative at zero must therefore satisfy

$$\sum_k (q_k - a_k)(z_k - q_k) \geq 0. \quad (6.118)$$

When the reference itself is feasible, take $z = t$. Reversing both signs gives $\sum_k (a_k - q_k)(q_k - t_k) \geq 0$. The componentwise identity

$$(a_k - t_k)^2 = (a_k - q_k)^2 + (q_k - t_k)^2 \quad (6.119)$$

$$+ 2(a_k - q_k)(q_k - t_k) \quad (6.120)$$

then implies, after summing over coordinates,

$$\sum_k (q_k - t_k)^2 \leq \sum_k (a_k - t_k)^2 - \sum_k (a_k - q_k)^2. \quad (6.121)$$

The last sum is exactly the squared standardized projection. Returning to the physical-coordinate notation, if a mechanistic reference χ_{ref} belongs exactly to $\mathcal{C}$, the projection variational inequality gives

$$\|\hat{\chi} - \chi_{\text{ref}}\|_{D_\chi^{-1}}^2 \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_\chi^{-1}}^2 - \|\hat{u}\|_2^2. \quad (6.122)$$

Thus the projection cannot increase aggregate squared error in this particular metric to an exactly feasible target. It does not guarantee improvement in every component, an unweighted physical-unit score, or the operating objective.

In the hypothetical weighted two-component problem, take the feasible reference (3, 7). Raw squared error is $(-2 - 3)^2 + (9 - 7)^2/4 = 26$. Projected squared error is $(0 - 3)^2 + (10 - 7)^2/4 = 11.25$. Projection size squared is $2^2 + 0.5^2 = 4.25$. The bound reads $11.25 \leq 26 - 4.25 = 21.75$. The second component's absolute error increases from 2 to 3 even though the aggregate weighted error decreases. This is why the guarantee cannot be restated as improvement of every coordinate.

The fitted overflow equality usually differs from the mechanistic endpoint. A physically consistent mechanistic target can therefore lie outside $\mathcal{C}$. Define

$$\delta_{\text{feas}} = \|P_{\mathcal{C}}(\chi_{\text{ref}}) - \chi_{\text{ref}}\|_{D_\chi^{-1}}. \quad (6.123)$$

Let p be the closest feasible point to t in standardized coordinates. The previous segment argument now applies to $z = p$. Split the remaining inner product into a feasible-point part and a reference-distance part,

$$\sum_k (a_k - q_k)(q_k - t_k) = \sum_k (a_k - q_k)(q_k - p_k) \quad (6.124)$$

$$+ \sum_k (a_k - q_k)(p_k - t_k). \quad (6.125)$$

The first term is non-negative. The magnitude of the second is at most

$$\left[\sum_k (a_k - q_k)^2 \right]^{1/2} \left[\sum_k (p_k - t_k)^2 \right]^{1/2} = r_{\text{corr}} \delta_{\text{feas}}. \quad (6.126)$$

This is the scalar Cauchy–Schwarz inequality. It bounds how far an infeasible reference can weaken the projection argument. Let $r_{\text{corr}} = \|\chi_{\text{raw}} - P_{\mathcal{C}}(\chi_{\text{raw}})\|_{D_{\chi}^{-1}}$. Apply the projection variational inequality with the feasible comparison point $P_{\mathcal{C}}(\chi_{\text{ref}})$. Expanding the squared error and bounding the remaining inner product gives

$$\|P_{\mathcal{C}}(\chi_{\text{raw}}) - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 - r_{\text{corr}}^2 + 2r_{\text{corr}}\delta_{\text{feas}} \quad (6.127)$$

$$\leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 + \delta_{\text{feas}}^2. \quad (6.128)$$

Taking square roots therefore gives the more easily interpreted bound

$$\|P_{\mathcal{C}}(\chi_{\text{raw}}) - \chi_{\text{ref}}\|_{D_{\chi}^{-1}} \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}} + \delta_{\text{feas}}. \quad (6.129)$$

The distance includes any finite numerical infeasibility of the reference. The guarantee cannot be invoked by checking only the physical identities and ignoring the empirical equality. The weighted-projection discussion in Sturm and Silva (2025) likewise separates mass conservation from predictive improvement. Biological kinetics, nonlinear settling, and dynamic stability remain outside the projected set.

For a hypothetical illustration of the empirical-closure issue, let the total fixed by mass conservation still be 10 but also require the first component to equal a fitted value of 4. The feasible set then contains only $(4, 6)$. Suppose the mechanistic reference and raw prediction are both $(2, 8)$ with scales $(1, 2)$. They satisfy the physical total and non-negativity but do not satisfy the fitted equality. Raw error is zero. Projection introduces weighted error $\sqrt{(4 - 2)^2 + (6 - 8)^2/4} = \sqrt{5}$. The reference distance to the set is also $\sqrt{5}$, so the general bound is met exactly. Physical identities alone would have given the wrong premise for the stronger guarantee.

6.5.5 Control-dependent sensitivities

With fixed constraint matrices and an affine parameter-dependent right-hand side, a parametric quadratic projection can have a piecewise affine solution (Tøndel et al., 2003). That more restrictive setting is not assumed here. Hydraulic coefficients, inverse

flow ratios, the input features, and the exponential endpoint all depend on ϑ . The projected plant response is generally piecewise smooth where regularity holds. It need not be globally piecewise affine or differentiable at an active-set change.

The differentiation can first be seen in a scalar hypothetical projection,

$$\min_u \frac{1}{2}u^2, \quad b(\theta)u = e(\theta), \quad (6.130)$$

where θ is one control and $b(\theta) \neq 0$. Stationarity and feasibility are $u + b\lambda = 0$ and $bu = e$. Written together, they form the two-by-two system

$$\begin{bmatrix} 1 & b \\ b & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ e \end{bmatrix}. \quad (6.131)$$

Differentiate the two scalar equations with respect to θ . The product rule gives $u' + b'\lambda + b\lambda' = 0$ and $b'u + bu' = e'$. Moving known products to the right leaves

$$\begin{bmatrix} 1 & b \\ b & 0 \end{bmatrix} \begin{bmatrix} u' \\ \lambda' \end{bmatrix} = \begin{bmatrix} -b'\lambda \\ e' - b'u \end{bmatrix}. \quad (6.132)$$

The matrix is unchanged. Its right-hand side now contains derivatives of both coefficients and targets.

For $b = 1 + \theta$ and $e = \theta$, the solution is $u = \theta/(1 + \theta)$ and $\lambda = -\theta/(1 + \theta)^2$. At the hypothetical control $\theta = 1$, the values are $u = 0.5$ and $\lambda = -0.25$. The derivative system becomes

$$\begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} u' \\ \lambda' \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.5 \end{bmatrix}. \quad (6.133)$$

Its second row gives $u' = 0.25$ and its first gives $\lambda' = 0$. Direct differentiation of $u = \theta/(1 + \theta)$ gives the same $u' = 1/(1 + \theta)^2$. Ignoring b' would instead differentiate a problem whose constraint coefficient did not vary.

For a stable active set, stack equality rows and active inequality rows into $B(\vartheta)$ and their right-hand sides into $b_a(\vartheta)$. The stationary system is

$$\begin{bmatrix} I & B^\top \\ B & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda_a \end{bmatrix} = \begin{bmatrix} 0 \\ b_a \end{bmatrix}. \quad (6.134)$$

The componentwise product rule is now applied to $u + B^\top \lambda_a = 0$ and $Bu = b_a$,

$$\partial_j u + B^\top \partial_j \lambda_a = -(\partial_j B)^\top \lambda_a, \quad (6.135)$$

$$B \partial_j u = \partial_j b_a - (\partial_j B)u. \quad (6.136)$$

For a control ϑ_j , putting these derivative equations into one block system gives

$$\begin{bmatrix} I & B^\top \\ B & 0 \end{bmatrix} \begin{bmatrix} \partial_j u \\ \partial_j \lambda_a \end{bmatrix} = \begin{bmatrix} -(\partial_j B)^\top \lambda_a \\ \partial_j b_a - (\partial_j B)u \end{bmatrix}. \quad (6.137)$$

Since D_χ is fixed after fitting, the physical response derivative is $\partial_j \hat{\chi} = \partial_j \chi_{\text{raw}} + D_\chi \partial_j \hat{u}$. For the endpoint, first differentiate the scalar fitted logarithm, which gives $\partial_j (\hat{\beta}^\top \phi) = \hat{\beta}^\top \partial_j \phi$. Exponential differentiation then gives $\partial_j \hat{X}_{E,\log} = \hat{X}_{E,\log} \hat{\beta}^\top \partial_j \phi$. Applying the product rule to its flow-weighted value gives the endpoint derivative,

$$\partial_j (q_E \hat{X}_{E,\log}) = (\partial_j q_E) \hat{X}_{E,\log} + q_E \hat{X}_{E,\log} \hat{\beta}_{E,\gamma_E}^\top \partial_j \phi. \quad (6.138)$$

Ignoring this term would differentiate a different optimization problem.

These derivatives are used only when active rows are independent, the stationary matrix is well conditioned, active inequality multipliers are strictly positive, and independent perturbation checks pass. Near degeneracy, different active descriptions can correspond to the same unique response. Uniqueness of the projection therefore does not ensure a reliable classical derivative. A value-only outer method remains available when the derivative chain cannot be validated.

6.6 Treatment priorities, resource measures, and operating safeguards

6.6.1 A common engineering objective

The effluent-quality map in Table 3.1 gives

$$z_E = I_{\text{comp}} c_E = \text{col}(\text{COD}, \text{TN}, \text{TP}, \text{TSS}). \quad (6.139)$$

Let the four diagonal entries of D_T be $T_{\text{COD}}, T_{\text{TN}}, T_{\text{TP}}, T_{\text{TSS}}$. Dividing each effluent quantity by its own scale gives a dimensionless quality contribution. The scalar weighted sum is

$$\begin{aligned} j_Q = & \varpi_{Q,\text{COD}} \frac{z_{E,\text{COD}}}{T_{\text{COD}}} + \varpi_{Q,\text{TN}} \frac{z_{E,\text{TN}}}{T_{\text{TN}}} \\ & + \varpi_{Q,\text{TP}} \frac{z_{E,\text{TP}}}{T_{\text{TP}}} + \varpi_{Q,\text{TSS}} \frac{z_{E,\text{TSS}}}{T_{\text{TSS}}}. \end{aligned} \quad (6.140)$$

For positive fixed normalization D_T and quality priorities $\varpi_Q \geq 0$ with $\mathbf{1}^\top \varpi_Q = 1$, this sum is written compactly as

$$j_Q = \varpi_Q^\top D_T^{-1} z_E. \quad (6.141)$$

The four totals describe different treatment concerns. They do not identify which individual nutrient transformations occur. A nearly constant TN across a reactor can accompany conversion of ammonium to nitrite. A decrease in overflow TP can arise from particulate separation rather than an additional clarifier reaction. The nutrient-control guidance of U.S. Environmental Protection Agency (2009) and component-resolved treatment framework of X. Zhang et al. (2026) support inspecting species and solids flows alongside aggregate quality measures.

Normalizing a bounded setting subtracts its lower bound and divides by the full range. This gives zero at the lower bound and one at the upper bound. For example,

$H = 18$ h within 6–36 h gives $(18 - 6)/(36 - 6) = 0.4$. Aeration instead sums physical transfer capacities. Each product $V_i k_L a_i$ has units $\text{m}^3 \text{d}^{-1}$, so its fixed reference has the same units. Wasted solids are proportional to $Q_0 w X_U$. Dividing this rate by its fixed reference $Q_0 w^U X_U^{\text{ref}}$ cancels Q_0 . The resource terms are

$$j_H = \frac{H - H^L}{H^U - H^L}, \quad j_A = \frac{\sum_{i \in \mathcal{A}} V_i k_L a_i}{E_A^{\text{ref}}}, \quad (6.142)$$

$$j_I = \frac{r_I - r_I^L}{r_I^U - r_I^L}, \quad j_R = \frac{r_R - r_R^L}{r_R^U - r_R^L}, \quad (6.143)$$

$$j_W = \frac{w X_U}{w^U X_U^{\text{ref}}}. \quad (6.144)$$

The set $\mathcal{A}$ contains stages that can be aerated. The aeration reference $E_A^{\text{ref}} > 0$ is fixed before optimization. It is the maximum transfer-capacity numerator over the declared operating box. A fixed reference prevents an increase in selected volume from being canceled by a changing denominator.

The aeration proxy measures transfer capacity. Actual oxygen transfer also depends on the dissolved-oxygen deficit in Equation (6.15). The energy-management model of Miederer et al. (2025) distinguishes aeration and other energy demands. The treatment model of X. Zhang et al. (2026) makes the oxygen-transfer driving force explicit. The present dimensionless terms do not constitute a monetary cost calculation or a measurement of energy consumption.

Write the overall coefficients in the same order as the six terms. The complete scalar objective is

$$J = \varpi_1 j_Q + \varpi_2 j_H + \varpi_3 j_A + \varpi_4 j_I + \varpi_5 j_R + \varpi_6 j_W. \quad (6.145)$$

Its compact vector form is

$$J(\vartheta, \chi, \varpi) = \varpi^\top \text{col}(j_Q, j_H, j_A, j_I, j_R, j_W), \quad \varpi \geq 0, \quad \mathbf{1}^\top \varpi = 1. \quad (6.146)$$

The overall vector ϖ separates quality from capacity and resource priorities. The vector ϖ_Q separates priorities among the quality measures. These coefficients are choices about trade-offs. They are not percentages of the objective attained at an operating point.

For the equal-volume case,

$$j_A = \frac{H}{36 \text{ h}} \frac{a_3 + a_4 + a_5}{3}. \quad (6.147)$$

This follows from $V_i = Q_0 H / (24 n_r)$ and $k_L a_i = 47 a_i$. For hypothetical settings with all three aeration coordinates equal to one, reducing H from 36 h to 12 h reduces this proxy from one to one-third. The normalized aeration settings have stayed the same, but the total transfer capacity has changed with volume. A comparison based only on the aeration coordinates would miss this effect.

The baseline priority vectors are

$$\varpi = (0.50, 0.15, 0.20, 0.05, 0.05, 0.05)^\top, \quad \varpi_Q = \frac{1}{4}\mathbf{1}_4. \quad (6.148)$$

Quality has the largest individual coefficient. Aeration has the largest resource coefficient. The importance of oxygen supply and solids age to treatment and energy use is discussed by Leu et al. (2012). The exact coefficients remain case-study preferences. Changes in those preferences define separate operating questions and require separately reported decisions.

A hypothetical objective calculation can now be followed through all six terms. Use effluent values (40, 5, 1, 4) mg L⁻¹ and illustrative fixed quality scales (100, 10, 2, 20) mg L⁻¹. With equal quality priorities, $j_Q = (0.4+0.5+0.5+0.2)/4 = 0.4$. These illustrative scales are not estimated study quantities. Let $H = 18$ h, $(a_3, a_4, a_5) = (0.8, 0.6, 0.4)$, $r_I = 1$, $r_R = 0.5$, $w = 0.02$, and $X_U = 6,000$ g m⁻³.

The total biological volume is $Q_0H/24 = 7,500$ m³, so each equal tank has volume 1,500 m³. Aeration capacity is $1,500(47)(0.8 + 0.6 + 0.4) = 126,900$ m³ d⁻¹. Table 6.2 gives every normalized term and weighted contribution. Their sum is 0.353. This arithmetic illustrates the declared objective and does not report a selected operating decision.

Table 6.2: Hypothetical calculation of dimensionless objective contributions.

Term	Calculation	Value	Weighted value
j_Q	$(0.4 + 0.5 + 0.5 + 0.2)/4$	0.4	0.200
j_H	$(18 - 6)/(36 - 6)$	0.4	0.060
j_A	$126,900/423,000$	0.3	0.060
j_I	$(1 - 0)/(4 - 0)$	0.25	0.0125
j_R	$(0.5 - 0.25)/(1.25 - 0.25)$	0.25	0.0125
j_W	$(0.02)(6,000)/[(0.05)(15,000)]$	0.16	0.008

The overall priority of 0.50 for quality multiplies j_Q after its four internal quality weights have been applied. A quality priority of one-quarter is therefore not an overall objective coefficient of one-quarter. Also, a resource reference need not be a hard maximum. In particular, j_W can exceed one if its numerator exceeds the declared reference. A denominator used for normalization does not itself impose feasibility.

6.6.2 Solids retention time and clarifier loading

Stored solids and lost solids have different units. For two reactors, the stored mass would be $V_1X_1 + V_2X_2 + M_{cl}$ in grams. Its loss rate would be $Q_EX_E + Q_0wX_U$ in grams per day. Dividing mass by mass per day gives days. The same accounting applies to all five reactors.

External solids leave in the overflow and waste-sludge streams,

$$\dot{M}_X = Q_0(q_E X_E + w X_U). \quad (6.149)$$

Return sludge does not count as an external loss. With $X_i = t_X^\top c_i$, the whole-plant solids retention time is

$$\theta_X = \frac{\sum_{i=1}^{n_r} V_i X_i + M_{cl}}{\dot{M}_X}, \quad \dot{M}_X \geq \epsilon_X > 0. \quad (6.150)$$

Its numerator includes the clarifier inventory. Omitting that mass would underestimate solids age for the same external loss rate. A wasting ratio alone also does not determine solids age because $w X_U$ and the stored mass both matter (Alex et al., 2008).

For a hypothetical accounting example, use $Q_0 = 10,000 \text{ m}^3 \text{ d}^{-1}$, $w = 0.02$, $X_E = 10$, and $X_U = 6,000 \text{ g m}^{-3}$. The two external loss contributions are

$$Q_0 q_E X_E = 10,000(0.98)(10) = 98,000 \text{ g d}^{-1}, \quad (6.151)$$

$$Q_0 w X_U = 10,000(0.02)(6,000) = 1,200,000 \text{ g d}^{-1}. \quad (6.152)$$

Their sum is $1,298,000 \text{ g d}^{-1}$. At a total reactor volume of $7,500 \text{ m}^3$, assigning every reactor $3,000 \text{ g TSS m}^{-3}$ gives $22,500,000 \text{ g}$ in the reactors. If the clarifier inventory is $7,500,000 \text{ g}$, the total inventory is $30,000,000 \text{ g}$. The illustrative solids retention time is $30,000,000/1,298,000 = 23.11 \text{ d}$. This ratio concerns the stated mass accounting and does not assert that these concentrations solve the mechanistic equations.

Clarifier surface loading divides the overflow volume rate by area. Clarifier solids loading divides the feed solids mass rate by area. Since one gram is 10^{-3} kilograms, its conversion appears before the area division. The descriptive clarifier loading measures are

$$\ell_{\text{surface}} = \frac{Q_0 q_E}{A_{cl}}, \quad \ell_{\text{solids}} = \frac{10^{-3} Q_0 q_C X_N}{A_{cl}}. \quad (6.153)$$

The factor 10^{-3} converts grams to kilograms. The surface quantity is in m d^{-1} and the solids quantity is in $\text{kg m}^{-2} \text{ d}^{-1}$. These quantities and θ_X are descriptive measures in the baseline comparison. They are not additional optimization limits.

For the same hypothetical flow with $r_R = 0.5$, $q_C = 1.5$, $A_{cl} = 1,500 \text{ m}^2$, and $X_N = 3,000 \text{ g m}^{-3}$, the surface loading is $9,800/1,500 = 6.53 \text{ m d}^{-1}$. The feed solids rate is $10,000(1.5)(3,000) = 45,000,000 \text{ g d}^{-1}$, or $45,000 \text{ kg d}^{-1}$. Dividing by area gives $30 \text{ kg m}^{-2} \text{ d}^{-1}$. Using underflow instead of clarifier-feed solids in this formula would calculate a different loading quantity.

The common engineering safeguards are

$$X_U \leq X_U^{\max}, \quad X_N \geq X_F^{\min} > 0, \quad \dot{M}_X \geq \epsilon_X > 0. \quad (6.154)$$

The positive feed-solids limit keeps particulate outlet reconstruction defined. The positive external-loss limit keeps solids retention time defined. The underflow bound represents the admitted thickening or solids-handling range. Their numerical values are

plant-specific problem inputs selected before search. An underflow limit should be based on accepted sludge-handling conditions. The feed and loss floors should prevent undefined calculations without excluding physically relevant low-solids operation. These engineering inputs are distinct from the fixed $X_U^{\text{ref}} = 15,000 \text{ g m}^{-3}$ used only to normalize the wasted-solids objective.

6.6.3 Trust diagnostics without a mechanistic-accuracy guarantee

The imposed physical balances alone admit states that do not satisfy all mechanistic equations. The empirical closure can also exclude mechanistic states. An optimizer might exploit an admitted response that does not solve the full kinetics or settling equations. The projection diagnostic first squares each standardized movement, averages these squares over the response coordinates, and takes the square root. For the hypothetical two-component projection $u = (2, 0.5)$, it is $\sqrt{(4 + 0.25)/2} = 1.458$. Dividing by the number of coordinates makes its interpretation comparable across response dimensions.

Feature leverage uses the feature vector and the regularized fitting geometry. In a hypothetical intercept-plus-one-feature example with three standardized feature values $(-\sqrt{3/2}, 0, \sqrt{3/2})$ and $\gamma = 1$,

$$\Phi^T \Phi + 3\gamma R_\phi = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}. \quad (6.155)$$

Its inverse is $\text{diag}(1/3, 1/6)$. At query feature $(1, z)$, leverage is $1/3 + z^2/6$. It is $1/3$ at the center, $1/2$ at $z = 1$, and $11/6$ at $z = 3$. The largest development value in this teaching example is $7/12$. These numbers illustrate weaker joint feature support away from the observed range. They do not supply the plant's calibrated threshold.

The split diagnostic compares each particulate underflow flow with the flow expected from a common recovery factor. In a hypothetical two-particulate example with unit TSS weights, let the feed component concentrations be $(40, 60)$, so $X_N = 100$. An underflow component-flow vector $(20, 30)$ has total 50 and gives numerators $100(20) - 50(40) = 0$ and $100(30) - 50(60) = 0$. Changing the outlet allocation to $(25, 25)$ preserves total underflow solids but gives numerators 500 and -500 . With illustrative fixed split scales of 500, the standardized values are 1 and -1 , and their root mean square is one. Total solids consistency therefore need not establish composition consistency.

The reactor diagnostic first evaluates the kinetic balances, then divides each residual by its fixed balance scale, and finally computes their root mean square. In the hypothetical two-component reactor, a projected state $(9, 6)$ preserves the upstream total of $(12, 3)$. With $d = 2$ and the reaction rate held at 8 for this illustration, its

two residuals are $2(12 - 9) - 8 = -2$ and $2(3 - 6) + 8 = 2$. A balance scale of 10 for each component gives standardized residuals $(-0.2, 0.2)$ and a diagnostic of 0.2. The invariant is satisfied even though the kinetic equations are not.

These four fixed diagnostics are written compactly as

$$\kappa_{\text{corr}} = \frac{\|D_{\chi}^{-1}(\hat{\chi} - \chi_{\text{raw}})\|_2}{\sqrt{n_{\chi}}}, \quad (6.156)$$

$$\kappa_{\text{lev}} = \phi^{\top} (\Phi^{\top} \Phi + n_{\text{dev}} \gamma R_{\phi})^{-1} \phi, \quad (6.157)$$

$$\kappa_{\text{split}} = \left[\frac{1}{F_X} \sum_{f \in X} \left\{ \frac{X_N g_{U,f} - (t_X^{\top} g_U) c_{n_r,f}}{d_{\text{split},f}} \right\}^2 \right]^{1/2}, \quad (6.158)$$

$$\kappa_{\text{reac}} = \frac{\|D_{\text{bal}}^{-1} f_{\text{sm,reactor}}(\vartheta, \hat{\chi})\|_2}{\sqrt{n_r F}}. \quad (6.159)$$

The first measures standardized projection size. A large value identifies a raw response that requires a large movement to satisfy the imposed constraints. It does not establish that the projected response is inaccurate. The projection is constrained and reported separately from the engineering objective.

The second diagnostic is regularized feature leverage. It measures input support within the fitted quadratic feature map. Sahigara et al. (2012) compare applicability-domain assessments and motivate checking where a statistical model is being used. Feature leverage does not detect every form of response error. A control vector inside its coordinate bounds may still have weak support in the joint feature space.

The third diagnostic checks the common particulate-recovery assumption in Equation (6.48). If every particulate component has the same recovery, its numerator vanishes. The projection enforces component mass conservation and densification but does not enforce that composition exactly. This diagnostic therefore checks a relation that can hold in the reduced mechanistic mapping while remaining outside the projection equalities.

The fourth diagnostic checks smooth reactor kinetics at the projected concentrations. It includes hydraulic transport, net biological conversion, and oxygen transfer on separate scales. It cannot supply a clarifier-layer residual because the surrogate has no internal layer concentrations. Its smoothing width is 10^{-8} under the definitions given below.

A row scale is based on the sizes of its separate physical terms before cancellation. Suppose a hypothetical row has terms $(100, -100)$ in one development state and $(200, -200)$ in another. Its net balance is zero in both states, but its terms have substantial magnitude. Their mean squared magnitude is $(100^2 + 100^2 + 200^2 + 200^2)/4 = 25,000$, giving a root mean square of 158.11. Scaling only the canceled net balance would lose this information. For a constraint row with p terms b_{ij} over n_{dev} development states,

its fixed scale is

$$d_b = \max \left\{ 1, \left[\frac{1}{n_{\text{dev}} p} \sum_{i=1}^{n_{\text{dev}}} \sum_{j=1}^p b_{ij}^2 \right]^{1/2} \right\}. \quad (6.160)$$

The same convention defines inequality and split scales. The overflow row uses the mechanistic overflow flow and the out-of-fold fitted endpoint flow as its two terms. The floor of one is in the corresponding numerical row units. It is not a concentration floor.

Projection, split, and reactor limits use nearest-rank 95th percentiles of grouped out-of-fold development diagnostics. If sorted values are $v_{(1)}, \dots, v_{(n_{\text{dev}})}$, the limit is $v_{(\lceil 0.95 n_{\text{dev}} \rceil)}$. The leverage limit is the maximum development leverage under the final fitted feature map. These quantities are fixed before optimization. They are empirical guardrails rather than confidence bounds. A diagnostic with zero limit remains constrained to zero and uses one only as a numerical row scale.

The surrogate decision problem is consequently

$$\begin{aligned} \min_{\vartheta} \quad & J(\vartheta, \hat{\chi}(\vartheta, c_{\text{in}}), \varpi), \\ \text{subject to} \quad & \vartheta \in \mathcal{V}, \\ & g_{\text{eng}}(\vartheta, \hat{\chi}) \leq 0, \\ & g_{\text{trust}}(\vartheta, \chi_{\text{raw}}, \hat{\chi}) \leq 0, \\ & \hat{\chi} \text{ satisfies Equation (6.100).} \end{aligned} \quad (6.161)$$

The lower problem is a convex quadratic projection. The upper problem is generally nonconvex. This distinction follows the nested-problem perspective described by Colson et al. (2007). A best retained feasible candidate is not a certified global optimum. The trust-region analysis of Alexandrov et al. (1998) explains why reliable approximation management needs more than a small average prediction error. The four fixed diagnostics here do not establish the fully linear approximation conditions of a convergent adaptive trust-region method.

6.7 Bounded products and the limits of a convex relaxation

Flow and concentration products help explain why process optimization remains nonlinear even when a surrogate is linear in its fitted coefficients. A bounded product provides a simple example. Let $x_L \leq x \leq x_U$, $y_L \leq y \leq y_U$, and let z represent the product xy . The four products

$$(x - x_L)(y - y_L), \quad (x_U - x)(y_U - y), \quad (x_U - x)(y - y_L), \quad (x - x_L)(y_U - y)$$

are non-negative. Expanding the first gives $xy - x_L y - y_L x + x_L y_L \geq 0$. Replacing xy by z gives one lower bound on z . Expanding the other three products gives the

remaining bounds. Together they form the McCormick relaxation (McCormick, 1976)

$$\begin{aligned}
 z &\geq x_L y + y_L x - x_L y_L, \\
 z &\geq x_U y + y_U x - x_U y_U, \\
 z &\leq x_U y + y_L x - x_U y_L, \\
 z &\leq x_L y + y_U x - x_L y_U.
 \end{aligned} \tag{6.162}$$

The four inequalities are linear in the variables x , y , and z . Every exact product within the stated bounds satisfies them. Their intersection nevertheless permits points that do not satisfy $z = xy$.

For a hypothetical pair with $0 \leq x, y \leq 1$, the bounds become $z \geq 0$, $z \geq x + y - 1$, $z \leq x$, and $z \leq y$. At $x = y = 0.5$, they permit $0 \leq z \leq 0.5$. The exact product is only $z = 0.25$. A favorable objective obtained at another permitted value of z is therefore not, by itself, an attainable process response. Tighter variable bounds can improve the relaxation, but they do not generally replace the exact product relation. Process-network optimization uses this distinction when constructing bounds and assessing solutions (Rubio-Castro et al., 2013; L. Yang et al., 2014).

This example separates a convex relaxation from the physical projection used here. The relaxation enlarges a nonlinear feasible set to obtain an optimization bound. The projection selects a response satisfying the declared mass conservation and non-negativity conditions near a predicted response. Neither operation alone proves that a control vector is a global optimum of the full mechanistic plant. The present fixed-configuration plant retains the surrogate-assisted and direct mechanistic searches described below. It does not substitute a mixed-integer network-design formulation for those searches.

6.8 A direct mechanistic comparator

6.8.1 The full-state constrained optimization problem

The direct route retains the controls and full reduced mechanistic state. It reconstructs $\chi(\vartheta, c_{\text{in}}, y)$ and evaluates the same engineering objective. Its finite-volume clarifier equations require a positive feed-solids denominator at solver iterates. An auxiliary variable $\widetilde{X}_F \geq X_F^{\text{min}}$ supplies that denominator, with equality $\widetilde{X}_F = t_X^\top c_{n_r}$ imposed at feasibility.

In a hypothetical one-reactor, two-component, three-layer illustration, the full state is $(c_S, c_X, s_1, s_2, s_3)$. It has five entries rather than the nine entries of the reduced statistical response. A simple conversion rate kc_S gives two reactor equations,

$$0 = d(m_S - c_S) - kc_S, \tag{6.163}$$

$$0 = d(m_X - c_X) + kc_S. \tag{6.164}$$

The three layer equations are those obtained from the interface-flow balances above, divided by their own volumes. The direct problem imposes all five equations rather than replacing the two reactor equations by only their sum fixed by mass conservation. Its three-layer envelope contains two scalar inequalities, $s_1 - s_2 \leq 0$ and $s_2 - s_3 \leq 0$. Four layers would require $(s_1 - s_2, s_1 - s_3, s_2 - s_4, s_3 - s_4) \leq 0$. This ordering explains how the envelope is assembled for ten layers.

The auxiliary denominator also has a concrete role. An infeasible trial can have zero feed solids while a nonzero layer concentration is being adjusted. Division by the actual feed value would then be undefined. An independently bounded $\widetilde{X}_F$ keeps the trial calculation defined. The equality $\widetilde{X}_F = X_N$ must subsequently hold, so this device does not allow a zero-feed trial to be accepted as a feasible physical state. It provides a usable calculation during the search, not an alternative physical feed concentration.

Let f_{sm} collect all $n_r F$ reactor equations and all L smoothed layer equations. Let

$$g_{\text{env}}(y) = \text{col}(s_1 \mathbf{1}_{L-2} - s_{2:L-1}, s_{2:L-1} - s_L \mathbf{1}_{L-2}). \quad (6.165)$$

The direct nonlinear program is

$$\begin{aligned} \min_{\vartheta, y, \widetilde{X}_F} \quad & J(\vartheta, \chi(\vartheta, c_{\text{in}}, y, \widetilde{X}_F), \varpi), \\ \text{subject to} \quad & f_{\text{sm}}(\vartheta, c_{\text{in}}, y, \widetilde{X}_F) = 0, \\ & \widetilde{X}_F = t_X^\top c_{n_r}, \quad \widetilde{X}_F \geq X_F^{\min}, \\ & \vartheta \in \mathcal{V}, \quad y \geq 0, \quad g_{\text{env}}(y) \leq 0, \\ & g_{\text{eng,red}}(\vartheta, \chi) \leq 0. \end{aligned} \quad (6.166)$$

The reduced engineering vector removes the feed-solids row already imposed by the auxiliary equality and lower bound. The auxiliary does not alter a feasible physical state. It keeps the outlet mapping defined while a nonlinear solver moves through infeasible trial states.

6.8.2 Smooth switches and continuation

The exact kinetic and settling functions contain minima, maxima, positive parts, possible zero denominators, and a receiver switch. Derivative-based optimization benefits from smooth approximations to these operations. The chemical-process optimization treatment of Buzzi-Ferraris and Manenti (2013) provides the general derivative and conditioning context. The exact identities

$$\max(a, b) = \frac{a + b + |a - b|}{2}, \quad \min(a, b) = \frac{a + b - |a - b|}{2} \quad (6.167)$$

show where the nondifferentiable absolute value enters. Replacing $|r|$ by $\sqrt{r^2 + \delta^2}$ removes its corner for every $\delta > 0$. The positive part uses the same replacement in $(v + |v|)/2$. A denominator regularization instead replaces $1/b$ by $b/(b^2 + \delta^2)$. These

are different operations chosen for different numerical difficulties. For positive width $\delta = \epsilon S$, define

$$\max_{\delta}(a, b) = \frac{1}{2} \left[a + b + \sqrt{(a - b)^2 + \delta^2} \right], \quad (6.168)$$

$$\min_{\delta}(a, b) = \frac{1}{2} \left[a + b - \sqrt{(a - b)^2 + \delta^2} \right], \quad (6.169)$$

$$[v]_{+, \delta} = \frac{1}{2} \left[v + \sqrt{v^2 + \delta^2} \right], \quad (6.170)$$

$$\left(\frac{a}{b} \right)_{\delta} = \frac{ab}{b^2 + \delta^2}. \quad (6.171)$$

For negative v , the positive-part expression is evaluated as $\delta^2/[2(\sqrt{v^2 + \delta^2} - v)]$ to avoid subtraction of nearly equal values. These approximations change the equations near a switch. They do not make the exact and smooth models equivalent.

For hypothetical values $a = 2$, $b = 1$, and $\delta = 0.1$, the smooth maximum is $(3 + \sqrt{1.01})/2 = 2.00249$ and the smooth minimum is 0.99751. The exact values are 2 and 1. For $v = -0.2$, the smooth positive part is $(-0.2 + \sqrt{0.05})/2 = 0.01180$, while the exact positive part is zero. At $b = 0$, the regularized quotient $ab/(b^2 + \delta^2)$ is defined and equals zero. At $a = 1$ and $b = 0.1$, it is 5 for $\delta = 0.1$, compared with the exact quotient 10. These differences illustrate why smaller widths and an independent exact-model evaluation are needed.

The stable positive-part expression follows by rationalization,

$$\frac{v + \sqrt{v^2 + \delta^2}}{2} = \frac{(v + \sqrt{v^2 + \delta^2})(\sqrt{v^2 + \delta^2} - v)}{2(\sqrt{v^2 + \delta^2} - v)} \quad (6.172)$$

$$= \frac{\delta^2}{2(\sqrt{v^2 + \delta^2} - v)}. \quad (6.173)$$

The numerator becomes $v^2 + \delta^2 - v^2 = \delta^2$. This form avoids canceling a negative v against a nearly equal square root.

For the receiver switch, first measure where the receiver concentration lies across its transition band. Subtracting the lower edge $X_t - h$ and dividing by the band width $2h$ maps that band to $[0, 1]$. For receiver concentration s , the smooth switch is zero below $X_t - h$, one above $X_t + h$, and

$$6\zeta^5 - 15\zeta^4 + 10\zeta^3, \quad \zeta = \frac{s - X_t + h}{2h} \quad (6.174)$$

within the transition band. This polynomial matches the values and first two derivatives at the band endpoints. It gives smooth interpolation between the two gravitational-flux rules.

For the illustrative band $X_t = 3,000$ and $h = 10 \text{ g m}^{-3}$, concentrations 2,990, 3,000, and 3,010 correspond to $\zeta = 0, 0.5, 1$. The switch values are 0, 0.5, and 1. At 2,995, $\zeta = 0.25$ and the switch is 0.10352. At 3,005, it is 0.89648. If the unrestricted gravitational flux is J_1 and the receiver-limited flux is J_2 , interpolation gives $(1 - \sigma)J_1 + \sigma J_2$ for

switch value σ . The switch therefore changes a flux rule gradually rather than only replacing a reported concentration.

Each argument scale S is its development-set root mean square with a floor of one. Reactor arguments are pooled across stages. They include total oxidized nitrogen, fermentable substrate plus acetate, the hydrolysis denominator, the polyphosphate and stored-carbon denominators, and remaining storage capacity. Clarifier arguments include $s - f_{\text{ns}}X_N$ and gravitational solids flux. The velocity scale is 250 m d^{-1} . These scales stay fixed throughout continuation.

The direct route reduces smoothing through three stages,

$$(\epsilon, h) = (10^{-6}, 10), (10^{-7}, 3), (10^{-8}, 1), \quad (6.175)$$

where h is in g TSS m^{-3} . Each stage initializes from the preceding stage. Continuation helps a solver approach sharper switches while preserving a usable starting state. The final smooth candidate still requires an exact-model evaluation.

6.8.3 Jacobian structure and globalization

The Jacobian entry J_{jk} is the partial derivative of residual f_j with respect to state coordinate y_k . It describes how one balance changes when one coordinate changes while the others are held fixed. For a hypothetical two-reactor soluble conversion with rate $kc_{i,S}$, soluble pass-through in the recycle gives

$$m_S = \frac{c_{\text{in},S} + (r_I + r_R)c_{2,S}}{q_P}. \quad (6.176)$$

The two soluble residuals are $f_{1,S} = d_1(m_S - c_{1,S}) - kc_{1,S}$ and $f_{2,S} = d_2(c_{1,S} - c_{2,S}) - kc_{2,S}$. Their two-by-two state block is

$$\frac{\partial(f_{1,S}, f_{2,S})}{\partial(c_{1,S}, c_{2,S})} = \begin{bmatrix} -(d_1 + k) & d_1(r_I + r_R)/q_P \\ d_2 & -(d_2 + k) \end{bmatrix}. \quad (6.177)$$

The off-diagonal upper-right entry is created by recycle. Dropping it would treat the mixer as independent of the final reactor. For hypothetical values $d_1 = d_2 = 2$, $k = 1$, $r_I = 1$, $r_R = 0.5$, and $q_P = 2.5$, the block is $\begin{bmatrix} -3 & 1.2 \\ 2 & -3 \end{bmatrix}$. Its entries have units d^{-1} because the residual has concentration-per-day units and its derivative is with respect to concentration.

Finite differences permit an independent check of those entries. Increasing $c_{1,S}$ by 0.01 g m^{-3} changes $f_{1,S}$ by $-0.03 \text{ g m}^{-3} \text{ d}^{-1}$, giving $-0.03/0.01 = -3$. Increasing $c_{2,S}$ by the same amount changes m_S by 0.006 and $f_{1,S}$ by 0.012 , giving 1.2 . These values are exact for the hypothetical linear teaching equations. In a nonlinear residual, the check instead uses a small step h ,

$$J_{jk} \approx \frac{f_j(y + he_k) - f_j(y - he_k)}{2h}. \quad (6.178)$$

Here e_k changes only coordinate k . Near a non-negative boundary, a central step can leave the allowed state region. A second-order forward alternative is

$$J_{jk} \approx \frac{-3f_j(y) + 4f_j(y + he_k) - f_j(y + 2he_k)}{2h}. \quad (6.179)$$

The step has the physical unit of y_k . Scaling coordinates makes its relative size comparable across different component bases.

For three layers at fixed feed conditions, only neighboring concentrations enter each interface flux. Their layer Jacobian therefore has the pattern

$$J_{\text{layer}} = \begin{bmatrix} \star & \star & 0 \\ \star & \star & \star \\ 0 & \star & \star \end{bmatrix}, \quad (6.180)$$

where a star denotes an entry that may be nonzero. Feed solids and outlet reconstruction add connections to reactor coordinates. The zeros describe the declared dependency structure and are not approximations made to reduce computation.

The direct residual is sparse in its mechanistic state. A reactor balance depends on its own component vector and the preceding stage. Recycle reconstruction adds dependencies on final-reactor and underflow quantities. A layer balance depends on nearby layer concentrations, feed solids, and hydraulic controls. The inventory and reporting maps are linear aggregations. These dependencies provide a structured Jacobian rather than a dense arbitrary response map.

Scaling is applied before testing Jacobian conditioning or comparing residual sizes. Different component bases include oxygen, COD, nitrogen, phosphorus, solids, and charge equivalents. An unscaled norm could be dominated by a large solids balance and conceal a nutrient-balance defect. Analytic derivatives or differentiated smooth equations supply the optimization Jacobian. Independent finite differences are retained for derivative checks. Sparse linear algebra exploits the local reactor and layer connections without omitting recycle terms.

The direct search uses the interior-point filter method studied by Wächter and Biegler (2006). Its line search manages progress in objective and constraint violation. This numerical globalization controls acceptance of local steps. It does not make a nonconvex plant objective globally convex or establish a global optimum. The surrogate route similarly separates numerical search progress from the final feasibility and stationarity audits.

6.9 Case-study domain and fixed parameters

6.9.1 Topology and free controls

The case study has five equal-volume reactors and ten clarifier layers. Reactors 1 and 2 are unaerated. Reactors 3–5 have independent aeration controls. The free-control

vector is

$$\vartheta = (H, a_3, a_4, a_5, r_I, r_R, w)^\top \in \mathbb{R}^7. \quad (6.181)$$

For every stage, $h_i = H/5$. The unaerated settings are $a_1 = a_2 = 0$, the aerated transfer coefficients are $k_L a_i = 47a_i \text{ d}^{-1}$, and $\xi_i = 0$. These are case-study conditions rather than structural restrictions on the connected formulation.

Table 6.3: Plant geometry and operating domain.

Quantity	Value or interval
Fresh-influent flow Q_0	10,000 $\text{m}^3 \text{ d}^{-1}$
Reactors n_r , layers L , feed layer ℓ_F	5, 10, 5
Reactor volume fraction f_i^V	1/5 for every stage
Clarifier area and total depth	1,500 m^2 and 4.0 m
Clarifier total and individual layer volumes	6,000 m^3 and 600 m^3
Nominal total HRT H	6–36 h
Each of a_3, a_4, a_5	0–1 independently
Mixed-liquor recycle r_I	0–4
Return-sludge ratio r_R	0.25–1.25
Waste-sludge ratio w	0.001–0.05
Aeration reference E_A^{ref}	423,000 $\text{m}^3 \text{ d}^{-1}$
Wasted-solids reference X_U^{ref}	15,000 g TSS m^{-3}

The equal volumes provide a compact unaerated-to-aerated train. The layer depth is 0.4 m. The operating intervals preserve both positive clarifier outlet flow ratios. Together with 20 influent coordinates, the seven controls give 27 statistical inputs, 406 features, and 161 reduced-response coordinates. The mechanistic state has 110 coordinates. The projection has 76 equality rows and 12 physical inequality rows. Adding response non-negativity gives 173 inequality rows.

Changing the number of reactors or free controls changes the statistical dimensions. Changing the layer count changes the mechanistic dimension and inventory mapping but does not add coordinates to the aggregate surrogate response. In either case, physical and predictive validation must be repeated. A small aggregate response does not establish insensitivity to the clarifier discretization.

6.9.2 Fresh-influent intervals

Table 6.4 defines independent intervals for the plant experiment. The nominal influent is their componentwise midpoint. The concentration bases and quality-map weights are those in Table 3.1. The plant alkalinity interval is 1.6–5.2 mol charge equivalents m^{-3} . It belongs to this connected-plant design and is not interchangeable with an interval specified for an isolated-reactor experiment.

Table 6.4: Independent fresh-influent intervals for the connected plant.

Component	Lower	Upper	Constituent basis
S_O	0	0.5	oxygen
S_F	20	180	COD
S_A	5	80	COD
S_{NH4}	12	55	nitrogen
S_{NO2}	0	3	nitrogen
S_{NO3}	0	8	nitrogen
S_{N2}	0	2	nitrogen
S_{PO4}	2	18	phosphorus
S_I	10	90	COD
S_{ALK}	1.6	5.2	charge equivalents
X_I	20	120	COD
X_S	60	280	COD
X_H	15	100	COD
X_{PAO}	5	60	COD
X_{PP}	2	20	phosphorus
X_{PHA}	1	30	COD
X_{AOB}	0.5	8	COD
X_{NOB}	0.5	8	COD
X_{MeP}	0	12	solids
X_{MeOH}	0	12	solids

All concentration values are in grams of the stated constituent per m^3 , apart from alkalinity. Numerically, g m^{-3} equals mg L^{-1} . Independent interval sampling is an experimental design. It does not assert that all sampled influent combinations occur with equal frequency at a real plant.

6.9.3 Stoichiometric, kinetic, and settling parameters

The constants in Tables 6.5–6.8 define the plant case. They are fixed parameters rather than fitted kinetic estimates. A parameter's concentration basis follows the associated component. Symbols with several process subscripts on the same row have equal values. The two nitrifier populations remain separate, with different growth rates and oxygen half-saturation constants.

Table 6.5: Fixed yields, constituent contents, and conversion factors.

Parameter	Value	Unit or interpretation
Y_H, Y_{PAO}	0.625	g biomass COD per g substrate COD
Y_{PHA}	0.20	g stored COD per g phosphorus
Y_{PO4}	0.40	g phosphorus per g stored COD
Y_{AOB}	0.18	g biomass COD per g nitrogen
Y_{NOB}	0.08	g biomass COD per g nitrogen
f_{SI}	0	soluble inert fraction
f_{XI}	0.10	particulate inert fraction
$i_{N,SI}$	0.01	g nitrogen per g COD
$i_{N,SF}, i_{N,XS}$	0	g nitrogen per g COD
$i_{N,XI}$	0.02	g nitrogen per g COD
$i_{N,BM}$	0.07	g nitrogen per g biomass COD
$i_{P,SI}, i_{P,SF}, i_{P,XS}$	0	g phosphorus per g COD
$i_{P,XI}$	0.01	g phosphorus per g COD
$i_{P,BM}$	0.02	g phosphorus per g biomass COD
$i_{P,MeP}$	1/4.87	g phosphorus per g metal-phosphate solids
γ_{OH}	3.45	g metal-hydroxide solids per g phosphorus

Table 6.6: Fixed hydrolysis and heterotrophic rate and saturation constants.

Parameter	Value	Unit
K_H	3.0	d^{-1}
$\eta_{\text{hyd},\text{NO}_3}, \eta_{\text{hyd},\text{NO}_2}$	0.6	dimensionless
$\eta_{\text{hyd},\text{fe}}$	0.4	dimensionless
$K_{\text{O},\text{hyd}}, K_{\text{O},\text{H}}, K_{\text{O},\text{PAO}}$	0.20	g oxygen m^{-3}
$K_{\text{NO}_3,\text{hyd}}, K_{\text{NO}_2,\text{hyd}}, K_{\text{NO}_x,\text{hyd}}$	0.5	g nitrogen m^{-3}
K_X	0.10	g substrate COD per g biomass COD
μ_H	6.0	d^{-1}
q_{fe}	3.0	d^{-1}
b_H	0.40	d^{-1}
$\eta_{\text{H},\text{NO}_3}, \eta_{\text{H},\text{NO}_2}$	0.9	dimensionless
K_F, K_{fe}, K_A	4.0	g COD m^{-3}
$K_{\text{NO}_3,\text{H}}, K_{\text{NO}_2,\text{H}}, K_{\text{NO}_x,\text{H}}$	0.5	g nitrogen m^{-3}
$K_{\text{NH}_4,\text{H}}, K_{\text{NH}_4,\text{PAO}}, K_{\text{NH}_4,\text{NOB}}$	0.05	g nitrogen m^{-3}
$K_{\text{PO}_4,\text{H}}, K_{\text{PO}_4,\text{PAO}}, K_{\text{PO}_4,\text{nit}}$	0.01	g phosphorus m^{-3}
$K_{\text{ALK},\text{H}}, K_{\text{ALK},\text{PAO}}$	0.10	mol equivalents m^{-3}

6.10 Mechanistic sampling, fitting, and prediction assessment

6.10.1 Separate simulation streams and acceptance-conditioned data

The joint domain has seven control coordinates and twenty influent coordinates. Strength-one Latin-hypercube sampling places one point in each stratum of every coordinate. Independent random stratum orderings prevent a shared ordering from forcing artificial coordinate correlations. Open-unit jitter places a point inside each selected stratum. The comparison of sampling designs by McKay et al. (1979) provides the basis for this coverage-oriented design.

The study design attempts 8,000 development candidates and 2,000 holdout candidates from separate seeded streams. Rejected candidates are retained in the accounting and are not replaced. The two-start physical acceptance procedure determines which candidates enter a statistical partition. These attempted design sizes do not imply any accepted sample total.

The sampling arithmetic can be understood with four strata in one hypothetical coordinate. Each stratum has width $1/4$ in the unit interval. Number the strata from zero to three. Selecting stratum two and a within-stratum fraction of 0.6 gives $(2 + 0.6)/4 = 0.65$. For an illustrative physical interval from 10 to 20, this becomes

Table 6.7: Fixed storage, nitrifier, and chemical rate and saturation constants.

Parameter	Value	Unit
q_{PHA}	5.0	g stored COD per g biomass COD per d
q_{PP}	0.60	g phosphorus per g biomass COD per d
μ_{PAO}	0.56	d ⁻¹
$\eta_{PAO,NO3}$	0.07	dimensionless
$\eta_{PAO,NO2}$	0.90	dimensionless
$b_{PAO}, b_{PP}, b_{PHA}, b_{AOB}$	0.20	d ⁻¹
$K_{NO3,PAO}, K_{NO2,PAO}, K_{NOx,PAO}$	0.5	g nitrogen m ⁻³
K_{PS}	0.20	g phosphorus m ⁻³
K_{PP}	0.01	g phosphorus per g biomass COD
K_{max}	0.34	g phosphorus per g biomass COD
K_{IPP}	0.02	g phosphorus per g biomass COD
K_{PHA}	0.01	g stored COD per g biomass COD
μ_{AOB}	1.81	d ⁻¹
μ_{NOB}	1.52	d ⁻¹
b_{NOB}	0.17	d ⁻¹
$K_{O,AOB}$	0.74	g oxygen m ⁻³
$K_{O,NOB}$	1.75	g oxygen m ⁻³
$K_{NH4,AOB}, K_{NO2,NOB}$	0.5	g nitrogen m ⁻³
$K_{ALK,nit}, K_{ALK,PRE}, K_{ALK,chem}$	0.50	mol equivalents m ⁻³
k_{PRE}	1.0	m ³ per g solids per d
k_{RED}	0.60	d ⁻¹

Table 6.8: Fixed settling and oxygen-transfer parameters.

Quantity	Value
Maximum settling velocity v_{max}	250 m d ⁻¹
Theoretical settling velocity v_0	474 m d ⁻¹
Hindered-settling coefficient r_h	0.000576 m ³ per g TSS
Low-concentration coefficient r_p	0.00286 m ³ per g TSS
Nonsettleable fraction f_{ns}	0.00228
Receiver threshold X_t	3,000 g TSS m ⁻³
Oxygen saturation S_O^*	8.5 g oxygen m ⁻³
Maximum aerated-stage k_La	47 d ⁻¹

Table 6.9: Randomization and fitting settings specified before assessment.

Setting	Value
Development Latin-hypercube seed	100042
Holdout Latin-hypercube seed	100043
Influent-scenario Latin-hypercube seed	314159
Grouped-fold permutation seed	271828
Number of fitting folds	5
Raw-response and logarithmic ridge grids	$\{10^{-8}, 10^{-7}, \dots, 10^2\}$
Standard-deviation divisor	fitting-set size
Near-constant coordinate cutoff	$10^{-12} \max(1, \max_i v_i)$
Low-overflow group	at or below development 25th percentile
High-overflow group	at or above development 90th percentile

$10 + 0.65(20 - 10) = 16.5$. Applying independent permutations to the other coordinates retains one observation in each marginal stratum without forcing the same combinations in every coordinate.

The exact jitter expression uses the same idea at a much finer resolution. The floor operation $\lfloor v \rfloor$ means the largest integer no greater than v . Dividing U by 2^{11} and taking its floor yields an integer from zero to $2^{53} - 1$. Adding 0.5 selects the midpoint of the corresponding fine stratum. Division by 2^{53} places that midpoint inside the unit interval. In exact arithmetic, the extreme values are 2^{-54} and $1 - 2^{-54}$. Both are strictly inside the interval. Endpoint checks remain necessary when values are represented with finite numerical precision.

For an exact numerical randomization convention, each coordinate uses a descending Fisher–Yates permutation of its strata. The jitter from an unsigned 64-bit random integer U is

$$\zeta_U = \frac{\lfloor U/2^{11} \rfloor + 0.5}{2^{53}} \in (0, 1). \quad (6.182)$$

This definition excludes exact stratum boundaries. Accepted states retain their candidate ordering before the grouped-fold permutation is applied.

Conditioning on mechanistic acceptance changes the population represented by the data. Regions with frequent integration, balance, branch, or stability failures may contribute few accepted states. The fitted model therefore represents the accepted portion of the design. A prediction score on that portion is not a reliability estimate for the complete rectangular operating domain.

6.10.2 Two initial states and the nonsmoothed steady solve

The first initial state assigns c_{in} to every reactor and fresh-influent TSS to every clarifier layer. The second retains the influent soluble concentrations and multiplies the particulate concentrations in reactors 1–5 by (1.5, 2, 2.5, 3, 3.5). Its final-reactor TSS sets a layer profile with multipliers

$$(0.002, 0.005, 0.01, 0.03, 0.10, 0.50, 1.25, 2.25, 3.25, 4.00) \quad (6.183)$$

from surface to hopper. The different solids distributions probe sensitivity to the initial biomass and storage state. Agreement from two starts is a finite numerical check. It is not a proof of a unique steady state for all possible initial conditions.

The positive-state transformation follows from the scalar chain rule. Treat y_f as the numerical concentration in its assigned native unit and introduce $\eta_f = \log y_f$ for $y_f > 0$. The logarithm therefore acts on the numerical coordinate, with one native unit as its implicit reference. Differentiating its inverse gives

$$y_f = \exp(\eta_f),$$

$$\frac{dy_f}{dt} = \exp(\eta_f) \frac{d\eta_f}{dt} = y_f \frac{d\eta_f}{dt}.$$

The original component balance requires $dy_f/dt = f_f(y)$. Equating the two expressions and dividing by positive y_f gives $d\eta_f/dt = f_f(y)/y_f$. The balance is transformed rather than replaced by another physical process.

For a hypothetical concentration of two native units and a balance rate of -0.4 native units per day, the transformed rate is $-0.4/2 = -0.2$ per day. A simple half-day illustrative step in logarithmic coordinates gives

$$\eta_{\text{new}} = \log 2 + 0.5(-0.2), \quad y_{\text{new}} = \exp(\eta_{\text{new}}) = 2 \exp(-0.1) \approx 1.8097.$$

This example explains the algebra of a positive coordinate. The prescribed numerical integration retains the backward differentiation method and its error controls. Strict positivity of the exponential in exact arithmetic does not certify an accurate time step or an accepted steady state.

Time integration uses the original nonsmoothed equations. A logarithmic state transformation keeps integrated coordinates positive. Before this transformation alone, an initial coordinate smaller than 10^{-8} is raised to 10^{-8} in its own physical unit. For $y_f = \exp(\eta_f)$, the transformed balance is $\dot{\eta}_f = f_f(y)/y_f$. This initialization safeguard is distinct from clipping a final predicted concentration.

The integration horizon contains two nested choices. First, the wasting coordinate is bounded below by 0.001 for this horizon calculation. Second, its reciprocal term is compared with a minimum horizon of 400 days. At a hypothetical $w = 0.01$, the inner maximum is 0.01, so the reciprocal term is $50/0.01 = 5000$ and the horizon is 5000 days. At $w = 0.20$, that term is 250, so the horizon is 400 days. At $w = 0$, the inner

floor gives $50/0.001 = 50,000$ days. These arithmetic examples illustrate the declared rule and are not claims about actual convergence durations.

The values zero and 0.20 illustrate the branches of the horizon rule and lie outside the case-study wasting interval. They are not admitted operating candidates.

The backward differentiation method provides an integration route for stiff equations (Curtiss & Hirschfelder, 1952). The horizon is

$$T = \max\{400, 50/\max(w, 0.001)\} \text{ d.} \quad (6.184)$$

The calculation can stop early when its scaled residual falls below 10^{-9} . Dynamic relaxation supplies an initial state for algebraic steady calculation, consistent with the continuation rationale of Kelley and Keyes (1998) and its activated sludge use in E. F. Selerio Jr. (2026). A bounded least-squares polishing calculation is applied only if the integrated endpoint fails the physical audit. Solver convergence and physical acceptance use separate tolerances.

Table 6.10: Steady-state generation and physical acceptance settings.

Setting	Value
Integration relative tolerance	10^{-7}
Transformed integration absolute tolerance	10^{-9}
Maximum integration step	$T/100$
Polishing evaluation limit	5,000
Polishing step, function, and gradient tolerances	10^{-9} each
Term-scaled balance tolerance	10^{-6}
State non-negativity tolerance	10^{-10}
Reaction-rate non-negativity tolerance	10^{-12}
Two-start scaled state agreement	10^{-6}
Rightmost reduced-state Jacobian eigenvalue	at most -10^{-8} d^{-1}
Eigenvalue agreement between two step sizes	10^{-6} d^{-1}
Matrix mass conservation and rank audit tolerance	10^{-10}

A residual first adds the signed physical terms in a balance. For hypothetical incoming, consumption, and outgoing contributions of 100, -60 , and -39.99999 in one consistent component-rate unit, the remaining imbalance is

$$100 - 60 - 39.99999 = 0.00001.$$

The largest absolute term is 100. Dividing the absolute imbalance by that scale gives $0.00001/100 = 10^{-7}$. The floor of one in the scale prevents division by a very small reference when all terms are near zero. The resulting dimensionless ratio assesses cancellation relative to the terms in that component equation. It is evaluated separately for each balance rather than obtained by adding balances with different constituent units.

For a balance with separately evaluated terms b_j , the acceptance residual is

$$r_{\text{bal}} = \frac{|\sum_j b_j|}{\max(1, \max_j |b_j|)}. \quad (6.185)$$

Reactor generation scales repeat $\max(1, c_{\text{in},f}^U)$ for each component. Layer scales use $\max(1, X_N)$. State agreement is tested in these scaled coordinates.

Local stability asks what happens to a small perturbation about an equilibrium. A first-order expansion of the dynamic equations has the form $\dot{\delta y} = J_{\text{dyn}} \delta y$, where entry (f, j) of J_{dyn} is the derivative of balance f with respect to state j . In a hypothetical independent two-coordinate example,

$$\begin{bmatrix} \dot{\delta y}_1 \\ \dot{\delta y}_2 \end{bmatrix} = \begin{bmatrix} -0.2 & 0 \\ 0 & 0.05 \end{bmatrix} \begin{bmatrix} \delta y_1 \\ \delta y_2 \end{bmatrix}.$$

The scalar equations are $\dot{\delta y}_1 = -0.2\delta y_1$ and $\dot{\delta y}_2 = 0.05\delta y_2$. Their perturbations evolve as $\delta y_1(0) \exp(-0.2t)$ and $\delta y_2(0) \exp(0.05t)$ when time is in days. One decays and the other grows. The two diagonal eigenvalues are -0.2 and 0.05 per day, so the rightmost eigenvalue is positive and this hypothetical equilibrium fails the stability condition.

For a coupled system, an eigenvector gives a combination of coordinates that behaves as one local mode. The real part of its eigenvalue determines growth or decay. The rightmost eigenvalue means the largest real part among all modes. This is a local test because the linear expansion describes small perturbations. Checking derivatives with two finite-difference steps tests numerical sensitivity of the estimated modes, while the stated negative threshold tests their local decay.

Reduced-state local stability is assessed by the rightmost eigenvalue of the dynamic Jacobian. Finite differences use normalized steps $\epsilon_{\text{mach}}^{1/3}$ and $2\epsilon_{\text{mach}}^{1/3}$. Central differences are used away from the non-negative boundary. Second-order forward differences are used near it. Both eigenvalue estimates must pass the upper-bound and agreement checks. This is stability evidence for the reduced reactor-and-aggregate-layer model. It does not establish stability of unrepresented species profiles or of a real plant.

Both starts must provide finite states, satisfy non-negativity and rate checks, pass component and invariant mass conservation checks, obey the operating domain and layer envelope, and meet the layer-flux residual tolerance. Generation also requires identical branch-classification tuples between the two starts. These tuples record the active kinetic and settling branches. A state that meets a residual test but fails another physical check is not admitted as a training target.

6.10.3 Fold-local fitting and two separately selected penalties

Every component and location from one accepted plant state stays in the same fold. This grouping prevents the same simulated plant from contributing training information and an assessment target within a fold. Fold sizes differ by at most one. Centers,

input scales, quadratic-feature scales, and response scales are recomputed inside each fitting fold. The holdout contributes none of these quantities. The information-leakage analysis of Kaufman et al. (2012) explains why assessment information must be kept outside fitting choices.

The one-standard-error rule can be followed with a hypothetical five-fold comparison. Suppose the candidate with the smallest mean score has fold scores 0.40, 0.42, 0.38, 0.41, and 0.39. Their mean is 0.40. The sum of squared deviations from that mean is

$$0^2 + 0.02^2 + (-0.02)^2 + 0.01^2 + (-0.01)^2 = 0.001.$$

The sample standard deviation is $\sqrt{0.001/4} \approx 0.01581$. The standard-error summary is $0.01581/\sqrt{5} \approx 0.00707$. The admissible mean-score threshold is consequently $0.40 + 0.00707 = 0.40707$.

If a stronger hypothetical penalty has mean score 0.406, it lies within the threshold. A still stronger candidate with mean score 0.430 lies outside. The rule retains the largest candidate penalty within the threshold. It does not select the numerical score 0.40707 as a penalty. This example explains the regularization choice and does not give fitting outcomes for the plant. The standard-error summary used by this selection rule is not a confidence guarantee for a new prediction.

Each candidate penalty is assessed by the standardized root mean square error of the complete raw response. The largest penalty within one standard error of the smallest mean score is selected. The rule favors stronger regularization when the score does not clearly distinguish the smaller penalty (Hastie et al., 2009). Projected errors do not choose the raw-response penalty. The logarithmic model uses the same folds, grid, and one-standard-error rule but is scored on its logarithmic target. Its penalty is selected separately.

The augmented ridge equations are solved with column-pivoted orthogonal-triangular factorization and audited using singular-value decomposition. A coordinate whose standard deviation meets the near-constant cutoff in Table 6.9 is rejected rather than assigned an arbitrary scale. Positive ridge penalties and the unpenalized intercept give a unique coefficient solution under the stated fitting design. The fitted penalties and coefficient matrices are retained as estimation quantities. They cannot be inferred from the candidate grid alone.

After selection, both models are refitted once on all accepted development states. Grouped out-of-fold predictions supply trust calibration and projection assessment. The overflow equality for an out-of-fold state uses the corresponding out-of-fold logarithmic model, not the final fit that has seen that state. The final models supply holdout predictions and optimization evaluations.

The overflow target must be strictly positive before logarithmic fitting. Predicted logarithms and their back-transformations must be finite. The development-set low-overflow and high-overflow strata are fixed by the percentiles in Table 6.9. Separate

stratum assessment can reveal an error pattern concealed by a central concentration range. The imbalanced-regression analysis of Shahbazi et al. (2026) motivates attention to sparse or extreme response ranges. It does not establish imbalance as the cause of an error in this plant experiment.

6.10.4 Scientific admission and projection audits

Admission of the connected surrogate requires finite and independently audited out-of-fold projections for all accepted development states. The complete raw-response normalized root mean square error and the raw clarifier-inventory normalized error must each be below one. Because errors use development response scales, the threshold requires aggregate error below one scale unit. It is a case-specific admission criterion. It is not a universal standard of surrogate quality.

Every projection is assessed for equality feasibility, physical inequalities, non-negativity, dual feasibility, stationarity, and complementarity. The overflow-closure residual is also reported separately. A failed admission criterion remains a failure rather than triggering an unreported refit or changed penalty selection rule.

Table 6.11: Projection solve and independent audit settings.

Setting	Value
Solver absolute and relative tolerances	10^{-12} each
Initial iteration limit	100,000
Polishing	enabled
Retry values of algorithm parameter ρ_{alg}	0.01 and 10
Retry iteration limit	200,000 each
Adaptive ρ_{alg} during retries	disabled
Scaled equality and inequality tolerances	10^{-8} each
Dual, stationarity, and complementarity tolerances	10^{-8} each
Physical non-negativity tolerance	10^{-10}
Active-set identification tolerance	10^{-7}

6.10.5 Prediction metrics and their denominators

The error calculations can be expanded with two hypothetical observations of one concentration coordinate. Let the reference values be 2 and 6 and the predictions be 3 and 4 in the same native unit. Subtracting reference from prediction gives residuals

$3 - 2 = 1$ and $4 - 6 = -2$. Their different summaries are

$$\begin{aligned}\text{mean squared error} &= \frac{1^2 + (-2)^2}{2} = 2.5, \\ \text{root mean square error} &= \sqrt{2.5} \approx 1.5811, \\ \text{mean absolute error} &= \frac{|1| + |-2|}{2} = 1.5, \\ \text{bias} &= \frac{1 + (-2)}{2} = -0.5.\end{aligned}$$

The mean squared error has squared concentration units. The other three quantities have concentration units. Bias retains the signs and can be small when positive and negative errors cancel.

For the same illustrative values, the reference mean is $(2 + 6)/2 = 4$. The squared error of the two predictions is $1^2 + (-2)^2 = 5$. Always predicting the reference mean instead would give squared error $(2 - 4)^2 + (6 - 4)^2 = 8$. Their ratio is $5/8$, so the coefficient of determination is $1 - 5/8 = 0.375$. The denominator describes variation of the assessment reference. It is not a training scale or a concentration limit. The general formulas record these same operations over all assessed observations.

For a response coordinate f and assessment state i , define the signed error $a_{if} = p_{if} - t_{if}$, where p is predicted and t is mechanistic. Physical-unit root mean square error, mean absolute error, and bias are

$$\left(\text{mean}_i a_{if}^2\right)^{1/2}, \quad \text{mean}_i |a_{if}|, \quad \text{mean}_i a_{if}. \quad (6.186)$$

Normalized joint-response metrics replace a_{if} by $a_{if}/(D_\chi)_{ff}$ and state whether they pool errors across coordinates. Coordinatewise coefficients of determination use

$$R_f^2 = 1 - \frac{\sum_i (p_{if} - t_{if})^2}{\sum_i (t_{if} - \bar{t}_f)^2}. \quad (6.187)$$

A constant assessment target has an undefined denominator and is recorded accordingly. An arithmetic mean of coordinate scores is distinct from a score calculated on pooled observations.

Normalization changes the numerical comparison by assigning a denominator to each coordinate. In the preceding hypothetical error example, suppose the training response scale is two native units. Dividing residuals 1 and -2 by two gives 0.5 and -1 . Their standardized root mean square error is

$$\sqrt{\frac{0.5^2 + (-1)^2}{2}} = \sqrt{0.625} \approx 0.7906.$$

If a descriptive holdout report instead uses the observed range $6 - 2 = 4$, its normalized residuals are 0.25 and -0.5 . Their root mean square summary is approximately 0.3953. The predictions have not changed. The denominator has changed. This explains why

a training-scale admission score and a holdout-range reporting score must be identified separately.

Location-quality summaries cover the mixer, five reactors, overflow, and underflow. Outlet flows are divided by their own positive flow ratios before applying the quality map. For location ℓ and composite j , define the holdout reporting range and normalized error as

$$r_{\ell j} = \max_i t_{i\ell j} - \min_i t_{i\ell j}, \quad e_{i\ell j} = \frac{p_{i\ell j} - t_{i\ell j}}{r_{\ell j}}. \quad (6.188)$$

For positive ranges, an aggregate score pools the 8×4 location-composite coordinates. A location score pools four composites and a composite score pools eight locations. The displayed R^2 summaries average the corresponding coordinatewise scores. The holdout ranges serve only reporting. They do not enter fitting, admission, objective normalization, or trust calibration.

The overflow model receives concentration-scale and logarithmic assessment separately. The concentration report includes root mean square error, mean absolute error, bias, normalized error, and R^2 . The logarithmic report includes root mean square error and bias. Strict positivity of Equation (6.79) does not replace these assessments.

The experimental design includes knowledge from exploratory simulation when specifying its response, closure, and acceptance rules. The separate holdout stream therefore supports descriptive post-selection assessment. It is not independent external confirmation. Holdout and optimization responses do not change the fitted coefficients, preprocessing, priorities, trust limits, engineering safeguards, or numerical settings. This separation controls later information leakage while preserving the appropriate limits on interpretation (Kaufman et al., 2012).

6.11 Operating searches and convergence evidence

6.11.1 A common baseline and an explicit basin-sensitivity experiment

A normalized control measures a fraction of its permitted interval. Consider an illustrative interval from 10 to 30 physical units. A control value of 15 lies five units above its lower bound in an interval of width twenty, so its normalized value is $5/20 = 0.25$. The midpoint 20 maps to 0.5, and the upper bound 30 maps to one. Conversely, a normalized value of 0.75 maps back to $10 + 0.75(30 - 10) = 25$. The general forward and inverse relations therefore use the same interval width,

$$\vartheta_j = \vartheta_j^L + (\vartheta_j^U - \vartheta_j^L)\eta_j.$$

This transformation allows operating variables with different units to use comparable numerical step sizes. It does not assign them equal importance in the engineering objective.

The baseline search for each route starts at the center of the normalized operating box. For each coordinate,

$$\eta_j = \frac{\vartheta_j - \vartheta_j^L}{\vartheta_j^U - \vartheta_j^L} \in [0, 1]. \quad (6.189)$$

The nominal influent is the midpoint of the fresh-influent intervals. The baseline explores one initial basin. Its interpretation remains local even if the numerical search terminates successfully.

For seven controls, the normalized center has seven entries equal to 0.5. Moving only the first coordinate upward by one quarter of its interval makes that coordinate 0.75. Moving it downward makes that coordinate 0.25. The six other entries remain 0.5 in both cases. Writing out the entries gives

$$\begin{aligned} \eta_{\text{first},+} &= (0.75, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5)^\top, \\ \eta_{\text{first},-} &= (0.25, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5)^\top, \\ e_1 &= (1, 0, 0, 0, 0, 0, 0)^\top. \end{aligned}$$

The unit coordinate vector e_1 records which coordinate moves. Two signs for each of seven coordinates give fourteen displaced starts. Adding the center gives fifteen. The compact expression below writes this same collection without listing every vector.

A planned multistart sensitivity experiment evaluates basin dependence separately from this baseline. It uses the same 15 starting control vectors for both routes,

$$\eta^{(0)} = \frac{1}{2}\mathbf{1}_7, \quad \eta^{(j,\pm)} = \frac{1}{2}\mathbf{1}_7 \pm \frac{1}{4}e_j, \quad j = 1, \dots, 7. \quad (6.190)$$

Each start receives the same route-specific budget and audit rules. A direct start obtains its mechanistic initial state from an independently accepted solve at those fixed controls. Failed initial states remain failures. The best independently validated feasible candidate is retained, with all starting-point outcomes and total search work reported. This planned sensitivity analysis does not replace the single-start baseline or claim exhaustive basin coverage.

The distinction matters when interpreting a faster search. An algorithm that visits fewer basins can finish quickly while missing a better candidate. The expensive-function benchmarking study of Blik et al. (2023) motivates keeping objective quality, evaluation work, and computational time as separate comparison dimensions.

6.11.2 Surrogate search and derivative fallback

The initial surrogate outer method is sequential least squares programming, based on the constrained optimization approach of Kraft (1988). Each trial obtains a fully audited joint projection. The active-set derivative chain is used only after the rank, conditioning, multiplier, and independent derivative checks pass. An active-set change requires reconstruction of the projection and its derivatives.

When that chain cannot be validated, a deterministic value-only quadratic-approximation method continues within the normalized bounds (Ragoneau, 2022). It uses the same audited projection and does not substitute an approximate lower solve. Finite differences are used for independent derivative audits rather than as an unqualified derivative replacement. The best feasible visited point is solved again from an independent projection initialization before retention.

Table 6.12: Surrogate search, sensitivity, and finite-poll settings.

Setting	Value
Initial outer iteration limit	250
Initial outer function tolerance	10^{-10}
Active slack tolerance	10^{-7}
Required active inequality multiplier	greater than 10^{-8}
Domain-aware active-set perturbation	10^{-6}
Stationary-system condition number times machine precision	at most 10^{-8}
Sensitivity residual and scaled state reproduction	10^{-8} each
Independent derivative steps	10^{-6} and 5×10^{-7}
Derivative relative-error tolerance	10^{-5}
Upper feasibility and stationarity tolerance	10^{-6} each
Value-only fallback iteration and evaluation limits	250 each
Initial and final fallback trust radii	0.25 and 10^{-6}
Fallback feasibility tolerance	10^{-6}
Coarse and fine poll radii	10^{-3} and 10^{-4}
Coarse and fine direction counts	14 and 106
Absolute and relative sufficient-decrease tolerances	10^{-8} each
Direction-rank tolerance	10^{-10}
Certification evaluation limit	10,000
Ray growth factor and maximum probes	2 and 16
Objective relative tie tolerance	10^{-10}

6.11.3 First-order evidence and finite-resolution polling

A candidate receives first-order stationarity evidence only when the derivative chain, lower projection certificate, and upper constrained-stationarity audit pass. This is a statement about the declared surrogate optimization problem at that candidate. It does not establish a local minimum, a global optimum, or stationarity of the original mechanistic objective.

Coordinate and coupled directions can have different feasibility near an active constraint. Consider a hypothetical two-control problem with $x + y = 1$, non-negative

controls, and objective $J = x$. At $(0.5, 0.5)$, any nonzero move in x alone or y alone violates the equality. A diagonal move that lowers x and raises y by the same amount preserves it. For a unit direction $(-1, 1)^\top / \sqrt{2}$ and illustrative step radius 0.1, the trial is

$$\begin{aligned}x_{trial} &= 0.5 - 0.1/\sqrt{2} \approx 0.4293, \\y_{trial} &= 0.5 + 0.1/\sqrt{2} \approx 0.5707, \\x_{trial} + y_{trial} &= 1.\end{aligned}$$

The objective decreases by approximately 0.0707. An axis-only poll would contain no feasible nonzero trial in this example, even though a feasible improving direction exists.

In seven coordinates, each of the seven axes has a positive and a negative direction, giving fourteen. There are $7(6)/2 = 21$ unordered coordinate pairs. Each pair admits four sign combinations, giving $4(21) = 84$ pairwise directions. Eight regular-simplex directions supplement them. A regular simplex is the higher-dimensional counterpart of an equilateral triangle. Its directions distribute around a center rather than following only the coordinate axes. The fine direction count is $14 + 84 + 8 = 106$. These finite collections test particular moves. They do not enumerate every feasible direction.

If the derivative evidence is unavailable, the convergence procedure performs complete feasible-direction polls at normalized radii 10^{-3} and 10^{-4} . The coarse poll contains the 14 signed coordinate directions. The fine poll adds 84 signed pairwise diagonals and eight regular-simplex directions from a Helmert construction. The pairwise count is $4\binom{7}{2}$. Coupled directions can reveal feasible improvements that an axis-only test misses near a constraint.

Each complete poll must contain a feasible nonzero displacement. A rank calculation is diagnostic because active constraints can reduce the dimension of the feasible cone. An accepted improvement triggers a new complete poll. A fine-radius move also triggers revalidation at the coarse radius. After an improving step, up to 16 search-only probes grow along the same ray by successive factors of two. These probes may accelerate search but supply no convergence certificate.

Absence of sufficient decrease in both complete revalidated polls gives evidence only for the tested directions and radii. An incomplete poll, a failed calculation, or an exhausted certification budget leaves stationarity unresolved. The limiting direct-search analysis of Audet and Dennis (2006) requires suitable refining directions and an asymptotic argument. Two finite polls do not provide that argument. Neither convergence tier establishes global optimality.

6.11.4 Direct continuation, recovery, and endpoint checks

The direct route uses the interior-point filter algorithm of Wächter and Biegler (2006) through the three smoothing stages. Its endpoint audit independently checks all smooth reactor and layer equations, inequalities, non-negativity, multipliers, and

stationarity. The full-state audit is separate from the final solver status.

If the initial direct search fails, one recovery may start from the same influent's certified surrogate controls and repeat the same smoothing sequence. Recovery is not attempted after an initially successful search or from an uncertified surrogate candidate. Because this recovery uses a surrogate decision, it is a dependent rescue calculation rather than independent basin exploration. Its status and computational cost are reported separately.

Table 6.13: Direct continuation and exact-evaluation qualification settings.

Setting	Value
Interior-point iteration limit per smoothing stage	2,500
Convergence and constraint tolerances	10^{-8} each
Dual and complementarity tolerances	10^{-6} each
Endpoint equality acceptance	10^{-8}
Endpoint inequality and stationarity acceptance	10^{-6} each
Endpoint active-set tolerance	10^{-7}
Receiver-switch ambiguity distance from X_t	1 g TSS m^{-3}
Other branch ambiguity distances	$10^{-6}S$
Two-start agreement under both prescribed state scales	10^{-6}
Engineering feasibility tolerance after exact evaluation	10^{-6}

A validated feasible candidate can be retained when its search budget is exhausted or stationarity remains unresolved. Its convergence qualification remains visible. A case with no validated feasible candidate remains a search failure. A favorable success flag alone is insufficient for either route.

6.12 Independent mechanistic verification and fair comparison

The two searches use different response calculations but must be assessed on the same original-model basis. Figure 6.4 separates the search, native-candidate audit, independent physical evaluation, and comparison eligibility. A failure at one stage does not imply that every other stage is undefined or successful.

The common evaluation stage supplies treatment responses at fixed controls rather than returning information to the fitted models. A valid physical response can still violate a retained engineering safeguard and remain useful as a diagnostic. Separating these outcomes prevents numerical convergence, physical validity, and decision eligibility from being treated as the same property.

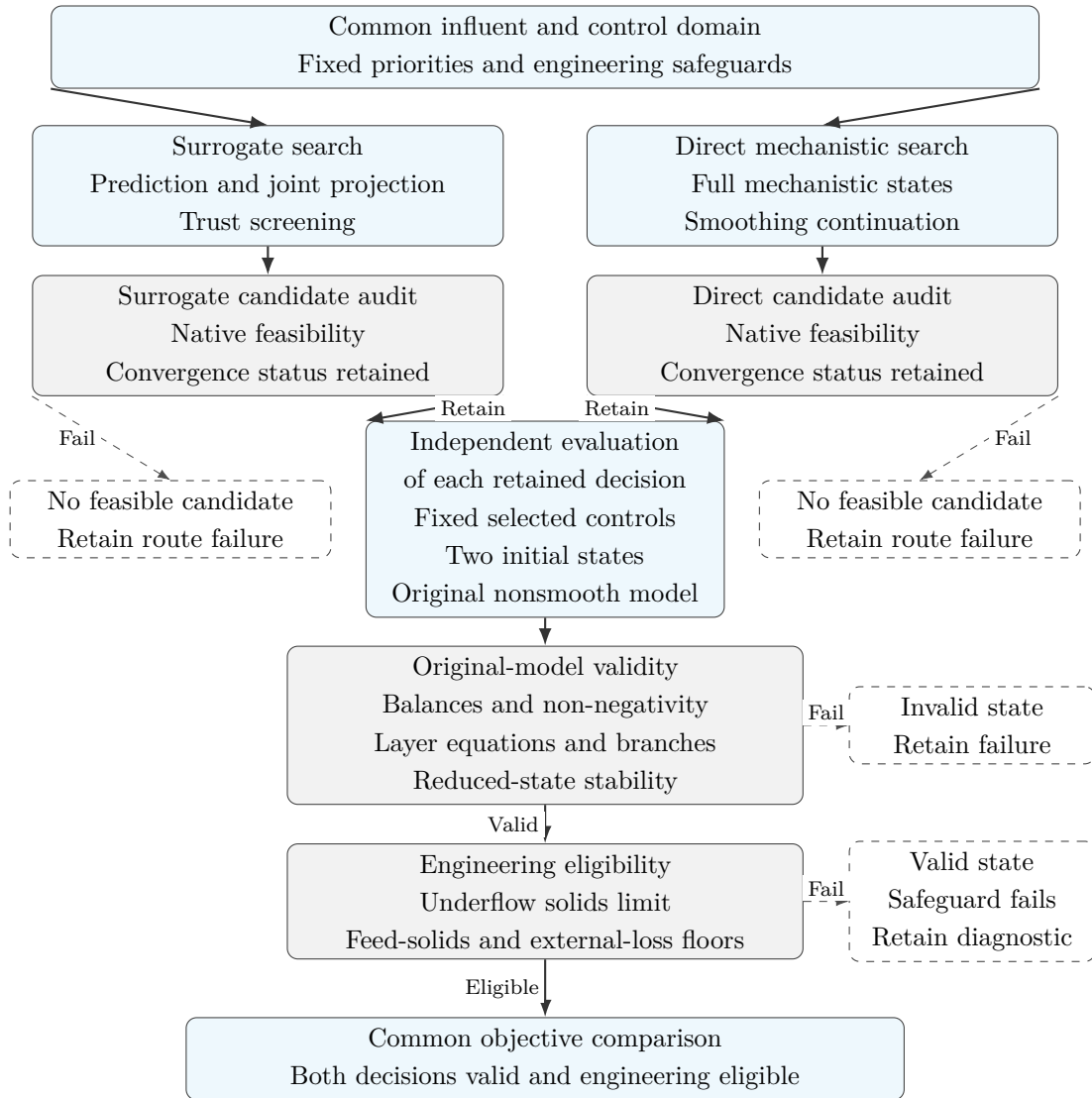


Figure 6.4: Conceptual paired-search and verification workflow. Each retained control vector receives its own independent original-model evaluation. The surrogate and smooth-direct native responses do not supply the reference treatment outcome. Original-model validity and engineering eligibility are distinct checks. Only eligible pairs enter the common-objective comparison, while failed or qualified cases remain in the accounting.

6.12.1 Evaluation at fixed selected controls

Every available selected decision is evaluated with its controls and influent held fixed. Both initial states are solved on the original nonsmoothed reactor and ten-layer clarifier equations. This calculation does not use the statistical prediction as its treatment outcome. It also does not use the smooth direct state as proof of an exact steady state. The role of an original-model check follows the feasibility and validation requirements discussed by Bhosekar and Ierapetritou (2018).

Exact evaluation uses the physical checks in Table 6.10. Two-start state agreement is tested under both generation scales and fixed state scales. The fixed scales are development-set coordinatewise median absolute deviations with a floor of one. A valid evaluation must pass mass conservation, non-negativity, layer residuals, the domain, the endpoint envelope, and the reduced-state stability test.

Branch tuples must agree unless an evaluation start is boundary-ambiguous. Ambiguity bands cover the receiver threshold, zero storage capacity, zero settling offset, zero or maximum settling velocity, and equal adjacent gravitational fluxes. An ambiguity flag qualifies the state-level branch comparison. It does not prove that every mismatching branch falls within its own ambiguity band. Agreement of the physical states and all other audits is still required. Smooth stationarity or smooth-reference equivalence near such a boundary requires that qualification.

This verification is especially relevant near a prescribed nutrient limit. The aeration-scheduling framework of Ren et al. (2026) explicitly considers influent disturbances and effluent ammonium safeguards. Its safety setting is not identical to the present TN or TP objective. It reinforces the need to evaluate the particular effluent quantity and limit that governs a decision. A non-negative or mass conservation-projected concentration can still misclassify compliance.

6.12.2 Native and common-reference objectives

Let route S denote the connected surrogate and route M the direct smooth mechanistic search. Their native responses are $\chi_{\text{nat},S} = \hat{\chi}$ and $\chi_{\text{nat},M} = \chi_{\text{sm},M}$. The independently solved responses are $\chi_{\text{ref},S}$ and $\chi_{\text{ref},M}$. Raw and projected surrogate predictions are also evaluated, where defined, at both retained control vectors.

Route-specific normalization can change an objective even when the physical response is identical. Consider an illustrative two-quality case with positive development standard deviations 0.5 and 2 in the corresponding native units. A scale with a floor of one gives denominators 1 and 2. The unfloored scale retains 0.5 and 2. For a hypothetical response of 0.8 and 4, their standardized coordinates are

$$\begin{aligned}\text{floored scale} &= (0.8/1, 4/2) = (0.8, 2), \\ \text{unfloored scale} &= (0.8/0.5, 4/2) = (1.6, 2).\end{aligned}$$

With equal quality weights, the means are 1.4 and 1.8. This difference arises entirely from the normalization. Writing the diagonal matrices makes the componentwise denominators explicit,

$$D_{T,S}^{\text{toy}} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad D_{T,M}^{\text{toy}} = \begin{bmatrix} 0.5 & 0 \\ 0 & 2 \end{bmatrix}.$$

These matrices illustrate the two general definitions. The comparison of selected decisions uses one common reference normalization so that such a numerical choice is not mistaken for better treatment.

For positive development-set effluent standard deviations σ_j , native quality normalization is

$$D_{T,S} = \text{diag}(\max(1, \sigma_j)), \quad D_{T,M} = \text{diag}(\sigma_j). \quad (6.191)$$

The unit floor is in the physical unit of each quality coordinate. The common reference objective uses $D_{T,M}$ for both decisions. Actual fitted standard deviations are estimation quantities and are not specified as theoretical constants.

Two distinct subtractions are needed in decision assessment. In a hypothetical surrogate route with native objective 0.90 and independent reference objective 1.10, the native-minus-reference difference is $0.90 - 1.10 = -0.20$. This negative sign means the native calculation presents a smaller objective. It does not mean that the independently evaluated decision improves on another decision.

For an illustrative direct-route decision with native objective 0.95 and reference objective 0.98, its native-minus-reference difference is -0.03 . Comparing the independently evaluated decisions instead gives $1.10 - 0.98 = 0.12$. On the shared objective basis, this positive difference means the surrogate-selected decision has the larger objective among these two choices. The compact expressions distinguish within-route prediction differences from between-route decision differences.

The native-to-reference and route-to-route differences are

$$\Delta J_{\text{nat-ref},k} = J_{\text{nat},k}(\vartheta_k, \chi_{\text{nat},k}) - J_{\text{ref}}(\vartheta_k, \chi_{\text{ref},k}), \quad k \in \{S, M\}, \quad (6.192)$$

$$\Delta J_{S-M} = J_{\text{ref}}(\vartheta_S, \chi_{\text{ref},S}) - J_{\text{ref}}(\vartheta_M, \chi_{\text{ref},M}). \quad (6.193)$$

The first can reflect both response approximation and route normalization. The second compares independently evaluated decisions using a common objective basis. Neither is called regret because a global optimal objective is unavailable.

An objective comparison and a control comparison also have different denominators. A hypothetical objective difference of 0.12 with reference direct objective 0.98 has direct-relative percentage $100(0.12)/0.98 \approx 12.24\%$. The symmetric scaled difference uses the denominator $\max(1, 1.10, 0.98) = 1.10$ and gives $0.12/1.10 \approx 0.1091$. The first uses one route as its reference. The second uses a symmetric magnitude scale with a floor.

For an illustrative control interval from 10 to 30, two selected settings of 25 and 20 differ by five physical units. Their normalized difference is $5/(30 - 10) = 0.25$.

If this is the only differing coordinate among seven, the root mean square difference is $\sqrt{0.25^2/7} \approx 0.0945$, while the maximum absolute difference is 0.25. The average spreads one change over all seven coordinates. The maximum keeps the largest change visible.

Standardized response errors at each decision retain the raw, projected, and native quantities separately. A symmetric relative objective difference divides ΔJ_{S-M} by $\max(1, |J_S|, |J_M|)$. A direct-relative percentage divides $100\Delta J_{S-M}$ by J_M only when that denominator is nonzero. Normalized control differences are

$$d_j = \frac{\vartheta_{S,j} - \vartheta_{M,j}}{\vartheta_j^U - \vartheta_j^L}, \quad d_{\text{rms}} = \left(\frac{1}{7} \sum_{j=1}^7 d_j^2 \right)^{1/2}, \quad d_{\text{max}} = \max_j |d_j|. \quad (6.194)$$

These summaries distinguish similar objective values from similar controls.

Removal follows from the material entering and leaving on the stated concentration basis. For a hypothetical fresh-influent concentration of 200 mg L^{-1} and an effluent concentration of 2 mg L^{-1} , the concentration reduction is $200 - 2 = 198 \text{ mg L}^{-1}$. Dividing by 200 gives a fraction of 0.99, and multiplying by one hundred gives 99%. Algebraically,

$$100 \frac{z_{\text{in},j} - z_{E,j}}{z_{\text{in},j}} = 100 \left(1 - \frac{z_{E,j}}{z_{\text{in},j}} \right).$$

This step requires a positive influent denominator. It compares concentrations on one reporting basis. A component inventory balance additionally uses the appropriate flow rates and represented outlets.

Selected-decision removal for composite j is

$$\mathcal{R}_j = 100 \left(1 - \frac{z_{E,j}}{z_{\text{in},j}} \right), \quad (6.195)$$

when the fresh-influent denominator is positive. Predicted-to-reference removal differences use percentage points. Concentration percentage errors instead divide by the mechanistic effluent concentration. For a hypothetical influent of 200 mg L^{-1} and effluent values of 2 and 4 mg L^{-1} , the effluent prediction error is 100% relative to 2. The removal difference is only one percentage point. Reporting both makes their different denominators clear.

6.12.3 Validity, engineering eligibility, and failures

A paired objective comparison requires both routes to supply natively feasible candidates, valid exact evaluations, and satisfaction of the common retained engineering safeguards after evaluation. A valid physical evaluation that violates an underflow, feed-solids, or external-loss safeguard still retains its diagnostic objective and treatment response. It does not enter the eligible paired comparison. Solids retention time and loading measures are reported for every finite evaluation and do not add eligibility thresholds.

Table 6.14: Distinct questions retained in failure and qualification accounting.

Assessment	Question answered
Generation acceptance	Did both original-model starts pass the required physical and numerical checks?
Projection feasibility	Did the fitted closure and physical constraints admit an audited joint response?
Native engineering feasibility	Did the search candidate satisfy its declared optimization constraints?
Convergence qualification	Did derivative audits or complete finite polls supply their stated evidence?
Exact-model validity	Did independent original-model evaluation converge to an accepted state?
Retained engineering feasibility	Did the exact response pass the common safeguards?
Paired eligibility	Were both decisions available and admissible on the common reference basis?

These status fields are not collapsed into one success label. Missing or nonfinite quantities, an undefined feature scale, a failed factorization, or an inconsistent numerical audit remain computational failures. A finite state that fails a scientific threshold remains a diagnostic state with the failed criterion identified. Generation rejection, optimization failure, projection infeasibility, and branch qualification have different causes and denominators.

6.12.4 Influent scenarios and time accounting

Ten influent vectors from a separate twenty-dimensional Latin hypercube define the scenario experiment. They use the seed in Table 6.9 and do not enter fitting, calibration, or tuning. Both routes are attempted for the nominal influent and every scenario. Available selected decisions receive exact evaluation even when the other route fails. These cases examine sensitivity within a specified set. They do not estimate population-level reliability under arbitrary disturbances.

Optimization time is the duration of the route’s initial search. For the surrogate, it includes regression, projection, and any value-only fallback inside that search. For the direct route, it includes the initial three-stage smoothing sequence. Development sampling, fitting, convergence certification, conditional recovery, and exact-model evaluation are outside this interval. Failed initial attempts remain in the time accounting. Scenario aggregates use the ten influent cases and keep the nominal case separate. Both routes use the same processor allocation and thread count.

Search time is accompanied by evaluated points, iterations, projection retries, fall-

back use, completed smoothing stages, recovery status, and convergence qualification. The benchmark concerns identified by Blik et al. (2023) make these workload quantities useful when comparing expensive-function methods. Time alone cannot establish equal decision quality or equal completeness of a search.

A complete workflow account additionally includes development, certification, recovery, and verification. These costs are reported separately rather than silently absorbed into a search-speed claim. Repeated use can amortize development effort, but the relevant number of later decisions depends on the full per-decision cost and on acceptable mechanistic decision quality. A reduction in the selected search interval therefore does not by itself establish a reduction in the complete computational task.

The effect of repeated use can be shown with hypothetical total costs. Suppose development requires 100 seconds, and each later surrogate-assisted decision requires three seconds in total for search, qualification, and verification. Suppose a directly evaluated decision requires six seconds for the corresponding complete task. For twenty decisions, the costs are $100 + 20(3) = 160$ seconds and $20(6) = 120$ seconds. For fifty decisions, they are 250 and 300 seconds. These numbers illustrate amortization and are not measured computational performance.

More generally, let n_{dec} be the number of later decisions, T_{dev} the development cost, and T_S and T_M the complete per-decision costs. Expanding the comparison gives

$$\begin{aligned} T_{dev} + n_{dec}T_S &< n_{dec}T_M, \\ T_{dev} &< n_{dec}(T_M - T_S), \\ n_{dec} &> \frac{T_{dev}}{T_M - T_S} \quad \text{when } T_M > T_S. \end{aligned}$$

The illustrative threshold is $100/(6 - 3) = 33\frac{1}{3}$, so at least thirty-four complete decisions are needed to cross it. If the complete per-decision cost is not smaller, a positive development cost cannot be recovered by this timing argument. The calculation also assumes that the compared decisions meet the required quality and feasibility conditions. Cost arithmetic alone does not establish that condition.

The resulting evaluation keeps four types of evidence separate. Statistical assessment concerns approximation of accepted mechanistic states. Projection audits concern the imposed network and inventory constraints. Search audits concern local numerical evidence for the declared optimization problem. Original-model verification concerns treatment and feasibility at the selected controls. This separation allows a connected surrogate to be studied as a decision aid while preserving the limits of its empirical closure and aggregate clarifier representation.

Chapter 7

Integrated Assessment and Research Boundaries

7.1 A common basis for evaluating the prediction chain

The research links a mechanistic component representation, statistical approximation, physical enforcement, interpretation, and operating choice. Each part introduces a specific question about credibility. The mechanistic representation defines the reference system. The statistical predictor approximates its response. The projection restricts the returned state to the imposed feasible set. Interpretation describes selected parts of the prediction map. Optimization uses the returned response to select controls. Evidence is needed at each point where the intended claim changes.

The raw benchmark and the projection benchmark share a particular prediction. This permits a paired assessment of what changes when enforcement is applied. A comparison between two independently trained models answers a different question because it also contains differences in fitting and hyperparameter selection. The structured regression adds another distinction. Its raw head, affine projection, and final constrained output are separately assessable objects. Reporting only the final output would leave the contribution of the learned structure unclear.

The connected plant adds several linked outputs. Its projection must be assessed across the mixer, reactor train, clarifier outlets, and solids inventory. A change that improves one location can worsen another while preserving joint constraints. Location-specific errors therefore remain necessary alongside a plant aggregate. This follows the general principle that approximation quality must be assessed in the quantities used by the intended decision (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019).

Table 7.1: Claims and the evidence required to assess them.

Claim	Assessment evidence	Limit of the inference
Accurate approximation	Held-out component and composite errors with defined scales	Agreement concerns the selected reference model and tested domain
Physical admissibility	Original equality residuals, non-negativity checks, and inventory bounds	Unimposed kinetic and dynamic relations remain unverified
Sample efficiency	Common learning curves and variation across partitions	The conclusion depends on the sampling and fitting budget
Transparent structure	Named blocks, physical-unit coefficients, and sensitivity calculations	Coefficients are predictive associations rather than identified causal laws
Reliable operating choice	Independent mechanistic solve at selected controls	A local search does not establish a global optimum
Computational benefit	Separate development, search, prediction, projection, and verification costs	Search-time savings alone do not establish a lower total cost

7.2 Examples that separate the assessment questions

7.2.1 Mass conservation and prediction error in the same calculation

Consider a hypothetical two-component prediction problem in which both components use one material basis. The first reference concentration is 6 and the second is 4 in arbitrary concentration units. Their sum is $6 + 4 = 10$. The raw prediction gives 8 for the first component and 3 for the second. Its sum is $8 + 3 = 11$, so its inventory residual is $11 - 10 = 1$. Both coordinates are non-negative, but the state is inconsistent with mass conservation under the specified equality.

Suppose a hypothetical projection for mass conservation lowers the first prediction by one unit and leaves the second unchanged. The projected sum is $7 + 3 = 10$. This is one admissible projection and is not asserted to be the nearest projection under every

metric. The component errors can be calculated directly,

$$\begin{aligned}\text{raw errors} &= (8 - 6, 3 - 4) = (2, -1), \\ \text{projected errors} &= (7 - 6, 3 - 4) = (1, -1), \\ \text{raw squared distance} &= 2^2 + (-1)^2 = 5, \\ \text{projected squared distance} &= 1^2 + (-1)^2 = 2.\end{aligned}$$

The corresponding Euclidean distances are $\sqrt{5} \approx 2.236$ and $\sqrt{2} \approx 1.414$. Their values follow from comparison with the reference state. The mass conservation test instead uses the predicted sums. In compact component notation, the raw state is $(8, 3)^\top$, the reference state is $(6, 4)^\top$, and the chosen projected state is $(7, 3)^\top$. This calculation illustrates how feasibility and error are assessed on the same prediction without treating either calculation as a measured treatment outcome.

7.2.2 Why the two physical requirements are simultaneous

Now let the first hypothetical concentration be 11 and the second be -1 . Their sum is $11 + (-1) = 10$, so the inventory equality is satisfied. The second component fails non-negativity. Replacing it by zero changes the sum to $11 + 0 = 11$. To return to the non-negative part of the line defined by mass conservation, both coordinates can be changed. The state with first concentration 10 and second concentration zero has sum ten and passes both checks.

The individual adjustments make the distinction explicit,

$$\begin{aligned}\text{raw state} &= (11, -1)^\top, \\ \text{clipped state} &= (11 + 0, -1 + 1)^\top = (11, 0)^\top, \\ \text{jointly feasible state} &= (11 - 1, -1 + 1)^\top = (10, 0)^\top.\end{aligned}$$

Clipping changes only the second coordinate. The joint projection also compensates in the first coordinate. An equality projection alone can retain a negative component, while clipping alone can violate an equality. The final assessment therefore checks both properties on the same returned state (Haddad et al., 2010; Kircher & Votsmeier, 2025).

Figure 7.1 places the three states on the line defined by mass conservation and its non-negative segment. Their positions show which requirement each state satisfies.

The raw point is on the extended line defined by mass conservation but below the concentration boundary. The clipped point reaches a non-negative second coordinate while leaving that line. The jointly projected point lies at the end of the thick feasible segment. The geometry therefore makes the two simultaneous tests visible without replacing their numerical checks.

Separate marginal summaries can also be misleading. If one prediction satisfies only mass conservation and another satisfies only non-negativity, neither is jointly feasible.

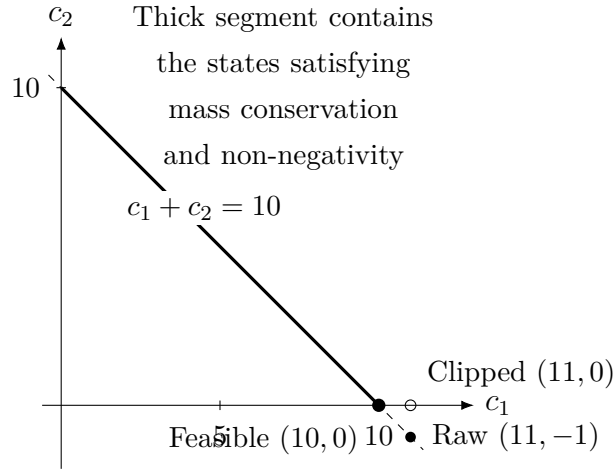


Figure 7.1: Hypothetical two-component geometry showing why clipping a negative concentration can violate a sum fixed by mass conservation.

For a hypothetical collection of two such states, each marginal compliance frequency is one half, while the joint compliance frequency is zero. The correct denominator is the number of states checked, and both conditions are tested within each state before counting joint compliance.

7.2.3 Objective comparisons at selected decisions

A hypothetical decision comparison uses two operating choices with surrogate-predicted dimensionless objectives of 0.90 and 0.95. Smaller values are preferred, so prediction favors the first choice by $0.95 - 0.90 = 0.05$. Suppose independent mechanistic evaluation instead gives 1.10 and 0.98. On that common reference, the first choice has a larger objective by $1.10 - 0.98 = 0.12$. Relative to the second reference value, the difference is $100(0.12)/0.98 \approx 12.24\%$.

Both projected surrogate states could satisfy every imposed balance while this ordering changes. The value 0.12 compares the two tested choices. It is not a gap from a known global optimum. These hypothetical arithmetic steps explain why physical consistency and average prediction error cannot establish the ordering of selected decisions (Alexandrov et al., 1998; Pedrozo et al., 2025).

7.2.4 Following sensitivity through projection

Interpretation also depends on the complete map. A positive quadratic aeration coefficient in a raw driver does not imply that every final effluent quantity increases with aeration. Coupling transforms the driver into component predictions. The reporting map combines components. The active output constraints can alter local derivatives again. Physical-unit interpretation must therefore identify whether it concerns a driver coefficient, a raw prediction derivative, or a derivative of the final projected response

(Brunton et al., 2016; Shen et al., 2023).

A simplified calculation isolates the effect of projection. Let a dimensionless operating input be a , and let two hypothetical raw concentration equations be

$$c_{raw,1}(a) = 4 + a, \quad c_{raw,2}(a) = 3 + 2a.$$

The raw first-component slope is one and the second-component slope is two. At $a = 2$, the concentrations are 6 and 7, with a total of 13. Suppose the required total is ten and the projection minimizes the sum of squared adjustments with equal weights. If the first adjustment is d_1 and the second is d_2 , mass conservation requires $d_1 + d_2 = -3$. Substitute $d_2 = -3 - d_1$ into the squared-adjustment objective and differentiate,

$$\begin{aligned} d_1^2 + d_2^2 &= d_1^2 + (-3 - d_1)^2, \\ \frac{d}{dd_1} [d_1^2 + (-3 - d_1)^2] &= 4d_1 + 6 = 0. \end{aligned}$$

Both adjustments are -1.5 , giving a state satisfying mass conservation with concentrations 4.5 and 5.5. For a general a , the excess total is $(4 + a) + (3 + 2a) - 10 = 3a - 3$. Equal-weight projection subtracts one half of that excess from each coordinate,

$$\begin{aligned} c_{aff,1}(a) &= 4 + a - \frac{3a - 3}{2} = 5.5 - 0.5a, \\ c_{aff,2}(a) &= 3 + 2a - \frac{3a - 3}{2} = 4.5 + 0.5a. \end{aligned}$$

The projected slopes are -0.5 and 0.5 . The first changes sign even though its raw slope is positive. The sum of the projected slopes is zero, consistent with a total that does not depend on a .

Non-negativity introduces another stage. The affine first component reaches zero at $a = 11$. For $a > 11$, the affine state lies outside the segment satisfying mass conservation and non-negativity. In this equal-weight two-component example, the nearest jointly feasible state is then $(0, 10)^\top$. Both local output slopes are zero while that boundary remains active. The slope changes at the transition point. This calculation illustrates why a raw coefficient, an affine sensitivity, and a final constrained sensitivity have distinct meanings. It does not identify an aeration response of the activated sludge model.

Figure 7.2 plots the hypothetical first-component equations and the slopes calculated from them. The left panel shows the concentration returned at each stage. The right panel shows the corresponding local rate of change with respect to the operating input.

Before the boundary becomes active, the affine and checked curves coincide. Their negative first-component slope differs from the positive raw slope because mass conservation redistributes the operating response between the two components. After the boundary becomes active, the checked concentration stays at zero and its local slope is zero. The checked map has no single classical derivative at the transition itself. These

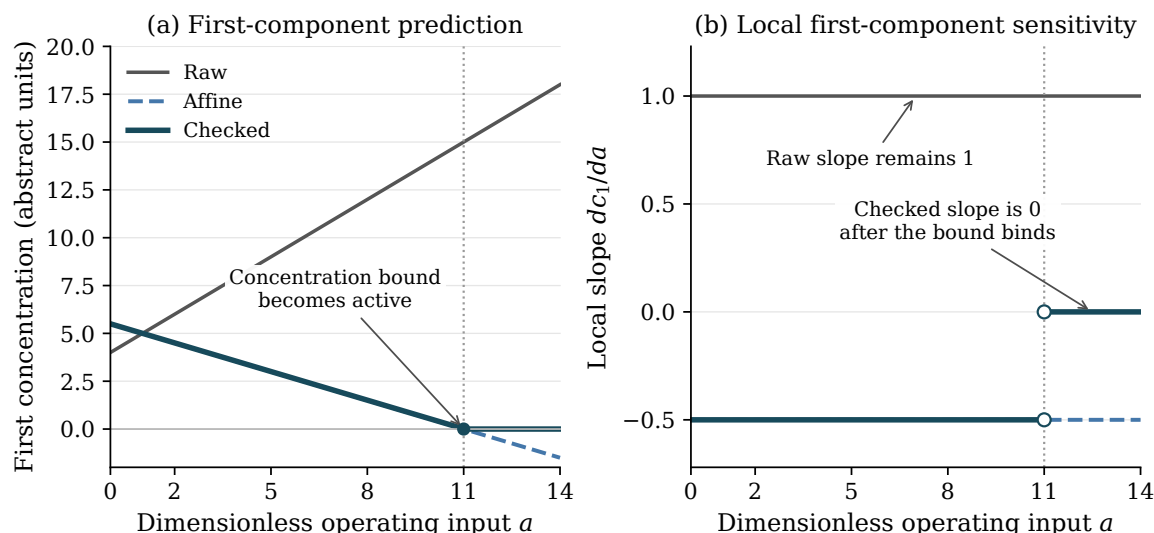


Figure 7.2: Hypothetical first-component response and sensitivity before and after mass conservation and non-negativity enforcement. Open circles mark the two one-sided checked slopes at the active-boundary transition.

visual relationships belong to the analytical illustration and do not represent fitted treatment behavior.

7.3 Data separation and the status of operating evidence

All quantities learned from data require a declared fitting reference. Input transformations, output scales, hyperparameters, regularization, overflow solids fitting, and trust calibration belong to the development information. The final assessment information is reserved for evaluating the specified procedure. This separation must remain consistent when the procedure contains a learned model followed by a constrained calculation (Kaufman et al., 2012).

The standalone reactor benchmark uses nested cross-validation to separate model selection from outer assessment. Its beyond-domain samples are simulated separately and are not used to tune the fitted predictor. The structured-regression protocol uses its declared training and holdout partitions and repeated learning-curve assessment. These designs answer related questions with different units of evidence. Their scores should not be pooled into one model ranking as if they arose from a common assessment sample.

The connected plant design distinguishes development, holdout prediction assessment, and operating verification. It also recognizes that engineering choices can be informed by earlier inspection of a simulated response. Where an analysis setting follows such inspection, mechanistic evaluation of the selected controls provides con-

ditional evidence for that declared setting. It does not become a fully prospective assessment merely because another solve is performed. A later independent evaluation with frozen choices is needed for a stronger prospective claim (Kaufman et al., 2012; X. Zhang et al., 2026).

Failure accounting is part of this evidence. A failed mechanistic solve, an infeasible projection, an operating search that fails a stopping criterion, and a control that violates the original engineering constraints are different events. Omitting them changes the apparent domain and the apparent computational cost. Acceptance rules and failure counts must therefore accompany empirical assessment when it is undertaken.

7.4 Limits of physical guarantees

The reactor invariant constraints follow from the stoichiometric matrix and the compatibility of the external forcing with those invariants. They apply to the declared component representation and reactor boundary. They do not establish that every feasible concentration vector is a reachable reactor steady state. The feasible set generally contains many states that satisfy inventories and non-negativity without satisfying the nonlinear rate equations.

The plant constraints extend this logic to connected unit outputs. Mixing, flow-weighted clarifier balances, soluble-component continuity, and solids inventory bounds address specified physical relationships. Their satisfaction does not reconstruct the full layered clarifier or prove dynamic stability. The direct reference equations remain necessary for judging the selected operating controls.

Numerical feasibility also needs a declared tolerance. Exact equalities describe the mathematical formulation. Finite arithmetic and iterative solvers return approximate numerical states. A solver termination message is insufficient without residual checks in the original units or in a clearly stated scaled form. The same principle applies to the direct mechanistic comparator (Boyd & Vandenberghe, 2004; Buzzi-Ferraris & Manenti, 2013).

7.5 Limits of generalization and interpretation

Simulated training data reflect the selected process model, parameter values, and input design. A wide numerical range does not guarantee coverage of every important joint condition. A steady-state input description does not encode all dynamic or historical influences. The distinct standalone and plant alkalinity ranges also prevent an automatic transfer claim between those domains. Domain checks need to refer to the actual training representation.

Interpretation is conditional on the fitted predictor. Second-order blocks support transparent inspection, but correlated inputs and the coupling variables can produce

nonunique coefficient combinations. Resampling can assess stability of structural patterns. It cannot turn a predictive coefficient into a causal biological constant. Sensitivity analysis should identify the input unit, the operating point, and the projection stage to which it applies (Brunton et al., 2016; Machado et al., 2014).

A projected point prediction also lacks an automatic uncertainty guarantee. Model ensembles, probabilistic reconciliation, and hard-constrained probabilistic learning provide other approaches to uncertainty (D. Hansen et al., 2024; Lakshminarayanan et al., 2017; Utkarsh et al., 2025). Their existence does not make a deterministic projection a calibrated distribution. Uncertainty remains a separate extension requiring its own assumptions and validation.

7.6 Engineering interpretation of the planned assessment

The intended assessment can identify which prediction procedures are suitable for particular uses. A rapid unconstrained approximation may be useful for exploratory analysis if its limitations are visible. A projected component state can support screening that requires declared balances and non-negativity. An interpretable constrained procedure can support inspection of named response terms. A connected plant surrogate can support an operating search when selected controls are independently checked.

The assessment must retain these different uses when interpreting empirical evidence. A smaller error does not settle every engineering requirement. A stronger physical guarantee does not settle every accuracy question. A faster search does not settle every cost question. The methodological contribution lies in making each requirement explicit and connecting it to evidence that can assess it. Treatment outcomes and empirical performance remain questions for the subsequent evaluation.

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